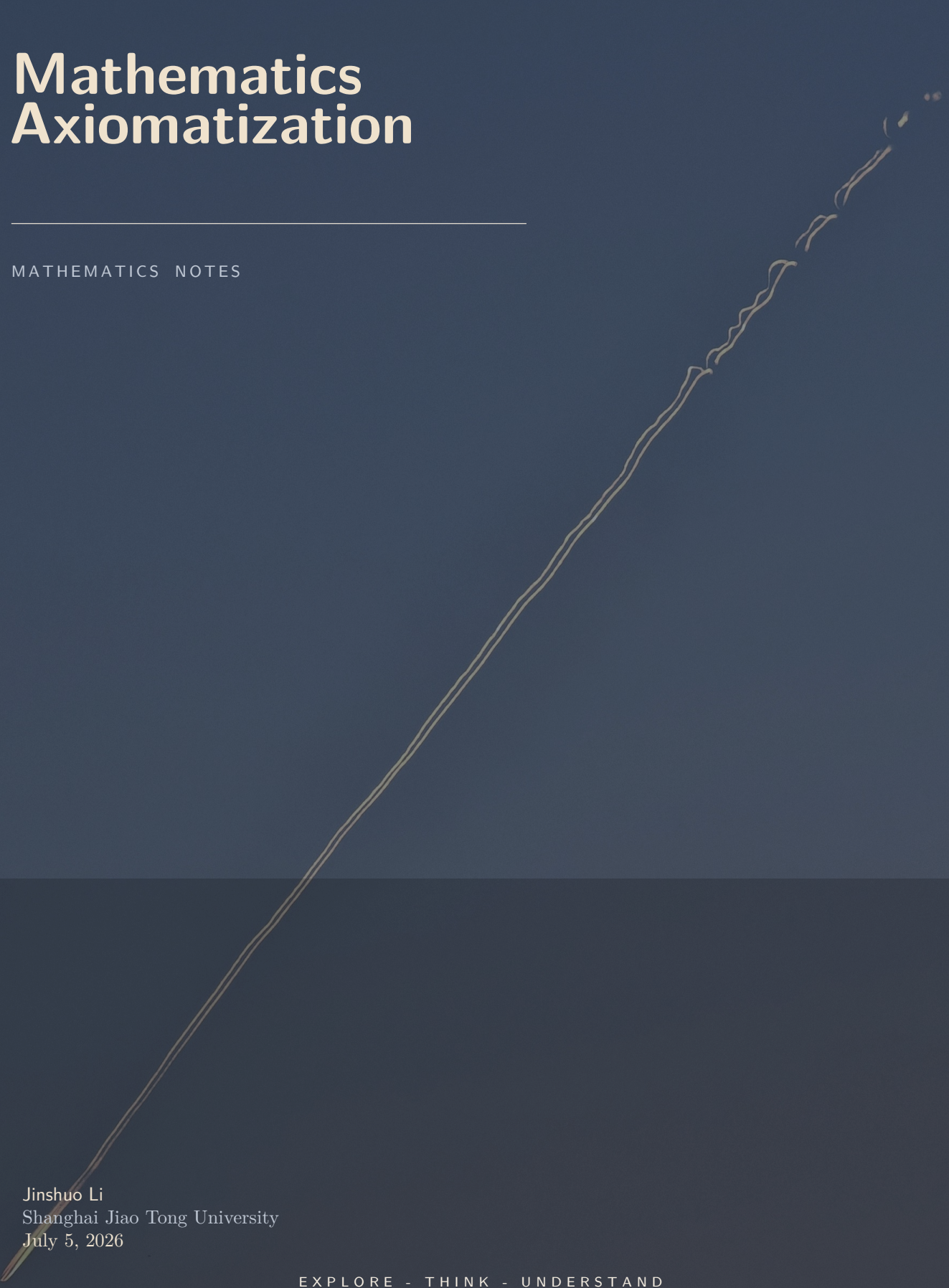


Mathematics Axiomatization



MATHEMATICS NOTES

Jinshuo Li
Shanghai Jiao Tong University
July 5, 2026

EXPLORE - THINK - UNDERSTAND

Contents

Preface	ii
How to Use This Book	iv
I Optional Foundations	1
Chapter 0: Fundamentals	2
0.1 Propositional Logic	2
0.1.1 Naïve introduction to propositional logic	2
0.1.2 Functional Completeness	5
0.2 Predicate Logic	7
0.3 Chapter Workshop	9
0.4 Exercises	9
1 Sets, Relations, and Axiomatic Foundations	10
1.1 The Naïve Set Theory	10
1.2 The Axiomatic Set Theory	12
1.2.1 The ZFC Axioms System	12
1.3 Extensions of Axiomatic Set Theory	13
1.3.1 Ordered Pairs and Cartesian Product	13
1.3.2 Relations and Their Special Types	14
1.3.3 A Brief Example: Measure	15
1.3.4 Another Example: Closure of Relation	19
1.3.5 Mapping and Function	19
1.3.6 Ordered Structure	21
1.4 Chapter Workshop	22
1.5 Exercises	22
II Core Undergraduate Mathematics	23
2 Mathematical Analysis I	24
2.1 Extension of the Number System	24
2.1.1 Peano Axioms and Natural Numbers	25
2.1.2 Integers and Rational Numbers	26
2.1.3 Real Numbers and Complex Numbers	27
2.2 Sequence Limit and The Properties of Real Numbers	30
2.2.1 Definitions and Basic Properties	31
2.2.2 Convergence Criteria and the Properties of the Real Number System	34
2.3 Derivatives and Related Theorem	37
2.3.1 Derivatives and Differentials	37

2.3.2	Mean Value Theorems and L'Hôpital's Rule	40
2.3.3	Taylor Expansion	44
2.4	Integration	47
2.4.1	Indefinite Integration	47
2.4.2	Definite Integration	53
2.5	Improper Integrals	59
2.5.1	Improper Integrals of the First Kind (Infinite Intervals)	59
2.5.2	Improper Integrals of the Second Kind (Unbounded Functions)	60
2.5.3	General Theory of Convergence	60
2.5.4	Convergence Tests	61
2.6	Chapter Workshop	65
2.7	Exercises	65
3	Linear Algebra	67
3.1	Linear Equations and Matrices	68
3.1.1	Systems of Linear Equations	68
3.1.2	Matrix Algebra	69
3.1.3	Solving Systems of Linear Equations	76
3.2	Determinants	81
3.2.1	The Determinant of a Matrix	81
3.2.2	Properties of Determinants	83
3.2.3	Cramer's Rule and Adjoint Formula	85
3.2.4	The Invertible Matrix Theorem (IMT)	86
3.3	Vectors in $\mathbb{R}^n$	86
3.3.1	Vectors and Operations	86
3.3.2	Dot Product, Norm, and Orthogonality	87
3.3.3	Linear Combinations and Span	88
3.3.4	The Matrix Equation $A\mathbf{x} = \mathbf{b}$	89
3.3.5	Linear Independence	90
3.4	Linear Transformations	91
3.4.1	Matrix Transformations	91
3.4.2	Linearity	92
3.4.3	Geometric Transformations in $\mathbb{R}^2$	93
3.5	Abstract Linear Spaces and Subspaces	93
3.5.1	The Formal Definition	93
3.5.2	Examples of Linear Spaces	94
3.5.3	Subspaces	94
3.5.4	Null Spaces and Column Spaces	95
3.5.5	Basis and Dimension	96
3.5.6	Eigenvalues and Eigenvectors	98
3.5.7	Diagonalization	99
3.5.8	Orthogonal Bases and the Gram-Schmidt Process	100
3.5.9	Symmetric Matrices and Quadratic Forms	100
3.6	Conclusions	102
3.6.1	Interpretations of Rank	102
3.6.2	The Rank-Nullity Theorem	102
3.6.3	A Unifying Viewpoint: The Vector-Space Axioms	104
3.7	Chapter Workshop	105
3.8	Exercises	105
4	Abstract Algebra	107
4.1	Operations and Algebraic Structures	107
4.2	Groups	108
4.2.1	Definition and Basic Examples	109

4.2.2	Elementary Consequences of the Axioms	110
4.2.3	Subgroups and Generated Subgroups	111
4.2.4	Cosets and Lagrange's Theorem	111
4.2.5	Normal Subgroups and Quotient Groups	112
4.2.6	Homomorphisms and Isomorphism Theorems	113
4.2.7	Group Actions	114
4.2.8	Sylow Theorems	115
4.3	Rings	116
4.3.1	Definition and Examples	116
4.3.2	Units, Zero Divisors, and Domains	117
4.3.3	Ideals and Quotient Rings	117
4.3.4	Ring Homomorphisms	118
4.3.5	Prime and Maximal Ideals	119
4.3.6	Polynomial Rings and Divisibility	119
4.4	Fields	120
4.4.1	Definitions and First Examples	120
4.4.2	Field Extensions	121
4.4.3	Algebraic Elements and Minimal Polynomials	121
4.4.4	Splitting Fields and Algebraic Closure	122
4.4.5	Finite Fields	122
4.5	Galois Theory	123
4.5.1	Field Automorphisms	123
4.5.2	Normality and Separability	123
4.5.3	The Fundamental Theorem	124
4.5.4	Solvability by Radicals	124
4.6	Modules	125
4.6.1	Definitions and Examples	125
4.6.2	Module Homomorphisms and Exact Sequences	125
4.6.3	Free Modules and Failure of Bases	126
4.6.4	Finitely Generated Modules over PIDs	126
4.7	Concrete Group Models and Classification Principles	127
4.7.1	Cyclic Groups	127
4.7.2	Permutations and Cycle Decomposition	128
4.7.3	Direct Products	129
4.8	Euclidean Domains, PIDs, and Unique Factorization	129
4.9	The Chinese Remainder Theorem and Interpolation	131
4.10	Irreducibility and Constructing Field Extensions	132
4.11	Conclusion	133
4.12	Chapter Workshop	133
4.13	Exercises	133
5	Mathematical Analysis II	135
5.1	Series	135
5.1.1	Numerical Series	135
5.1.2	Upper and Lower Limits	136
5.1.3	Function Series	144
5.1.4	Power Series	149
5.2	Limits and Continuity in Euclidean Space	153
5.2.1	The Structure of Euclidean Space $\mathbb{R}^n$	153
5.2.2	Basic Topology of $\mathbb{R}^n$	154
5.2.3	Limits of Functions of Several Variables	156
5.2.4	Continuity	157
5.2.5	Properties of Continuous Functions on Compact Sets	157
5.2.6	Connectedness and the Intermediate Value Theorem	158

5.3	Differential Calculus of Multivariable Functions	159
5.3.1	Partial Derivatives	159
5.3.2	Differentiability and the Total Differential	160
5.3.3	Vector-Valued Functions and Jacobian Matrices	162
5.3.4	The Gradient and Directional Derivatives	163
5.3.5	Chain Rules for Multivariable Functions	163
5.3.6	Taylor Formula for Functions of Two Variables	164
5.3.7	Implicit Functions	164
5.3.8	Geometric Applications	166
5.3.9	Extrema of Multivariable Functions	167
5.4	Multiple Integrals	168
5.4.1	Jordan Area and Double Integrals	168
5.4.2	Multiple Integrals	170
5.4.3	Basic Properties of Double Integrals	170
5.5	Evaluation of Multiple Integrals	171
5.5.1	Reduction of Double Integrals	171
5.5.2	Changing the Order of Integration	172
5.5.3	Change of Variables	172
5.5.4	Triple Integrals	173
5.5.5	Cylindrical and Spherical Coordinates	173
5.6	Line and Surface Integrals	174
5.6.1	Arc Length and First-Type Line Integrals	174
5.6.2	Second-Type Line Integrals	174
5.6.3	Path Independence and Potential Functions	174
5.6.4	Surface Area	175
5.6.5	First-Type Surface Integrals	175
5.6.6	Second-Type Surface Integrals	176
5.7	Green, Gauss, and Stokes Formulas	176
5.7.1	Green's Formula	176
5.7.2	Gauss' Formula	177
5.7.3	Stokes' Formula	177
5.7.4	Gradient, Divergence, Curl, and Conservative Fields	177
5.8	Parameterized Integrals	178
5.8.1	Proper Parameterized Integrals	178
5.8.2	Improper Parameterized Integrals	179
5.9	Euler Integrals	180
5.9.1	The Beta Function	180
5.9.2	The Gamma Function	180
5.9.3	Relation Between Beta and Gamma	181
5.10	Fourier Series	181
5.10.1	Trigonometric Orthogonality	181
5.10.2	Fourier Coefficients and Fourier Series	182
5.10.3	Complex Form	182
5.10.4	Bessel Inequality and the Least-Squares View	183
5.10.5	Riemann–Lebesgue Lemma	183
5.10.6	Dirichlet Kernel and Pointwise Convergence	183
5.10.7	What Fourier Series Really Adds	184
5.11	Conclusion	184
5.12	Chapter Workshop	184
5.13	Exercises	185

III Spaces and Complex Structure 186

6 Topology 187

6.1	From Metric Spaces to Topological Spaces	187
6.1.1	Open Sets in Metric Spaces	187
6.2	Topological Spaces	188
6.2.1	Definition of a Topology	188
6.2.2	Fundamental Examples	189
6.2.3	Comparing Topologies	190
6.3	Bases and Subbases	190
6.3.1	Basis for a Topology	191
6.3.2	Subbasis	191
6.3.3	Countability Axioms	191
6.4	Closed Sets, Closure, Interior, and Boundary	192
6.4.1	Closure and Interior	192
6.4.2	Limit Points and Density	193
6.5	Constructing New Spaces	193
6.5.1	Subspace Topology	193
6.5.2	Product Topology	194
6.5.3	Quotient Topology	194
6.6	Separation Axioms	195
6.7	Continuity and Homeomorphism	196
6.7.1	Continuous Functions	196
6.7.2	Homeomorphisms	196
6.8	Connectedness	197
6.8.1	Definitions	197
6.8.2	Continuous Images of Connected Spaces	197
6.8.3	Path Connectedness	198
6.9	Compactness	198
6.9.1	Open Covers	198
6.9.2	Compactness in $\mathbb{R}^n$	199
6.9.3	Continuous Images of Compact Spaces	199
6.10	Metric Spaces Revisited	200
6.11	Separation Axioms Beyond the Hausdorff Condition	200
6.12	Components, Local Connectedness, and the Topologist's Sine Curve	202
6.13	Compactness Beyond Closed and Bounded Sets	203
6.13.1	The Finite Intersection Property	204
6.13.2	Sequential Compactness in Metric Spaces	204
6.13.3	Finite Products and the First Form of Tychonoff's Theorem	205
6.14	Products and Quotients as Universal Constructions	206
6.15	Conclusion	207
6.16	Chapter Workshop	207
6.17	Exercises	207

7 Complex Analysis 209

7.1	Complex Numbers and the Complex Plane	209
7.1.1	Algebraic Definition	210
7.1.2	Geometric Interpretation	210
7.1.3	Euler's Formula	211
7.2	Topology of the Complex Plane	212
7.3	Complex Differentiability	212
7.3.1	Definition of Complex Derivative	213
7.3.2	Holomorphic Functions	213
7.4	Cauchy-Riemann Equations	214

7.4.1	Geometric Meaning	215
7.5	Elementary Holomorphic Functions	215
7.5.1	Exponential and Trigonometric Functions	215
7.5.2	Complex Logarithm	216
7.6	Complex Integration	216
7.6.1	Curves and Contour Integrals	217
7.6.2	Primitives	217
7.7	Cauchy's Theorem and Cauchy's Integral Formula	218
7.8	Power Series and Analyticity	219
7.8.1	Power Series	219
7.8.2	Holomorphic Implies Analytic	220
7.9	Zeros and Singularities	220
7.9.1	Zeros of Holomorphic Functions	220
7.9.2	Isolated Singularities	221
7.10	Laurent Series and Residues	221
7.10.1	Laurent Series	221
7.10.2	Residues	222
7.11	Major Theorems and Consequences	223
7.12	A More Complete Form of Cauchy's Theory	224
7.12.1	Goursat's Theorem	224
7.12.2	Winding Number and the General Integral Formula	225
7.13	Singularities: Criteria and Local Behavior	226
7.14	Residue Calculus for Real Integrals	227
7.14.1	Rational Functions on the Real Line	228
7.14.2	Trigonometric Integrals and the Unit Circle	228
7.15	The Argument Principle and Rouché's Theorem	229
7.16	Maximum Principles, Open Mapping, and Schwarz's Lemma	230
7.17	Geometric Visualizations	231
7.18	Conclusion	232
7.19	Chapter Workshop	233
7.20	Exercises	233

IV Discrete and Probabilistic Models 235

8	Applied Mathematics: Graph Theory	236
8.1	Introduction to Graph Theory	236
8.1.1	Basic Concepts and Classification	236
8.2	Paths and Circuits	242
8.2.1	Eulerian Circuits	242
8.2.2	Hamiltonian Cycles	243
8.3	Trees and Forests	244
8.3.1	Definitions and Characterization	244
8.3.2	Rooted Trees	245
8.3.3	Spanning Trees	245
8.4	Random Graphs	252
8.4.1	Basic Properties of Random Graphs	252
8.4.2	Important Theorems in Random Graph Theory	254
8.4.3	Algorithmic Generation of Random Graphs	254
8.4.4	Applications of Random Graphs	254
8.4.5	Conclusion and Future Directions	255
8.5	Chapter Workshop	255
8.6	Exercises	255

9 Applied Mathematics: Probability Theory	257
9.1 Classical Probability and Counting	257
9.1.1 Random Experiments and Sample Spaces	258
9.1.2 Classical Definition of Probability	258
9.1.3 Basic Counting Principles	258
9.2 Conditional Probability and Independence	259
9.2.1 Conditional Probability	259
9.2.2 Independence	260
9.2.3 Bayes' Theorem	261
9.3 Discrete Random Variables	261
9.3.1 Definition	262
9.3.2 Expectation and Variance in the Discrete Case	262
9.3.3 Common Discrete Distributions	263
9.4 Continuous Random Variables	264
9.4.1 Density Functions	264
9.4.2 Common Continuous Distributions	264
9.5 The Measure-Theoretic Foundation	265
9.5.1 Why Classical Probability Is Not Enough	265
9.5.2 Sigma-Algebras	265
9.5.3 Probability Spaces	266
9.6 Random Variables as Measurable Functions	267
9.6.1 Measurability	267
9.6.2 Expectation as Integral	267
9.7 Joint Distributions and Independence	268
9.7.1 Joint Distributions	268
9.7.2 Independence of Random Variables	268
9.7.3 Covariance and Correlation	269
9.8 Conditional Expectation	269
9.8.1 Conditioning on Events	269
9.8.2 Conditioning on Random Variables	270
9.9 Inequalities in Probability	270
9.10 Generating Functions and Characteristic Functions	271
9.10.1 Probability Generating Functions	271
9.10.2 Moment Generating Functions	272
9.10.3 Characteristic Functions	272
9.11 Modes of Convergence	272
9.12 Limit Theorems	273
9.12.1 Law of Large Numbers	274
9.12.2 Central Limit Theorem	276
9.13 Borel-Cantelli Lemmas	278
9.14 Introduction to Stochastic Processes	279
9.15 Distribution Functions and Transformations	280
9.15.1 Functions of One Random Variable	281
9.16 Joint Densities, Convolution, and Conditioning	281
9.17 Convergence Theorems for Expectations	283
9.18 Conditional Expectation as Information	284
9.19 Martingales and Stopping Times	285
9.20 Markov Chains	286
9.21 Poisson Processes and Brownian Motion	287
9.21.1 The Poisson Process	287
9.21.2 Brownian Motion	288
9.22 Conclusion	289
9.23 Chapter Workshop	289
9.24 Exercises	290

Preface

Mathematics begins with intuition, but it becomes reliable only when its assumptions and arguments are made explicit. This manuscript grew from an attempt to understand that transition: how familiar calculations are reorganized into definitions, theorems, proofs, algorithms, and models that can be reused in unfamiliar settings.

The book is best described as a *guided second-pass reference*. It is written primarily for university students who have already encountered at least some calculus and linear algebra and now want to connect those courses with abstract algebra, topology, complex analysis, graph theory, and probability. It is not intended to replace a first textbook in every subject. Instead, it supplies a coherent map, substantial proofs of central results, worked examples, applications, and exercises that help a reader move between computational skill and structural understanding.

The opening chapters on logic and set theory are optional foundations, not prerequisites for every reader. A student whose immediate goal is calculus may begin with Mathematical Analysis I. A reader interested in computation and data may begin with Linear Algebra, then move to Graph Theory and Probability. The foundational chapters remain available for readers who want to examine how mathematical language and constructions are formalized, but the architecture of the book no longer requires every reader to pass through ZFC before doing analysis.

The level of detail varies because the subjects serve different roles. Analysis and linear algebra receive extended development because they provide techniques used throughout the rest of the book. Later chapters are selective introductions, but they are not intended to be theorem catalogues. Each chapter now states learning objectives, develops representative worked examples, connects the theory to scientific or computational practice, and ends with exercises and selected guidance.

Proofs are included whenever the mechanism of the argument is central to the subject. In particular, limit arguments, compactness arguments, quotient constructions, convergence theorems, and probabilistic truncation methods are written out rather than hidden behind the phrase “standard proof.” Results whose complete proofs require machinery far beyond the scope of the chapter are explicitly identified as such. This distinction is important: a concise proof is not the same as a missing proof, and an advanced theorem should not be presented as elementary merely because its statement is short.

Applications are used to clarify structure rather than to decorate it. Linear algebra is connected to least squares, data compression, and principal component analysis; abstract algebra to coding and cryptography; multivariable analysis to optimization and physical fields; topology to continuity of data representations and quotient constructions; complex analysis to signal processing and potential theory; graph theory to routing, networks, and graph-based learning; and probability to statistical inference, simulation, and stochastic processes. These examples are deliberately modest: their purpose is to show how the mathematical language enters a real model, not to replace a specialized applied course.

The exposition distinguishes personal interpretation from mathematical convention. Phrases such as “the axiom of linear algebra” are replaced by more precise claims: the vector-space axioms are one especially powerful organizing viewpoint, but computational, geometric, and operator-theoretic approaches are equally legitimate. Readers are encouraged to compare formulations and to ask which viewpoint is most effective for a given problem.

A productive way to use this book is to read with pencil and paper. Pause after a definition and construct examples and non-examples. Before reading a proof, identify the hypotheses and predict where each one will be used. Work the chapter exercises without looking immediately at the hints. For computational topics, verify a small case numerically. Mathematics becomes durable when an argument can be reconstructed rather than merely recognized.

This remains a developing manuscript. Its purpose is not to present one final route through undergraduate mathematics, but to provide a precise and usable route whose choices are visible to the reader.

Jinshuo Li
Shanghai Jiao Tong University
2026.7.3, in Shanghai

How to Use This Book

What kind of book is this?

This is neither an encyclopedic textbook nor a private formula sheet. It is a structured set of undergraduate mathematics notes designed for two uses:

1. a **guided review**, when the reader has seen a subject before but wants clearer connections, proofs, and examples;
2. a **reference map**, when the reader needs to locate a definition, theorem, technique, or application and understand how it fits into a larger theory.

A complete first course still benefits from lectures, problem classes, and a specialized textbook. The exercises in this manuscript are intended to make reading active, not to imitate an entire semester of assessed work.

Recommended routes

Standard undergraduate route

Begin with Chapter 2, Mathematical Analysis I, and Chapter 3, Linear Algebra. Continue to Chapters 4–7 according to interest. Treat Chapters 0–1 as optional reference material.

Foundations route

Read Chapters 0–1 before Chapter 2. This route is appropriate when the construction of number systems, formal logic, and axiomatic set theory are themselves part of the objective.

Computation and data route

Read Chapter 3, then Chapters 8 and 9. Return to Chapters 2 and 5 for optimization, approximation, integration, and Fourier analysis.

Structure route

Read Chapters 3, 4, 6, and 7. This route emphasizes vector spaces, algebraic structures, topological invariants, and holomorphic structure.

Dependency map

Most chapters are locally self-contained, but the following dependencies are useful:

Analysis II	←	Analysis I,
Abstract Algebra	←	basic Linear Algebra,
Topology	←	sets, functions, and elementary analysis,
Complex Analysis	←	Analysis I and basic topology,
Probability	←	calculus; measure theory sections are optional on first reading.

Graph Theory can be read almost independently.

Proof and scope conventions

A theorem followed by a **proof** is intended to contain a complete argument at the level of the chapter. A result described as an **advanced theorem** is included for orientation when a full proof would require substantial additional machinery. No omitted argument should be mistaken for an elementary exercise.

Definitions are followed, whenever useful, by examples, non-examples, or a computational interpretation. Application boxes show how a concept enters a model in science, engineering, or data analysis. They are not claims that the application is exhausted by the mathematics shown there.

How to work the exercises

The exercises are arranged roughly from direct verification to synthesis. A good minimum routine is:

1. reproduce one proof from memory;
2. solve two computational exercises;
3. solve one conceptual or counterexample exercise;
4. explain one application in your own words.

Selected hints and answers are intentionally brief. They indicate a route or a check, not a substitute for a written solution.

Note

The foundational material is available when it clarifies a later construction, but it should not become an entrance fee for learning calculus, linear algebra, graph theory, or probability. Start where your mathematical question begins, and move backward only when a dependency genuinely matters.

Part I

Optional Foundations

Chapter 0: Fundamentals

Learning objectives

By the end of this chapter, you should be able to:

- translate ordinary mathematical statements into propositional and predicate logic;
- construct truth tables and test logical equivalence;
- distinguish implication, converse, inverse, and contrapositive;
- read quantified statements and write their negations correctly;
- recognize the logical patterns used in direct proof, contradiction, and induction.

Note

This chapter is optional. Readers already comfortable with logical connectives, quantifiers, and standard proof patterns may proceed directly to Chapter 2 and return here as needed.

All mathematical insights undergo a process from “naïve understanding” to “rigorous formulation”. Mathematics itself is the same; all axiomatized languages cannot be separated from the empiricist’s naïve descriptive language.

To introduce certain special mathematical symbols that will be frequently used in the future, I have specially added a Chapter Zero before Chapter One, intended to present parts of the knowledge concerning mathematical language and mathematical logic. In this chapter, I will also introduce knowledge related to axioms; however, in the future, we will not necessarily define axioms so rigorously every time.

0.1 Propositional Logic

0.1.1 Naïve introduction to propositional logic

Before we start the main part of the propositional logic, let’s start with a question: what is a proposition? Take a look at the sentences below:

1. The equation $x^2 + 1 = 0$ has a real-number solution;
2. There are infinitely many prime numbers;
3. Every even integer $n \geq 4$ is a sum of two prime numbers (The famous Goldbach Conjecture);
4. What time is it?
5. This statement is false;

In the sentences above, the first 3 sentences share the same properties: 1) they are sentences or assertions that declare facts. 2) They are either true or false, but not both. However, the 4th sentence is not a

declarative sentence, and we can't judge the correctness of the 5th sentence.

So, to study mathematics, we need the sentences that assert one or several facts. Sometimes, whether it is true or not is not that important once it can be determined. It doesn't mean that other types of expressions are not vital. We value this kind of expression because all the axioms, definitions, and theorems are written in this form.

Definition 0.1.1 (Proposition). Propositions are sentences or assertions that declare facts and are either true or false, but not both.

To make propositions more actionable, we use **propositional variables** to represent different propositions. Commonly used propositional variables are letters like $p, q, r, \dots$

Sometimes we need to connect different propositional variables to form a new proposition. And we need to use words like “and”, “or”, “imply” to explain their relationships.

Definition 0.1.2 (Logical operators/connectives). Logical operators/connectives are marks that are used to connect propositional variables, forming compound propositions.

Here are some common operators:

$\neg$ negation (“not”)

$\wedge$ conjunction (“and”)

$\vee$ disjunction (“or”)

$\rightarrow$ (**or** $\Rightarrow$) conditional (“imply”)

$\leftrightarrow$ biconditional (“if and only if” (“iff”))

$\oplus$ exclusive-or

Definition 0.1.3 (Compound proposition). A compound proposition is built from propositional variables or constants through logical operators.

Definition 0.1.4 (Atomic proposition). An atomic proposition is a proposition that can't be divided any further. It is the most basic building block of logical expressions and is considered an indivisible semantic unit.

If we want to determine whether an atomic proposition is true or not, we need to use correlated knowledge of different fields in mathematics. But how can we determine the truth of a compound proposition? We can use truth tables!

Definition 0.1.5 (Truth table). A truth table is a mathematical table used in logic to compute the functional values of logical expressions based on their inputs.

Negation

p	$\neg p$
T	F
F	T

Conjunction

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Conditional (ImPLY)

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Biconditional (iff)

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Exclusive-or

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Just like numerical computation, logical operators also have precedence. Here is the order of precedence for logical operators (from highest to lowest):

$$() \quad [] \quad \neg \quad \wedge \quad \vee \quad \oplus \quad \rightarrow \quad \leftrightarrow$$

Apart from this, operators that appear first are of higher precedence. In short, we compute higher-precedence logical operators first, then lower ones.

Using these laws, we can define the truth value of different propositions. But there exist some sub-classes of compound propositions:

Tautology A tautology is a compound proposition that is always true, no matter what the truth values of the propositional variables are.

Contradiction A compound proposition that is always false, regardless of the truth values of propositional variables.

Contingency A compound proposition that is neither a tautology nor a contradiction. It can be true or false, depending on the value of the variables.

Based on the definitions above, now we can rigorously define what logical equivalence is.

Definition 0.1.6 (Logical Equivalence). We can say compound propositions p, q are **logically equivalent** if $p \leftrightarrow q$ is a tautology. If p and q are logically equivalent, we denote it as $p \equiv q$.

This definition means that for different truth values of every atomic proposition, p and q come out with the same truth value. In further study, we will know that several of the operators above will be enough to express all the propositions.

Here are a few frequently used examples. These can be used directly.

Identity Laws:	$p \wedge T \equiv p$ $p \vee F \equiv p$
Domination Laws:	$p \vee T \equiv T$ $p \wedge F \equiv F$
Absorption Laws:	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$
Negation Laws:	$p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$
Commutative Laws:	$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$
Associative Laws:	$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
Distributive Laws:	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
De Morgan's Laws:	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$

Definition 0.1.7 (Satisfiability). A compound proposition is **satisfiable** if it is true under some truth assignment for propositional variables. A truth assignment that makes a compound proposition true is called a **solution**. If a compound proposition is not satisfiable, we say it is **unsatisfiable**.

For now, we can use a truth table and logical equivalence to check whether a compound proposition is satisfiable. For readers majoring in computer science, they can use special algorithms to solve the so-called “Boolean Satisfiability Problem”, like heuristic searching, etc.

0.1.2 Functional Completeness

According to previous study, we know that there are connectives for different variables to form compound propositions. However, do we really need that many connectives to represent all the relationships and compound propositions? The answer is absolutely NO! In some cases, we only need two connectives to reach the so-called state of functional completeness!

Definition 0.1.8 (Disjunctive Normal Form). A compound proposition is in **disjunctive normal form (DNF)** if it is a disjunction of conjunctions of propositional variables or their negations.

Theorem 0.1.1

Every compound proposition is logically equivalent to some compound proposition in disjunctive normal form.

Proof. It's not difficult to prove it. According to previous content, we know that we can make a truth table for every compound proposition. According to the truth table, we can rewrite it in the form of disjunctive normal form. For example, we have the truth table of the exclusive-or connective:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

The content of the middle two lines (where the result is T) represents the solution to this compound proposition. Then we can write a proposition like this:

$$p \oplus q \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$$

Every part of the compound proposition connected with the disjunction represents a solution of the satisfiable problem of the proposition. That means if we can make a truth table for all the compound propositions, we can rewrite them in the form of disjunctive normal form. From this perspective, the theorem is quite clear. $\square$

Let's move further: can we rewrite the compound proposition in other forms? The answer is obviously YES!

Definition 0.1.9 (Conjunctive Normal Form). A compound proposition is in **conjunctive normal form (CNF)** if it is a conjunction of disjunctions of propositional variables or their negations.

Theorem 0.1.2

Every compound proposition is logically equivalent to some compound proposition in conjunctive normal form.

Proof. We will use the truth table again to prove this theorem. Assume we have a proposition A . Firstly, we use Theorem 0.1.1 to rewrite $\neg A$ into the form of disjunctive normal form. Next, we take the negation of $\neg A$, and using De Morgan's Law, we can now rewrite A in the form of conjunctive normal form. $\square$

Theorem 0.1.1 and Theorem 0.1.2 are quite a marvel. In circuit design, since we cannot design a circuit component for every logical connector, we must use as few circuit components as possible to express all logical expressions. And according to these theorems, we no longer need to use that many connectives. Only conjunction, disjunction, and negation are necessary to construct logically equivalent propositions to any propositions in the world.

Definition 0.1.10 (Functionally Complete). A collection of logical operators is **functionally complete** if any compound proposition is logically equivalent to a compound proposition that involves only the logical operators in the collection.

Thus, we can declare that the collection $\{\neg, \vee, \wedge\}$ is functionally complete. Also, according to De Morgan's Law:

$$\begin{aligned} p \wedge q &\equiv \neg(\neg p \vee \neg q) \\ p \vee q &\equiv \neg(\neg p \wedge \neg q) \end{aligned}$$

We found that conjunction can be expressed using disjunction and negation, and disjunction can be expressed using conjunction and negation. Which means that the collections $\{\neg, \vee\}$ and $\{\neg, \wedge\}$ are functionally complete.

Remark 0.1.1

The most striking fact is that there exists a connective that is functionally complete in itself (e.g., NAND or NOR)! But we won't use it in the future, so you can search by yourselves.

0.2 Predicate Logic

In many cases, using propositional logic is enough to cover the usage. But there is a kind of statement that can't be expressed using the tool of propositional logic. In such cases, we need to use predicate logic to formulate logical expressions.

Definition 0.2.1 (Predicate). A **predicate** is in the form of a statement with variables. A predicate is also known as a **propositional function**. A predicate can be denoted as $P(x, y, z, \dots)$, and x, y, z are known as the variables.

Definition 0.2.2 (Quantifiers). We use **quantifiers** to indicate the quantity of elements being referred to. We have two types of quantifiers, $\forall$ and $\exists$.

$\forall$ The **universal quantifier** $\forall$ means “for all”.

$\exists$ The **existential quantifier** $\exists$ means “there exists”.

For example, the statement “Every SJTUer is a talent” can be written in the form of predicate logic: $\forall x(\text{SJTUer}(x) \rightarrow \text{Talent}(x))$.

Variables in predicates all have restricted domains; they are the **domains** of predicates. For all the values in the domain, we can rewrite the predicates in the form of propositions. If the domain is finite, $\{x_1, x_2, \dots, x_n\}$, then:

$$\forall x P(x) \equiv P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$$

$$\exists x P(x) \equiv P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$$

But if the domain is infinite, we can only express the logic in the form of predicate logic.

- $\forall x P(x)$ stands for “For every x in domain, $P(x)$ ”. If there is a value a in the domain such that $P(a)$ is false, the statement is false. a is called a **counterexample**.
- $\exists x P(x)$ stands for “There exists an x , $P(x)$ ”. If for all values a in the domain, $P(a)$ is false, the statement is false.

However, not all sentences with predicates and quantifiers can form a predicate logic. For example, $\forall x(P(x) \wedge Q(y))$ is not a statement with predicate logic. Because as the value of y varies, we can't decide whether the sentence is true or false, which makes its truth value unknown.

So, how can we make sure a sentence is a predicate logic? In the case of $\forall x(P(x) \wedge Q(y))$, it is y that causes trouble. And x , which is connected with a quantifier, did not cause any inconvenience. So, we call variables that are bounded only if they follow a quantifier, and they are called the **bounded variables**. Otherwise, they are the **free variables**. If every variable in a sentence is bounded, it's a proposition.

Just like propositional logic, predicate logic can also be connected using logical connectives. And there are also a few arithmetic laws.

De Morgan's Laws for Quantifiers

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

Asymmetric Associative Laws

$$\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$$

$$\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$$

Note

The associative laws are asymmetric. The following two assertions are not necessarily valid. In some cases, they can be false.

$$\forall x \exists y P(x, y) \neq \exists y \forall x P(x, y)$$

Definition 0.2.3 (Logical Equivalence in Predicate Logic). If ϕ and ψ are statements with predicates and quantifiers, but without free variables, then we say that ϕ and ψ are **logically equivalent**, written as $\phi \equiv \psi$, if, no matter which concrete predicates (for the predicate symbols) and domains (for variables) are given, the truth values of ϕ, ψ coincide.

Also, we can define universal validity (just like tautology in propositional logic), and predicate logic also has satisfiability. As this chapter only provides a brief introduction, it will not be elaborated further here.

In this chapter, we came to know what logic is in the form of mathematics. This overview of propositional and predicate logic establishes the formal basis for clear reasoning. Logic is the indispensable scaffolding of rational thought—the *a priori framework* that precedes all specific knowledge. Its importance doesn't need much explanation.

Keywords: propositional logic, predicate logic, logical equivalence

Reference: Discrete Mathematics and Its Applications (Eighth Edition), Kenneth H. Rosen, McGraw-Hill Education.

0.3 Chapter Workshop

Worked example: Negating a quantified statement

Consider the statement

$$\forall x \in \mathbb{R} \exists n \in \mathbb{N} \quad n > x.$$

Its negation is obtained by reversing each quantifier and negating the final predicate:

$$\exists x \in \mathbb{R} \forall n \in \mathbb{N} \quad n \leq x.$$

The original statement is the Archimedean property; its negation asserts the existence of a real upper bound for all natural numbers. Notice that merely inserting the word “not” at the beginning does not produce a usable mathematical formula.

0.4 Exercises

1. Construct a truth table for $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$.
2. Negate: “For every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for every $n \geq N$, $|a_n - L| < \varepsilon$.”
3. Determine whether $(p \vee q) \wedge \neg p$ logically implies q .
4. Write the converse and contrapositive of: “If an integer is divisible by 4, then it is even.” State which forms are true.

Selected hints and answers

1. The two columns agree in all four rows; this is the logical equivalence underlying proof by contrapositive.
2. The negation is: there exists $\varepsilon > 0$ such that for every N there exists $n \geq N$ with $|a_n - L| \geq \varepsilon$.
3. Yes. The first conjunct says at least one of p, q is true, while the second rules out p .
4. Converse: even $\Rightarrow$ divisible by 4 (false). Contrapositive: not even $\Rightarrow$ not divisible by 4 (true).

Chapter 1

Sets, Relations, and Axiomatic Foundations

Learning objectives

By the end of this chapter, you should be able to:

- work fluently with sets, Cartesian products, relations, functions, and orders;
- distinguish equivalence relations from partial and total orders;
- construct quotient sets and interpret fibers, images, and inverse images;
- explain why naive set formation leads to paradoxes and what role ZFC plays;
- decide when axiomatic set theory is relevant and when ordinary set notation is sufficient.

Note

The ZFC material is a foundations module, not a prerequisite for ordinary calculus or linear algebra. Readers interested primarily in those subjects may read the sections on sets, functions, and relations, then continue to Chapter 2.

Set theory supplies a common language for modern mathematics, but it is not the only possible starting point and it should not be confused with the working content of every subject. In most undergraduate calculations, naive set notation is sufficient. Axiomatic set theory becomes important when we ask which collections are legitimate sets, how constructions are justified, or why unrestricted set formation leads to paradoxes. This chapter therefore separates practical set language from the optional study of its formal foundations.

1.1 The Naïve Set Theory

Actually, if you are a student majoring in engineering or applied mathematics, learning naïve set theory is enough for you to cover the math you need during your work and study. But we need to know that the naïve theory is basically an intuitive definition, which is not included as a part of the modern axiomatic set theory. Although we say that the naïve set theory is a really clear and useful definition of the concept “set”, we have to declare that the naïve set theory is incomplete because naïve set theory itself cannot resolve Russell’s paradox. We will introduce the paradox that causes the third mathematical crisis to readers later.

Definition 1.1.1 (Set). A **set** is an unordered collection of distinct objects, called **elements** or **members** of the set. A set is said to *contain* its elements. We write $a \in A$ to denote that a is an element of the set A . The notation $a \notin A$ means a is not an element of the set.

The concept of set is quite simple. A set can be the container of anything, like numbers, points, other sets, even an apple can be a set. The only requirement is that the elements in a set must be distinct. For example, the set $\{1, 2, 2, 3\}$ is actually equal to the set $\{1, 2, 3\}$ because the element 2 is repeated.

Note

An object can only be in one of the two states: 'belonging to A ' or 'not belonging to A '; it cannot be in both states at the same time, nor can it be in neither state.

Definition 1.1.2 (Subset). We say that the set A is a **subset** of B , written $A \subseteq B$, if every element of A is an element of B . Additionally, A is a **proper subset** of B , written $A \subset B$, if $A \subseteq B$ and $A \neq B$.

The concept of subset is quite straightforward. If all the elements in set A can be found in set B , then A is a subset of B . For example, $\{1, 2\} \subseteq \{1, 2, 3\}$, and $\{1, 2\} \subset \{1, 2, 3\}$. However, in later study, we will find that the concept of subset creates contradictions in naïve set theory, which is the root cause of Russell's paradox.

Definition 1.1.3 (Equality). Two sets A and B are **equal** (i.e., they are the same set), written $A = B$, if they contain the same members.

Set equality can be determined using the definition of subset. If $A \subseteq B$ and $B \subseteq A$, then $A = B$. This is the formal definition of set equality.

Here are several common set operations, shown below:

Definition 1.1.4 (Set Union). The **union** of sets A and B is denoted as $A \cup B$, which contains all the elements of A together with the elements of B .

Definition 1.1.5 (Set Intersection). The **intersection** of sets A and B is denoted as $A \cap B$, which contains the elements that are the members of both A and B .

Definition 1.1.6 (Set difference). The **difference** of the set A w.r.t B , written as $A - B$ (or $A \setminus B$), is the set consisting of those elements of A that are not in B .

Definition 1.1.7 (Empty set). The **empty set**, denoted $\emptyset$, is the set that contains no elements.

Note that the empty set is a subset of every set. And there is only one empty set in the whole universe of sets. Which means that if two sets have no elements, they are equal.

Definition 1.1.8 (Power set). The **power set** $\mathcal{P}(x)$ (aka 2^x): the set consisting of all subsets of x .

According to the definition above, we can easily find several conclusions below:

Corollary 1.1.1

1. $\emptyset \subseteq A$ for any set A .
2. Assume that the number of elements in the finite set x is $\text{card}(x)$, then $\text{card}(\mathcal{P}(x)) = 2^{\text{card}(x)}$.
3. If the set x is empty, then $\text{card}(\mathcal{P}(\emptyset)) = 1$. (This inference can be seen as an extension of the inference 1.1.2).

Russell's Paradox

Russell's Paradox makes the naïve set theory incomplete. Russell assumed there exists a set X that contains sets that don't include themselves.

$$X = \{x \mid x \notin x\}$$

And then he found X doesn't belong to set X nor does it belong to set X .

$$X \in X \implies X \notin X$$

$$X \notin X \implies X \in X$$

So, does X belong to X or not? Thus, this creates a contradiction, because X either belongs to X or does not belong to X , which is determined by the nature of naïve set theory itself. However, the properties of the naïve set theory allow the existence of set X , which creates a contradiction.

In order to mend such a flaw present in naive set theory, countless mathematicians devoted themselves tirelessly and proposed the ZFC axiomatic system, which ultimately became the core of modern set theory.

1.2 The Axiomatic Set Theory

1.2.1 The ZFC Axioms System

ZFC stands for **Zermelo-Fraenkel Set theory with the axiom of choice**, which includes 9 different axioms (numbered from ZF1 to ZF8 and AC). We will now introduce them to the readers one by one.

Principles:

- Either $a \in A$ or $a \notin A$ but not both.
- A formal language is required for constructing meaningful statements.
- Every object is a set, and every set is an object.

ZF1 (Axiom of Extensionality) If X and Y have the same elements, then $X = Y$.

$$\forall X \forall Y (\forall u (u \in X \leftrightarrow u \in Y) \rightarrow X = Y)$$

(ZF1 defines the “=” in set theory)

ZF2 (Axiom of the Unordered Pair) For any a and b , there exists a set $\{a, b\}$ that contains exactly a and b . (Also called Axiom of Pairing)

$$\forall a \forall b \exists Z \forall u (u \in Z \leftrightarrow (u = a \vee u = b))$$

(ZF2 constructs the unordered pairs and allows the existence of ordered pairs.)

ZF3 (Axiom of Subsets) Assume ϕ is a property with parameter p , then for any X and p , there exists a Set $Y = \{u \in X \mid \phi(u, p)\}$ that contains all those $u \in X$ that have the property ϕ . (also called Axiom of Separation or Axiom of Comprehension)

$$\forall X \forall p \exists Y \forall u (u \in Y \leftrightarrow (u \in X \wedge \phi(u, p)))$$

(ZF3 is the key axiom that prevents the situation Russell’s paradox described. Assume $X = \{x \in C \mid x \notin x\}$. According to the axiom of subsets, X must be constructed from an existing set C . This form of definition automatically excluded X from being a “set of all sets”.)

ZF4 (Axiom of the Sum Set) For any X , there exists a set $Y = \bigcup X$, the union of all elements of X . (Also called Axiom of Union)

$$\forall X \exists Y \forall u (u \in Y \leftrightarrow \exists Z (Z \in X \wedge u \in Z))$$

(ZF4 allows the construction of a union set, allowing mathematicians to construct bigger sets and form more complex mathematical structures.)

ZF5 (Axiom of the Power Set) For any X , there exists a set $Y = \mathcal{P}(X)$, the set of all subsets of X .

$$\forall X \exists Y \forall u (u \in Y \leftrightarrow u \subseteq X)$$

(ZF5 defines the concept of a power set.)

ZF6 (Axiom of Infinity) There exists an infinite set.

$$\exists S(\emptyset \in S \wedge \forall x(x \in S \rightarrow x \cup \{x\} \in S))$$

(ZF6 allows the existence of infinite sets, which is the basis of the set of $\mathbb{N}$.)

ZF7 (Axiom of Replacement) If F is a function, then for any X , there exists a set $Y = F[X] = \{F(x) \mid x \in X\}$.

$$\forall X[(\forall x \in X \exists! y \phi(x, y)) \rightarrow \exists Y \forall y(y \in Y \leftrightarrow \exists x \in X \phi(x, y))]$$

(ZF7 allows mathematicians to construct new sets using existing sets and a function. It can imply ZF3.)

ZF8 (Axiom of Foundation) Every non-empty set A contains a member that is disjoint from A .

$$\forall A(A \neq \emptyset \rightarrow \exists x(x \in A \wedge x \cap A = \emptyset))$$

(ZF8 avoids infinite nesting of sets ($A \in A$) and is a powerful aid to ZF3 when refuting Russell's paradox. If $A \in A$, construct $B = \{A\}$. Then A is the only element in B . By ZF8, $A \cap B = \emptyset$. But $A \in B$ and $A \in A$, so $A \in A \cap B$, which means $A \cap B \neq \emptyset$. This is a contradiction.)

AC (Axiom of Choice) Every family of nonempty sets has a choice function.

$$\forall \mathcal{F}[(\emptyset \notin \mathcal{F}) \rightarrow \exists f : \mathcal{F} \rightarrow \bigcup \mathcal{F}(\forall A \in \mathcal{F}(f(A) \in A))]$$

(AC is fundamental as it guarantees the ability to make infinitely many simultaneous, non-constructive choices, which is indispensable for proving a vast number of crucial theorems across diverse fields of mathematics.)

Although Gödel's incompleteness theorems tell us that the ZFC axiom system cannot be proven to be consistent, most mathematicians believe that ZFC is consistent because it has not yet produced any fundamental contradiction that threatens the integrity of mathematics. Therefore, it can be regarded as a reliable foundation for modern mathematics.

Of course, other axiomatic systems exist, like the NBG (von Neumann–Bernays–Gödel) axiomatic system. But for practical purposes, ZFC is enough. From the ZFC axiomatic system, we know that no set includes everything.

1.3 Extensions of Axiomatic Set Theory

1.3.1 Ordered Pairs and Cartesian Product

1.3.1.1 Ordered Pairs

Now, with the help of ZFC axiomatic set theory, we can define ordered pairs:

Definition 1.3.1 (Ordered Pair). We define the ordered pair (a, b) as:

$$(a, b) := \{\{a\}, \{a, b\}\}$$

$\{a\} \in \{\{a\}, \{a, b\}\}$ guarantees the order of the pair (a, b) .

Corollary 1.3.1

1. $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.
2. $(a, a) = \{\{a\}, \{a, a\}\} = \{\{a\}\}$.

1.3.1.2 Cartesian Product

Definition 1.3.2 (Cartesian Product). The Cartesian Product of two sets A and B is defined as:

$$A \times B := \{(a, b) \in 2^{2^{x \cup y}} \mid a \in A \wedge b \in B\}$$

Furthermore, we can define the n -ary Cartesian product as below:

$$X_1 \times X_2 \times \cdots \times X_n := \{(x_1, x_2, \dots, x_n) \mid x_i \in X_i \text{ for } i = 1, \dots, n\}$$

Cartesian Product is widely used in Mathematical Analysis, Analytic Geometry, and Group Theory, etc.

1.3.2 Relations and Their Special Types

1.3.2.1 Relations

Now, based on the concept of Cartesian Product, we can define Relations.

Definition 1.3.3 (Relation). Suppose we are given two sets X and Y . A **relation** R from X to Y is a subset of the Cartesian Product $X \times Y$.

$$R \subseteq X \times Y$$

If $X = Y$, we can say that R is a relation on A . We write xRy for $(x, y) \in R$.

For all relations, there are descriptions unique to themselves, which we denote as $P(x, y)$. Then a relation can be described as $R = \{(x, y) \in X \times Y \mid P(x, y)\}$.

Remark 1.3.1

Mark that the Y is the range of the relation, and X is the domain of the relation. This fact may contradict our common sense of the relation.

$$\text{dom}R := \{x \in \cup \cup R \mid \exists y(x, y) \in R\}$$

$$\text{ran}R := \{y \in \cup \cup R \mid \exists x(x, y) \in R\}$$

1.3.2.2 Some special types of relations

Here are a few properties that can be used to classify different relations on a set A .

Reflexive R is reflexive if and only if $\forall a \in A((a, a) \in R)$.

Antireflexive (or Irreflexive) R is antireflexive if and only if $\forall a \in A((a, a) \notin R)$.

Symmetric R is symmetric if and only if $\forall a, b \in A((a, b) \in R \rightarrow (b, a) \in R)$.

Antisymmetric R is antisymmetric if and only if $\forall a, b \in A[(a, b) \in R \wedge (b, a) \in R \rightarrow a = b]$.

Transitive R is transitive if and only if $\forall a, b, c \in A[(a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R]$.

Using these properties, we can define a very important concept in mathematics: the equivalence relation.

Definition 1.3.4 (Equivalence Relation). Suppose R is a relation on A . If R simultaneously possesses reflexivity, symmetry, and transitivity, then R is an **equivalence relation** on A .

We can use an equivalence relation to classify a set.

Definition 1.3.5 (Equivalence Class). We define the **equivalence class** $[a]$ of an element $a \in A$ by:

$$[a] = \{x \in A \mid xRa\}$$

We can completely classify a set using equivalence classes. For example, in SJTU, we can classify all students by their nationality. This relation satisfies reflexivity, symmetry, and transitivity, so it's an equivalence relation.

Similarly, we can define order relations.

Definition 1.3.6 (Partial Order). Suppose R is a relation on A . If R simultaneously possesses reflexivity, antisymmetry, and transitivity, then R is a **partial order relation** on A . A set A with a partial order R is a **partially ordered set (poset)**, denoted (A, R) or $(A, \preceq)$.

Definition 1.3.7 (Total/Linear Order). Suppose R is a partial order relation on A . If for all $a, b \in A$, either aRb or bRa is true, then R is a **total order relation** on A .

Definition 1.3.8 (Strict Partial Order). Suppose R is a relation on A . If R simultaneously possesses antireflexivity and transitivity, then R is a **strict partial order relation** on A . (Note: antireflexivity and transitivity imply asymmetry).

We can also define inverse and composite relations.

Mind that the domain and range of the inverse relation are swapped compared to the original relation.

Definition 1.3.9 (Inverse Relation). We define the **inverse relation** $R^{-1} \subseteq Y \times X$:

$$R^{-1} = \{(y, x) \mid (x, y) \in R\}$$

In more formal language:

$$R^{-1} = \{(x, y) \in \text{ran}R \times \text{dom}R \mid yRx\}$$

Definition 1.3.10 (Composite Relation). Let $R \subseteq X \times Y$ and $S \subseteq Y \times Z$. We define the **composite relation** $S \circ R \subseteq X \times Z$ by:

$$S \circ R = \{(x, z) \mid \exists y \in Y((x, y) \in R \wedge (y, z) \in S)\}$$

Theorem 1.3.1

1. $(R^{-1})^{-1} = R$
2. $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$
3. $(R \circ (S \cup T)) = (R \circ S) \cup (R \circ T)$

However, mind that $R \circ (S \cap T) \neq (R \circ S) \cap (R \circ T)$. Here is a counterexample: We define $U = \{a, b, c, d\}$, $R = \{(a, b), (a, c)\}$, $S = \{(b, d)\}$, $T = \{(c, d)\}$. We know that $R \circ (S \cap T) = \emptyset$ because $S \cap T = \emptyset$. But $(R \circ S) \cap (R \circ T) = \{(a, d)\}$, which is a counterexample.

Using the concept of equivalence relation, we can “divide” a set precisely.

Definition 1.3.11 (Partition). Assume A is a nonempty set. A collection P of non-empty subsets of A ($P \subseteq \mathcal{P}(A)$) is a **partition** on A if:

1. No set in P is empty: $\forall S \in P (S \neq \emptyset)$.
2. The union of sets in P is A : $\bigcup_{S \in P} S = A$.
3. The sets in P are pairwise disjoint: $\forall S_1, S_2 \in P (S_1 \neq S_2 \rightarrow S_1 \cap S_2 = \emptyset)$.

Theorem 1.3.2

Every partition of a set A corresponds to an equivalence relation on A , and vice versa.

1.3.3 A Brief Example: Measure

Here comes a question: can we assign a non-negative value representing the “length” of a subset of $\mathbb{R}$?

Definition 1.3.12 (Measure). A **measure** μ on a collection $\mathcal{M}$ (of subsets of a set X) is a function $\mu : \mathcal{M} \rightarrow [0, \infty]$ such that for any pairwise-disjoint countable-infinite sequence of sets $\{A_i\}_{i=1}^{\infty}$ in $\mathcal{M}$ (i.e., $A_i \cap A_j = \emptyset$ whenever $i \neq j$), if their union is also in $\mathcal{M}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

This key property is called **countable additivity**.

Does there exist a measure m on $\mathcal{P}(\mathbb{R})$ (i.e., all subsets of the real numbers) that satisfies the following conditions (our naïve understanding of "length"):

1. $m([0, 1]) = 1$.
2. Translation invariance: $m(A + x) = m(A)$ for any $A \subseteq \mathbb{R}$ and $x \in \mathbb{R}$, where $A + x = \{a + x \mid a \in A\}$. Which means that the same measure apply to all the set with the same structure on the axis.
3. Countable additivity.

Unfortunately, the answer is no. We construct the **Vitali Set**, which can't be measured following these conditions.

Proof of the existence of a non-measurable set (Vitali Set). Define an equivalence relation $\sim$ on $[0, 1]$ by $x \sim y \iff x - y \in \mathbb{Q}$. This is an equivalence relation. Let V be the set of its equivalence classes. According to the Axiom of Choice, we can choose exactly one element from each equivalence class, altogether to form a set of representatives M . We can assume that $M \subseteq [0, 1]$. Let $\mathbb{Q}^* = \mathbb{Q} \cap [-1, 1]$. For each $q \in \mathbb{Q}^*$, define $M_q = M + q = \{m + q \mid m \in M\}$. We have the following conclusions:

1. All M_q (for $q \in \mathbb{Q}^*$) are pairwise disjoint. (If $y \in M_q \cap M_r$, then $y = m_1 + q = m_2 + r$. This means $m_1 - m_2 = r - q \in \mathbb{Q}$, so $m_1 \sim m_2$. By construction of M , this implies $m_1 = m_2$, so $q = r$.)
2. $[0, 1] \subseteq \bigcup_{q \in \mathbb{Q}^*} M_q$. (For any $x \in [0, 1]$, let $m \in M$ be the representative x is equivalent to, so $x - m = q \in \mathbb{Q}$. Since $x, m \in [0, 1]$, $q \in [-1, 1]$. Thus $x = m + q \in M_q$.)
3. $\bigcup_{q \in \mathbb{Q}^*} M_q \subseteq [-1, 2]$. (Since $M \subseteq [0, 1]$ and $\mathbb{Q}^* \subseteq [-1, 1]$.)

Now, let's try to measure M . Assume $m(M) = c$. By translation invariance, $m(M_q) = m(M) = c$ for all $q \in \mathbb{Q}^*$. By countable additivity (since $\mathbb{Q}^*$ is countable):

$$m\left(\bigcup_{q \in \mathbb{Q}^*} M_q\right) = \sum_{q \in \mathbb{Q}^*} m(M_q) = \sum_{q \in \mathbb{Q}^*} c$$

From our inclusions:

$$\begin{aligned} m([0, 1]) &\leq m\left(\bigcup_{q \in \mathbb{Q}^*} M_q\right) \leq m([-1, 2]) \\ &1 \leq \sum_{q \in \mathbb{Q}^*} c \leq 3 \end{aligned}$$

If $c = 0$, then $1 \leq 0$, a contradiction. If $c > 0$, then $1 \leq \infty$, which is not a contradiction, but $\sum c \leq 3$ implies $\infty \leq 3$, a contradiction. Thus, M cannot be assigned a measure $m(M)$, and is **non-measurable**. $\square$

Additional Knowledge: The Definition of Lebesgue Measure

Remark 1.3.2

Note: This section provides supplementary material on the definition of the Lebesgue measure. It presents both the original constructive approach and the modern, abstract definition. This content is provided for a deeper historical and theoretical context and can be considered optional for the first reading.

1. Lebesgue's Original Idea & Construction

The original idea, as developed by Henri Lebesgue, is a constructive process for defining the measure of a set $E \subset \mathbb{R}^n$. It starts with simple sets (intervals) and then approximates more complex sets from the outside.

Definition 1.3.13 (Lebesgue Outer Measure and Measurability). The **Lebesgue outer measure** $m^*(E)$ of any set $E \subset \mathbb{R}^n$ is defined by covering E with a **countable** collection of n -dimensional intervals (or cubes) and taking the infimum of the total volume of such coverings.

$$m^*(E) = \inf \left\{ \sum_{k=1}^{\infty} \ell(I_k) : E \subset \bigcup_{k=1}^{\infty} I_k \right\}$$

where $\{I_k\}$ is a countable collection of n -dimensional intervals, and $\ell(I_k)$ is the product of the lengths of its sides (its volume).

A set E is called **Lebesgue measurable** if for every $\epsilon > 0$, there exists an open set $O \supset E$ such that the outer measure of the difference $m^*(O \setminus E) < \epsilon$. This is Carathéodory's criterion, a later improvement that perfectly captures the idea that a measurable set can be "approximated closely" by open sets.

For a measurable set E , its Lebesgue measure $m(E)$ is simply defined as its outer measure: $m(E) = m^*(E)$.

2. Modern (Improved) Definition via σ -Algebras

The modern approach, which is more abstract and powerful, defines the Lebesgue measure as the completion of a measure defined on a specific σ -algebra. This is the standard definition found in most modern textbooks on measure theory.

Definition 1.3.14 (Lebesgue Measure via σ -Algebra). Let $\mathcal{B}(\mathbb{R}^n)$ be the Borel σ -algebra on $\mathbb{R}^n$. The **Lebesgue σ -algebra**, denoted $\mathcal{L}$, is the completion of $\mathcal{B}(\mathbb{R}^n)$ with respect to the Lebesgue measure.

The **Lebesgue measure** is the unique measure

$$m : \mathcal{L} \rightarrow [0, \infty]$$

satisfying the following properties:

1. **Translation Invariance:** For any $A \in \mathcal{L}$ and $x \in \mathbb{R}^n$, $m(A + x) = m(A)$.
2. **Normalization:** The measure of the unit cube is 1: $m([0, 1]^n) = 1$.
3. **Countable Additivity:** For any countable collection $\{E_i\}_{i=1}^{\infty}$ of pairwise disjoint Lebesgue measurable sets, $m(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} m(E_i)$.

Equivalently, it is the unique extension of the pre-measure defined on the algebra of elementary sets (finite unions of intervals) to the full Lebesgue σ -algebra, via Carathéodory's extension theorem.

3. The Vitali Set: An Example of a Non-Measurable Set

An important consequence of the properties of the Lebesgue measure is the existence of sets that are not Lebesgue measurable. The most famous example is the **Vitali set**, constructed by Giuseppe Vitali in 1905.

Definition 1.3.15 (Construction of the Vitali Set). Consider the interval $[0, 1] \subset \mathbb{R}$. Define an equivalence relation $\sim$ on $[0, 1]$ by:

$$x \sim y \iff x - y \in \mathbb{Q}.$$

This partitions $[0, 1]$ into equivalence classes. Using the Axiom of Choice, we select exactly one element from each equivalence class to form a set $V \subset [0, 1]$. This set V is called a **Vitali set**.

Theorem 1.3.3: The Vitali Set is Not Lebesgue Measurable

The Vitali set V is not Lebesgue measurable.

Proof. The proof proceeds by contradiction. Assume V is Lebesgue measurable.

Consider the rational numbers in $[-1, 1]$, denoted $\mathbb{Q} \cap [-1, 1]$. For each $q \in \mathbb{Q} \cap [-1, 1]$, define the translation:

$$V_q = V + q = \{v + q : v \in V\}.$$

These sets V_q are pairwise disjoint. If $v_1 + q_1 = v_2 + q_2$ for $v_1, v_2 \in V$ and $q_1, q_2 \in \mathbb{Q}$, then $v_1 - v_2 = q_2 - q_1 \in \mathbb{Q}$, which implies $v_1 \sim v_2$. Since V contains exactly one element from each equivalence class, $v_1 = v_2$ and thus $q_1 = q_2$.

By translation invariance of the Lebesgue measure, if V is measurable, then each V_q is measurable and $m(V_q) = m(V)$.

Now, observe that:

$$[0, 1] \subset \bigcup_{q \in \mathbb{Q} \cap [-1, 1]} V_q \subset [-1, 2].$$

If V is measurable with $m(V) = 0$, then by countable additivity:

$$m\left(\bigcup_q V_q\right) = \sum_q m(V_q) = 0,$$

which contradicts $m([0, 1]) = 1$.

If $m(V) > 0$, then:

$$m\left(\bigcup_q V_q\right) = \sum_q m(V_q) = \infty,$$

which contradicts $m([-1, 2]) = 3$.

Therefore, our assumption that V is measurable must be false. The Vitali set V is not Lebesgue measurable. We can't find any countable intervals to cover the Vitali set. $\square$

Remark 1.3.3

The existence of non-measurable sets like the Vitali set demonstrates that the Lebesgue measure cannot be extended to all subsets of $\mathbb{R}^n$ while preserving translation invariance and countable additivity. This result relies on the Axiom of Choice, and indeed, it can be shown that in models of set theory without the Axiom of Choice, all subsets of $\mathbb{R}$ can be Lebesgue measurable.

1.3.4 Another Example: Closure of Relation

Having defined the fundamental properties of relations, we now address a natural question: if a relation R on a set A lacks a certain property, what is the *smallest* relation containing R that *does* possess that property? This leads to the concept of the **closure** of a relation.

Definition 1.3.16 (Closure). Let P be a property of relations (such as reflexivity, symmetry, or transitivity). The P -**closure** of a relation R on a set A is the smallest relation S on A such that:

1. $R \subseteq S$
2. S has the property P

There are three important types of closure:

Reflexive Closure $R' = R \cup I_A$, where $I_A = \{(a, a) \mid a \in A\}$ is the identity relation on A .

Symmetric Closure $R' = R \cup R^{-1}$.

Transitive Closure $R^* = \bigcup_{n=1}^{\infty} R^n$, where $R^1 = R$ and $R^{n+1} = R^n \circ R$.

Proof that R^ is the Transitive Closure.* We must show R^* is transitive and is the smallest such relation containing R .

1. **(Transitivity)** Let $(x, y) \in R^*$ and $(y, z) \in R^*$. By definition, $(x, y) \in R^m$ for some $m \geq 1$ and $(y, z) \in R^n$ for some $n \geq 1$. By definition of composition, this implies $(x, z) \in R^{m+n}$. Since $m+n \geq 1$, $(x, z) \in \bigcup_{k=1}^{\infty} R^k = R^*$. Thus R^* is transitive.
2. **(Minimality)** Let T be any transitive relation such that $R \subseteq T$. We must show $R^* \subseteq T$. We know $R^1 = R \subseteq T$. Assume $R^k \subseteq T$ (inductive hypothesis). Let $(a, c) \in R^{k+1} = R^k \circ R$. Then $\exists b$ such that $(a, b) \in R^k$ and $(b, c) \in R$. By the inductive hypothesis, $(a, b) \in T$. Since $R \subseteq T$, $(b, c) \in T$. Because T is transitive, $(a, b) \in T \wedge (b, c) \in T \implies (a, c) \in T$. Thus $R^{k+1} \subseteq T$. By induction, $R^n \subseteq T$ for all $n \geq 1$. Therefore, $R^* = \bigcup_{n=1}^{\infty} R^n \subseteq T$. This shows R^* is the smallest transitive relation containing R .

□

1.3.5 Mapping and Function

In mathematical expressions, we use the language of set theory to define what a map is.

Definition 1.3.17 (Map / Function). A **map** (or **function**) f from A to B , denoted $f : A \rightarrow B$, is a relation $f \subseteq A \times B$ such that for every $x \in A$, there is a *unique* object $y \in B$ such that $(x, y) \in f$. We write this unique y as $f(x)$.

- The set A is called the **domain** of f .
- The set B is called the **codomain** of f .
- The set $\{f(x) \mid x \in A\} \subseteq B$ is called the **range** or **image** of f .

The uniqueness requirement is: $\forall x \in A \forall y_1, y_2 \in B [(x, y_1) \in f \wedge (x, y_2) \in f \rightarrow y_1 = y_2]$.

Definition 1.3.18 (Injection). A function $f : A \rightarrow B$ is an **injection** (or one-to-one) if and only if:

$$\forall x_1, x_2 \in A (f(x_1) = f(x_2) \rightarrow x_1 = x_2)$$

This means that each element in the domain corresponds to a unique element in the codomain.

Definition 1.3.19 (Surjection). A function $f : A \rightarrow B$ is a **surjection** (or onto) if and only if:

$$\forall y \in B \exists x \in A (f(x) = y)$$

This means that the codomain is equal to the range.

Definition 1.3.20 (Bijection). A function $f : A \rightarrow B$ is a **bijection** if and only if it is both an injection and a surjection.

Definition 1.3.21 (Inverse Function). Let $f : A \rightarrow B$ be a function. A function $g : B \rightarrow A$ is called the **inverse function** (or inverse map) of f if and only if it satisfies the following two conditions:

1. $g \circ f = id_A$ (where id_A is the identity map on A)
2. $f \circ g = id_B$ (where id_B is the identity map on B)

The inverse map of f , if it exists, is unique and denoted by f^{-1} . A function has an inverse if and only if it is a bijection.

Definition 1.3.22 (Function Composition). Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be two functions. The **composition** of g and f is a new function denoted $g \circ f : A \rightarrow C$ defined as:

$$(g \circ f)(x) = g(f(x)) \quad \text{for all } x \in A$$

1.3.5.1 Function (Special Types)

Definition 1.3.23 (Real Function). If the domain $X \subseteq \mathbb{R}$ and the codomain $Y \subseteq \mathbb{R}$, the mapping is called a **real function of one variable**, denoted $y = f(x)$.

Piecewise Function A piecewise function is a function that is defined by multiple sub-functions, each of which applies to a certain interval or region of the main function's domain.

$$f(x) = \begin{cases} f_1(x) & \text{if condition 1} \\ f_2(x) & \text{if condition 2} \\ \vdots & \vdots \end{cases}$$

Implicit Function An implicit function is a function that is defined by an equation relating its variables, e.g., $F(x, y) = 0$, rather than by an explicit formula $y = f(x)$.

Parametric Function A parametric function describes a curve by expressing the coordinates of the points on the curve as functions of a third variable, called a parameter, t .

$$\begin{cases} x = f(t) \\ y = g(t) \end{cases} \quad \text{for } t \text{ in some interval } I$$

Definition 1.3.24 (Basic Elementary Functions). Basic elementary functions are a finite set of fundamental functions:

1. Constant Functions: $f(x) = c$.
2. Power Functions: $f(x) = x^\alpha$.
3. Exponential Functions: $f(x) = a^x$ (where $a > 0, a \neq 1$).
4. Logarithmic Functions: $f(x) = \log_a x$.
5. Trigonometric Functions: $\sin x, \cos x, \tan x$, etc.
6. Inverse Trigonometric Functions: $\arcsin x, \arccos x$, etc.

Definition 1.3.25 (Elementary Function). An **elementary function** is any function that can be obtained from the basic elementary functions by performing a finite number of the following operations:

- Arithmetic Operations: Addition, subtraction, multiplication, division.

- Composition: The operation of function composition.

Definition 1.3.26 (Operation). An **operation** is a function from a set to itself. More specifically, an **n-ary operation** ω on a set X is a function $\omega : X^n \rightarrow X$.

1.3.6 Ordered Structure

1.3.6.1 Well-order

In previous parts, we introduced several ordered relations. Now we will define the concept and property of well-order. This is the last part of chapter 1, and well-order can be seen as a bridge to the next chapter.

Definition 1.3.27 (Well-ordered Set). A totally ordered set $(W, \leq)$ is called a **well-ordered set** if it satisfies that every non-empty subset has a least element.

$$\forall S \subseteq W (S \neq \emptyset \rightarrow \exists s \in S \forall x \in S (s \leq x))$$

Theorem 1.3.4

The set of natural numbers $\mathbb{N}$ under the usual order $\leq$ is well-ordered. (This is the **Well-Ordering Principle**).

Proof. Consider an arbitrary non-empty subset S of $\mathbb{N}$. Start checking each natural number from 0 onwards. The first number that belongs to S is the least element. This process must terminate in a finite number of steps, because if no such element were ever found, it would imply that S is empty, contradicting the assumption. Thus, $\mathbb{N}$ under the usual order is a well-ordered set. $\square$

It seems that there is a very close connection between the well-ordering principle and the set of natural numbers. And this principle of “starting from the least element” is seemingly very symmetrical to Mathematical Induction (MI). We will reveal their relationships in the next chapter.

Keywords: Set, Axiom, ZFC Axiomatic Set theory, Ordered Pairs, Cartesian Product, Relations, Well-ordered Sets.

Reference: Discrete Mathematics and Its Applications (Eighth Edition), Kenneth H. Rosen, McGraw-Hill Education.

1.4 Chapter Workshop

Worked example: From an equivalence relation to a quotient set

On $\mathbb{Z}$ define $a \sim b$ when $a - b$ is divisible by 3. Reflexivity follows from $a - a = 0$; symmetry from $3 \mid (a - b) \Rightarrow 3 \mid (b - a)$; and transitivity from addition. The quotient set has three equivalence classes:

$$\mathbb{Z}/\sim = \{[0], [1], [2]\}.$$

Addition of classes is well defined because replacing a and b by congruent representatives does not change the class of $a + b$. This elementary construction is the model for quotient groups, quotient rings, and many identification spaces in topology.

Application: Relations in databases

A database table may be viewed as a finite relation, while a key constraint asserts that a coordinate projection is injective on the stored rows. Joins are built from Cartesian products followed by selection conditions. This viewpoint explains why set operations and maps are not merely foundational notation: they are the algebra of structured data.

1.5 Exercises

1. Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.
2. Let $f : X \rightarrow Y$. Prove $f^{-1}(U \cap V) = f^{-1}(U) \cap f^{-1}(V)$, and give an example showing that $f(A \cap B) = f(A) \cap f(B)$ can fail.
3. On $\mathbb{R}$ define $x \sim y$ if $x - y \in \mathbb{Z}$. Describe the quotient classes geometrically.
4. Decide whether divisibility is a partial order on the positive integers.
5. Explain in one paragraph why Russell's paradox challenges unrestricted set formation but does not prevent ordinary finite set calculations.

Selected hints and answers

1. Use element chasing: x belongs to the left side exactly when $x \in A$, $x \notin B$, and $x \notin C$.
2. Inverse images preserve all unions, intersections, and complements. For the failure of direct images, use a non-injective map.
3. Each class is $x + \mathbb{Z}$; choosing representatives in $[0, 1)$ identifies the quotient with a circle after the endpoints are glued.
4. Yes: divisibility is reflexive, antisymmetric, and transitive on positive integers, but it is not a total order.
5. The paradox concerns allowing every predicate to define a set. ZFC restricts set formation; finite sets used in elementary work can be constructed without the problematic unrestricted principle.

Part II

Core Undergraduate Mathematics

Chapter 2

Mathematical Analysis I

Learning objectives

By the end of this chapter, you should be able to:

- explain how completeness distinguishes the real numbers from the rationals;
- prove and compute limits of sequences and functions;
- use derivatives, mean-value theorems, Taylor expansion, and L'Hopital's rule with their hypotheses visible;
- construct and evaluate Riemann and improper integrals;
- select convergence tests and justify limiting operations rigorously.

Note

For a first pass focused on calculus, the construction of the number systems may be skimmed. The operational core begins with sequence limits, while the earlier sections explain why the real-number framework supports those arguments.

We now embark on a great undertaking: building a rigorous framework for calculus. Our journey through logic and set theory has provided the tools; the ordered structures of Chapter 1 have revealed a critical insight—the rational numbers, though dense, are incomplete. They possess gaps, like the legendary irrational $\sqrt{2}$, which defy representation as a ratio.

This chapter confronts that insufficiency. We will construct the real number system, a complete, continuous tapestry woven to fill these voids. Its cornerstone is the Completeness Axiom, which guarantees that bounded sets have precise bounds, a property the rationals fatally lack.

Upon this unshakable foundation, we will erect the central pillars of analysis: the precise theory of limits, the formal definition of continuity, and the powerful machinery of the derivative and integral in the next chapter. This is the transition from intuitive calculation to profound understanding—from calculus to Analysis.

2.1 Extension of the Number System

If asked about the foundation of mathematical analysis, the answer is clear: the axiom of real numbers. In this part, we will start from Peano Axioms to form the axioms of natural numbers. And for practical purposes, we extend it to integers and rational numbers. Then, to support theorems of calculus, we construct the axioms of real numbers and complex numbers.

Before we officially start this part, we need to clarify three simple principles:

Motivation Principle (Solving a Limitation)

Each expansion is driven by the need to perform an operation that is not always possible within the smaller number system. The primary goal is to achieve **closure** under this new operation.

Embedding Principle (Preserving the Original Structure)

The smaller, original number system must be isomorphic to a subsystem of the new, larger number system. This is achieved by constructing an **injective embedding** that preserves all the essential operations (like addition and multiplication) and properties of the original system.

Minimality Principle (The "Smallest" Extension)

The new number system should be the "smallest" or "most economical" extension that satisfies the first two principles. It should introduce *only* the elements necessary to solve the limitation, without any superfluous structure.

2.1.1 Peano Axioms and Natural Numbers

Peano axioms define natural numbers using 5 axioms:

Definition 2.1.1 (Peano's Axioms). A set $\mathbb{N}$ is called the **natural number set** if it satisfies the following properties (with a "successor" function $S : \mathbb{N} \rightarrow \mathbb{N}$):

1. $0 \in \mathbb{N}$. (Zero is a natural number.)
2. $\forall a \in \mathbb{N}(S(a) \in \mathbb{N})$. (If a is a natural number, the successor of a is a natural number.)
3. $\forall a \in \mathbb{N}(S(a) \neq 0)$. (Zero is not the successor of any natural number.)
4. $\forall a, b \in \mathbb{N}(S(a) = S(b) \rightarrow a = b)$. (Two numbers whose successors are equal are themselves equal.)
5. $\forall K \subseteq \mathbb{N}[(0 \in K \wedge \forall n(n \in K \rightarrow S(n) \in K)) \rightarrow K = \mathbb{N}]$. (If a set S of numbers contains zero and also the successor of every number in S , then every number is in S . This is the **Axiom of Induction**.)

We denote the successor of a as $a + 1$.

The definition of the natural numbers has a very strong relationship with the concept of mathematical induction and the well-ordered set. In fact, they are logically equivalent (assuming the other axioms).

- Peano axioms ensure the validity of mathematical induction (Axiom 5 is MI).
- Peano axioms ensure the validity of the well-ordering principle.

Here, we will prove the logical equivalence of the well-ordering principle (WOP) and mathematical induction (MI).

MI implies Well-Ordering Principle. To prove: every non-empty subset of natural numbers has a least element. Assume, for contradiction, that there exists a non-empty subset $A \subseteq \mathbb{N}$ that has *no* least element. Define another set $B = \{n \in \mathbb{N} \mid n < a \text{ for all } a \in A\}$. (B is the set of all numbers strictly smaller than everything in A).

Base Case: $0 \in B$? If $0 \notin B$, then there exists some $a \in A$ such that $0 \geq a$. Since $a \in \mathbb{N}$, this implies $a = 0$. So $0 \in A$. Since 0 is the smallest natural number, it would be the least element of A , contradicting the assumption that A has no least element. Thus, $0 \in B$.

Inductive Step: Assume $k \in B$ and prove $S(k) \in B$. Inductive Hypothesis: $k \in B$, meaning $k < a$ for all $a \in A$. If $S(k) \notin B$, then there exists some $a \in A$ such that $S(k) \geq a$. From the hypothesis, $k < a$. Combined with $S(k) \geq a$, and since numbers are discrete, the only possibility is $a = S(k)$. So $S(k) \in A$.

Furthermore, since all numbers smaller than $S(k)$ (like k) are in B (and thus not in A), $S(k)$ would be the least element of A . This contradicts the assumption that A has no least element. Therefore, $S(k) \in B$.

Conclusion: By the principle of Mathematical Induction (Axiom 5), $B = \mathbb{N}$. Since A is non-empty, take any element $a \in A$. Because $B = \mathbb{N}$, $a \in B$. By the definition of B , this means $a < a$, which is a contradiction. The initial assumption that A has no least element must be false. Therefore, every non-empty subset of $\mathbb{N}$ has a least element. $\square$

Well-Ordering Principle implies MI. (This part is left for the readers to practice.) $\square$

Cardinality

The concept “amount” is clear for finite sets. But for an infinite set, how can we measure the “size” of the set?

Definition 2.1.2 (Cardinality / Equinumerosity). Cardinality is an intrinsic property of sets. Two sets A and B are said to be **equinumerous** or have the same **cardinality**, denoted $A \approx B$ or $|A| = |B|$, if there exists a bijection $f : A \rightarrow B$.

Definition 2.1.3. For two sets A and B , if we can find an injection $f : A \rightarrow B$ but no bijection, we said the set A is strictly smaller than set B , denoted $A \prec B$ or $|A| < |B|$. If there is an injection $f : A \rightarrow B$, we write $A \preceq B$ or $|A| \leq |B|$.

Theorem 2.1.1: Schröder-Bernstein Theorem

If $A \preceq B$ and $B \preceq A$, then $A \approx B$.

Definition 2.1.4 (Countable Set). A set is **countable** if either it is finite or it can be made in one-to-one correspondence with the set of natural numbers $\mathbb{N}$. If a set is countably infinite, we define its cardinality as $\aleph_0$ (aleph-nought).

Theorem 2.1.2: Cantor's Theorem

For every non-empty set A , $A \prec \mathcal{P}(A)$. (That is, $|A| < |\mathcal{P}(A)|$).

(This implies there is no “largest” infinity.)

In section 2.3, we will know that the cardinality of the real number set $\mathbb{R}$ is $|\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = 2^{\aleph_0}$, which is called the **Continuum**.

The Continuum Hypothesis (CH) Some students might be interested whether there exists any size of a set between $\aleph_0$ and $2^{\aleph_0}$. The Continuum Hypothesis (CH) is a famous conjecture that proposes there is no set S such that $\aleph_0 < |S| < 2^{\aleph_0}$.

So can we determine whether it's true or false? It was proven (by Kurt Gödel and Paul Cohen) that CH is “undecidable” within the standard ZFC foundation of mathematics. This means it can neither be proven true nor false using the accepted axioms, revealing a fundamental limitation of that system.

2.1.2 Integers and Rational Numbers

Why do we need integers and rational numbers? Why are natural numbers not enough?

Definition 2.1.5 (Closure of operation). Closure of operation refers to the property that when an operation is performed on members of a set, the result is always a member of the same set.

In the set of natural numbers $\mathbb{N}$, operations like addition and multiplication are closed. But for subtraction ($3 - 5 = ?$) and division ($3/5 = ?$), $\mathbb{N}$ itself is not enough. Thus, we need to extend the number system.

Construction of Integers ($\mathbb{Z}$)

Definition 2.1.6. The relation R on $\mathbb{N} \times \mathbb{N}$ is defined as:

$$(a, b)R(c, d) \iff a + d = b + c$$

(This is the formal way of saying $a - b = c - d$).

Theorem 2.1.3

The relation R is an equivalence relation on $\mathbb{N} \times \mathbb{N}$.

Proof. (Left for readers to practice: check reflexivity, symmetry, transitivity.) $\square$

Definition 2.1.7 (Integers $\mathbb{Z}$). The **integers**, denoted as $\mathbb{Z}$, is the set of equivalence classes of $\mathbb{N} \times \mathbb{N}$ w.r.t the equivalence relation R . We denote the equivalence class $[(a, b)]$ as $a - b$.

We can now define operations on $\mathbb{Z}$:

Definition 2.1.8 (Operations on $\mathbb{Z}$). • **Addition:** $[(a, b)] + [(c, d)] = [(a + c, b + d)]$

• **Negation:** $-[(a, b)] = [(b, a)]$

• **Subtraction:** $[(a, b)] - [(c, d)] = [(a, b)] + (-[(c, d)]) = [(a, b)] + [(d, c)] = [(a + d, b + c)]$

Because $a, b, c, d \in \mathbb{N}$, and $\mathbb{N}$ is closed under addition, $a + d$ and $b + c$ also belong to $\mathbb{N}$. This means that subtraction is closed for integers. The construction of $\mathbb{Z}$ from $\mathbb{N}$ via ordered pairs is a perfect example of the principles of number system expansion.

Construction of Rational Numbers ($\mathbb{Q}$) Though $\mathbb{Z}$ is closed under subtraction, it is not closed under division. That's why we still need to extend from integers to rational numbers.

Definition 2.1.9. Let $\mathbb{Z}^* = \mathbb{Z} - \{0\}$. The relation R on $\mathbb{Z} \times \mathbb{Z}^*$ is defined by:

$$(a, b)R(c, d) \iff ad = bc$$

(This is the formal way of saying $a/b = c/d$).

It's clear that this relation R is an equivalence relation.

Definition 2.1.10 (Rational Numbers $\mathbb{Q}$). The set of **rational numbers**, denoted $\mathbb{Q}$, is the set of equivalence classes of $\mathbb{Z} \times \mathbb{Z}^*$ w.r.t the equivalence relation R . We denote the class $[(a, b)]$ as a/b .

Density of Rational Numbers There is a very different property of rational numbers compared to integers. The rational number set is a **dense order set**. We can't always find an intermediate number between two integers (e.g., 1 and 2), but we can always find a rational intermediate value between two rational numbers.

Proof. For two rational numbers a and b with $a < b$, their average $m = (a + b)/2$ is also a rational number, and $a < m < b$. Thus, we found an intermediate number between them. $\square$

This property (the density of $\mathbb{Q}$ in $\mathbb{R}$) is essential for constructing and understanding the real number system. In topology, $\mathbb{Q}$ is a countable dense subset of $\mathbb{R}$, making $\mathbb{R}$ a separable space.

2.1.3 Real Numbers and Complex Numbers

The rational numbers $\mathbb{Q}$, while dense, are not *complete*. They contain "gaps". For example, the set $A = \{x \in \mathbb{Q} \mid x^2 < 2\}$ has upper bounds in $\mathbb{Q}$ (e.g., 1.5), but it has no *least* upper bound *within* $\mathbb{Q}$. The "number" $\sqrt{2}$ is missing. We construct the real numbers $\mathbb{R}$ to fill these gaps.

2.1.3.1 Construction of Real Numbers by Dedekind Cuts

Definition 2.1.11 (Dedekind Cut). A **Dedekind cut** is a pair (A, B) of subsets of $\mathbb{Q}$ satisfying:

1. A and B are non-empty and form a partition of $\mathbb{Q}$ (i.e., $A \cup B = \mathbb{Q}$ and $A \cap B = \emptyset$).
2. Every element of A is less than every element of B .
3. A has no greatest element.

The set A is called the **lower class** and B the **upper class**. A real number is defined as a Dedekind cut.

For example, the real number $\sqrt{2}$ is represented by the cut where:

- $A = \{x \in \mathbb{Q} \mid x < 0 \text{ or } x^2 < 2\}$
- $B = \{x \in \mathbb{Q} \mid x > 0 \text{ and } x^2 > 2\}$

Definition 2.1.12 (Order on Real Numbers). For two real numbers $\alpha = (A_1, B_1)$ and $\beta = (A_2, B_2)$, we define $\alpha < \beta$ if $A_1 \subset A_2$ (proper subset). We define $\alpha = \beta$ if $A_1 = A_2$.

Definition 2.1.13 (Addition of Real Numbers). Let $\alpha = (A_1, B_1)$ and $\beta = (A_2, B_2)$ be real numbers. Define:

- $A = \{a_1 + a_2 \mid a_1 \in A_1, a_2 \in A_2\}$
- $B = \mathbb{Q} \setminus A$

Then $\alpha + \beta$ is defined as the cut (A, B) .

Definition 2.1.14 (Multiplication of Positive Real Numbers). For positive real numbers $\alpha = (A_1, B_1)$ and $\beta = (A_2, B_2)$ (where "positive" means they contain some positive rationals in their lower classes), define:

- $A = \{a_1 a_2 \mid a_1 \in A_1, a_2 \in A_2, a_1 > 0, a_2 > 0\} \cup \{q \in \mathbb{Q} \mid q \leq 0\}$
- $B = \mathbb{Q} \setminus A$

Then $\alpha \cdot \beta$ is defined as the cut (A, B) . For other sign combinations, we adjust the definition accordingly.

Definition 2.1.15 (Additive Inverse). For a real number $\alpha = (A, B)$, define its additive inverse $-\alpha$ by:

- $A' = \{-b \mid b \in B, b \text{ is not the smallest element of } B\}$
- $B' = \mathbb{Q} \setminus A'$

Then $-\alpha = (A', B')$.

These operations make $\mathbb{R}$ an ordered field. The multiplicative inverse can be defined similarly for non-zero elements.

Now we can define the fundamental concepts of analysis:

Definition 2.1.16 (Upper Bound). Let $S \subseteq \mathbb{R}$. A number u is an **upper bound** of S if $s \leq u$ for all $s \in S$. A number l is a **lower bound** of S if $l \leq s$ for all $s \in S$.

Definition 2.1.17 (Supremum). The **supremum** (or **least upper bound**) of S , denoted $\sup S$, is the smallest upper bound of S . That is:

1. $s \leq \sup S$ for all $s \in S$. ($\sup S$ is an upper bound.)
2. If v is any upper bound of S , then $\sup S \leq v$. (It is the *least* upper bound.)

The definition of **infimum** (or **greatest lower bound**), denoted $\inf S$, is similar.

Theorem 2.1.4: Completeness Axiom

Every non-empty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

(The case for the infimum is analogous: every non-empty subset of $\mathbb{R}$ that is bounded below has an infimum in $\mathbb{R}$.)

Proof. In the Dedekind cut construction, the supremum of a bounded set S of real numbers is given by the union of the lower classes of all elements in S . More precisely, if $S = \{\alpha_i = (A_i, B_i)\}$ is bounded above, then:

$$\sup S = \left(\bigcup_i A_i, \mathbb{Q} \setminus \bigcup_i A_i \right)$$

This pair forms a Dedekind cut and satisfies the definition of supremum. $\square$

The completeness axiom is fundamental to analysis and distinguishes $\mathbb{R}$ from $\mathbb{Q}$. It ensures that limits of Cauchy sequences exist, continuous functions attain their maximum and minimum on closed intervals, and many other essential properties. From now on we can claim that every convergent real sequence converges to a real number. This is what the **Completeness Axiom** means in analysis.

Definition 2.1.18 (Archimedean Property). The real numbers satisfy the **Archimedean property**: for any $x \in \mathbb{R}$, there exists a natural number n such that $n > x$. Equivalently, for any $\epsilon > 0$, there exists $n \in \mathbb{N}$ such that $1/n < \epsilon$.

Theorem 2.1.5: Density of Rationals

Between any two distinct real numbers, there exists a rational number. That is, $\mathbb{Q}$ is **dense** in $\mathbb{R}$.

While $\mathbb{R}$ solves the completeness problem of $\mathbb{Q}$, it is not **algebraically closed**. There are polynomial equations with real coefficients that have no real solutions, such as $x^2 + 1 = 0$. This motivates the extension to complex numbers.

Construction of Complex Numbers We now construct a new number field which is the biggest field we ever known in human history.

Definition 2.1.19 (Complex Numbers). The set of **complex numbers**, denoted $\mathbb{C}$, consists of all expressions of the form $a + bi$, where $a, b \in \mathbb{R}$ and i is the **imaginary unit** satisfying $i^2 = -1$. For $z = a + bi \in \mathbb{C}$:

- a is called the **real part**, denoted $\Re(z)$
- b is called the **imaginary part**, denoted $\Im(z)$

Two complex numbers $a + bi$ and $c + di$ are equal if and only if $a = c$ and $b = d$.

Definition 2.1.20 (Operations on Complex Numbers). For complex numbers $z = a + bi$ and $w = c + di$, we define:

- **Addition:** $z + w = (a + c) + (b + d)i$
- **Multiplication:** $zw = (ac - bd) + (ad + bc)i$

With these operations, $\mathbb{C}$ forms a field. The real numbers $\mathbb{R}$ can be identified with the subset $\{a + 0i : a \in \mathbb{R}\}$ of $\mathbb{C}$.

Definition 2.1.21 (Complex Conjugate and Modulus). For $z = a + bi \in \mathbb{C}$:

- The **complex conjugate** is $\bar{z} = a - bi$
- The **modulus** (or absolute value) is $|z| = \sqrt{a^2 + b^2}$

Theorem 2.1.6: Properties of Conjugate and Modulus

For $z, w \in \mathbb{C}$:

1. $\overline{z + w} = \bar{z} + \bar{w}$
2. $\overline{zw} = \bar{z} \cdot \bar{w}$
3. $z\bar{z} = |z|^2$
4. $|zw| = |z||w|$
5. $|z + w| \leq |z| + |w|$ (Triangle Inequality)

Theorem 2.1.7: Fundamental Theorem of Algebra

Every non-constant polynomial with complex coefficients has at least one complex root. Equivalently, $\mathbb{C}$ is algebraically closed.

This theorem is profound: while we extended $\mathbb{R}$ to $\mathbb{C}$ to solve the equation $x^2 + 1 = 0$, we actually obtained a number system where every polynomial equation has a solution.

Definition 2.1.22 (Polar Form of Complex Numbers). Any complex number $z = a + bi \neq 0$ can be written in **polar form** as:

$$z = r(\cos \theta + i \sin \theta)$$

where $r = |z| > 0$ is the modulus and θ is the **argument** of z , satisfying $\tan \theta = b/a$.

Using Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$, we can write $z = re^{i\theta}$.

Theorem 2.1.8: De Moivre's Theorem

For any integer n and complex number $z = r(\cos \theta + i \sin \theta)$:

$$z^n = r^n(\cos(n\theta) + i \sin(n\theta))$$

This theorem simplifies computations with powers and roots of complex numbers.

The complex numbers provide a powerful framework for many areas of mathematics, physics, and engineering. They allow us to:

- Solve all polynomial equations
- Represent periodic phenomena using complex exponentials
- Analyze signals and systems in electrical engineering
- Study fluid dynamics and electromagnetism
- Develop the mathematical foundation of quantum mechanics

2.2 Sequence Limit and The Properties of Real Numbers

As long as we finished the content about real numbers, we will now move on to the the core of the mathematical analysis: the **limit**. Why I claim that the limit is the core concept of mathematical analysis? In the following content you will realize that almost all the concept has some kind of connection with limit. I can say that the limit is the basis of many theory. And the language of limit represents a dynamic, approaching, and rigorous mathematical mindset. Let us begin.

2.2.1 Definitions and Basic Properties

2.2.1.1 The Definition of Limits

Definition 2.2.1 (limit of a sequence). A sequence $\{a_n\}$ converges to a real number A if for all $\epsilon > 0$, there exists an integer N such that $|a_n - A| < \epsilon$ if $n \geq N$. The number A is the limit of the sequence and we write:

$$\lim_{n \rightarrow \infty} a_n = A$$

Naively speaking, if the sequence $\{a_n\}$ is a **convergent sequence** and A is the limit of the sequence, the value of a_n become arbitrarily close to a finite number A . You can get any value that is anyhow closer to the convergence A , once you pick a big enough n .

In a more commonly used language, if we pick an open interval in the real number line, whose center is a and radius is ϵ , written as $(a - \epsilon, a + \epsilon)$. We call this kind of intervals the neighborhood, denoted as $O(a, \epsilon)$:

$$O(a, \epsilon) = \{x | a - \epsilon < x < a + \epsilon\}$$

And what the definition said is that for all terms after a_n fall within the $O(a, \epsilon)$. Since the neighborhood is contractive, the sequence eventually converges to a .

However, in the contrary, if a sequence $\{a_n\}$ is not convergent, we say it is a **divergent sequence**. Rigorously speaking, if for all $\epsilon > 0$, $N \in \mathbb{N}^*$ and $A \in \mathbb{R}$, there exists at least one n_0 , $|a_{n_0} - A| > \epsilon$.

For those sequences converges to 0, we call those sequences **infinitesimal**.

Remark 2.2.1

When we talk about infinitesimal, what we are discussing about is a sequence rather than a simple number. Be clear that infinitesimal is not a number.

2.2.1.2 Properties of Limits

After we define what is limit, let's take a look at it's properties.

Theorem 2.2.1: the uniqueness of limit

The limit of a convergent sequence must be unique.

Proof. Assume there exists a sequence $\{a_n\}$ that converges to two different values a and b , $a \neq b$. According to the definition of limit:

$$\forall \epsilon > 0, \exists N_1, n > N_1 : |x_n - a| < \epsilon/2$$

$$\forall \epsilon > 0, \exists N_2, n > N_2 : |x_n - b| < \epsilon/2$$

Pick $N = \max\{N_1, N_2\}$, according to the triangle inequality, then $\forall n > N$ we have:

$$|a - b| = |a - x_n + x_n - b| \leq |x_n - a| + |x_n - b| < \epsilon$$

Since ϵ can get arbitrarily close to 0, we know that $a = b$ □

Theorem 2.2.2

A convergent sequence must be bounded.

Remark 2.2.2

However, the contrapositive is not always true. A bounded sequence may not be convergent. Consider $a_n = (-1)^n$, the sequence $\{a_n\}$ bounded from -1 to 1 , but the sequence won't converge to any value. In the future may be we can add a stronger condition to make the $\{a_n\}$ convergent. We will introduce it in future content.

Proof. Assume $\{a_n\}$ is a convergent sequence, the limit is a . According to the definition of limit, we pick $\epsilon = 1$, thus $\exists N, \forall n > N : |x_n - a| < 1$, then $a - 1 < x_n < a + 1$

let $m = \max\{a_1, a_2, \dots, a_N, a - 1\}$, $M = \max\{a_1, a_2, \dots, a_N, a + 1\}$

Then we have $m \leq x_n \leq M$, which means that the sequence $\{x_n\}$ is bounded. $\square$

Theorem 2.2.3: isotonicity

Assume we have two sequence $\{x_n\}$ and $\{y_n\}$, they converges to two different limits a and b , and $a < b$. There exists a N , $\forall n > N$, $x_n < y_n$

Proof. For two sequence $\{x_n\}$ and $\{y_n\}$ that converge to two different values a and b . Let's assume $a > b$ According to the definition of limit, we take $\epsilon = \frac{a-b}{2}$, and we have:

$$\exists N_1, \forall n > N_1, |x_n - a| < \epsilon$$

$$\exists N_2, \forall n > N_2, |y_n - b| < \epsilon$$

Thus we have: $x_n > (a+b)/2 > y_n$, pick $N = \max\{N_1, N_2\}$, then we have $\forall n > N, x_n > y_n$. Finally we can claim that the limits have the property of isotonicity. $\square$

Theorem 2.2.4: the calculations' law of limits

Assume that there exist two limits:

$$\lim_{n \rightarrow \infty} x_n = a, \lim_{n \rightarrow \infty} y_n = b$$

And we have:

- $\lim_{n \rightarrow \infty} (\alpha x_n + \beta y_n) = \alpha a + \beta b$, for two constants α and β .
- $\lim_{n \rightarrow \infty} x_n y_n = ab$
- $\lim_{n \rightarrow \infty} (\frac{x_n}{y_n}) = \frac{a}{b}, (b \neq 0)$

We recommend the readers finish the proof above themselves.

After we finished the definition and some basic properties of the convergent sequence, we will now move on to define a special kind of not-convergent sequence: **the infinity**.

2.2.1.3 Infinity

Definition 2.2.2 (Infinity). If we have a sequence $\{x_n\}$, for every given G , $G > 0$, we can find $N \in \mathbb{N}$, $\forall n > N$, we have $|x_n| > G$, we call the sequence $\{x_n\}$ is a infinity, denoted as:

$$\lim_{n \rightarrow \infty} x_n = \infty$$

Just like infinitesimal, infinity is also a sequence rather than a number. But in some conditions, we will deal with it as if it is a number.

If a infinity start to be positive from some point, we call this form of infinity the **positive infinity**. We can define what is negative infinity likewise. We denote them specially like this:

$$\lim_{n \rightarrow \infty} a_n = +\infty, (\lim_{n \rightarrow \infty} a_n = -\infty)$$

Theorem 2.2.5

The infinity has special relationship with the infinitesimal: The sequence $\{x_n\} (x_n \neq 0)$ is infinity iff $\{\frac{1}{x_n}\}$ is infinitesimal.

Using the definition of the limit will be enough to prove this theorem. We'll skip this part here.

Theorem 2.2.6

Assume $\{x_n\}$ is a infinity, if when $n > N_0$, $|y_n| \geq \delta > 0$, then $\{x_n y_n\}$ is infinity.

Corollary 2.2.1

Assume $\{x_n\}$ is infinity, $\lim_{n \rightarrow \infty} y_n = b \neq 0$, then $\{x_n y_n\}$ and $\{\frac{x_n}{y_n}\}$ are both infinity.

2.2.1.4 Stolz Theorem

With the help of the definitions of the limit, we can calculate various kinds of limits using algebraic techniques and the definition of limit. But when we face certain forms of limit like $\frac{0}{0}$ and $\frac{\infty}{\infty}$, they are especially tricky to deal with. But with the help of **Stolz theorem** we are going to introduced now, it will be much easier to deal with them (in some occasions).

Definition 2.2.3 (Increasing Sequence). If a sequence $\{x_n\}$ satisfies: $x_n \leq x_{n+1}$, $n = 1, 2, 3 \dots$, we will call it the **monotone increasing Sequence**. Similarly, if $\forall n \in \mathbb{N}$, $x_n < x_{n+1}$, we call it the **strict monotone increasing sequence**.

Theorem 2.2.7

Let $\{y_n\}$ be a **strict monotone increasing positive infinity**, and:

$$\lim_{n \rightarrow \infty} \frac{x_n - x_{n-1}}{y_n - y_{n-1}} = a$$

Then we have that:

$$\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = a$$

Proof. Let's consider the condition when $a = 0$. Because $\lim_{n \rightarrow \infty} \frac{x_n - x_{n-1}}{y_n - y_{n-1}} = 0$, according to the definition of limit, we know that:

$$\forall \epsilon > 0, \exists N_1, \forall n > N_1 : |x_n - x_{n-1}| < \epsilon(y_n - y_{n-1})$$

Because $\{y_n\}$ is infinity, we can let $y_{N_1} > 0$ obviously. For the inequality above, we take everything from N_1 to n , and then we add them together, we have:

$$|x_n - x_{N_1}| \leq |x_n - x_{n-1}| + |x_{n-1} - x_{n-2}| + \dots + |x_{N_1+1} - x_{N_1}|$$

$$< \epsilon(y_n - y_{n-1}) + \cdots + \epsilon(y_{N_1+1} - y_{N_1}) = \epsilon(y_n - y_{n-1})$$

Divide both side of the inequality by y_n , and we have:

$$\left| \frac{x_n}{y_n} - \frac{x_{N_1}}{y_n} \right| \leq \epsilon \left(1 - \frac{y_{N_1}}{y_n} \right) \leq \epsilon$$

And, for a fixed N_1 , we can pick $N > N_1$, $\forall n > N : \left| \frac{x_{N_1}}{y_n} \right| < \epsilon$, then we have:

$$\left| \frac{x_n}{y_n} \right| < \epsilon + \left| \frac{x_{N_1}}{y_n} \right| < 2\epsilon$$

For other conditions: if a is a bounded value, and $a \neq 0$, let $x_n' = x_n - ay_n$, and with the help of the proof above, we can reach the conclusion.

When $a = +\infty$, We take the reciprocal of $\frac{x_n}{y_n}$. Similarly, it is not difficult to reach a conclusion.

□

We will see something very similar to the **Stolz theorem**, that is the **L'Hospital law**. There are many similarity between functions and sequences. We will see more examples in future chapters.

2.2.2 Convergence Criteria and the Properties of the Real Number System

Before we discuss more advanced concepts in analysis (such as derivatives and integrals), we must first firmly establish a fundamental property of the real number system $\mathbb{R}$: **Completeness**. It is this property that distinguishes the real numbers $\mathbb{R}$ from the rational numbers $\mathbb{Q}$ and serves as the bedrock for all important theorems in analysis.

2.2.2.1 The Completeness of the Real Number System

The completeness of the real number system can be expressed in several equivalent ways. Let's take a review. (The proof is in section 2.1.3)

Theorem 2.2.8: The Completeness Axiom

Every non-empty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

This axiom, while seemingly simple, directly leads to the first major convergence criterion in analysis.

2.2.2.2 The Monotone Convergence Theorem

We previously defined monotone increasing sequences. The Completeness Axiom guarantees that a bounded monotone sequence must converge.

Theorem 2.2.9: Monotone Convergence Theorem

A monotone sequence (either increasing or decreasing) that is bounded must converge.

- (i) If $\{x_n\}$ is a monotone increasing and bounded above, then $\lim_{n \rightarrow \infty} x_n = \sup\{x_n\}$.
- (ii) If $\{x_n\}$ is a monotone decreasing and bounded below, then $\lim_{n \rightarrow \infty} x_n = \inf\{x_n\}$.

Proof. We will prove (i); the proof for (ii) is analogous. Let $\{x_n\}$ be a monotone increasing sequence that is bounded above. Let $S = \{x_n \mid n \in \mathbb{N}\}$ be the set of its terms. By hypothesis, S is non-empty and bounded above. By the Completeness Axiom, S must have a supremum. Let $a = \sup S$.

We will now prove that $\lim_{n \rightarrow \infty} x_n = a$. According to the definition of a limit, we must show:

$$\forall \epsilon > 0, \exists N \in \mathbb{N}, \forall n > N : |x_n - a| < \epsilon$$

This is equivalent to $a - \epsilon < x_n < a + \epsilon$.

1. First, by the definition of a supremum, a is an upper bound for S , so $x_n \leq a$ for all n . It is clear that $x_n < a + \epsilon$.
2. Next, consider $a - \epsilon$. By the definition of a supremum, a is the *least* upper bound, which means $a - \epsilon$ (being smaller than a) *cannot* be an upper bound for S .
3. Since $a - \epsilon$ is not an upper bound, there must exist some element x_N in S such that $x_N > a - \epsilon$.
4. Because $\{x_n\}$ is monotone increasing, for any $n > N$, we have $x_n \geq x_N$.
5. Combining (1), (3), and (4), we have:

$$\forall n > N : a - \epsilon < x_N \leq x_n \leq a < a + \epsilon$$

This implies $\forall n > N : |x_n - a| < \epsilon$.

Therefore, $\lim_{n \rightarrow \infty} x_n = a$. □

2.2.2.3 The Cauchy Convergence Criterion

The Monotone Convergence Theorem is powerful, but it requires the sequence to be monotone. For the general case, we need a criterion for convergence that does not depend on monotonicity, nor on knowing the value of the limit beforehand. This is the Cauchy Criterion.

Definition 2.2.4 (Cauchy Sequence). A sequence $\{x_n\}$ is called a **Cauchy Sequence** if:

$$\forall \epsilon > 0, \exists N \in \mathbb{N}, \forall m, n > N : |x_m - x_n| < \epsilon$$

Intuitively, a Cauchy sequence is one whose terms become arbitrarily close to each other in the "tail" of the sequence.

Before proving the Cauchy Criterion, we need a key lemma, which is itself an important consequence of the Completeness Axiom. And we also need to introduce a key concept called **subsequence** to introduced the Bolzano- Weierstrass Theorem.

Definition 2.2.5 (Subsequence). Assume we have a strict increasing monotone sequence $\{k_n\}$ and a "mother" sequence $\{a_n\}$:

$$k_1, k_2, \dots, k_n, k_{n+1}, \dots$$

And $\forall i \in \mathbb{N}, k_i \in \mathbb{N}$, then we call the sequence:

$$a_{k_1}, a_{k_2}, a_{k_3}, \dots, a_{k_n}, a_{k_{n+1}}, \dots$$

a subsequence of the sequence $\{a_n\}$

Theorem 2.2.10: Bolzano-Weierstrass Theorem

Every bounded sequence in $\mathbb{R}$ must contain a convergent subsequence.

Proof. (Proof Sketch) Let $\{x_n\}$ be a bounded sequence, with all its terms contained in a closed interval $[a, b]$. We bisect $[a, b]$ into two subintervals $[a, \frac{a+b}{2}]$ and $[\frac{a+b}{2}, b]$. At least one of these must contain infinitely many terms of $\{x_n\}$. We choose such an interval and call it $I_1 = [a_1, b_1]$. Next, we bisect I_1 and again select a subinterval, $I_2 = [a_2, b_2]$, that contains infinitely many terms. We repeat this process, obtaining a **nest of closed intervals** $\{I_k = [a_k, b_k]\}$ such that:

1. $I_1 \supset I_2 \supset I_3 \supset \dots$
2. The length of I_k , $len(I_k) = b_k - a_k = (b - a)/2^k \rightarrow 0$ as $k \rightarrow \infty$.

By the **Nested Intervals Property** of $\mathbb{R}$ (an equivalent form of completeness), there exists a unique real number c such that $c \in \bigcap_{k=1}^{\infty} I_k$.

Now, we construct a subsequence $\{x_{n_k}\}$ that converges to c :

- Choose $x_{n_1} \in I_1$.
- Since I_2 has infinitely many terms, we can choose $x_{n_2} \in I_2$ such that $n_2 > n_1$.
- ...
- Having chosen $x_{n_{k-1}} \in I_{k-1}$, we can choose $x_{n_k} \in I_k$ such that $n_k > n_{k-1}$ (as I_k has infinitely many terms).

This gives us a subsequence $\{x_{n_k}\}$. Since $c \in I_k$ and $x_{n_k} \in I_k$, we have:

$$|x_{n_k} - c| \leq len(I_k) = \frac{b - a}{2^k}$$

As $k \rightarrow \infty$, $\frac{b-a}{2^k} \rightarrow 0$. By the Squeeze Theorem, $\lim_{k \rightarrow \infty} |x_{n_k} - c| = 0$, which means $\lim_{k \rightarrow \infty} x_{n_k} = c$. $\square$

Now we can prove the Cauchy Criterion.

Theorem 2.2.11: Cauchy Convergence Criterion

A sequence in $\mathbb{R}$ converges if and only if it is a Cauchy sequence.

Proof. $(\Rightarrow)$ **Convergent** $\implies$ **Cauchy** Assume $\lim_{n \rightarrow \infty} x_n = L$. By the definition of a limit, $\forall \epsilon > 0, \exists N, \forall n > N : |x_n - L| < \frac{\epsilon}{2}$. Now, take any $m, n > N$. By the triangle inequality:

$$\begin{aligned} |x_m - x_n| &= |(x_m - L) + (L - x_n)| \leq |x_m - L| + |x_n - L| \\ |x_m - x_n| &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \end{aligned}$$

Thus, $\{x_n\}$ is a Cauchy sequence.

$(\Leftarrow)$ **Cauchy** $\implies$ **Convergent** This direction relies critically on the completeness of $\mathbb{R}$.

1. **Step 1: Prove that a Cauchy sequence is bounded.** Let $\epsilon = 1$. By the Cauchy definition, $\exists N_1, \forall m, n > N_1 : |x_m - x_n| < 1$. Fix $m = N_1 + 1$. Then $\forall n > N_1 : |x_n - x_{N_1+1}| < 1$, which implies $x_{N_1+1} - 1 < x_n < x_{N_1+1} + 1$. This shows the "tail" of the sequence (terms with $n > N_1$) is bounded. The "head" of the sequence, $\{x_1, x_2, \dots, x_{N_1}\}$, is a finite set and is thus bounded. Therefore, the entire sequence $\{x_n\}$ is bounded.
2. **Step 2: Apply the Bolzano-Weierstrass Theorem.** Since $\{x_n\}$ is bounded, it has a convergent subsequence $\{x_{n_k}\}$ by Bolzano-Weierstrass. Let:

$$\lim_{k \rightarrow \infty} x_{n_k} = L$$

3. **Step 3: Prove the entire sequence $\{x_n\}$ converges to L .** We must show $\lim_{n \rightarrow \infty} x_n = L$. $\forall \epsilon > 0$:

- Since $\{x_n\}$ is Cauchy, $\exists N_2, \forall m, n > N_2 : |x_m - x_n| < \frac{\epsilon}{2}$.
- Since $\lim_{k \rightarrow \infty} x_{n_k} = L$, $\exists K, \forall k > K : |x_{n_k} - L| < \frac{\epsilon}{2}$.

We need to find an N such that $\forall n > N : |x_n - L| < \epsilon$. Let's choose $N = N_2$. Then, we pick a single index n_k from the subsequence such that $k > K$ and $n_k > N_2$. (This is always possible since $n_k \rightarrow \infty$ as $k \rightarrow \infty$).

Now, for any $n > N_2$, we have:

$$|x_n - L| = |(x_n - x_{n_k}) + (x_{n_k} - L)| \leq |x_n - x_{n_k}| + |x_{n_k} - L|$$

Since $n > N_2$ and $n_k > N_2$, the first term is $< \frac{\epsilon}{2}$ by the Cauchy condition. Since $k > K$, the second term is $< \frac{\epsilon}{2}$ by the subsequence convergence.

Thus, $\forall n > N_2 : |x_n - L| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$. This proves $\lim_{n \rightarrow \infty} x_n = L$.

□

2.3 Derivatives and Related Theorem

2.3.1 Derivatives and Differentials

Having studied limits of sequences and functions, we now turn to the central concept of differential calculus: the derivative. The derivative is the tool for studying the *rate of change* of a function.

2.3.1.1 The Concept of the Derivative

Definition 2.3.1 (Derivative). Let the function f be defined in some neighborhood of a point x_0 . If the limit

$$\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

exists, we say the function f is **differentiable** at x_0 , and this limit is called the **derivative** of f at x_0 . It is denoted by $f'(x_0)$, $\frac{df}{dx}(x_0)$, or $y'|_{x=x_0}$.

Letting $\Delta y = f(x_0 + \Delta x) - f(x_0)$, the derivative can also be written as $\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$.

Geometric Meaning: $f'(x_0)$ is the slope of the tangent line to the curve $y = f(x)$ at the point $(x_0, f(x_0))$.

Physical Meaning: If $s(t)$ is the displacement as a function of time, then $s'(t)$ is the instantaneous velocity.

Differentiability is a stronger condition than continuity.

Theorem 2.3.1: Differentiability implies Continuity

If a function f is differentiable at x_0 , then f must be continuous at x_0 .

Proof. We want to prove $\lim_{x \rightarrow x_0} f(x) = f(x_0)$, which is equivalent to proving $\lim_{x \rightarrow x_0} [f(x) - f(x_0)] = 0$. Let $x = x_0 + \Delta x$, so $x \rightarrow x_0$ is equivalent to $\Delta x \rightarrow 0$.

$$\lim_{x \rightarrow x_0} [f(x) - f(x_0)] = \lim_{\Delta x \rightarrow 0} [f(x_0 + \Delta x) - f(x_0)]$$

We use the trick of multiplying and dividing by Δx (for $\Delta x \neq 0$):

$$= \lim_{\Delta x \rightarrow 0} \left[\frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \cdot \Delta x \right]$$

By the product rule for limits:

$$= \left(\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \right) \cdot \left(\lim_{\Delta x \rightarrow 0} \Delta x \right)$$

Since f is differentiable at x_0 , the first limit exists and is equal to $f'(x_0)$. The second limit is clearly 0.

$$= f'(x_0) \cdot 0 = 0$$

Thus $\lim_{x \rightarrow x_0} f(x) = f(x_0)$, so f is continuous at x_0 .

□

2.3.1.2 Uniform Continuity

A stronger and often more useful concept is uniform continuity.

Definition 2.3.2 (Uniform Continuity). A function $f : D \rightarrow \mathbb{R}$ is **uniformly continuous** on D if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for **all** $x, y \in D$:

$$|x - y| < \delta \implies |f(x) - f(y)| < \epsilon$$

The key difference: In standard continuity, δ can depend on both ϵ and the point x_0 . In uniform continuity, δ depends *only* on ϵ and works for the entire domain simultaneously.

Theorem 2.3.2: Heine-Cantor Theorem

If a function f is continuous on a **closed and bounded** interval $[a, b]$, then f is uniformly continuous on $[a, b]$.

Example 2.3.1

$f(x) = x^2$ is uniformly continuous on $[0, 1]$ but *not* uniformly continuous on $[0, \infty)$. As x gets larger, we need a smaller and smaller δ to keep the change in $f(x)$ bounded, so no single δ works for the whole infinite domain.

Example 2.3.2: Continuous but not Differentiable

The converse is false. The function $f(x) = |x|$ is continuous at $x = 0$, but not differentiable. We check the derivative at $x = 0$:

$$\lim_{\Delta x \rightarrow 0} \frac{f(0 + \Delta x) - f(0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{|\Delta x|}{\Delta x}$$

We check the left-hand and right-hand limits:

- Right-hand limit: $\lim_{\Delta x \rightarrow 0^+} \frac{|\Delta x|}{\Delta x} = \lim_{\Delta x \rightarrow 0^+} \frac{\Delta x}{\Delta x} = 1$
- Left-hand limit: $\lim_{\Delta x \rightarrow 0^-} \frac{|\Delta x|}{\Delta x} = \lim_{\Delta x \rightarrow 0^-} \frac{-\Delta x}{\Delta x} = -1$

Since the left and right limits are not equal, the limit does not exist. $f(x) = |x|$ is not differentiable at $x = 0$.

Using the definition to calculate the differentiation is complex. It's worth remembering the expression of the differentiation of some commonly used functions.

2.3.1.3 Differentiation Rules

Here are some basic rules for differentiation:

- Sum: $(u \pm v)' = u' \pm v'$.
- Product: $(uv)' = u'v + uv'$.
- Quotient: $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$.
- Composition: $f'[g(x)] = f'(u) \cdot g'(x), u = g(x)$.

We (omit here) the proofs for basic differentiation rules (sum, product, quotient), but we will provide a rigorous proof for the Chain Rule.

Theorem 2.3.3: The Chain Rule

Let $u = g(x)$ be differentiable at x , and let $y = f(u)$ be differentiable at $u = g(x)$. Then the composite function $y = f(g(x))$ is differentiable at x , and

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad \text{or} \quad (f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$

Proof. (A rigorous proof) Let $u_0 = g(x_0)$. Since $y = f(u)$ is differentiable at u_0 , we define an auxiliary function $\phi(u)$:

$$\phi(u) = \begin{cases} \frac{f(u) - f(u_0)}{u - u_0} & \text{if } u \neq u_0 \\ f'(u_0) & \text{if } u = u_0 \end{cases}$$

Because $\lim_{u \rightarrow u_0} \phi(u) = \lim_{u \rightarrow u_0} \frac{f(u) - f(u_0)}{u - u_0} = f'(u_0) = \phi(u_0)$, the function $\phi(u)$ is continuous at $u = u_0$.

For all u (including $u = u_0$), we have $f(u) - f(u_0) = \phi(u)(u - u_0)$. Let $u = g(x_0 + \Delta x)$. Then $u - u_0 = g(x_0 + \Delta x) - g(x_0) = \Delta u$.

$$f(g(x_0 + \Delta x)) - f(g(x_0)) = \phi(g(x_0 + \Delta x)) \cdot (g(x_0 + \Delta x) - g(x_0))$$

Divide both sides by Δx (for $\Delta x \neq 0$):

$$\frac{f(g(x_0 + \Delta x)) - f(g(x_0))}{\Delta x} = \phi(g(x_0 + \Delta x)) \cdot \frac{g(x_0 + \Delta x) - g(x_0)}{\Delta x}$$

Now we take the limit as $\Delta x \rightarrow 0$:

$$\lim_{\Delta x \rightarrow 0} \frac{f(g(x_0 + \Delta x)) - f(g(x_0))}{\Delta x} = \lim_{\Delta x \rightarrow 0} \phi(g(x_0 + \Delta x)) \cdot \lim_{\Delta x \rightarrow 0} \frac{g(x_0 + \Delta x) - g(x_0)}{\Delta x}$$

The left side is the definition of $(f \circ g)'(x_0)$. On the right side, the second term is $g'(x_0)$. For the first term, since g is differentiable at x_0 , it is continuous at x_0 . Thus, as $\Delta x \rightarrow 0$, $g(x_0 + \Delta x) \rightarrow g(x_0) = u_0$. And since $\phi(u)$ is continuous at u_0 , we have $\lim_{\Delta x \rightarrow 0} \phi(g(x_0 + \Delta x)) = \phi(u_0) = f'(u_0) = f'(g(x_0))$. Therefore,

$$(f \circ g)'(x_0) = f'(g(x_0)) \cdot g'(x_0)$$

□

2.3.1.4 Higher Derivative

Definition 2.3.3 (Second Derivative). Likewise, if the differentiation of a function is differentiable, and we differentiate the differentiation, we get the second derivative of the function, and we call it differentiable for second order.

Similarly, we can define what is n -th derivative of the function $f(x)$, and we call the $f(x)$ n -th differentiable if the n -th derivative exists. The n -th derivative of $f(x)$ can be denoted as: $f^{(n)}(x)$ or $f^{(n)}$.

Theorem 2.3.4: Leibniz theorem

If u and v are two functions that are differentiable up to n times, the n -th derivative of their product can be expressed as:

$$(uv)^{(n)} = \sum_{r=0}^n C_n^r \cdot u^{(r)} \cdot v^{(n-r)}$$

The Leibniz formula solves the higher-order derivative of a product. For the derivatives of other operations, they can be easily derived from the preceding content.

2.3.1.5 The Differential

The derivative $f'(x_0)$ is a number, representing the rate of change. The differential provides a linear approximation.

Definition 2.3.4 (Differential). Let $y = f(x)$ be differentiable at x_0 . The increment Δy can be expressed as:

$$\Delta y = f(x_0 + \Delta x) - f(x_0) = f'(x_0)\Delta x + o(\Delta x)$$

where $\lim_{\Delta x \rightarrow 0} \frac{o(\Delta x)}{\Delta x} = 0$. We call the **linear principal part** of Δy , $f'(x_0)\Delta x$, the **differential** of f at x_0 , denoted dy .

$$dy = f'(x_0)\Delta x$$

By convention, we define the differential of the independent variable dx to be equal to the increment $dx = \Delta x$. Therefore, the differential can be written as:

$$dy = f'(x_0)dx$$

This also provides the notation $f'(x) = \frac{dy}{dx}$, the derivative as a ratio of differentials.

Geometric Meaning:

- $\Delta y = f(x_0 + \Delta x) - f(x_0)$ is the **actual change** in y along the curve.
- $dy = f'(x_0)dx$ is the **change in y along the tangent line**.

When Δx is small, $dy \approx \Delta y$. This provides the basis for linear approximation: $f(x_0 + \Delta x) \approx f(x_0) + f'(x_0)\Delta x$.

2.3.2 Mean Value Theorems and L'Hôpital's Rule

The Mean Value Theorems are the bridge connecting the derivative of a function to its values, and they are the theoretical foundation for applications of differential calculus.

2.3.2.1 Mean Value Theorems

We begin with a necessary lemma.

Theorem 2.3.5: Fermat's Theorem

Let the function $f(x)$ satisfy at x_0 :

1. f has a local extremum (max or min) at x_0 .
2. f is differentiable at x_0 .

Then $f'(x_0) = 0$.

Proof. Assume f has a local maximum at x_0 . Then in some neighborhood $(x_0 - \delta, x_0 + \delta)$, $f(x) \leq f(x_0)$ for all x .

- For $x \in (x_0, x_0 + \delta)$, we have $x - x_0 > 0$ and $f(x) - f(x_0) \leq 0$. Thus, the difference quotient $\frac{f(x) - f(x_0)}{x - x_0} \leq 0$. The right-hand derivative $f'_+(x_0) = \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0} \leq 0$.
- For $x \in (x_0 - \delta, x_0)$, we have $x - x_0 < 0$ and $f(x) - f(x_0) \leq 0$. Thus, the difference quotient $\frac{f(x) - f(x_0)}{x - x_0} \geq 0$. The left-hand derivative $f'_-(x_0) = \lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0} \geq 0$.

Since f is differentiable at x_0 , $f'(x_0) = f'_+(x_0) = f'_-(x_0)$. The only number that is both ≤ 0 and ≥ 0 is 0. Therefore, $f'(x_0) = 0$. $\square$

Theorem 2.3.6: Rolle's Theorem

Let the function $f(x)$ satisfy:

1. f is continuous on the closed interval $[a, b]$;
2. f is differentiable on the open interval (a, b) ;
3. $f(a) = f(b)$.

Then there exists at least one point $\xi \in (a, b)$ such that $f'(\xi) = 0$.

Proof. • **Case 1:** $f(x)$ is a constant function on $[a, b]$. Then $f(x) = f(a)$ for all $x \in [a, b]$. In this case, $f'(x) = 0$ for all $x \in (a, b)$. We can choose any $\xi \in (a, b)$.

- **Case 2:** $f(x)$ is not a constant function. Since f is continuous on the closed interval $[a, b]$, by the **Extreme Value Theorem**, f must attain an absolute maximum M and an absolute minimum m on $[a, b]$. Since f is not constant, at least one of M or m must be different from $f(a)$ (and $f(b)$). Assume $M > f(a)$. Let $f(\xi) = M$ (where $\xi \in [a, b]$). Because $f(a) = f(b) < M$, ξ cannot be a or b . Thus, $\xi \in (a, b)$. At this point ξ , $f(x)$ attains a local (and global) maximum. By Fermat's Theorem, $f'(\xi) = 0$. (If $m < f(a)$, the same logic applies to the point ξ where the minimum occurs).

□

Theorem 2.3.7: Lagrange's Mean Value Theorem

Let the function $f(x)$ satisfy:

1. f is continuous on the closed interval $[a, b]$;
2. f is differentiable on the open interval (a, b) .

Then there exists at least one point $\xi \in (a, b)$ such that

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}$$

or $f(b) - f(a) = f'(\xi)(b - a)$.

Geometric Meaning: There is at least one point $\xi \in (a, b)$ where the slope of the tangent line is equal to the slope of the secant line connecting $(a, f(a))$ and $(b, f(b))$.

Proof. The proof technique involves constructing an auxiliary function that satisfies Rolle's Theorem. Let $g(x)$ be the equation of the secant line connecting $(a, f(a))$ and $(b, f(b))$:

$$g(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a)$$

Now, construct the auxiliary function $h(x) = f(x) - g(x)$.

$$h(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a)$$

We check if $h(x)$ satisfies the conditions of Rolle's Theorem:

1. $h(x)$ is the difference of $f(x)$ and a linear function. Since both are continuous on $[a, b]$, $h(x)$ is continuous on $[a, b]$.
2. Similarly, $h(x)$ is differentiable on (a, b) .
3. $h(a) = f(a) - f(a) - \frac{f(b) - f(a)}{b - a}(a - a) = 0$.

$$4. h(b) = f(b) - f(a) - \frac{f(b)-f(a)}{b-a}(b-a) = f(b) - f(a) - (f(b) - f(a)) = 0.$$

$h(a) = h(b) = 0$. $h(x)$ satisfies all conditions for Rolle's Theorem. Thus, $\exists \xi \in (a, b)$ such that $h'(\xi) = 0$. We compute $h'(x)$:

$$h'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}$$

Setting $h'(\xi) = 0$:

$$f'(\xi) - \frac{f(b) - f(a)}{b - a} = 0$$

This gives $f'(\xi) = \frac{f(b)-f(a)}{b-a}$. □

Theorem 2.3.8: Cauchy's Mean Value Theorem

Let functions $f(x)$ and $g(x)$ satisfy:

1. f, g are continuous on the closed interval $[a, b]$;
2. f, g are differentiable on the open interval (a, b) ;
3. $g'(x) \neq 0$ for all $x \in (a, b)$.

Then there exists at least one point $\xi \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}$$

Proof. (Note: By (3) and Rolle's Theorem, $g(a) \neq g(b)$, otherwise $g'(\xi) = 0$ would hold for some ξ , which is forbidden.) We again construct an auxiliary function $h(x)$ for Rolle's Theorem:

$$h(x) = [f(b) - f(a)](g(x) - g(a)) - [g(b) - g(a)](f(x) - f(a))$$

(This form is chosen to ensure $h(a) = h(b) = 0$)

1. $h(x)$ is continuous on $[a, b]$.
2. $h(x)$ is differentiable on (a, b) .
3. $h(a) = [f(b) - f(a)](g(a) - g(a)) - [g(b) - g(a)](f(a) - f(a)) = 0$.
4. $h(b) = [f(b) - f(a)](g(b) - g(a)) - [g(b) - g(a)](f(b) - f(a)) = 0$.

By Rolle's Theorem, $\exists \xi \in (a, b)$ such that $h'(\xi) = 0$. We compute $h'(x)$:

$$h'(x) = [f(b) - f(a)]g'(x) - [g(b) - g(a)]f'(x)$$

Setting $h'(\xi) = 0$:

$$\begin{aligned} [f(b) - f(a)]g'(\xi) - [g(b) - g(a)]f'(\xi) &= 0 \\ [f(b) - f(a)]g'(\xi) &= [g(b) - g(a)]f'(\xi) \end{aligned}$$

Since $g'(x) \neq 0$, we know $g'(\xi) \neq 0$. We also know $g(a) \neq g(b)$. We can safely divide:

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}$$

□

2.3.2.2 L'Hôpital's Rule

Cauchy's Mean Value Theorem is the key to proving L'Hôpital's Rule, which is used to evaluate indeterminate forms of type $\frac{0}{0}$ and $\frac{\infty}{\infty}$.

Theorem 2.3.9: L'Hôpital's Rule ($\frac{0}{0}$ form)

Let c be a real number (or $\pm\infty$). Let f, g be differentiable on a (punctured) neighborhood of c , with $g'(x) \neq 0$. If

1. $\lim_{x \rightarrow c} f(x) = 0$ and $\lim_{x \rightarrow c} g(x) = 0$;
2. $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} = L$ (where L can be a finite value or $\pm\infty$).

Then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = L$$

Proof. We prove the case for $x \rightarrow c^+$ where c is a finite real number. We can define $f(c) = 0$ and $g(c) = 0$ (since the limits are 0), making f and g continuous at c . Now, for any x in a right-neighborhood of c , the functions f and g are continuous on $[c, x]$ and differentiable on (c, x) . By Cauchy's Mean Value Theorem, there exists a $\xi_x \in (c, x)$ such that

$$\frac{f(x) - f(c)}{g(x) - g(c)} = \frac{f'(\xi_x)}{g'(\xi_x)}$$

Since $f(c) = 0$ and $g(c) = 0$, this simplifies to:

$$\frac{f(x)}{g(x)} = \frac{f'(\xi_x)}{g'(\xi_x)}$$

As $x \rightarrow c^+$, since $\xi_x \in (c, x)$, we must also have $\xi_x \rightarrow c^+$. By condition (2), $\lim_{\xi_x \rightarrow c^+} \frac{f'(\xi_x)}{g'(\xi_x)} = L$. Therefore,

$$\lim_{x \rightarrow c^+} \frac{f(x)}{g(x)} = \lim_{\xi_x \rightarrow c^+} \frac{f'(\xi_x)}{g'(\xi_x)} = L$$

The proofs for $x \rightarrow c^-$ and $x \rightarrow \infty$ are similar. □

Theorem 2.3.10: L'Hôpital's Rule ($\frac{\infty}{\infty}$ form)

Let c be a real number (or $\pm\infty$). Let f, g be differentiable on a (punctured) neighborhood of c , with $g'(x) \neq 0$. If

1. $\lim_{x \rightarrow c} |f(x)| = \infty$ and $\lim_{x \rightarrow c} |g(x)| = \infty$;
2. $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} = L$ (where L can be a finite value or $\pm\infty$).

Then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = L$$

Proof. We prove the right-hand limit for a finite value L ; the left-hand case is identical. Limits at $\pm\infty$ follow after the substitution $x = 1/t$ or $x = -1/t$. The cases $L = \pm\infty$ are treated at the end.

Fix $\varepsilon > 0$. Because

$$\frac{f'(x)}{g'(x)} \rightarrow L,$$

there is a punctured interval $I = (c, c + \delta)$ on which

$$\left| \frac{f'(x)}{g'(x)} - L \right| < \eta, \quad \left| \frac{f'(x)}{g'(x)} \right| \leq |L| + 1,$$

where $\eta > 0$ will be chosen below. Because derivatives have the intermediate-value property and g' never vanishes on I , the derivative g' has a constant sign there; hence g is strictly monotone. Fix one point $y \in I$.

For $x \in (c, y)$, the denominator $g(x) - g(y)$ is nonzero, and Cauchy's mean value theorem gives a point $\xi_x \in (x, y)$ such that

$$A_x := \frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(\xi_x)}{g'(\xi_x)}.$$

Hence $|A_x - L| < \eta$ and $|A_x| \leq |L| + 1$.

Rewrite the desired ratio as

$$\frac{f(x)}{g(x)} = A_x \left(1 - \frac{g(y)}{g(x)}\right) + \frac{f(y)}{g(x)}.$$

Subtracting L yields

$$\frac{f(x)}{g(x)} - L = (A_x - L) - A_x \frac{g(y)}{g(x)} + \frac{f(y)}{g(x)}.$$

Since $|g(x)| \rightarrow \infty$, the last two terms tend to 0. Choose $\eta = \varepsilon/3$, and then choose x sufficiently close to c that

$$(|L| + 1) \left| \frac{g(y)}{g(x)} \right| < \frac{\varepsilon}{3}, \quad \left| \frac{f(y)}{g(x)} \right| < \frac{\varepsilon}{3}.$$

The triangle inequality now gives

$$\left| \frac{f(x)}{g(x)} - L \right| < \varepsilon.$$

Thus $f(x)/g(x) \rightarrow L$.

If $L = +\infty$, fix $M > 0$ and choose the interval so that $f'(x)/g'(x) > 2M + 1$. The same mean-value argument gives $A_x > 2M + 1$, while

$$1 - \frac{g(y)}{g(x)} \rightarrow 1, \quad \frac{f(y)}{g(x)} \rightarrow 0.$$

For x sufficiently close to c , the displayed decomposition implies $f(x)/g(x) > M$. Hence the ratio tends to $+\infty$. The case $L = -\infty$ follows by applying the same argument to $-f$. $\square$

Example 2.3.3

(1) Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ ($\frac{0}{0}$ form)

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} \stackrel{L'H}{=} \lim_{x \rightarrow 0} \frac{(\sin x)'}{(x)'} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = \cos 0 = 1$$

(2) Evaluate $\lim_{x \rightarrow +\infty} \frac{\ln x}{x}$ ($\frac{\infty}{\infty}$ form)

$$\lim_{x \rightarrow +\infty} \frac{\ln x}{x} \stackrel{L'H}{=} \lim_{x \rightarrow +\infty} \frac{(\ln x)'}{(x)'} = \lim_{x \rightarrow +\infty} \frac{1/x}{1} = \lim_{x \rightarrow +\infty} \frac{1}{x} = 0$$

Remark 2.3.1

Caution: The condition $\lim \frac{f'(x)}{g'(x)} = L$ is **sufficient** but not **necessary**. If $\lim \frac{f'(x)}{g'(x)}$ does not exist, we cannot conclude that $\lim \frac{f(x)}{g(x)}$ does not exist.

2.3.3 Taylor Expansion

The differential provides a *linear* approximation of a function $f(x)$ near a point x_0 (the tangent line). Taylor's Theorem generalizes this idea, providing a method to approximate a function with a polynomial of any arbitrary degree n .

2.3.3.1 Taylor Polynomials

We seek an n -th degree polynomial $P_n(x)$ that "best" approximates $f(x)$ near x_0 . We do this by forcing the polynomial's value and its first n derivatives to match those of $f(x)$ at x_0 .

$$P_n^{(k)}(x_0) = f^{(k)}(x_0) \quad \text{for } k = 0, 1, \dots, n$$

Let the polynomial have the form:

$$P_n(x) = c_0 + c_1(x - x_0) + c_2(x - x_0)^2 + \dots + c_n(x - x_0)^n$$

We determine the coefficients c_k :

- $P_n(x_0) = c_0 \implies c_0 = f(x_0)$
- $P'_n(x) = c_1 + 2c_2(x - x_0) + 3c_3(x - x_0)^2 + \dots$
- $P'_n(x_0) = c_1 \implies c_1 = f'(x_0)$
- $P''_n(x) = 2c_2 + 3 \cdot 2c_3(x - x_0) + \dots$
- $P''_n(x_0) = 2c_2 \implies c_2 = \frac{f''(x_0)}{2!}$
- $P'''_n(x_0) = 3 \cdot 2 \cdot 1c_3 \implies c_3 = \frac{f'''(x_0)}{3!}$

By induction, we find $P_n^{(k)}(x_0) = k!c_k$, which gives $c_k = \frac{f^{(k)}(x_0)}{k!}$.

Definition 2.3.5 (Maclaurin Polynomial). When the center is $x_0 = 0$, the Taylor polynomial is called the **Maclaurin polynomial**.

2.3.3.2 Taylor's Theorem and the Remainder

The Taylor polynomial $P_n(x)$ is an approximation of $f(x)$. The error of this approximation is called the remainder.

Definition 2.3.6 (Remainder). The **remainder** $R_n(x)$ is defined as the difference:

$$R_n(x) = f(x) - P_n(x)$$

Thus, $f(x) = P_n(x) + R_n(x)$.

Taylor's Theorem gives us a precise formula for this remainder.

Theorem 2.3.11: Taylor's Theorem with Lagrange Remainder

Let f be a function such that $f^{(n+1)}$ (the $(n+1)$ -th derivative) exists on an open interval I containing x_0 . Then for any $x \in I$, there exists a number ξ (xi) strictly between x and x_0 such that

$$f(x) = P_n(x) + R_n(x)$$

where the **Lagrange form of the remainder** is

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}$$

Proof. This proof is a clever application of Rolle's Theorem. Fix x and x_0 . For simplicity, let $x = b$. We are looking for $R_n(b) = f(b) - P_n(b)$. We want to find a constant K such that

$$f(b) = P_n(b) + K(b - x_0)^{n+1}$$

This means $K = \frac{f(b) - P_n(b)}{(b - x_0)^{n+1}}$. We must show that $K = \frac{f^{(n+1)}(\xi)}{(n+1)!}$ for some $\xi \in (x_0, b)$.

Define an auxiliary function $g(t)$ on the interval $[x_0, b]$:

$$g(t) = f(t) - P_n(t) - K(t - x_0)^{n+1}$$

where $P_n(t) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (t - x_0)^k$.

We check the values of $g(t)$ and its derivatives at $t = x_0$:

- $g(x_0) = f(x_0) - P_n(x_0) - K(x_0 - x_0)^{n+1} = f(x_0) - f(x_0) - 0 = 0$.
- $g'(t) = f'(t) - P'_n(t) - (n+1)K(t - x_0)^n$. $P'_n(t) = \sum_{k=1}^n \frac{f^{(k)}(x_0)}{(k-1)!} (t - x_0)^{k-1}$. $P'_n(x_0) = f'(x_0)$. So, $g'(x_0) = f'(x_0) - f'(x_0) - 0 = 0$.
- In general, for $k \leq n$, $P_n^{(k)}(x_0) = f^{(k)}(x_0)$. $g^{(k)}(t) = f^{(k)}(t) - P_n^{(k)}(t) - \frac{(n+1)!}{(n+1-k)!} K(t - x_0)^{n+1-k}$. So, $g^{(k)}(x_0) = f^{(k)}(x_0) - f^{(k)}(x_0) - 0 = 0$.

We have $g(x_0) = g'(x_0) = \dots = g^{(n)}(x_0) = 0$.

Now we check $g(t)$ at $t = b$:

$$g(b) = f(b) - P_n(b) - K(b - x_0)^{n+1}$$

By our definition of K , $g(b) = 0$.

We are ready to apply Rolle's Theorem:

1. We have $g(x_0) = 0$ and $g(b) = 0$. By Rolle's Theorem, $\exists \xi_1 \in (x_0, b)$ s.t. $g'(\xi_1) = 0$.
2. We have $g'(x_0) = 0$ and $g'(\xi_1) = 0$. By Rolle's Theorem (applied to g'), $\exists \xi_2 \in (x_0, \xi_1)$ s.t. $g''(\xi_2) = 0$.
3. ...
4. We have $g^{(n)}(x_0) = 0$ and $g^{(n)}(\xi_n) = 0$. By Rolle's Theorem (applied to $g^{(n)}$), $\exists \xi \in (x_0, \xi_n)$ s.t. $g^{(n+1)}(\xi) = 0$. Note that $\xi \in (x_0, \xi_n) \subset \dots \subset (x_0, b)$.

Finally, we compute $g^{(n+1)}(t)$:

$$g^{(n+1)}(t) = f^{(n+1)}(t) - P_n^{(n+1)}(t) - \frac{d^{n+1}}{dt^{n+1}} [K(t - x_0)^{n+1}]$$

Since $P_n(t)$ is a polynomial of degree n , $P_n^{(n+1)}(t) = 0$. The $(n+1)$ -th derivative of $K(t - x_0)^{n+1}$ is $K \cdot (n+1)!$.

$$g^{(n+1)}(t) = f^{(n+1)}(t) - 0 - K(n+1)!$$

At $t = \xi$, we know $g^{(n+1)}(\xi) = 0$:

$$f^{(n+1)}(\xi) - K(n+1)! = 0$$

Solving for K : $K = \frac{f^{(n+1)}(\xi)}{(n+1)!}$. Substituting this back into $R_n(b) = K(b - x_0)^{n+1}$ (and replacing b with x):

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}$$

□

2.3.3.3 Taylor and Maclaurin Series

If the remainder $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$, then the function $f(x)$ can be represented by its infinite series.

Definition 2.3.7 (Taylor Series). If $\lim_{n \rightarrow \infty} R_n(x) = 0$ for x in an interval I , then $f(x)$ is equal to its **Taylor Series** on I :

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

If $x_0 = 0$, this is called the **Maclaurin Series**.

Example 2.3.4: Common Maclaurin Series

1. $f(x) = e^x$ $f^{(k)}(x) = e^x$ for all k . So $f^{(k)}(0) = e^0 = 1$.

$$e^x = \sum_{k=0}^{\infty} \frac{1}{k!} x^k = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

The remainder is $R_n(x) = \frac{e^\xi}{(n+1)!} x^{n+1}$. For any fixed x , $\frac{x^{n+1}}{(n+1)!} \rightarrow 0$ as $n \rightarrow \infty$. Thus, this series converges to e^x for all $x \in \mathbb{R}$.

2. $f(x) = \sin x$ $f(0) = 0, f'(0) = 1, f''(0) = 0, f'''(0) = -1, f^{(4)}(0) = 0, \dots$ (Pattern: $0, 1, 0, -1, \dots$)

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} \quad (\text{for all } x)$$

3. $f(x) = \cos x$ $f(0) = 1, f'(0) = 0, f''(0) = -1, f'''(0) = 0, f^{(4)}(0) = 1, \dots$ (Pattern: $1, 0, -1, 0, \dots$)

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} \quad (\text{for all } x)$$

4. $f(x) = \frac{1}{1-x}$ (**Geometric Series**) $f^{(k)}(x) = k!(1-x)^{-(k+1)}$. So $f^{(k)}(0) = k!$.

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} \frac{k!}{k!} x^k = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + x^3 + \dots$$

This series is the geometric series, and it converges to $f(x)$ only when $|x| < 1$.

2.4 Integration

2.4.1 Indefinite Integration

Think about one question: if we have a differentiation of a function, how can we find its primitive function(s)?

We need to answer this question because in many cases, we need to figure out the primitive function from the differentiation. Considering a condition when population growth in the absence of predators or resource restrictions. In this case, the population growth rate will be proportional to the population size. In mathematical expressions, we can denote it like this:

$$\begin{cases} p'(t) = \lambda p(t) \\ p(t_0) = p_0 \end{cases}$$

If we want to know the expression of $p(t)$, we need to use the knowledge and methods of integration. But because it's too simple, many readers can simply guess the answer: that is $p(t) = p_0 \cdot e^{\lambda x}$. But how about a more complex condition? Considering the model of logistic growth:

$$\begin{cases} \frac{dN(t)}{dt} = \frac{rN(t) \cdot (K - N(t))}{K} \\ N(0) = N_0 \end{cases}$$

Guesses won't be enough to get the answers. Thus, we need to know how to do the work of integration.

Definition 2.4.1. If in a specific interval, the function $F(x)$ and $f(x)$ satisfy the following relationship:

$$F'(x) = f(x)$$

Or equivalently,

$$d[F(x)] = f(x) \cdot dx$$

Then we call $F(x)$ is **one of the antiderivative** in this interval.

The reason why we said "one of" is because the antiderivative of a function is **not unique**. For example, if a function $F(x)$ is the antiderivative of $f(x)$, then $\forall [F(x) + C]$ C is a constant, $F(x) + C$ is also the antiderivative of $f(x)$. So we can say that there are infinity many antiderivatives of a function once it is integrable, and if we know one of the antiderivative of the function, we can use $G(x) = F(x) + C$ to represent all the primitive functions of the $f(x)$.

Definition 2.4.2. All the antiderivative of a function $f(x)$ is called the **indefinite integration** of this function. denoted as $\int f(x)dx$. The sign $\int$ is called the integral sign, $f(x)$ is called the integrand, and x is called the variable of integration.

In fact, the process of finding antiderivative is to find the primitive function of the derivative. And integration is the inverse operation of differentiation. And based on the table of derivative of commonly used function in section 2.3.1, we can deduce the integration of the commonly used functions.

One of the most important properties of the integration is its linear properties. We can express it in such way:

Theorem 2.4.1

If $f(x)$ and $g(x)$ are both integrable, then for every constant k_1 and k_2 , the function $k_1f(x) + k_2g(x)$ is also integrable. and we have:

$$\int [k_1f(x) + k_2g(x)]dx = k_1 \int f(x)dx + k_2 \int g(x)dx$$

This is called the linear property of integration.

However, since our primitive purpose is to find ways to figure out the antiderivative of a function, using the definition won't be enough to cover all the needs of figure out the antiderivative of a function. So we will introduce several methods to help us figure out the antiderivative.

2.4.1.1 Integration By Substitutions

Substitution is one of the most popular methods in analysis. When figuring out the antiderivative of the function $f(x)$, we can use this protocol.

Definition 2.4.3 (Integration by Substitution (First Part)). If the integrand can be transformed into the form: $f(x) = g(h(x)) \cdot h'(x)$, and the integration of $g(u)$ is easy to know. Then we have:

$$\int f(x)dx = \int g(h(x)) \cdot h'(x) \cdot dx = \int g(h(x)) \cdot d(h(x)) = G(h(x)) + C$$

(We denote the integration of the function $g(x)$ as $G(x)$)

Definition 2.4.4 (Integration by Substitution (Second Part)). We can also perform substitution in a different manner. To evaluate $\int f(x)dx$, we can introduce a new variable t by setting $x = \phi(t)$, where $\phi(t)$ is a function with a continuous derivative and an inverse $t = \phi^{-1}(x)$.

If we substitute $x = \phi(t)$, then $dx = \phi'(t)dt$. The integration becomes:

$$\int f(x)dx = \int f(\phi(t)) \cdot \phi'(t)dt$$

If we can find the antiderivative of the right-hand side, say $H(t)$, we can then substitute back $t = \phi^{-1}(x)$ to express the final answer in terms of x .

$$\int f(x)dx = H(t) + C = H(\phi^{-1}(x)) + C$$

This method is particularly useful when the integrand $f(x)$ contains expressions that can be simplified by such a substitution.

The second method of substitution gives rise to several powerful, standardized techniques.

Trigonometric Substitution This method is used to eliminate square roots of quadratic expressions, specifically of the forms $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, and $\sqrt{x^2 - a^2}$.

- **For integrands containing $\sqrt{a^2 - x^2}$:** We use the substitution $x = a \sin(\theta)$, with $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. Then $dx = a \cos(\theta)d\theta$. The expression becomes:

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2(\theta)} = \sqrt{a^2(1 - \sin^2(\theta))} = \sqrt{a^2 \cos^2(\theta)} = a \cos(\theta)$$

(Note: $\cos(\theta) \geq 0$ in the specified range).

- **For integrands containing $\sqrt{a^2 + x^2}$:** We use the substitution $x = a \tan(\theta)$, with $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Then $dx = a \sec^2(\theta)d\theta$. The expression becomes:

$$\sqrt{a^2 + x^2} = \sqrt{a^2 + a^2 \tan^2(\theta)} = \sqrt{a^2(1 + \tan^2(\theta))} = \sqrt{a^2 \sec^2(\theta)} = a \sec(\theta)$$

(Note: $\sec(\theta) > 0$ in the specified range).

- **For integrands containing $\sqrt{x^2 - a^2}$:** We use the substitution $x = a \sec(\theta)$, with $0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$. Then $dx = a \sec(\theta) \tan(\theta)d\theta$. The expression becomes:

$$\sqrt{x^2 - a^2} = \sqrt{a^2 \sec^2(\theta) - a^2} = \sqrt{a^2(\sec^2(\theta) - 1)} = \sqrt{a^2 \tan^2(\theta)} = a \tan(\theta)$$

(Note: $\tan(\theta) \geq 0$ in the specified range).

After integration, one must convert the result from θ back to x using the original substitution, often by drawing a right-angled triangle.

Tangent Half-Angle Substitution (Weierstrass Substitution) This substitution is very powerful for integrals which are rational functions of $\sin(x)$ and $\cos(x)$. We introduce the substitution $t = \tan(x/2)$. Using trigonometric identities, we can express $\sin(x)$, $\cos(x)$, and dx in terms of t :

$$\begin{aligned}\sin(x) &= \frac{2 \tan(x/2)}{1 + \tan^2(x/2)} = \frac{2t}{1 + t^2} \\ \cos(x) &= \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)} = \frac{1 - t^2}{1 + t^2}\end{aligned}$$

From $t = \tan(x/2)$, we have $dt = \frac{1}{2} \sec^2(x/2)dx = \frac{1}{2}(1 + \tan^2(x/2))dx = \frac{1}{2}(1 + t^2)dx$. This gives:

$$dx = \frac{2}{1+t^2}dt$$

This substitution converts any rational function of $\sin(x)$ and $\cos(x)$ into a rational function of t , which can then be integrated using methods like partial fraction decomposition.

Remark 2.4.1

In trigonometric substitution, there are some commonly used equations:

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

2.4.1.2 Integration By Parts

Integration by Parts is another fundamental technique, derived from the product rule for differentiation.

Definition 2.4.5 (Integration by Parts). Let $u = u(x)$ and $v = v(x)$ be differentiable functions. The product rule for differentiation states:

$$\frac{d}{dx}(u(x)v(x)) = u'(x)v(x) + u(x)v'(x)$$

Integrating both sides with respect to x :

$$\begin{aligned} \int \frac{d}{dx}(u(x)v(x))dx &= \int u'(x)v(x)dx + \int u(x)v'(x)dx \\ u(x)v(x) &= \int v(x)u'(x)dx + \int u(x)v'(x)dx \end{aligned}$$

Rearranging this gives the integration by parts formula. In the more compact differential notation, let $u = u(x)$ and $v = v(x)$, so $du = u'(x)dx$ and $dv = v'(x)dx$. The formula is:

$$\int u dv = uv - \int v du$$

The key to this method is to split the integrand into two parts, u and dv , such that the new integral, $\int v du$, is simpler to solve than the original. The real magic of using integration by parts is to use the u' to eliminate something.

Example: Let's compute $\int x \cos(x)dx$.

We must choose u and dv . A good choice is:

- Let $u = x$ (because its derivative, du , is simpler)
- Let $dv = \cos(x)dx$ (because it is easy to integrate)

Now we compute du (by differentiating u) and v (by integrating dv):

- $du = dx$
- $v = \int \cos(x)dx = \sin(x)$ (We omit the constant of integration until the final step)

Applying the formula $\int u dv = uv - \int v du$:

$$\int x \cos(x)dx = x \cdot \sin(x) - \int \sin(x)dx = x \sin(x) + \cos(x) + C$$

Tips for Choosing u and dv :

- **The LIATE Rule:** A helpful mnemonic for choosing u is the acronym **LIATE**, which stands for:
 - **L:** Logarithmic functions (e.g., $\ln(x)$)
 - **I:** Inverse trigonometric functions (e.g., $\arctan(x)$, $\arcsin(x)$)
 - **A:** Algebraic functions (e.g., x^2 , $x^3 + 1$)
 - **T:** Trigonometric functions (e.g., $\sin(x)$, $\cos(x)$)
 - **E:** Exponential functions (e.g., e^x , 2^x)

You should choose u as the function that appears first in this list. The remaining part of the integrand becomes dv . This heuristic works because functions at the top of the list (like $\ln(x)$) generally become simpler upon differentiation, while functions at the bottom (like e^x) are easy to integrate.

- **Repeated Application:** Sometimes, integration by parts must be applied more than once. For example, to solve $\int x^2 e^x dx$, you would first set $u = x^2$, which would lead to a new integral involving xe^x . You would then apply integration by parts a second time to solve that integral.
- **The "Boomerang" Technique:** For integrals like $\int e^x \cos(x) dx$, applying integration by parts twice (using consistent choices for u and dv) will result in the original integral appearing on the right-hand side of the equation. You can then algebraically solve for the value of the integral.

Also, For functions in forms like $\int P(x)e^x dx$, $\int P(x) \sin x dx$, $\int P(x) \cos x dx$, We can apply the integration by parts again and again. Then we will have that:

$$\int uv^{(n+1)} dx = uv^{(n)} - u'v^{(n-1)} + u''v^{(n-2)} - u^{(3)}v^{(n-3)} + \dots + (-1)^n u^{(n)}v + (-1)^{n+1} \int u^{n+1} v dx$$

This formula is especially useful if u is a **polynomial function**.

2.4.1.3 General Method of Integrating the Rational Functions

There exists a form of function, that have really nice property and we can figure out the integration of every functions in that form. That is the rational function.

Definition 2.4.6 (Rational Functions). Rational functions is a class of function, which is the fraction of two real coefficients polynomials, we can denote it as:

$$R(x) = \frac{P(x)}{Q(x)}$$

$P(x), Q(x)$ are both real coefficients polynomials. If the power of $P(x)$ is smaller than $Q(x)$, then we say $R(x)$ is a proper fraction. Else, we claim the $R(x)$ is a improper fraction.

And according to the **Fundamental Theorem of Algebra**, which we will introduce in later chapter, we know that the $Q(x)$ has the same amount of solutions of the number of the power.

Thus we can rewrite the $Q(x)$ in form of:

$$Q(x) = k(x-a)^\alpha(x-b)^\beta \cdots (x^2+px+q)^\mu(x^2+rx+s)^\delta \cdots$$

And now we will introduce a lemma, readers can try to prove it themselves.

Theorem 2.4.2

If we have a proper fraction $R(x) = \frac{P(x)}{Q(x)}$. and $Q(x)$ has factorization like above, then we can assert that:

$$R(x) = \sum_{j=1}^{\alpha} \frac{A_j}{(x-a)^j} + \sum_{j=1}^{\beta} \frac{B_j}{(x-b)^j} + \cdots + \sum_{j=1}^{\mu} \frac{2K_jx + L_j}{(x^2 + px + q)^j} + \sum_{j=1}^{\delta} \frac{2M_jx + N_j}{(x^2 + rx + s)^j} + \cdots$$

This factorization is unique for all proper rational fractions.

And it's easy to deduce that:

$$\int \frac{mx + n}{x^2 + px + q} dx = \frac{m}{2} \ln |x^2 + px + q| + \frac{2n - mp}{\sqrt{4q - p^2}} \arctan \frac{2x + p}{\sqrt{4q - p^2}} + C (q > \frac{p^2}{4})$$

$$\int \frac{mx + n}{x^2 + px + q} dx = \frac{m}{2} \ln |x^2 + px + q| + \frac{2n - mp}{2\sqrt{p^2 - 4q}} \ln \left| \frac{2x + p - \sqrt{p^2 - 4q}}{2x + p + \sqrt{p^2 - 4q}} \right| + C (q < \frac{p^2}{4})$$

Although the forms of factorization is complex, but they are all integrable.

- $\int \frac{A}{(x-a)^n} dx = \frac{A}{1-n} (x-a)^{1-n} + C, (n \neq 1)$
- $\int \frac{A}{x-a} dx = A \ln |x-a| + C$
- $\int \frac{Mx+N}{x^2+px+q} dx = \int \frac{\frac{M}{2}(2x+p) + N - \frac{M}{2}p}{(x+\frac{p}{2})^2 - \frac{p^2}{4} + q} dx$
- $I_r = \int \frac{1}{(x^2+px+q)^r} dx =, I_r = \frac{2x+p}{(4q-p^2)(r-1)(x^2+px+q)^{r-1}} + \frac{2(2r-3)}{(4q-p^2)(r-1)} I_{r-1}$

2.4.1.4 Integration Table

Integration by substitutions and parts are vital when facing different kinds of integration. But to accelerate the integration process, here we present some commonly used integration formula. We strongly suggest the readers to prove them one by one. They are practical for cultivating mathematical mindset.

$$\int \frac{1}{a^2+x^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

$$\int \frac{1}{x^2-a^2} dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$$

$$\int \frac{1}{a^2-x^2} dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$$

$$\int \frac{x}{a^2+x^2} dx = \frac{1}{2} \ln(a^2+x^2) + C$$

$$\int \frac{1}{\sqrt{a^2-x^2}} dx = \arcsin\left(\frac{x}{a}\right) + C$$

$$\int \frac{1}{\sqrt{x^2+a^2}} dx = \ln|x+\sqrt{x^2+a^2}| + C$$

$$\int \frac{1}{\sqrt{x^2-a^2}} dx = \ln|x+\sqrt{x^2-a^2}| + C$$

$$\int \frac{1}{x\sqrt{x^2-a^2}} dx = \frac{1}{a} \operatorname{arcsec} \left| \frac{x}{a} \right| + C$$

$$\int \sqrt{a^2-x^2} dx = \frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \arcsin\left(\frac{x}{a}\right) + C$$

$$\int \frac{1}{x\sqrt{a^2-x^2}} dx = \frac{1}{a} \ln \left| \frac{a-\sqrt{a^2-x^2}}{x} \right| + C$$

$$\int \sqrt{x^2+a^2} dx = \frac{x}{2} \sqrt{x^2+a^2} + \frac{a^2}{2} \ln|x+\sqrt{x^2+a^2}| + C$$

$$\int \frac{1}{x\sqrt{a^2+x^2}} dx = -\frac{1}{a} \ln \left| \frac{a+\sqrt{a^2+x^2}}{x} \right| + C$$

$$\int \sqrt{x^2-a^2} dx = \frac{x}{2} \sqrt{x^2-a^2} - \frac{a^2}{2} \ln|x+\sqrt{x^2-a^2}| + C$$

$$\int \sin^2 x dx = \frac{x}{2} - \frac{\sin(2x)}{4} + C$$

$$\int \cos^2 x dx = \frac{x}{2} + \frac{\sin(2x)}{4} + C$$

$$\int \tan^2 x dx = \tan x - x + C$$

$$\int \arccos x dx = x \arccos x - \sqrt{1-x^2} + C$$

$$\int \arctan x dx = x \arctan x - \frac{1}{2} \ln(1+x^2) + C$$

$$\int \operatorname{arcsec} x dx = x \operatorname{arcsec} x - \ln|x+\sqrt{x^2-1}| + C$$

$$\int \operatorname{arccot} x dx = x \operatorname{arccot} x + \frac{1}{2} \ln(1+x^2) + C$$

2.4.2 Definite Integration

In previous content, we learned about the definition and calculation technique about indefinite integration. Here we are going to talk about definite integration. Unlike indefinite integration, definite integration have more intuitive geometric meaning. And when facing real world problems, definite integration have more direct connections with the physical background. Before we start to construct a rigorous theory and definition for definite integration, let's take a look at its geometric meaning first.

The geometric idea behind the definite integral is to compute the area under a curve $y = f(x)$ and above the x -axis, from $x = a$ to $x = b$. For a function that is non-negative and continuous, this "area" is an intuitive concept. The powerful idea of integration is to approximate this curved region by a collection of simple shapes—typically rectangles—whose areas are easy to calculate.

This approximation is achieved by first **partitioning** the interval $[a, b]$ into n subintervals. While a uniform partition is often used for simplicity, the general theory requires that our method must work for an **arbitrary partition**, not just a regular one. On each subinterval, we construct a rectangle that approximates the area under the curve on that small segment. The key point is that the height of this rectangle is determined by the function's value at some **sample point** within the subinterval. The choice of this sample point (e.g., left endpoint, right endpoint, midpoint, or any point in between) is also arbitrary in the general formulation.

The total area of these rectangles, known as a **Riemann sum**, provides an approximation to the true area under the curve:

$$S_n = \sum_{i=1}^n f(c_i) \Delta x_i.$$

Intuitively, as we take thinner and thinner rectangles (i.e., as the maximum subinterval width, called the **norm** of the partition, approaches zero), the approximation becomes more accurate. The area under the curve is then defined as the **limit** of these Riemann sums, provided that this limit exists.

Crucially, for this definition to be meaningful and well-defined, the limit must converge to the same value **regardless** of how we choose the partition and the sample points. This requirement of independence from

the arbitrary choices is what leads to the rigorous standard definition. For continuous functions, it can be proven that this is indeed the case, unifying the intuitive geometric concept with a precise analytical foundation.

That is the intuitive perspective of the definite integration. Now we can construct a rigorous definition.

Definition 2.4.7. Assume $f(x)$ is a bounded function defined on interval $[a, b]$, pick divisional points $\{x_i\}_{i=0}^n$ randomly on interval $[a, b]$, which forms a partition:

$$P : a = x_0 < x_1 < x_2 < \cdots < x_n = b$$

And we pick $\forall \xi_i \in [x_{i-1}, x_i]$, define the length of each intervals as $\Delta x_i = x_i - x_{i-1}$, and define $\lambda = \max_{1 \leq i \leq n}(\Delta x_i)$. If:

$$\lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i$$

exists, and the value is independent from the partition, then we claim that the function $f(x)$ is Riemann Integral on interval $[a, b]$. The limit is called the definite integration on $[a, b]$, denote as:

$$I = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i = \int_a^b f(x) dx$$

a is called the lower limit of integral, and the b is called the upper limit of integral.

In the definition above, we require $a < b$. When $a > b$, we define:

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

If $a = b$, we define the integral equals to 0.

When no confusion is likely to arise, a function that is Riemann integrable is generally simply referred to as integrable.

Example 2.4.1

Discuss the integrability of the Dirichlet function $D(x)$ on $[0, 1]$.

Proof. Due to the density of $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$, any partition subinterval $[x_{i-1}, x_i]$ contains both rational and irrational numbers. Choosing rational ξ_i yields the Riemann sum $\sum D(\xi_i) \Delta x_i = 1$. Alternatively, choosing irrational ξ_i yields $\sum D(\xi_i) \Delta x_i = 0$. Since the limit $\lim_{\lambda \rightarrow 0} \sum D(\xi_i) \Delta x_i$ depends on the choice of ξ_i , it does not exist. Thus, $D(x)$ is not Riemann integrable. $\square$

2.4.2.1 Equivalent Integration Conditions

Using the definition to test whether a function is integrable is complex and sometimes inoperable. So the question is: can we find a method that is logical equivalent to the definition but is more useful? The answer is obviously yes!

Just take a look at the definition:

$$\int_a^b f(x) dx = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i$$

The key process and properties of the limit is the random selection of the value of ξ_i . So we consider the extreme condition: the supremum and the infimum of the function $f(x)$ in interval $[x_{i-1}, x_i]$, denoted as M_i and m_i .

Then we shall have two limits:

$$M = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n M_i \cdot \Delta x_i$$

$$m = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n m_i \cdot \Delta x_i$$

If both of the limits M and m are convergent and they converge to the same value, we can claim that the definite integration exists, because:

$$\lim_{\lambda \rightarrow 0} \sum_{i=1}^n m_i \cdot \Delta x_i = m \leq \int_a^b f(x) dx = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i \leq M = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n M_i \cdot \Delta x_i$$

Then we can replace the random selection of ξ_i with the supremum and the infimum of the function $f(x)$, which is more operable compared to the original definition.

Now we will present a rigorous expression of the idea above, and give a proof of it.

Definition 2.4.8 (Darboux Sum). For a partition P and each of its interval(s) $[x_{i-1}, x_i]$, we make the following denotions:

$$M_i = \sup\{f(x)|x \in [x_{i-1}, x_i]\}, m_i = \inf\{f(x)|x \in [x_{i-1}, x_i]\}$$

Obviously they are related to the choice of the partition. After we choose the partition P , we define:

$$\overline{S}(P) = \sum_{i=1}^n M_i \cdot \Delta x_i, \underline{S}(P) = \sum_{i=1}^n m_i \cdot \Delta x_i$$

The $\overline{S}(P)$ is called the **Darboux upper sum**, and the $\underline{S}(P)$ is called the **Darboux lower sum**.

It is very obvious that:

$$\underline{S}(P) \leq \sum_{i=1}^n f(\xi_i) \Delta x_i \leq \overline{S}(P)$$

The next step is to prove that if the limits $\lim_{\lambda \rightarrow 0} \overline{S}(P)$ and $\lim_{\lambda \rightarrow 0} \underline{S}(P)$ exist and converge to the same value, then the definite integral exist. ($\lambda = \max\{\Delta x_i\}$)

Theorem 2.4.3

Adding points to the original partition forms a new partition; the Darboux upper sum does not increase, and the Darboux lower sum does not decrease.

Proof. Assume $\overline{S}(P)$ and $\underline{S}(P)$ correspond to certain partition P , and $P : \{x_i\}_{i=1}^n$. And with a new divisional point added, we have a new partition P' , whose Darboux upper sum and Darboux lower sum are $\overline{S}(P')$ and $\underline{S}(P')$. What we need to prove is that:

$$\overline{S}(P') \leq \overline{S}(P), \underline{S}(P) \leq \underline{S}(P')$$

Assume the added point x' falls into the interval (x_{i-1}, x_i) , we denoted that:

$$M_i = \sup\{f(x)|x \in (x_{i-1}, x_i)\}, M'_i = \sup\{f(x)|x \in (x_{i-1}, x')\}, M''_i = \sup\{f(x)|x \in (x', x_i)\}$$

Because $(x_{i-1}, x') \in (x_{i-1}, x_i)$, $(x', x_i) \in (x_{i-1}, x_i)$, then we have:

$$M'_i \leq M_i, M''_i \leq M_i$$

$$M'_i(x' - x_{i-1}) + M''_i(x_i - x') \leq M_i(x_i - x_{i-1})$$

Adding one divisional point won't interrupt other intervals, so now we have $\bar{S}(P') \leq \bar{S}(P)$. Likewise, we can prove that $\underline{S}(P) \leq \underline{S}(P')$

□

Now we can deduce that $m(b-a) \leq \underline{S}(P_2) \leq \bar{S}(P_1) \leq M(b-a)$.

According to The Monotone Convergence Theorem in 2.2.2.2, we can claim that the limits of $\lim_{\lambda \rightarrow 0} \bar{S}(P)$ and $\lim_{\lambda \rightarrow 0} \underline{S}(P)$ exists.

We denoted that:

$$\lim_{\lambda \rightarrow 0} \bar{S}(P) = L, \lim_{\lambda \rightarrow 0} \underline{S}(P) = l$$

And now we are going to prove that $L = \inf\{\bar{S}(P) | \bar{S}(P) \in \bar{\mathbf{S}}\}$, $l = \sup\{\underline{S}(P) | \underline{S}(P) \in \underline{\mathbf{S}}\}$ are established for all bounded function $f(x)$.

Theorem 2.4.4: Darboux Theorem

$$\lim_{\lambda \rightarrow 0} \bar{S}(P) = \inf\{\bar{S}(P) | \bar{S}(P) \in \bar{\mathbf{S}}\}$$

$$\lim_{\lambda \rightarrow 0} \underline{S}(P) = \sup\{\underline{S}(P) | \underline{S}(P) \in \underline{\mathbf{S}}\}$$

Proof. We shall only present the proof of the Darboux upper sum. The situation of the Darboux lower sum is likewise. The basic idea is to use the $\epsilon - \delta$ language, select a Darboux upper sum that satisfy the limit's condition, and prove that $\forall P, \lambda = \max_{1 \leq i \leq n}(\Delta x_i) < \delta$, we have that $0 \leq \bar{S}(P) - L < \epsilon$. Now, assume we have partition P' that satisfy $0 \leq \bar{S}(P') - L < \frac{\epsilon}{2}$. And:

$$P' : a = x'_0 < x'_1 < x'_2 < \cdots < x'_p = b$$

We pick $\delta = \min\{\Delta x'_1, \Delta x'_2, \dots, \Delta x'_p, \frac{\epsilon}{2(p-1)(M-m)}\}$. Now assume we have another partition P that satisfy $\lambda = \max_{1 \leq i \leq n}(\Delta x_i) < \delta$:

$$P : a = x_0 < x_1 < x_2 < \cdots < x_n = b$$

And its Darboux upper sum is $\bar{S}(P)$, we insert $P' = \{x'_j\}_{j=0}^p$ into $P = \{x_i\}_{i=0}^n$ and form a new partition P^* . Likewise, we denote its Darboux upper sum as $\bar{S}(P^*)$. According to previous lemma, we have that:

$$\bar{S}(P^*) - \bar{S}(P') \leq 0$$

For all the interval (x_{i-1}, x_i) , we have at most $p-1$ intervals that have divisional points inserted. For other intervals, there won't be any changes. For the intervals that are inserted, take use of the notations previously used in the proof, we have:

$$M_i(x_i - x_{i-1}) - [M'_i(x'_j - x_{i-1}) + M''_i(x_i - x'_j)] \leq (M-m)(x_i - x_{i-1}) < (M-m)\delta$$

So now we have that:

$$0 \leq \bar{S}(P) - \bar{S}(P^*) < (p-1)(M-m)\delta \leq \frac{\epsilon}{2}$$

So after all, we conclude:

$$0 \leq \bar{S}(P) - L = [\bar{S}(P) - \bar{S}(P^*)] + [\bar{S}(P^*) - \bar{S}(P')] + [\bar{S}(P') - L] < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

□

Now we have the necessary and sufficient condition of the integrable.

Theorem 2.4.5

The necessary and sufficient condition of bounded function $f(x)$ on interval $[a, b]$ is that: for every partition P , when $\lambda = \max_{1 \leq i \leq n} \Delta x_i \rightarrow 0$, and we have:

$$\lim_{\lambda \rightarrow 0} \overline{S}(P) = L = l = \lim_{\lambda \rightarrow 0} \underline{S}(P)$$

Proof. We now complete the proof of the theorem. Let f be a bounded function on $[a, b]$, and define:

$$L = \lim_{\lambda \rightarrow 0} \overline{S}(P), \quad l = \lim_{\lambda \rightarrow 0} \underline{S}(P).$$

By the Darboux Theorem, we have:

$$L = \inf \{ \overline{S}(P) \}, \quad l = \sup \{ \underline{S}(P) \}.$$

Necessity: If f is integrable on $[a, b]$, then there exists a number I such that for every $\epsilon > 0$, there exists $\delta > 0$ such that for any partition P with $\lambda < \delta$ and any choice of sample points $\xi_i \in [x_{i-1}, x_i]$, we have:

$$\left| \sum_{i=1}^n f(\xi_i) \Delta x_i - I \right| < \epsilon.$$

In particular, for any such partition P , we can choose sample points such that $f(\xi_i)$ is arbitrarily close to M_i , yielding:

$$|\overline{S}(P) - I| \leq \epsilon.$$

Similarly, by choosing sample points where $f(\xi_i)$ is arbitrarily close to m_i , we obtain:

$$|\underline{S}(P) - I| \leq \epsilon.$$

Hence, as $\lambda \rightarrow 0$, we have:

$$\overline{S}(P) \rightarrow I \quad \text{and} \quad \underline{S}(P) \rightarrow I,$$

which implies $L = l = I$.

Sufficiency: Conversely, suppose $L = l = I$. Then for every $\epsilon > 0$, there exists $\delta > 0$ such that for any partition P with $\lambda < \delta$, we have:

$$|\overline{S}(P) - I| < \epsilon \quad \text{and} \quad |\underline{S}(P) - I| < \epsilon.$$

For any Riemann sum $\sum_{i=1}^n f(\xi_i) \Delta x_i$ corresponding to P , we have:

$$\underline{S}(P) \leq \sum_{i=1}^n f(\xi_i) \Delta x_i \leq \overline{S}(P).$$

Therefore,

$$I - \epsilon < \underline{S}(P) \leq \sum_{i=1}^n f(\xi_i) \Delta x_i \leq \overline{S}(P) < I + \epsilon,$$

which implies:

$$\left| \sum_{i=1}^n f(\xi_i) \Delta x_i - I \right| < \epsilon.$$

Thus, f is integrable on $[a, b]$ with integral I .

This completes the proof of the theorem. □

From the theorem above, we can derive a more practical criterion involving the oscillation of the function. Define the oscillation on the subinterval $[x_{i-1}, x_i]$ as $\omega_i = M_i - m_i$. The condition $L = l$ is equivalent to:

$$\lim_{\lambda \rightarrow 0} \sum_{i=1}^n \omega_i \Delta x_i = 0$$

This criterion allows us to identify broad classes of functions that are Riemann integrable.

2.4.2.2 Classes of Integrable Functions

While the definition involving Riemann sums or Darboux sums is necessary for theoretical rigor, we do not use it to check every specific function. Instead, we rely on established theorems regarding the integrability of common function classes.

Theorem 2.4.6: Integrability of Continuous Functions

If $f(x)$ is continuous on the closed interval $[a, b]$, then $f(x)$ is Riemann integrable on $[a, b]$.

Theorem 2.4.7: Integrability of Monotonic Functions

If $f(x)$ is bounded and monotonic on the closed interval $[a, b]$, then $f(x)$ is Riemann integrable on $[a, b]$.

These theorems cover the vast majority of functions encountered in physical and engineering problems.

2.4.2.3 The Fundamental Theorem of Calculus

So far, we have defined the definite integral as a limit of sums. However, calculating limits of sums for complex functions is incredibly tedious. The connection between the definite integral (area) and the indefinite integral (antiderivative) is provided by the Newton-Leibniz Formula.

Theorem 2.4.8: Newton-Leibniz Formula

If $f(x)$ is continuous on $[a, b]$, and $F(x)$ is an antiderivative of $f(x)$ on $[a, b]$ (i.e., $F'(x) = f(x)$), then:

$$\int_a^b f(x) dx = F(b) - F(a)$$

This theorem transforms the problem of summation into a problem of finding an antiderivative, unifying the geometric concept of integration with the algebraic operation of differentiation.

When applying the Newton-Leibniz Formula, we must ensure that the function is continuous on the interval. If there are discontinuities, we need to check whether the function is still integrable using the criteria discussed earlier.

There is some technique when applying the Newton-Leibniz Formula, such as substitution and integration by parts, which are similar to those used in indefinite integration.

For example, how can we calculate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1+e^x} \cos^3 x dx$?

We spotted that the function is continuous on the interval and if we take $x = -t$, then we have:

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1+e^x} \cos^3 x dx = \int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \frac{e^t}{1+e^t} \cos^3(-t) dt = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{e^x}{1+e^x} \cos^3 x dx$$

Then we have:

$$I + I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1}{1+e^x} \cos^3 x \right) + \left(\frac{e^x}{1+e^x} \cos^3 x \right) dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^3 x dx$$

$$I + I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^3 x dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 - \sin^2 x) d(\sin x) = \left(\sin x - \frac{\sin^3 x}{3} \right) \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{4}{3}$$

Thus, $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1+e^x} \cos^3 x dx = \frac{2}{3}$.

2.5 Improper Integrals

In the discussion of the Riemann integral $\int_a^b f(x)dx$, we relied on two fundamental restrictions:

1. The integration interval $[a, b]$ is finite (a closed, bounded interval).
2. The function $f(x)$ is bounded on $[a, b]$.

However, many problems in mathematics, physics, and probability theory require us to relax these conditions. For example, calculating the escape velocity of a rocket requires integrating gravitational force over an infinite distance, or calculating the mean lifetime of a particle involves an integral from 0 to $+\infty$. Furthermore, some physically relevant functions approach infinity (blow up) at certain points. Integrals that violate either of these two conditions are called **Improper Integrals** (or Generalized Integrals). We define them using limits. First, we consider the case where the interval of integration is infinite. These are often called improper integrals of the **first kind**.

2.5.1 Improper Integrals of the First Kind (Infinite Intervals)

Definition 2.5.1. Let $f(x)$ be defined on the infinite interval $[a, +\infty)$ and be integrable on every finite subinterval $[a, u]$ where $u > a$. We define the improper integral of f over $[a, +\infty)$ as:

$$\int_a^{+\infty} f(x)dx = \lim_{u \rightarrow +\infty} \int_a^u f(x)dx$$

- If the limit exists and is a finite number, we say the improper integral **converges**.
- If the limit does not exist (including becoming infinite), we say the improper integral **diverges**.

Geometrically, this represents the area of an unbounded region. Even though the region extends infinitely to the right, the total area can still be finite if the curve approaches the x -axis sufficiently fast.

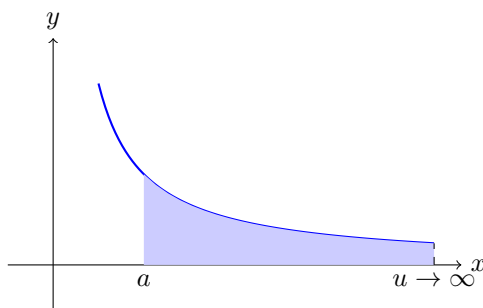


Figure 2.1: Visualizing an integral over an infinite interval

Similarly, we can define integrals for other infinite intervals:

$$\int_{-\infty}^b f(x)dx = \lim_{u \rightarrow -\infty} \int_u^b f(x)dx$$

$$\int_{-\infty}^{+\infty} f(x)dx = \int_{-\infty}^c f(x)dx + \int_c^{+\infty} f(x)dx$$

For the integral from $-\infty$ to $+\infty$ to converge, **both** constituent integrals must converge independently. The choice of the splitting point c does not affect the convergence.

2.5.1.1 Benchmark: The p -Integral (Type I)

To determine the convergence of complex functions, we compare them to the power function $1/x^p$.

Theorem 2.5.1

The integral $\int_1^{+\infty} \frac{1}{x^p} dx$:

- **Converges** if $p > 1$.
- **Diverges** if $p \leq 1$.

Proof. Evaluating $\int_1^u x^{-p} dx$:

- If $p = 1$, $\ln u \rightarrow \infty$ as $u \rightarrow \infty$.
- If $p \neq 1$, $\frac{u^{1-p}-1}{1-p}$. For convergence, we need $u^{1-p} \rightarrow 0$, which requires $1-p < 0 \implies p > 1$.

□

2.5.2 Improper Integrals of the Second Kind (Unbounded Functions)

These integrals occur on a finite interval $[a, b]$ where the integrand $f(x)$ becomes infinite (has a singularity) at one or more points.

Definition 2.5.2. If f is continuous on $[a, b)$ and discontinuous at b (e.g., $\lim_{x \rightarrow b^-} |f(x)| = \infty$), we define:

$$\int_a^b f(x) dx = \lim_{\epsilon \rightarrow 0^+} \int_a^{b-\epsilon} f(x) dx$$

2.5.2.1 Benchmark: The p -Integral (Type II)

Be careful: the convergence condition for singularities is the *reverse* of infinite intervals.

Theorem 2.5.2

For the interval $(0, 1]$, the integral $\int_0^1 \frac{1}{x^p} dx$:

- **Converges** if $p < 1$.
- **Diverges** if $p \geq 1$.

2.5.3 General Theory of Convergence

Before applying practical tests, we establish the rigorous necessary and sufficient conditions for convergence.

2.5.3.1 Cauchy Criterion

The Cauchy Criterion is fundamental because it allows us to prove convergence without knowing the limit's value.

Theorem 2.5.3: Cauchy Criterion for Improper Integrals

The integral $\int_a^{+\infty} f(x) dx$ converges **if and only if** for every $\epsilon > 0$, there exists an $M > a$ such that for all $u_2 > u_1 > M$:

$$\left| \int_{u_1}^{u_2} f(x) dx \right| < \epsilon$$

2.5.3.2 Absolute vs. Conditional Convergence

- **Absolute Convergence:** $\int |f(x)| dx$ converges.
- **Conditional Convergence:** $\int f(x) dx$ converges, but $\int |f(x)| dx$ diverges.

Theorem 2.5.4

If $\int_a^{+\infty} |f(x)| dx$ converges, then $\int_a^{+\infty} f(x) dx$ converges.

2.5.4 Convergence Tests

2.5.4.1 Direct Comparison Test

Theorem 2.5.5

Let $f(x)$ and $g(x)$ be continuous functions on $[a, +\infty)$ such that $0 \leq f(x) \leq g(x)$ for all $x \geq a$.

1. If $\int_a^{+\infty} g(x) dx$ converges, then $\int_a^{+\infty} f(x) dx$ also converges.
2. If $\int_a^{+\infty} f(x) dx$ diverges, then $\int_a^{+\infty} g(x) dx$ also diverges.

Intuition: If the area under the larger curve is finite, the area under the smaller curve must be finite. If the area under the smaller curve is infinite, the larger area must be infinite.

Example 2.5.1

Does $\int_1^{+\infty} \frac{\sin^2 x}{x^2} dx$ converge?

Solution: We know that $0 \leq \sin^2 x \leq 1$. Therefore:

$$0 \leq \frac{\sin^2 x}{x^2} \leq \frac{1}{x^2}$$

We know that $\int_1^{+\infty} \frac{1}{x^2} dx$ converges ($p = 2 > 1$). By the Direct Comparison Test, $\int_1^{+\infty} \frac{\sin^2 x}{x^2} dx$ converges.

2.5.4.2 Limit Comparison Test

Sometimes finding a direct inequality is difficult. The Limit Comparison Test is often more powerful.

Theorem 2.5.6

Let $f(x)$ and $g(x)$ be positive continuous functions on $[a, +\infty)$. If:

$$\lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = L$$

where $0 < L < +\infty$, then $\int_a^{+\infty} f(x)dx$ and $\int_a^{+\infty} g(x)dx$ either both converge or both diverge.

If $L = 0$ and $g(x)$ converges, then $f(x)$ converges. If $L = +\infty$ and $g(x)$ diverges, then $f(x)$ diverges.

Example 2.5.2

Analyze $\int_1^{+\infty} \frac{x}{1+x^3} dx$.

Solution: For large x , the term 1 is negligible, so $\frac{x}{1+x^3} \approx \frac{x}{x^3} = \frac{1}{x^2}$.

Let $f(x) = \frac{x}{1+x^3}$ and $g(x) = \frac{1}{x^2}$.

$$\lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow +\infty} \frac{x/(1+x^3)}{1/x^2} = \lim_{x \rightarrow +\infty} \frac{x^3}{1+x^3} = 1$$

Since $L = 1$ (finite and positive) and we know $\int_1^{+\infty} \frac{1}{x^2} dx$ converges, the original integral converges.

For positive functions, we use comparison tests. For oscillating functions, we need more advanced tools.

2.5.4.3 Cauchy's Limit Comparison Test (Order Analysis)

This is the most practical method for determining convergence. It formalizes the idea of comparing a function to $1/x^p$.

Theorem 2.5.7

Let $f(x)$ be a positive function defined on $[a, +\infty)$. Consider the limit of $f(x)$ multiplied by a test power x^p :

$$\lambda = \lim_{x \rightarrow +\infty} x^p f(x)$$

1. **Convergence Case:** If we can find a $p > 1$ such that $0 \leq \lambda < +\infty$, then $\int_a^{+\infty} f(x) dx$ **converges**. (Meaning: $f(x)$ goes to zero faster than $1/x$, roughly like $1/x^p$)
2. **Divergence Case:** If we can find a $p \leq 1$ such that $0 < \lambda \leq +\infty$, then $\int_a^{+\infty} f(x) dx$ **diverges**. (Meaning: $f(x)$ goes to zero slower than or equal to $1/x$)

Example 2.5.3

Test $\int_1^{+\infty} \frac{\ln x}{x^2} dx$. We suspect convergence because x^2 dominates. Let's compare with $p = 1.5$ (since $1 < 1.5 < 2$, giving us "room").

$$\lim_{x \rightarrow +\infty} x^{1.5} \frac{\ln x}{x^2} = \lim_{x \rightarrow +\infty} \frac{\ln x}{x^{0.5}} = 0 \quad (\text{by L'Hopital})$$

Since $p = 1.5 > 1$ and the limit is finite, the integral converges.

2.5.4.4 Dirichlet and Abel Tests (For Conditional Convergence)

These tests are used for integrals of the product form $\int_a^{+\infty} f(x)g(x) dx$, typically where one part oscillates and the other decays.

Theorem 2.5.8: Dirichlet's Test

The integral $\int_a^{+\infty} f(x)g(x) dx$ converges if:

1. $f(x)$ has bounded partial integrals: $\exists M, \forall u > a, |\int_a^u f(t) dt| \leq M$.
2. $g(x)$ is monotonic decreasing and $\lim_{x \rightarrow +\infty} g(x) = 0$.

Classic Application: $\int_0^{+\infty} \frac{\sin x}{x} dx$. Here $f(x) = \sin x$ (bounded integral) and $g(x) = 1/x$ (monotonic to 0).

Theorem 2.5.9: Abel's Test

The integral $\int_a^{+\infty} f(x)g(x) dx$ converges if:

1. $\int_a^{+\infty} f(x) dx$ converges (the integral itself).
2. $g(x)$ is monotonic and bounded.

2.5.4.5 Cauchy Principal Value (P.V.)

In some divergent integrals, the positive and negative areas might cancel each other out if limits are taken symmetrically. This value is called the Cauchy Principal Value.

Definition 2.5.3. 1. For singularities at $c \in (a, b)$:

$$\text{P.V.} \int_a^b f(x) dx = \lim_{\epsilon \rightarrow 0^+} \left(\int_a^{c-\epsilon} f(x) dx + \int_{c+\epsilon}^b f(x) dx \right)$$

2. For infinite intervals $(-\infty, +\infty)$:

$$\text{P.V.} \int_{-\infty}^{+\infty} f(x) dx = \lim_{R \rightarrow +\infty} \int_{-R}^R f(x) dx$$

Remark 2.5.1

Convergence $\implies$ **P.V. exists**, but the converse is false.

Example 2.5.4

Consider $f(x) = \frac{1}{x}$ on $[-1, 1]$.

- **Improper Integral:** $\int_{-1}^1 \frac{1}{x} dx = \int_{-1}^0 \frac{1}{x} dx + \int_0^1 \frac{1}{x} dx$. Since $\int_{\epsilon}^1 \frac{1}{x} dx = -\ln \epsilon \rightarrow \infty$, the standard integral **diverges**.
- **Principal Value:**

$$\text{P.V.} \int_{-1}^1 \frac{1}{x} dx = \lim_{\epsilon \rightarrow 0^+} \left(\int_{-1}^{-\epsilon} \frac{1}{x} dx + \int_{\epsilon}^1 \frac{1}{x} dx \right) = 0$$

(Due to the odd symmetry of the function).

2.5.4.6 Comprehensive Example: The Gamma Function

$$\Gamma(s) = \int_0^{+\infty} x^{s-1} e^{-x} dx$$

This integral requires analysis of both singularity and infinite bounds.

- **At 0:** If $s < 1$, x^{s-1} has a singularity. Since $e^{-x} \approx 1$, it behaves like $\int_0^1 \frac{1}{x^{1-s}} dx$. By the Type II p -test, this converges if $1 - s < 1 \implies s > 0$.
- **At $+\infty$:** e^{-x} decays faster than any power x^p grows. Using the Limit Comparison Test with $1/x^2$:

$$\lim_{x \rightarrow \infty} x^2(x^{s-1}e^{-x}) = 0$$

Thus, it converges for all s .

Conclusion: The integral converges for $s > 0$.

At the end of this chapter, let's use the knowledge we learned to solve an interesting problem: the Gabriel's Horn.

Considering we have a horn, whose cross section is a circle but longitudinal section is function $y = \frac{1}{x}, x \in (1, \infty)$.

Let's first calculate the volume of the horn:

$$V = \pi \int_1^{+\infty} \left(\frac{1}{x}\right)^2 dx = \pi \left[-\frac{1}{x}\right]_1^{+\infty} = \pi$$

We can see that the horn have finite volume.

Then calculate the surface area of the horn.

$$S = 2\pi \int_1^{+\infty} \frac{1}{x} \sqrt{1 + \frac{1}{x^4}} dx \geq 2\pi \int_1^{+\infty} \frac{1}{x} dx = +\infty$$

The integration product of the surface area is divergent.

Here comes a very interesting paradox. We can use finite many paint to fill the horn, but this horn of paint cannot coated the inner wall of the horn.

This consequence is contradicting to our naive understanding.

As we close this chapter, remember that this is not an end, but a gateway. The journey of mathematical analysis continues: in the **Chapter 4**, we will extend these foundations to higher dimensions, study curves and surfaces in greater depth, and encounter even more beautiful and unexpected results. The horn's call, echoing from the realm of the infinite, invites us to explore further.

In later chapters, we will move on to more advanced topics in analysis, such as multi-variables calculus, differential equations, complex analysis and functional analysis. Each of these areas builds upon the concepts we have developed here, and each offers its own unique insights and challenges. We encourage readers to continue their exploration, armed with the rigorous tools and deep understanding gained from this chapter.

References:

Mathematical Analysis (Third Edition), Chen, J., Higher Education Press
The Real Numbers and Real Analysis, Ethan D. Bloch, Springer

2.6 Chapter Workshop

Worked example: An explicit epsilon–delta estimate

Prove that $x^2 \rightarrow 4$ as $x \rightarrow 2$. We need to control

$$|x^2 - 4| = |x - 2||x + 2|.$$

First restrict $|x - 2| < 1$, which implies $1 < x < 3$ and therefore $|x + 2| < 5$. It is then enough to require $5|x - 2| < \varepsilon$. Thus

$$\delta = \min \left\{ 1, \frac{\varepsilon}{5} \right\}$$

works. The preliminary bound is not cosmetic: it converts the moving factor $|x + 2|$ into a fixed constant.

Application: Numerical error and conditioning

If a differentiable function is evaluated with a small input error Δx , then

$$\Delta f \approx f'(x)\Delta x.$$

The derivative therefore measures local sensitivity. For relative errors, the dimensionless condition number

$$\kappa_f(x) = \left| \frac{xf'(x)}{f(x)} \right|$$

predicts how many digits may be lost. This connects the differential to numerical analysis rather than treating it only as symbolic notation.

2.7 Exercises

1. Give an epsilon proof that $3x - 1 \rightarrow 5$ as $x \rightarrow 2$.
2. Determine whether $a_n = (-1)^n/n$ converges, and whether $\sum a_n$ converges absolutely.
3. Use the mean value theorem to prove $|\sin x - \sin y| \leq |x - y|$.
4. Evaluate $\lim_{x \rightarrow \infty} \frac{x + \sin x}{x}$ without L'Hopital's rule.
5. Find the second-order Taylor approximation of e^x at 0 and bound the remainder on $[-0.1, 0.1]$.
6. Determine for which p the improper integral $\int_1^\infty x^{-p} dx$ converges.
7. Prove that $x^2 \rightarrow 9$ as $x \rightarrow 3$ using an epsilon–delta argument.
8. Decide whether the sequence $b_n = \frac{n}{n+1}$ converges, and find its limit if it does.
9. Show that $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$ without using L'Hopital's rule.

10. Test the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ for convergence and absolute convergence.
11. Compute the first-order Taylor polynomial of $\ln(1+x)$ at 0 and estimate the error for $|x| \leq 0.1$.
12. Determine whether $\int_0^1 x^{-p} dx$ converges, and specify the values of p for which it does.

Selected hints and answers

1. Since $|(3x-1)-5| = 3|x-2|$, take $\delta = \varepsilon/3$.
2. The sequence tends to 0. The series is conditionally convergent by the alternating-series test, but not absolutely because $\sum 1/n$ diverges.
3. Apply the mean value theorem to $\sin t$ on the interval between x and y , then use $|\cos \xi| \leq 1$.
4. Write the ratio as $1 + \sin x/x$ and squeeze the second term.
5. $1 + x + x^2/2$; the Lagrange remainder has magnitude at most $e^{0.1}|x|^3/6$.
6. It converges exactly when $p > 1$.
7. Factor $|x^2 - 9| = |x-3||x+3|$ and first bound $|x+3|$ by restricting x near 3.
8. Rewrite $\frac{n}{n+1} = 1 - \frac{1}{n+1}$ and pass to the limit.
9. Use $\tan x = \sin x / \cos x$ and the known limits for $\sin x/x$ and $\cos x$.
10. Compare $1/n^2$ with a p -series, or use the integral test.
11. The polynomial is $1+x$; the remainder is controlled by a second derivative bound for $\ln(1+x)$ on $|x| \leq 0.1$.
12. It converges exactly when $p < 1$.

Chapter 3

Linear Algebra

Learning objectives

By the end of this chapter, you should be able to:

- solve linear systems and interpret row reduction structurally;
- connect matrices with linear maps, coordinates, rank, and nullity;
- construct bases and orthogonal projections in finite-dimensional spaces;
- analyze eigenvalues, diagonalization, quadratic forms, and singular values;
- formulate least-squares, data-compression, and dynamical problems in linear-algebraic language.

So long after we finished the first part of the analysis section, we will now move on to the new chapter: Linear Algebra. It's a completely new part of mathematics.

Unlike analysis and calculus, linear algebra requires more geometric understanding. Here, proof is still important. But what is more crucial for learners is to construct geometric intuition for the subject. We would like to claim that intuition is different from imagination. Though we can only imagine the three dimensional Euclidean space, but our intuition can bring us forward to higher dimension, and more abstract linear spaces.

In this part, we will start from the topic of solving Systems of Linear Equations, then we will move on to study a special kind of linear space: the Vector Space. Finally, we will move on to a more abstract part: the Linear Space, this equipped us with a new tool to study more general forms of algebraic structure and system.

Specifically, in the final part focusing on abstract Linear Spaces, we will delve into key concepts such as linear independence, basis, dimension, and linear transformations. Understanding these foundational ideas allows us to unify seemingly disparate mathematical objects—like functions, polynomials, and matrices—under a single, powerful framework. This abstraction is not just an academic exercise; it provides the essential language and tools for tackling complex problems in fields ranging from differential equations and data science to quantum mechanics and engineering optimization. The geometric intuition developed in the study of $\mathbb{R}^n$ will prove indispensable as we navigate these higher-dimensional, abstract realms, cementing linear algebra as one of the most fundamental and broadly applicable areas of modern mathematics.

In latter chapters, we will introduce another branch of Algebra, the abstract algebra. Unlike Linear algebra that focused more on multivariate equations, abstract algebra study the solution structure of higher-degree equations, which is an even more abstract part of mathematics. But even in abstract algebra we still need the knowledge about linear algebra. So now, let's get started.

3.1 Linear Equations and Matrices

3.1.1 Systems of Linear Equations

I believe most of the readers have seen systems of equations like this:

$$\begin{cases} x + y = 1 \\ x - y = 0 \end{cases}$$

They might have different numbers of variables and equations, but they all shared the same feature: **Each equation is linear**. Variables are raised only to the first power, with no products between variables (like xy), or nonlinear functions (e.g., $\sin(x)$ or $\sqrt{x}$).

For such kind of equation system, we call it the **system of linear equations**. Their general expression has m equations and n variables (also called an $m \times n$ system):

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

Here, $x_1, \dots, x_n$ are the **variables** (unknowns), the a_{ij} are the **coefficients**, and $b_1, \dots, b_m$ are the **constant terms**.

A **solution** to the system is a set of numbers $(s_1, s_2, \dots, s_n)$ that satisfies all m equations simultaneously when substituted for $(x_1, x_2, \dots, x_n)$. The set of all possible solutions is called the **solution set**.

Example 3.1.1: A 2x2 System

Consider the system:

$$\begin{cases} x_1 + x_2 = 3 \\ x_1 - x_2 = -1 \end{cases}$$

We can solve this using simple algebra.

1. **Substitution:** From the first equation, $x_2 = 3 - x_1$. Substitute this into the second: $x_1 - (3 - x_1) = -1$, which gives $2x_1 - 3 = -1$, so $2x_1 = 2$, and $x_1 = 1$. Then $x_2 = 3 - 1 = 2$. The unique solution is $(1, 2)$.
2. **Elimination:** Add the two equations: $(x_1 + x_2) + (x_1 - x_2) = 3 + (-1)$, which gives $2x_1 = 2$, so $x_1 = 1$. Subtract the second from the first: $(x_1 + x_2) - (x_1 - x_2) = 3 - (-1)$, which gives $2x_2 = 4$, so $x_2 = 2$.

The solution set is the single point $(1, 2)$.

In linear algebra, we are interested in three questions:

1. **Existence:** Does a solution exist? (Is the system **consistent**?)
2. **Uniqueness:** If a solution exists, is it the only one?
3. **Computation:** If solutions exist, how do we find them?

Geometrically, for a 2x2 system, each equation represents a line in the $\mathbb{R}^2$ plane. The solution set is the intersection of these lines.

- **Unique Solution:** The lines intersect at a single point.

- **No Solution:** The lines are parallel and distinct.
- **Infinitely Many Solutions:** The two equations represent the same line.

For a 3x3 system, each equation is a plane in $\mathbb{R}^3$. The solution set is the intersection of these three planes, which could be a point, a line, a plane, or empty.

The methods of substitution and elimination become extremely cumbersome for larger systems (e.g., 5 equations, 5 variables). We need a more systematic and efficient approach. This is where matrices come in.

3.1.2 Matrix Algebra

3.1.2.1 Matrix Notation and Special Matrices

The essence of our new method is to manipulate the equations without changing their solution set. We observe that all the information of the system is contained in the coefficients a_{ij} and the constant terms b_i . The variable names $x_1, x_2, \dots$ are just placeholders. We can therefore encode the entire system into a compact rectangular array called a **matrix**.

Definition 3.1.1. A **matrix** is a rectangular array of numbers, called **entries** or **elements**. A matrix with m rows and n columns is called an $m \times n$ matrix (read "m by n").

For the general linear system, we define two key matrices.

The **coefficient matrix** is:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

We can denote this matrix as $A = [a_{ij}]$. a_{ij} represents the element in the i -th row and j -th column.

The **augmented matrix**, which includes the constant terms, is:

$$(A \mid \mathbf{b}) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right)$$

Solving the system is now equivalent to manipulating this augmented matrix.

Example 3.1.2

The system

$$\begin{cases} x_1 - 2x_2 + x_3 = 0 \\ 2x_2 - 8x_3 = 8 \\ 5x_1 - 5x_3 = 10 \end{cases}$$

has the coefficient matrix

$$A = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ 5 & 0 & -5 \end{pmatrix}$$

and the augmented matrix

$$(A \mid \mathbf{b}) = \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{array} \right)$$

Matrix Terminology and Special Matrices

- **Size/Dimension:** A matrix A with m rows and n columns has size $m \times n$.
- **Square Matrix:** A matrix is **square** if its number of rows equals its number of columns ($m = n$).
- **Equality:** Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are **equal** if and only if they have the same size ($m \times n$) and all their corresponding entries are equal ($a_{ij} = b_{ij}$ for all i, j).
- **Principal Diagonal:** In a square matrix, the entries $a_{11}, a_{22}, \dots, a_{nn}$ form the **principal diagonal** (or main diagonal).
- **Auxiliary Diagonal:** In a square matrix, the diagonal from the upper right to the lower left:

$$a_{1n}, a_{2,n-1}, \dots, a_{n1}$$

is the **auxiliary diagonal**.

There are several special types of matrices:

1. **Null matrix (Zero matrix):** A matrix (of any size) where all entries are 0. It is often denoted $\mathbf{0}$ or $\mathbf{0}_{m \times n}$.

$$\mathbf{0} = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}$$

2. **Identity matrix:** A square matrix I_n (or just I) whose entries on the principal diagonal are all 1, and all other entries are 0.

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The identity matrix is the multiplicative identity: $AI = A$ and $IA = A$ (for compatible sizes).

3. **Diagonal matrix:** A square matrix where all entries *off* the principal diagonal are 0.

$$D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

4. **Upper triangular matrix:** A square matrix whose entries *below* the principal diagonal are all 0 ($a_{ij} = 0$ for $i > j$).

$$U = \begin{pmatrix} 1 & 4 & -1 \\ 0 & 2 & 7 \\ 0 & 0 & 3 \end{pmatrix}$$

5. **Lower triangular matrix:** A square matrix whose entries *above* the principal diagonal are all 0 ($a_{ij} = 0$ for $i < j$).

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 5 & 2 & 0 \\ -1 & 0 & 3 \end{pmatrix}$$

6. **Transpose:** The **transpose** of an $m \times n$ matrix A , denoted A^T (or A'), is the $n \times m$ matrix obtained by interchanging its rows and columns. That is, $(A^T)_{ij} = A_{ji}$.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \implies A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

7. **Symmetric matrix:** A square matrix A such that $A^T = A$. This means $a_{ij} = a_{ji}$ for all i, j .

$$S = \begin{pmatrix} 1 & 5 & -1 \\ 5 & 2 & 0 \\ -1 & 0 & 3 \end{pmatrix}$$

8. **Skew-symmetric matrix:** A square matrix A such that $A^T = -A$. This means $a_{ij} = -a_{ji}$ (and $a_{ii} = 0$).

$$K = \begin{pmatrix} 0 & 5 & -1 \\ -5 & 0 & 2 \\ 1 & -2 & 0 \end{pmatrix}$$

3.1.2.2 Matrix Operations

We can define algebraic operations on matrices.

Matrix Addition and Scalar Multiplication

Definition 3.1.2. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two matrices of the **same size** $m \times n$.

1. **Addition:** Their sum $A + B$ is the $m \times n$ matrix $C = [c_{ij}]$ where $c_{ij} = a_{ij} + b_{ij}$.
2. **Scalar Multiplication:** Let c be a scalar (a real number). The scalar multiple cA is the $m \times n$ matrix $D = [d_{ij}]$ where $d_{ij} = c \cdot a_{ij}$.

Matrix subtraction is defined as $A - B = A + (-1)B$.

Example 3.1.3

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 0 \\ -1 & 7 \end{pmatrix}$. Then

$$A + B = \begin{pmatrix} 1+5 & 2+0 \\ 3-1 & 4+7 \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 2 & 11 \end{pmatrix}$$

$$3A = \begin{pmatrix} 3(1) & 3(2) \\ 3(3) & 3(4) \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix}$$

Note that $A + \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ is **undefined** as the sizes do not match.

These operations obey familiar properties:

Property 3.1.1: Properties of Addition and Scalar Multiplication

Let A, B, C be $m \times n$ matrices and c, d be scalars.

1. $A + B = B + A$ (Commutativity of Addition)
2. $(A + B) + C = A + (B + C)$ (Associativity of Addition)
3. $A + \mathbf{0} = A$ (Additive Identity)
4. $A + (-A) = \mathbf{0}$ (Additive Inverse)
5. $c(A + B) = cA + cB$ (Distributivity)
6. $(c + d)A = cA + dA$ (Distributivity)
7. $c(dA) = (cd)A$
8. $1A = A$

Remark 3.1.1

These 8 properties, plus closure, are precisely the axioms of a **Vector Space**. The set $M_{m \times n}$ of all $m \times n$ matrices is a prime example of a vector space.

Matrix Multiplication This operation is more complex and profoundly important.

Definition 3.1.3 (Matrix Multiplication). Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Their **product** AB is an $m \times p$ matrix $C = [c_{ij}]$. The entry c_{ij} in the i -th row and j -th column of AB is computed by taking the **dot product** of the i -th row of A and the j -th column of B .

$$c_{ij} = (\text{Row } i \text{ of } A) \cdot (\text{Column } j \text{ of } B) = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

$$c_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$$

Crucial Note: The product AB is only defined if the **number of columns in A** equals the **number of rows in B**.

$$(m \times n) \cdot (n \times p) \rightarrow (m \times p)$$

Example 3.1.4

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ (2×3) and $B = \begin{pmatrix} 7 & 8 \\ 9 & 0 \\ 1 & 2 \end{pmatrix}$ (3×2). The product AB will be a 2×2 matrix.

$$AB = \begin{pmatrix} (1 \cdot 7 + 2 \cdot 9 + 3 \cdot 1) & (1 \cdot 8 + 2 \cdot 0 + 3 \cdot 2) \\ (4 \cdot 7 + 5 \cdot 9 + 6 \cdot 1) & (4 \cdot 8 + 5 \cdot 0 + 6 \cdot 2) \end{pmatrix} = \begin{pmatrix} (7 + 18 + 3) & (8 + 0 + 6) \\ (28 + 45 + 6) & (32 + 0 + 12) \end{pmatrix} = \begin{pmatrix} 28 & 14 \\ 79 & 44 \end{pmatrix}$$

Now let's compute BA . This will be a 3×3 matrix.

$$BA = \begin{pmatrix} 7 & 8 \\ 9 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} (7 \cdot 1 + 8 \cdot 4) & (7 \cdot 2 + 8 \cdot 5) & (7 \cdot 3 + 8 \cdot 6) \\ (9 \cdot 1 + 0 \cdot 4) & (9 \cdot 2 + 0 \cdot 5) & (9 \cdot 3 + 0 \cdot 6) \\ (1 \cdot 1 + 2 \cdot 4) & (1 \cdot 2 + 2 \cdot 5) & (1 \cdot 3 + 2 \cdot 6) \end{pmatrix} = \begin{pmatrix} 39 & 54 & 69 \\ 9 & 18 & 27 \\ 9 & 12 & 15 \end{pmatrix}$$

Property 3.1.2: Properties of Matrix Multiplication

Let A, B, C be matrices of compatible sizes and c be a scalar.

1. **Warning:** $AB \neq BA$ in general. Matrix multiplication is **not commutative**. (See example above).
2. $A(BC) = (AB)C$ (Associativity)
3. $A(B + C) = AB + AC$ (Left Distributivity)
4. $(A + B)C = AC + BC$ (Right Distributivity)
5. $c(AB) = (cA)B = A(cB)$
6. $I_m A = A = A I_n$ (Multiplicative Identity)

But we still want the multiplication to have such kind of property like $AB = BA$, we will introduce a special kind of matrix in later chapters that can satisfy this property.

The proves of the properties above are left for readers.

Remark 3.1.2

Remember, $\mathbf{A}\mathbf{b} = \mathbf{0}$ can not deduce $A = 0$ or $b = 0$. This is another counterexample against our common sense of algebraic calculations.

Definition 3.1.4. Assume we have a square matrix with n orders, and a number $m \in \mathbb{N}^*$, then we call the product of m copies of A "A to the m -th power", denoted as A^m . Specifically, we define $A^0 = I_n$.

The calculations of power is **BASICALLY** the same with the regular rules. Except those have requirement with the orders of multiplication. Like:

$$(A + B)^2 = A^2 + AB + BA + B^2$$

$$(A + B)(A - B) = A^2 - AB + BA - B^2$$

Definition 3.1.5. Assume $A = (a_{ij})_{m \times n}$, we define $A^T = (b_{kl})_{n \times m}$ the transpose of matrix A , iff $b_{kl} = a_{lk}$, ($k = 1, 2, \dots, n, l = 1, 2, \dots, m$)

Property 3.1.3: Properties of the Transpose

1. $(A^T)^T = A$
2. $(A + B)^T = A^T + B^T$
3. $(cA)^T = cA^T$
4. **(Reversal Property)** $(AB)^T = B^T A^T$
5. $(A^m)^T = (A^T)^m$

Proof of $(AB)^T = B^T A^T$. Let A be $m \times n$ and B be $n \times p$. AB is $m \times p$, so $(AB)^T$ is $p \times m$. B^T is $p \times n$ and A^T is $n \times m$, so $B^T A^T$ is also $p \times m$. They have the same size. Let $C = AB$. The (i, j) -entry of C^T is C_{ji} . By definition, $C_{ji} = \sum_{k=1}^n A_{jk} B_{ki}$. Now let $D = B^T A^T$. The (i, j) -entry of D is:

$$D_{ij} = \sum_{k=1}^n (B^T)_{ik} (A^T)_{kj}$$

By definition of transpose, $(B^T)_{ik} = B_{ki}$ and $(A^T)_{kj} = A_{jk}$. So, $D_{ij} = \sum_{k=1}^n B_{ki} A_{jk} = \sum_{k=1}^n A_{jk} B_{ki}$. Thus, $D_{ij} = C_{ji} = (C^T)_{ij}$. Since all entries are equal, $B^T A^T = (AB)^T$. $\square$

There is also another important value for square matrix, we will use it in later contents.

Definition 3.1.6 (The trace of matrix). Assume a square matrix A with n orders $A = (a_{ij})$, $\sum_{i=1}^n a_{ii}$ is called the trace of matrix, denoted as $tr(A)$

Property 3.1.4

1. $tr(A + B) = tr(A) + tr(B)$
2. $tr(kA) = ktr(A)$
3. $tr(AB) = tr(BA)$
4. $tr(A^T) = tr(A)$

Definition of Block Matrices

A block matrix is a matrix that is partitioned into smaller submatrices called **blocks**. This is done by drawing horizontal and vertical lines that divide the matrix into rectangular blocks. Partitioning allows us to view a large matrix as composed of smaller, more manageable parts.

If A is an $m \times n$ matrix, we can partition it as follows:

$$A = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1q} \\ A_{21} & A_{22} & \cdots & A_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ A_{p1} & A_{p2} & \cdots & A_{pq} \end{bmatrix}$$

Here, each A_{ij} is a submatrix (block) of A , and the dimensions of the blocks must be consistent: the number of columns in A_{ik} must equal the number of columns in A_{jk} for all i, j, k , and similarly for rows.

Operations with Block Matrices

Block Matrix Addition If two matrices A and B are partitioned into blocks with the **same dimensions** for corresponding blocks, they can be added block-wise:

$$A + B = \begin{bmatrix} A_{11} + B_{11} & A_{12} + B_{12} & \cdots \\ A_{21} + B_{21} & A_{22} + B_{22} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Each block A_{ij} and B_{ij} must have the same dimensions for the addition to be valid.

Block Matrix Scalar Multiplication Scalar multiplication is performed by multiplying each block by the scalar:

$$cA = \begin{bmatrix} cA_{11} & cA_{12} & \cdots \\ cA_{21} & cA_{22} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Block Matrix Multiplication If A is an $m \times n$ block matrix and B is an $n \times p$ block matrix, and the partitions are such that the number of column blocks of A equals the number of row blocks of B , then the product $C = AB$ can be computed block-wise:

$$C_{ij} = \sum_{k=1}^q A_{ik} B_{kj}$$

This requires that the number of columns in A_{ik} equals the number of rows in B_{kj} for each k . The resulting block C_{ij} is the sum of products of corresponding blocks.

Example: If $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ and $B = \begin{bmatrix} B_{11} \\ B_{21} \end{bmatrix}$, then:

$$AB = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} \\ A_{21}B_{11} + A_{22}B_{21} \end{bmatrix}$$

Block Matrix Transpose The transpose of a block matrix is obtained by transposing each block and then transposing the block structure:

$$A^\top = \begin{bmatrix} A_{11}^\top & A_{21}^\top & \cdots \\ A_{12}^\top & A_{22}^\top & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Note that the positions of the blocks are also transposed (e.g., the block in the (1,2) position becomes the block in the (2,1) position after transposition).

Block Diagonal Matrices A block diagonal matrix is a square block matrix where all off-diagonal blocks are zero matrices:

$$A = \begin{bmatrix} A_{11} & 0 & \cdots & 0 \\ 0 & A_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_{pp} \end{bmatrix}$$

Operations on block diagonal matrices simplify because the blocks can be handled independently (e.g., the inverse of a block diagonal matrix is the block diagonal matrix of the inverses, if they exist).

Advantages of Using Block Matrices

- Simplifies operations on large matrices by breaking them into smaller parts.
- Facilitates parallel computation.
- Helps in proving theoretical results by induction on block structures.
- Commonly used in numerical linear algebra for efficient algorithms.

Definition 3.1.7. An $n \times n$ square matrix A is **invertible** (or **non-singular**) if there exists an $n \times n$ matrix B such that

$$AB = I_n \quad \text{and} \quad BA = I_n$$

This matrix B is unique and is called the **inverse** of A , denoted A^{-1} . If no such matrix B exists, A is **singular** (or **non-invertible**).

Example 3.1.5: Inverse of a 2x2 Matrix

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

If $ad - bc = 0$, A is singular. The quantity $ad - bc$ is the **determinant** of A .

Property 3.1.5: Properties of Inverses

Let A and B be invertible $n \times n$ matrices.

1. $(A^{-1})^{-1} = A$
2. $(AB)^{-1} = B^{-1}A^{-1}$ (Note the reversal of order)
3. $(A^T)^{-1} = (A^{-1})^T$
4. $(cA)^{-1} = \frac{1}{c}A^{-1}$ (for $c \neq 0$)

Proof of $(AB)^{-1} = B^{-1}A^{-1}$. We just need to check the definition.

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = A(I)A^{-1} = AA^{-1} = I$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}(I)B = B^{-1}B = I$$

Since $B^{-1}A^{-1}$ satisfies the definition, it must be the inverse of AB . □

3.1.3 Solving Systems of Linear Equations

Now we return to our main problem: solving m equations in n variables. The general idea, **Gaussian Elimination**, is to transform the augmented matrix into a simpler form from which we can just read off the solution.

3.1.3.1 Elementary Row Operations (EROs)

We can manipulate the augmented matrix using operations that correspond to manipulating the original equations. These operations **do not change the solution set**.

Definition 3.1.8. The three **elementary row operations (EROs)** are:

1. **(Replacement)** Add to one row a multiple of another row. ($R_i \rightarrow R_i + cR_j$)
2. **(Interchange)** Interchange two rows. ($R_i \leftrightarrow R_j$)
3. **(Scaling)** Multiply all entries in a row by a non-zero constant. ($R_i \rightarrow cR_i, c \neq 0$)

Definition 3.1.9. Two matrices A and B are **row equivalent**, denoted $A \sim B$, if B can be obtained from A by a sequence of EROs.

Theorem 3.1.1

If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.

Justification. • (Replacement) $R_i \rightarrow R_i + cR_j$ corresponds to adding c times equation j to equation i . This is a reversible step (by $R_i \rightarrow R_i - cR_j$), and any solution to the original system will also be a solution to the new one, and vice-versa.

- (Interchange) $R_i \leftrightarrow R_j$ corresponds to swapping the order of two equations, which clearly does not affect the solution set.
- (Scaling) $R_i \rightarrow cR_i$ (with $c \neq 0$) corresponds to multiplying an equation by c . This is reversible (by $R_i \rightarrow \frac{1}{c}R_i$), so it does not change the solution set.

□

3.1.3.2 Row Echelon Form and Rank

The goal is to use EROs to simplify the matrix into a "staircase" form.

Definition 3.1.10. A matrix is in **Row Echelon Form (REF)** if it satisfies:

1. All nonzero rows are above any rows of all zeros.
2. Each **leading entry** (or **pivot**), which is the leftmost nonzero entry of a row, is in a column to the right of the leading entry of the row above it.
3. All entries in a column *below* a leading entry are zeros.

Definition 3.1.11. A matrix is in **Reduced Row Echelon Form (RREF)** if it is in REF and also satisfies:

1. The leading entry in each nonzero row is 1.
2. Each leading 1 is the *only* nonzero entry in its column.

Example 3.1.6**REF:**

$$\begin{pmatrix} \boxed{2} & 3 & 4 & 5 \\ 0 & \boxed{1} & 6 & 7 \\ 0 & 0 & 0 & \boxed{8} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

RREF:

$$\begin{pmatrix} \boxed{1} & 0 & -1 & 0 \\ 0 & \boxed{1} & 2 & 0 \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(Pivots are boxed.)

Theorem 3.1.2

Every matrix is row equivalent to a **unique** Reduced Row Echelon Form (RREF).

This algorithm to get to REF/RREF is the core of our solution method.

Definition 3.1.12. • **Gaussian Elimination** is the process of using EROs to transform a matrix into REF.

- **Gauss-Jordan Elimination** is the process of using EROs to transform a matrix into RREF.

The Algorithm (Gauss-Jordan Elimination) Let's solve a system completely.

$$\begin{cases} x_2 + 3x_3 = 4 \\ x_1 + x_2 + x_3 = 1 \\ 2x_1 + 3x_2 + 4x_3 = 7 \end{cases}$$

The augmented matrix is:

$$\left(\begin{array}{ccc|c} 0 & 1 & 3 & 4 \\ 1 & 1 & 1 & 1 \\ 2 & 3 & 4 & 7 \end{array} \right)$$

This matrix is already in RREF, so the system is

$$\begin{cases} x_1 - 5x_3 = 1 \\ x_2 + x_3 = 4 \end{cases}$$

Here x_1 and x_2 are basic variables, and x_3 is free. Let $x_3 = t$. Then

$$x_1 = 1 + 5t, \quad x_2 = 4 - t,$$

so

$$\mathbf{x} = \begin{pmatrix} 1 + 5t \\ 4 - t \\ t \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix} + t \begin{pmatrix} 5 \\ -1 \\ 1 \end{pmatrix}.$$

This is a line in $\mathbb{R}^3$ through $(1, 4, 0)$ and parallel to $(5, -1, 1)$.

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & -1 & 1 \end{array} \right) \quad (R_3 \rightarrow R_3 - R_2)$$

The matrix is now in **Row Echelon Form**. This completes Gaussian Elimination. We could stop here and use **back substitution**: From R_3 : $-x_3 = 1 \implies x_3 = -1$. From R_2 : $x_2 + 3x_3 = 4 \implies x_2 + 3(-1) = 4 \implies x_2 = 7$. From R_1 : $x_1 + x_2 + x_3 = 1 \implies x_1 + 7 + (-1) = 1 \implies x_1 = -5$. The unique solution is $(-5, 7, -1)$.

Step 2: (Backward Phase - Get to RREF) Continue from the REF. Scale all pivots to 1.

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & -1 \end{array} \right) \quad (R_3 \rightarrow -1 \cdot R_3)$$

Create zeros *above* the pivots, starting from the rightmost pivot.

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{array} \right) \quad (R_1 \rightarrow R_1 - R_3, R_2 \rightarrow R_2 - 3R_3)$$

Create zero above the second pivot.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{array} \right) \quad (R_1 \rightarrow R_1 - R_2)$$

This is the **Reduced Row Echelon Form**. The corresponding system is:

$$\begin{cases} x_1 = -5 \\ x_2 = 7 \\ x_3 = -1 \end{cases}$$

This immediately gives the solution.

Rank of a Matrix A key concept emerges from the echelon form.

Definition 3.1.13. The **rank** of a matrix A , denoted $\text{rank}(A)$, is the number of leading entries (pivots) in its row echelon form. This number is unique for any given matrix.

Property 3.1.6: Properties of Rank

Let A be an $m \times n$ matrix.

1. $\text{rank}(A) \leq \min(m, n)$.
2. $\text{rank}(A) = 0$ if and only if $A = \mathbf{0}$.
3. (**Major Theorem**) $\text{rank}(A) = \text{rank}(A^T)$. (The number of pivot rows equals the number of pivot columns).
4. $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$.
5. $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$.
6. $\text{rank}(A) + \text{rank}(B) - n \leq \text{rank}(AB)$.
7. If P, Q are invertible, $\text{rank}(PAQ) = \text{rank}(A)$. EROs are equivalent to multiplying by an invertible matrix on the left, so row operations do not change the rank.

In short, we can denote them as:

$$\min\{r(A), r(B)\} \geq r(AB) \geq r(A) + r(B) - n$$

$$r(A + B) \leq r(A, B) \leq r \begin{pmatrix} A & O \\ O & B \end{pmatrix}$$

3.1.3.3 Solution Sets of Linear Systems

The RREF of the *augmented* matrix tells us everything about the solution set. A **pivot column** is a column in the coefficient matrix A that contains a pivot in its RREF. Variables corresponding to pivot columns are called **basic variables**. Variables corresponding to non-pivot columns are called **free variables**.

Let $r = \text{rank}(A)$ for an $m \times n$ coefficient matrix A . We analyze the RREF of the augmented matrix $[A \mid \mathbf{b}]$.

1. **No Solution (Inconsistent System)** This occurs if the RREF of $[A \mid \mathbf{b}]$ has a row of the form $(0 \ 0 \ \cdots \ 0 \mid 1)$. This corresponds to the impossible equation $0x_1 + \cdots + 0x_n = 1$, or $0 = 1$. In terms of rank, this means the last column (the augmented column) is a pivot column. **Condition:** $\text{rank}(A) < \text{rank}([A \mid \mathbf{b}])$.
2. **A Solution Exists (Consistent System)** This occurs if the augmented column is *not* a pivot column. **Condition:** $\text{rank}(A) = \text{rank}([A \mid \mathbf{b}])$. Let this rank be r .
 - **Unique Solution:** The system has a unique solution if there are *no free variables*. This means every variable is a basic variable, so every column of A is a pivot column. **Condition:** $r = n$ (the number of variables).
 - **Infinitely Many Solutions:** The system has infinitely many solutions if there is *at least one free variable*. This means some columns of A are not pivot columns. **Condition:** $r < n$ (the number of variables). The $n - r$ free variables can be set to any arbitrary value (parameters), and the basic variables can be expressed in terms of them.

Parametric Vector Form When we have infinitely many solutions, we write the solution set in **parametric vector form**.

Example 3.1.7: Infinitely Many Solutions

Find the general solution to the system with augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 0 & -5 & 1 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

This matrix is already in RREF. The corresponding system is:

$$\begin{cases} x_1 - 5x_3 = 1 \\ x_2 + x_3 = 4 \\ 0 = 0 \end{cases}$$

The pivot columns are 1 and 2. So, x_1 and x_2 are **basic variables**. Column 3 is not a pivot column. So, x_3 is a **free variable**. We introduce a parameter, t , for the free variable. Let $x_3 = t$, where t can be any real number. Now, we express the basic variables in terms of the free variables:

$$x_1 = 1 + 5x_3 = 1 + 5t$$

$$x_2 = 4 - x_3 = 4 - t$$

$$x_3 = t$$

The general solution $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ is:

$$\mathbf{x} = \begin{pmatrix} 1 + 5t \\ 4 - t \\ t \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix} + t \begin{pmatrix} 5 \\ -1 \\ 1 \end{pmatrix}$$

3.1.3.4 Homogeneous and Non-homogeneous Systems

Definition 3.1.14. A system of linear equations is called **homogeneous** if it is of the form $A\mathbf{x} = \mathbf{0}$, where $\mathbf{0}$ is the zero vector (all $b_i = 0$).

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = 0 \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = 0 \end{cases}$$

A system $A\mathbf{x} = \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$ is called **non-homogeneous**.

A homogeneous system $A\mathbf{x} = \mathbf{0}$ is **always** consistent, because $\mathbf{x} = \mathbf{0}$ (the zero vector) is always a solution, known as the **trivial solution**. The important question is whether a **non-trivial solution** exists. This happens if and only if there is at least one free variable, which is equivalent to $\text{rank}(A) < n$ (the number of variables).

Theorem 3.1.3

The homogeneous system $A\mathbf{x} = \mathbf{0}$ has a non-trivial solution if and only if $\text{rank}(A) < n$.

Corollary 3.1.1

If A is $m \times n$ with $m < n$ (fewer equations than variables, a "wide" matrix), then $A\mathbf{x} = \mathbf{0}$ **always** has infinitely many solutions, because $\text{rank}(A) \leq m < n$.

There is a fundamental connection between the solution sets of the two systems.

Theorem 3.1.4: Structure of Solutions

Suppose the non-homogeneous system $A\mathbf{x} = \mathbf{b}$ is consistent and has a particular solution $\mathbf{x}_p$. Then the general solution $\mathbf{x}_g$ of $A\mathbf{x} = \mathbf{b}$ is the set of all vectors of the form

$$\mathbf{x}_g = \mathbf{x}_p + \mathbf{x}_h$$

where $\mathbf{x}_h$ is any solution to the corresponding homogeneous system $A\mathbf{x} = \mathbf{0}$.

Proof. Let $\mathbf{x}_g$ be any solution to $A\mathbf{x} = \mathbf{b}$. Let $\mathbf{x}_h = \mathbf{x}_g - \mathbf{x}_p$. Then $A\mathbf{x}_h = A(\mathbf{x}_g - \mathbf{x}_p) = A\mathbf{x}_g - A\mathbf{x}_p = \mathbf{b} - \mathbf{b} = \mathbf{0}$. So, $\mathbf{x}_h$ is a solution to the homogeneous system. This shows any solution $\mathbf{x}_g$ can be written in the form $\mathbf{x}_p + \mathbf{x}_h$. Conversely, let $\mathbf{x}_h$ be any homogeneous solution. Then $A(\mathbf{x}_p + \mathbf{x}_h) = A\mathbf{x}_p + A\mathbf{x}_h = \mathbf{b} + \mathbf{0} = \mathbf{b}$. So, $\mathbf{x}_p + \mathbf{x}_h$ is a solution to the non-homogeneous system. $\square$

Remark 3.1.3

Look back at our last example:

$$\mathbf{x} = \underbrace{\begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix}}_{\mathbf{x}_p} + t \underbrace{\begin{pmatrix} 5 \\ -1 \\ 1 \end{pmatrix}}_{\mathbf{x}_h}$$

Here $\mathbf{x}_p = (1, 4, 0)$ is one *particular solution* to $A\mathbf{x} = \mathbf{b}$. $\mathbf{x}_h = t(5, -1, 1)$ is the *general solution* to the corresponding homogeneous system $A\mathbf{x} = \mathbf{0}$. Geometrically, the solution set to $A\mathbf{x} = \mathbf{b}$ is a *translation* (by $\mathbf{x}_p$) of the solution set to $A\mathbf{x} = \mathbf{0}$.

3.2 Determinants

We now study a powerful tool associated with **square** matrices: the determinant. The determinant is a single number that reveals a wealth of information about a matrix, most notably whether it is invertible.

The calculation of determinants require familiarity and patience, and once we can find other ways to avoid using determinants, we shall do so.

3.2.1 The Determinant of a Matrix

For a 1×1 matrix $A = (a)$, $\det(A) = a$. For a 2×2 matrix, the determinant is simple:

$$\det(A) = \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

This value $ad - bc$ is non-zero if and only if the matrix is invertible.

For larger $n \times n$ matrices, we define the determinant recursively using **cofactor expansion**.

Definition 3.2.1. Let A be an $n \times n$ matrix.

- The **minor** M_{ij} of the entry a_{ij} is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting the i -th row and j -th column of A .
- The **cofactor** C_{ij} is given by $C_{ij} = (-1)^{i+j} M_{ij}$.

The "checkerboard" pattern of signs for $(-1)^{i+j}$ is
$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

Theorem 3.2.1: Cofactor Expansion

The determinant of an $n \times n$ matrix A can be found by expanding along **any** row i :

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in} = \sum_{j=1}^n a_{ij}C_{ij}$$

Alternatively, we can expand down **any** column j :

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj} = \sum_{i=1}^n a_{ij}C_{ij}$$

Example 3.2.1: Cofactor Expansion of 3x3

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. Let's expand along Row 1.

$$\det(A) = 1 \cdot C_{11} + 2 \cdot C_{12} + 3 \cdot C_{13}$$

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} = +1(5 \cdot 9 - 6 \cdot 8) = 45 - 48 = -3$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} = -1(4 \cdot 9 - 6 \cdot 7) = -(36 - 42) = 6$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} = +1(4 \cdot 8 - 5 \cdot 7) = 32 - 35 = -3$$

$$\det(A) = 1(-3) + 2(6) + 3(-3) = -3 + 12 - 9 = 0$$

Since $\det(A) = 0$, this matrix is **singular** (not invertible).

A more formal definition of determinants relies on the concept of negative sequence. And it is logically equivalent to the definition above so we won't present it again. But here is a more axiomatized definitions about determinant I would like to share with you:

Definition 3.2.2 (The axiomatic definition of determinants). The determinant is the unique function $\det: M_n(\mathbb{R}) \leftarrow \mathbb{R}$ satisfying the following three axioms:

1. **Multilinearity:** It is a linear function of each column when the other columns are held fixed.
2. **Alternating:** If two columns of the matrix are identical, then its determinant is zero. This also implies that swapping two columns changes the sign of the determinant.
3. **Normalization:** The determinant of the identity matrix is 1.

Another definition is more **modern**, it's about exterior product

Definition 3.2.3 (Exterior Product (Wedge Product)). Let V be a vector space over field $\mathbb{K}$. The **exterior product** (or **wedge product**) is a bilinear map:

$$\wedge : V \times V \rightarrow \Lambda^2(V)$$

satisfying:

1. **Anticommutativity:** $u \wedge v = -v \wedge u$ for all $u, v \in V$
2. **Nilpotence:** $v \wedge v = 0$ for all $v \in V$

The k -th exterior power $\Lambda^k(V)$ is spanned by elements of the form $v_1 \wedge v_2 \wedge \cdots \wedge v_k$ where $v_i \in V$.

Definition 3.2.4 (Determinant via Exterior Algebra). Let V be an n -dimensional vector space with basis $\{e_1, \dots, e_n\}$. A linear operator $T: V \rightarrow V$ induces $\Lambda^n T: \Lambda^n(V) \rightarrow \Lambda^n(V)$ on the top exterior power:

$$\Lambda^n T(e_1 \wedge \cdots \wedge e_n) = T(e_1) \wedge \cdots \wedge T(e_n)$$

Since $\Lambda^n(V)$ is 1-dimensional, there exists a unique scalar $\det(T) \in \mathbb{K}$ such that:

$$T(e_1) \wedge \cdots \wedge T(e_n) = \det(T) \cdot (e_1 \wedge \cdots \wedge e_n)$$

This scalar $\det(T)$ is called the **determinant** of T .

For matrix $A = (a_{ij})$ with column vectors $a_1, \dots, a_n \in \mathbb{R}^n$:

$$a_1 \wedge a_2 \wedge \cdots \wedge a_n = \det(A) \cdot (e_1 \wedge e_2 \wedge \cdots \wedge e_n)$$

Theorem 3.2.2

The exterior algebra definition implies the axiomatic definition of determinant.

Proof. We verify the three axioms:

1. **Multilinearity:** The wedge product is linear in each argument:

$$(\lambda u + \mu v) \wedge w = \lambda(u \wedge w) + \mu(v \wedge w)$$

Thus $(a_1, \dots, a_n) \mapsto a_1 \wedge \dots \wedge a_n$ is multilinear, and so is $\det(A)$.

2. **Alternating property:** If $a_i = a_j$ ($i \neq j$), then:

$$a_1 \wedge \dots \wedge a_i \wedge \dots \wedge a_j \wedge \dots \wedge a_n = 0$$

since $v \wedge v = 0$. Hence $\det(A) = 0$. Swapping columns introduces a sign change due to anticommutativity.

3. **Normalization:** For identity matrix I :

$$e_1 \wedge \dots \wedge e_n = \det(I) \cdot (e_1 \wedge \dots \wedge e_n) \Rightarrow \det(I) = 1$$

□

Corollary 3.2.1

The multiplicative property $\det(AB) = \det(A) \det(B)$ follows naturally.

Proof. Consider the composition on $\Lambda^n(V)$:

$$\Lambda^n(AB)(e_1 \wedge \dots \wedge e_n) = \Lambda^n A(\Lambda^n B(e_1 \wedge \dots \wedge e_n)) = \det(A) \det(B) (e_1 \wedge \dots \wedge e_n)$$

But also equals $\det(AB)(e_1 \wedge \dots \wedge e_n)$, so $\det(AB) = \det(A) \det(B)$.

□

Remark 3.2.1: Sarrus's Rule for 3x3

For 3×3 matrices *only*, there is a shortcut. Write the first two columns again to the right:

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{matrix} a & b \\ d & e \\ g & h \end{matrix}$$

Sum the products of the down-right diagonals and subtract the products of the up-right diagonals:

$$\det = (aei + bfg + cdh) - (gec + hfa + idb)$$

Using our example: $(1 \cdot 5 \cdot 9 + 2 \cdot 6 \cdot 7 + 3 \cdot 4 \cdot 8) - (7 \cdot 5 \cdot 3 + 8 \cdot 6 \cdot 1 + 9 \cdot 4 \cdot 2) = (45 + 84 + 96) - (105 + 48 + 72) = 225 - 225 = 0$. **Warning:** This *does not* work for 4x4 or larger.

3.2.2 Properties of Determinants

Calculating determinants via cofactors is computationally slow ($O(n!)$). A more efficient method ($O(n^3)$) uses row operations.

Theorem 3.2.3: Determinants and EROs

Let A be an $n \times n$ matrix.

1. **(Replacement)** If B is obtained from A by $R_i \rightarrow R_i + cR_j$, then $\det(B) = \det(A)$.
2. **(Interchange)** If B is obtained from A by $R_i \leftrightarrow R_j$, then $\det(B) = -\det(A)$.
3. **(Scaling)** If B is obtained from A by $R_i \rightarrow cR_i$, then $\det(B) = c \cdot \det(A)$.

This allows us to row-reduce A to an echelon form U (which is triangular) while keeping track of the changes.

Theorem 3.2.4

If A is a triangular matrix (upper or lower), its determinant is the product of its diagonal entries.

$$\det(A) = a_{11}a_{22} \cdots a_{nn}$$

Proof. Expand cofactors along the first row (if lower triangular) or first column (if upper triangular) repeatedly. $\square$

Example 3.2.2: Calculating det with EROs

$$A = \begin{pmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{pmatrix}$$

$$\det(A) = - \begin{vmatrix} 3 & -6 & 9 \\ 0 & 1 & 5 \\ 2 & 6 & 1 \end{vmatrix} \quad (R_1 \leftrightarrow R_2)$$

$$\det(A) = -3 \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 2 & 6 & 1 \end{vmatrix} \quad (R_1 \rightarrow \frac{1}{3}R_1, \text{ pull out } 3)$$

$$\det(A) = -3 \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 10 & -5 \end{vmatrix} \quad (R_3 \rightarrow R_3 - 2R_1)$$

$$\det(A) = -3 \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & -55 \end{vmatrix} \quad (R_3 \rightarrow R_3 - 10R_2)$$

The matrix is now triangular.

$$\det(A) = -3 \cdot (1 \cdot 1 \cdot -55) = 165$$

Property 3.2.1: More Properties of Determinants

Let A, B be $n \times n$ matrices.

1. **(Major Theorem)** A is invertible if and only if $\det(A) \neq 0$.
2. **(Multiplicative Property)** $\det(AB) = \det(A) \det(B)$.
3. $\det(A^T) = \det(A)$. (This implies all ERO properties also work for *columns*).
4. If A is invertible, $\det(A^{-1}) = \frac{1}{\det(A)}$.
5. $\det(cA) = c^n \det(A)$ (where A is $n \times n$).
6. If A has a zero row (or column), $\det(A) = 0$.
7. If A has two identical rows (or columns), $\det(A) = 0$.

Proof of $\det(A^{-1}) = 1/\det(A)$. $AA^{-1} = I$. $\det(AA^{-1}) = \det(I)$. $\det(A) \det(A^{-1}) = 1$. $\det(A^{-1}) = \frac{1}{\det(A)}$. (This requires $\det(A) \neq 0$, which is true since A is invertible). $\square$

3.2.3 Cramer's Rule and Adjoint Formula

Determinants provide explicit formulas for solving $A\mathbf{x} = \mathbf{b}$ and finding A^{-1} . While elegant, they are computationally *inefficient* for large matrices compared to elimination.

Theorem 3.2.5: Cramer's Rule

Let A be an invertible $n \times n$ matrix. For any $\mathbf{b}$ in $\mathbb{R}^n$, the unique solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ has entries given by

$$x_i = \frac{\det(A_i(\mathbf{b}))}{\det(A)}, \quad \text{for } i = 1, 2, \dots, n$$

where $A_i(\mathbf{b})$ is the matrix obtained from A by replacing its i -th column with the vector $\mathbf{b}$.

Example 3.2.3

Solve $\begin{cases} 2x_1 + 5x_2 = -1 \\ 3x_1 + 7x_2 = 4 \end{cases}$ $A = \begin{pmatrix} 2 & 5 \\ 3 & 7 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$ $\det(A) = 2(7) - 5(3) = 14 - 15 = -1$. $A_1(\mathbf{b}) = \begin{pmatrix} -1 & 5 \\ 4 & 7 \end{pmatrix}, \det(A_1(\mathbf{b})) = -7 - 20 = -27$. $A_2(\mathbf{b}) = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \det(A_2(\mathbf{b})) = 8 - (-3) = 11$. $x_1 = \frac{-27}{-1} = 27$. $x_2 = \frac{11}{-1} = -11$.

Definition 3.2.5 (Adjoint Matrix). Let $C = [C_{ij}]$ be the matrix of cofactors of A . The **adjoint** (or **adjugate**) of A , denoted $\text{adj}(A)$, is the **transpose** of the cofactor matrix.

$$\text{adj}(A) = C^T$$

Theorem 3.2.6: Inverse Formula

Let A be an invertible matrix. Then

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

Remark 3.2.2

This theorem explains the 2×2 inverse formula. For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$: $C_{11} = d, C_{12} = -c, C_{21} = -b, C_{22} = a$. Cofactor Matrix $C = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}$. Adjoint Matrix $\text{adj}(A) = C^T = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. $A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

3.2.4 The Invertible Matrix Theorem (IMT)

This is one of the most important theorems in linear algebra. It links all the major concepts we have seen so far for a **square** $n \times n$ matrix A .

Theorem 3.2.7: The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix. The following statements are equivalent (that is, if any one is true, they are all true, and if any one is false, they are all false).

1. A is an invertible matrix.
2. A is row equivalent to the identity matrix I_n .
3. The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution ($\mathbf{x} = \mathbf{0}$).
4. The columns of A form a linearly independent set.
5. The linear transformation $T(\mathbf{x}) = A\mathbf{x}$ is one-to-one.
6. The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for *each* $\mathbf{b}$ in $\mathbb{R}^n$.
7. The columns of A span $\mathbb{R}^n$.
8. The linear transformation $T(\mathbf{x}) = A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
9. There is an $n \times n$ matrix C such that $CA = I_n$.
10. There is an $n \times n$ matrix D such that $AD = I_n$.
11. $\det(A) \neq 0$.
12. $\text{rank}(A) = n$.
13. $\text{Nul}(A) = \{\mathbf{0}\}$ (The null space is the zero vector).
14. $\text{Col}(A) = \mathbb{R}^n$ (The column space is all of $\mathbb{R}^n$).
15. 0 is not an eigenvalue of A . (We will see this later).

This theorem is a powerful diagnostic tool. To check if a square matrix is invertible, we only need to verify *one* of these conditions. For example, checking if $\det(A) \neq 0$ is often the fastest way.

3.3 Vectors in $\mathbb{R}^n$

We now introduce a new and fundamental object: the vector. This allows us to re-interpret systems of equations in a powerful, geometric way.

3.3.1 Vectors and Operations

Geometrically, in two ($\mathbb{R}^2$) or three ($\mathbb{R}^3$) dimensions, we can think of a vector as an arrow with a specific length and direction.

Algebraically, we define a **vector** in $\mathbb{R}^n$ (read: "R-n") as an ordered n -tuple of real numbers. We typically

write it as a **column vector**:

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

The set $\mathbb{R}^n$ is the collection of all such n -dimensional vectors. $\mathbb{R}^2$ is the set of all vectors $\begin{pmatrix} x \\ y \end{pmatrix}$, which we identify with the 2D Cartesian plane. A vector $\mathbf{v} = (v_1, \dots, v_n)$ can also be written as a **row vector**, but column vectors are standard when working with matrix equations.

We define two fundamental operations on vectors. Let $\mathbf{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$ be vectors in $\mathbb{R}^n$ and let c be a real number (a **scalar**).

1. **Vector Addition:** $\mathbf{u} + \mathbf{v}$ is found by adding corresponding components:

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{pmatrix}$$

Geometrically, this corresponds to the **Parallelogram Law**.

2. **Scalar Multiplication:** $c\mathbf{v}$ is found by multiplying each component by c :

$$c\mathbf{v} = \begin{pmatrix} cv_1 \\ \vdots \\ cv_n \end{pmatrix}$$

Geometrically, this scales the length of the vector by $|c|$ and reverses its direction if $c < 0$.

These operations satisfy the 8 properties (associativity, commutativity, etc.) listed in Section 2.2.1, making $\mathbb{R}^n$ a prime example of a vector space.

3.3.2 Dot Product, Norm, and Orthogonality

Beyond addition and scaling, we can define a product that gives a scalar in $\mathbb{R}^n$.

Definition 3.3.1 (Dot Product). The **dot product** (or **inner product**) of $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$ is:

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n = \sum_{i=1}^n u_iv_i$$

Note: $\mathbf{u} \cdot \mathbf{v}$ is a **scalar**, not a vector. We can also write this using matrix multiplication: $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$.

Property 3.3.1: Properties of the Dot Product

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ (Commutative)
2. $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$ (Distributive)
3. $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v})$
4. $\mathbf{u} \cdot \mathbf{u} \geq 0$, and $\mathbf{u} \cdot \mathbf{u} = 0 \iff \mathbf{u} = \mathbf{0}$.

Definition 3.3.2 (Norm and Distance). 1. The **norm** (or **length**) of a vector $\mathbf{v}$ is:

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

2. A vector $\mathbf{u}$ with $\|\mathbf{u}\| = 1$ is called a **unit vector**.
3. **Normalizing** a vector $\mathbf{v} \neq \mathbf{0}$ means finding the unit vector in its direction: $\mathbf{u} = \frac{1}{\|\mathbf{v}\|} \mathbf{v}$.
4. The **distance** between $\mathbf{u}$ and $\mathbf{v}$ is $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.

However, when dealing with abstract vector spaces, we may not have a natural dot product. In such cases, we can define an **inner product** that satisfies the same properties as the dot product. We shall see how to do this in later sections.

Definition 3.3.3 (Orthogonality). Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are **orthogonal** (perpendicular) if their dot product is zero:

$$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$$

The zero vector $\mathbf{0}$ is orthogonal to every vector in $\mathbb{R}^n$.

Likewise, we can define orthogonality in inner product spaces and weighted dot product spaces by replacing the dot product with the inner product or weighted dot product.

Theorem 3.3.1: Pythagorean Theorem

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \text{ if and only if } \mathbf{u} \cdot \mathbf{v} = 0.$$

Proof. $\|\mathbf{u} + \mathbf{v}\|^2 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = \mathbf{u} \cdot \mathbf{u} + \mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{u} + \mathbf{v} \cdot \mathbf{v} = \|\mathbf{u}\|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2$. The equality holds iff $2(\mathbf{u} \cdot \mathbf{v}) = 0$, which means $\mathbf{u} \cdot \mathbf{v} = 0$. $\square$

The dot product also defines the angle between two vectors.

Theorem 3.3.2

For $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$,

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$.

This leads to a famous inequality:

Theorem 3.3.3: Cauchy-Schwarz Inequality

For all $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$,

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

Theorem 3.3.4: Triangle Inequality

For all $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$,

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$$

3.3.3 Linear Combinations and Span

This is one of the most important ideas in the entire subject.

Definition 3.3.4. A **linear combination** of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ in $\mathbb{R}^n$ is any vector $\mathbf{y}$ of the form:

$$\mathbf{y} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_p \mathbf{v}_p$$

where $c_1, \dots, c_p$ are any scalars (also called weights).

Example 3.3.1

In $\mathbb{R}^3$, let $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. $\mathbf{y} = 3\mathbf{v}_1 + (-2)\mathbf{v}_2 = 3\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - 2\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ 0 \end{pmatrix}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2$. But $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is *not* a linear combination of $\mathbf{v}_1, \mathbf{v}_2$.

Definition 3.3.5. The set of **all possible** linear combinations of $\mathbf{v}_1, \dots, \mathbf{v}_p$ is called the **Span** of these vectors, denoted $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$.

Geometrically, the span has a simple interpretation:

- $\text{Span}\{\mathbf{v}\}$ (for $\mathbf{v} \neq \mathbf{0}$) is the line through the origin and $\mathbf{v}$.
- $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ (for non-collinear $\mathbf{u}, \mathbf{v}$) is the plane containing the origin, $\mathbf{u}$, and $\mathbf{v}$.
- $\text{Span}\{\mathbf{0}\}$ is just the set $\{\mathbf{0}\}$, the origin.

3.3.4 The Matrix Equation $A\mathbf{x} = \mathbf{b}$

We can now connect our topics. Let A be an $m \times n$ matrix. We can view its columns as n vectors in $\mathbb{R}^m$:

$A = (\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n)$. Let $\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ be a vector in $\mathbb{R}^n$.

Definition 3.3.6 (Matrix-Vector Product). The product of the $m \times n$ matrix A and the $n \times 1$ vector $\mathbf{x}$, denoted $A\mathbf{x}$, is defined as the **linear combination of the columns of A using the entries of $\mathbf{x}$ as weights**:

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$$

This product results in an $m \times 1$ vector (a vector in $\mathbb{R}^m$).

Example 3.3.2

$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ -1 \end{pmatrix} = 5\begin{pmatrix} 1 \\ 3 \end{pmatrix} - 1\begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 5 \\ 15 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 11 \end{pmatrix}$ Note: This matches the row-column rule for matrix multiplication: $\begin{pmatrix} 1(5) + 2(-1) \\ 3(5) + 4(-1) \end{pmatrix} = \begin{pmatrix} 3 \\ 11 \end{pmatrix}$.

Now look at our original system of equations:

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m \end{cases}$$

The left side can be written as a vector equation:

$$x_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} + x_2 \begin{pmatrix} a_{12} \\ \vdots \\ a_{m2} \end{pmatrix} + \cdots + x_n \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}$$

Using our new definitions, this is precisely:

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$$

Which is identical to the **matrix equation**:

$$A\mathbf{x} = \mathbf{b}$$

This gives us three equivalent ways to view the same problem:

1. A system of m linear equations in n variables.
2. A vector equation $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$.
3. A matrix equation $A\mathbf{x} = \mathbf{b}$.

This is a profound re-interpretation! The question "Does the system $A\mathbf{x} = \mathbf{b}$ have a solution?" is identical to the question:

"Is the vector $\mathbf{b}$ a linear combination of the column vectors of A ?"

Or, more simply: **"Is $\mathbf{b}$ in $\text{Span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$?"**

Theorem 3.3.5

The equation $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{b}$ is in the span of the columns of A . This span is called the **Column Space** of A , denoted $\text{Col}(A)$.

3.3.5 Linear Independence

We now ask a related question. What if $\mathbf{b} = \mathbf{0}$? The equation $A\mathbf{x} = \mathbf{0}$ (or $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$) is the **homogeneous equation**. We know this *always* has the **trivial solution** $\mathbf{x} = \mathbf{0}$ (i.e., $x_1 = 0, \dots, x_n = 0$). But does it have *only* the trivial solution?

Definition 3.3.7. A set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is said to be **linearly independent** if the vector equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$$

has **only** the trivial solution ($c_1 = c_2 = \cdots = c_p = 0$).

The set is **linearly dependent** if there exist weights c_i , *not all zero*, such that the equation holds. This is called a **linear dependence relation**.

Example 3.3.3

Check if $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix} \right\}$ is linearly independent. We must solve $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0}$. This is the matrix equation $A\mathbf{c} = \mathbf{0}$ where $A = (\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3)$.

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 1 & 3 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad (R_3 \rightarrow R_3 - R_2)$$

c_3 is a free variable! So there are non-trivial solutions. Let $c_3 = t$. Then $c_2 = -3t$ and $c_1 = -2t$. For $t = 1$, we get $c_1 = -2, c_2 = -3, c_3 = 1$. This gives the linear dependence relation:

$$-2\mathbf{v}_1 - 3\mathbf{v}_2 + 1\mathbf{v}_3 = \mathbf{0} \quad \text{or} \quad \mathbf{v}_3 = 2\mathbf{v}_1 + 3\mathbf{v}_2$$

The set is **linearly dependent**.

Property 3.3.2: Secondary Conclusions on Independence

- A set of two vectors $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly dependent if and only if one is a scalar multiple of the other.
- A set is linearly dependent if and only if at least one vector in the set is a linear combination of the others.
- Any set containing the zero vector ($\{\mathbf{v}_1, \dots, \mathbf{0}, \dots, \mathbf{v}_p\}$) is linearly dependent.
- **(Key Theorem)** If a set contains *more vectors than entries* in each vector (e.g., p vectors in $\mathbb{R}^n$ where $p > n$), the set is **linearly dependent**.

Proof of $p > n$ implies dependent. Let the set be $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$. Form the $n \times p$ matrix

$$A = (\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_p)$$

We want to solve $A\mathbf{x} = \mathbf{0}$. This is a homogeneous system with n equations and p variables. Since $p > n$ (more variables than equations), there must be at least $p - n > 0$ free variables. The existence of free variables guarantees a non-trivial solution. Therefore, the columns are linearly dependent. $\square$

Connecting this to matrices, we see that:

- The columns of a matrix A are linearly independent if and only if the homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- This happens if and only if there are no free variables, i.e., $\text{rank}(A) = n$ (every column is a pivot column).

This forms several more lines of the Invertible Matrix Theorem.

3.4 Linear Transformations

The matrix-vector product $A\mathbf{x}$ can be viewed as an *action* or *function*. The matrix A *transforms* the vector $\mathbf{x}$ into a new vector $A\mathbf{x}$.

3.4.1 Matrix Transformations

A **transformation** (or function, or mapping) T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each vector $\mathbf{x}$ in $\mathbb{R}^n$ a vector $T(\mathbf{x})$ in $\mathbb{R}^m$.

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

- $\mathbb{R}^n$ is the **domain** of T .
- $\mathbb{R}^m$ is the **codomain** of T .
- $T(\mathbf{x})$ is the **image** of $\mathbf{x}$ under T .
- The set of all images $T(\mathbf{x})$ is the **range** of T .

An important class of transformations are matrix transformations. For an $m \times n$ matrix A , the transformation $T(\mathbf{x}) = A\mathbf{x}$ maps $\mathbb{R}^n \rightarrow \mathbb{R}^m$.

Example 3.4.1

Let $A = \begin{pmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{pmatrix}$. This A defines $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$. Let $\mathbf{x} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$. $T(\mathbf{x}) = A\mathbf{x} =$
 $\begin{pmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1(2) - 3(-1) \\ 3(2) + 5(-1) \\ -1(2) + 7(-1) \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \\ -9 \end{pmatrix}$. The image of $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ is $\begin{pmatrix} 5 \\ 1 \\ -9 \end{pmatrix}$.

3.4.2 Linearity

Matrix transformations $T(\mathbf{x}) = A\mathbf{x}$ have special properties that come from the properties of matrix multiplication:

1. $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$
2. $A(c\mathbf{u}) = c(A\mathbf{u})$

Definition 3.4.1. A transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **linear** if for all $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$ and all scalars c :

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (Preserves addition)
2. $T(c\mathbf{u}) = cT(\mathbf{u})$ (Preserves scalar multiplication)

These two rules imply $T(\mathbf{0}) = \mathbf{0}$ and the "superposition principle": $T(c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p) = c_1T(\mathbf{v}_1) + \cdots + c_pT(\mathbf{v}_p)$.

Theorem 3.4.1

Every matrix transformation $T(\mathbf{x}) = A\mathbf{x}$ is a linear transformation.

The more powerful fact is that the reverse is also true.

Theorem 3.4.2

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then there exists a **unique** $m \times n$ matrix A such that

$$T(\mathbf{x}) = A\mathbf{x} \quad \text{for all } \mathbf{x} \in \mathbb{R}^n$$

This matrix A is called the **standard matrix** for T and is given by:

$$A = (T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \cdots \quad T(\mathbf{e}_n))$$

where $\mathbf{e}_j = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$ (1 in j -th position) is the j -th standard basis vector for $\mathbb{R}^n$.

Proof. Any vector $\mathbf{x} \in \mathbb{R}^n$ can be written as $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$. Since T is linear:

$$T(\mathbf{x}) = T(x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n) = x_1T(\mathbf{e}_1) + \cdots + x_nT(\mathbf{e}_n)$$

This is a linear combination of the vectors $T(\mathbf{e}_j)$. By the definition of $A\mathbf{x}$, this is exactly:

$$T(\mathbf{x}) = \begin{pmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = A\mathbf{x}$$

□

3.4.3 Geometric Transformations in $\mathbb{R}^2$

This section allows us to find the matrix for geometric operations.

Example 3.4.2: Rotation

Find the standard matrix for $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that rotates a vector counter-clockwise by an angle θ . We just need to find $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$. $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Rotating this by θ gives $T(\mathbf{e}_1) = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$. $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Rotating this by θ gives $T(\mathbf{e}_2) = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}$. The standard matrix is $A = \begin{pmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$.

Example 3.4.3: Reflection

Find the standard matrix for $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that reflects a vector across the x -axis. $T(\mathbf{e}_1) = T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. $T(\mathbf{e}_2) = T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$. The standard matrix is $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Definition 3.4.2 (Kernel and Range). Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation.

- The **Kernel** of T , $\text{Ker}(T)$, is the set of all $\mathbf{x}$ in $\mathbb{R}^n$ such that $T(\mathbf{x}) = \mathbf{0}$.
- The **Range** of T , $\text{Range}(T)$, is the set of all $\mathbf{y}$ in $\mathbb{R}^m$ such that $\mathbf{y} = T(\mathbf{x})$ for some $\mathbf{x}$ in $\mathbb{R}^n$.

T is **one-to-one** if $\text{Ker}(T) = \{\mathbf{0}\}$. T is **onto** if $\text{Range}(T) = \mathbb{R}^m$.

If $T(\mathbf{x}) = A\mathbf{x}$, these are just our old subspaces:

- $\text{Ker}(T)$ is the solution set of $A\mathbf{x} = \mathbf{0}$. This is the **Null Space** of A , $\text{Nul}(A)$.
- $\text{Range}(T)$ is the set of all linear combinations of the columns of A . This is the **Column Space** of A , $\text{Col}(A)$.

3.5 Abstract Linear Spaces and Subspaces

In the previous sections, we studied $\mathbb{R}^n$ and its algebraic properties. We observed that matrices ($M_{m \times n}$) and polynomials ($\mathcal{P}_n$) also have similar properties (we can add them, scale them). We will now **abstract** these properties to define a more general concept.

3.5.1 The Formal Definition

Definition 3.5.1. A **Linear Space** (or **Vector Space**) V is a non-empty set of objects, called **vectors**, on which two operations are defined: vector addition ($\mathbf{u} + \mathbf{v}$) and scalar multiplication ($c\mathbf{u}$) (over a field F ,

usually $\mathbb{R}$). These operations must satisfy the following ten axioms for all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V and all scalars c, d in $\mathbb{R}$:

1. $\mathbf{u} + \mathbf{v}$ is in V . (Closure under addition)
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$. (Commutativity)
3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$. (Associativity of addition)
4. There is a **zero vector** $\mathbf{0}$ in V such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$.
5. For each $\mathbf{u}$ in V , there is an **additive inverse** $-\mathbf{u}$ in V such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.
6. $c\mathbf{u}$ is in V . (Closure under scalar multiplication)
7. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$. (Distributivity)
8. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$. (Distributivity)
9. $c(d\mathbf{u}) = (cd)\mathbf{u}$. (Associativity of multiplication)
10. $1\mathbf{u} = \mathbf{u}$. (Scalar identity element)

3.5.2 Examples of Linear Spaces

The power of this definition comes from the variety of sets that satisfy these axioms.

- **Example 1:** $\mathbb{R}^n$ As we've just seen, $\mathbb{R}^n$ with standard component-wise operations is our prototype vector space.
- **Example 2: The Space of Polynomials $\mathcal{P}_n$** Let $V = \mathcal{P}_n$ be the set of all polynomials of degree **at most** n . A "vector" in this space is a polynomial $\mathbf{p}(t) = a_0 + a_1t + \cdots + a_nt^n$. Standard polynomial addition and scalar multiplication satisfy all ten axioms. The "zero vector" is the zero polynomial, $\mathbf{0}(t) = 0$.
- **Example 3: The Space of Matrices $M_{m \times n}$** The set $V = M_{m \times n}$ of all $m \times n$ matrices, with standard matrix addition and scalar multiplication (as defined in Section 2.2), forms a vector space. The "zero vector" is the $m \times n$ zero matrix.
- **Example 4: The Space of Functions $C[a, b]$** Let $V = C[a, b]$ be the set of all *continuous* real-valued functions on an interval $[a, b]$. We define operations "pointwise": $(f + g)(x) = f(x) + g(x)$ $(cf)(x) = c \cdot f(x)$ Since the sum of continuous functions is continuous, and a scalar multiple is continuous, the set is closed. The "zero vector" is the constant function $f(x) = 0$. This forms a vector space.

3.5.3 Subspaces

Often, a vector space is contained inside a larger one.

Definition 3.5.2. A **subspace** of a vector space V is a subset H of V that satisfies three properties:

1. The zero vector of V is in H . ($\mathbf{0} \in H$)
2. H is closed under vector addition: For all $\mathbf{u}, \mathbf{v}$ in H , $\mathbf{u} + \mathbf{v}$ is in H .
3. H is closed under scalar multiplication: For all $\mathbf{u}$ in H and scalar c , $c\mathbf{u}$ is in H .

These three properties guarantee that H is itself a vector space (it inherits the other 7 axioms from V).

Example 3.5.1: A subspace

Let $V = \mathbb{R}^3$. Let H be the xy -plane, i.e., $H = \left\{ \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} \mid x, y \in \mathbb{R} \right\}$. Is H a subspace?

1. Is $\mathbf{0} \in H$? Yes, $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ has $z = 0$.
2. Let $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ 0 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ 0 \end{pmatrix}$ be in H . Is $\mathbf{u} + \mathbf{v} \in H$? $\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ 0 \end{pmatrix}$. Yes, its third component is 0.
3. Let $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ 0 \end{pmatrix}$ be in H . Is $c\mathbf{u} \in H$? $c\mathbf{u} = \begin{pmatrix} cu_1 \\ cu_2 \\ c \cdot 0 \end{pmatrix} = \begin{pmatrix} cu_1 \\ cu_2 \\ 0 \end{pmatrix}$. Yes, it is in H .

Thus, H is a subspace of $\mathbb{R}^3$.

Example 3.5.2: A non-subspace

Let $V = \mathbb{R}^2$. Let H be the first quadrant, $H = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \mid x \geq 0, y \geq 0 \right\}$.

1. $\mathbf{0} \in H$. (Pass)
2. H is closed under addition. (Pass: $x_1 + x_2 \geq 0, y_1 + y_2 \geq 0$)
3. Is H closed under scalar multiplication? Let $c = -1$ and $\mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \in H$. $c\mathbf{u} = -1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$. This is *not* in H .

H is **not** a subspace.

Theorem 3.5.1

If $\mathbf{v}_1, \dots, \mathbf{v}_p$ are in a vector space V , then $H = \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is **always** a subspace of V .

Proof. 1. $\mathbf{0} = 0\mathbf{v}_1 + \dots + 0\mathbf{v}_p$, so $\mathbf{0}$ is in the span.

2. Let $\mathbf{u} = c_1\mathbf{v}_1 + \dots + c_p\mathbf{v}_p$ and $\mathbf{v} = d_1\mathbf{v}_1 + \dots + d_p\mathbf{v}_p$. Then $\mathbf{u} + \mathbf{v} = (c_1 + d_1)\mathbf{v}_1 + \dots + (c_p + d_p)\mathbf{v}_p$, which is a linear combination, so it is in the span.

3. Let k be a scalar. $k\mathbf{u} = k(c_1\mathbf{v}_1 + \dots + c_p\mathbf{v}_p) = (kc_1)\mathbf{v}_1 + \dots + (kc_p)\mathbf{v}_p$, which is also in the span.

Thus, any span is a subspace. $\square$

3.5.4 Null Spaces and Column Spaces

There are two fundamental subspaces associated with any $m \times n$ matrix A .

Definition 3.5.3. • The **Null Space** of A , $\text{Nul}(A)$, is the set of all solutions to the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

$$\text{Nul}(A) = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\}$$

This is a subspace of $\mathbb{R}^n$.

- The **Column Space** of A , $\text{Col}(A)$, is the span of the columns of A .

$$\text{Col}(A) = \text{Span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\} = \{\mathbf{b} \in \mathbb{R}^m \mid \mathbf{b} = A\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\}$$

This is a subspace of $\mathbb{R}^m$.

$\text{Nul}(A)$ describes the structure of the homogeneous solution set. $\text{Col}(A)$ describes the set of all $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ is consistent.

3.5.5 Basis and Dimension

We now unify the ideas of spanning and linear independence.

Definition 3.5.4. A **basis** for a vector space V is a set of vectors $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ in V such that:

1. $\mathcal{B}$ is a linearly independent set.
2. $\mathcal{B}$ spans V (i.e., $\text{Span}\{\mathcal{B}\} = V$).

A basis is the "smallest" possible spanning set and the "largest" possible linearly independent set.

Example 3.5.3

The set of standard vectors $\mathcal{E} = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is the **standard basis** for $\mathbb{R}^n$. The set $\{1, t, t^2, \dots, t^n\}$ is the **standard basis** for $\mathcal{P}_n$.

Theorem 3.5.2

All bases for a vector space V have the same number of vectors.

Definition 3.5.5. The **dimension** of a non-zero vector space V , denoted $\dim(V)$, is the number of vectors in any basis for V . The dimension of the zero subspace $\{\mathbf{0}\}$ is defined to be 0.

Examples of Dimension:

- $\dim(\mathbb{R}^n) = n$.
- $\dim(\mathcal{P}_n) = n + 1$ (because of the $t^0 = 1$ term).
- $\dim(M_{m \times n}) = m \times n$.
- $C[a, b]$ is **infinite-dimensional**.

There is an interesting conclusion. The cardinality of $C[0, 1]$ equals to the cardinality of $\mathbb{R}$.

Proof. Let $D = \mathbb{Q} \cap [0, 1]$ be the countable dense set of rationals in $[0, 1]$.

Upper bound ($\#C[0, 1] \leq \mathfrak{c}$): Define $\Phi : C[0, 1] \rightarrow \mathbb{R}^{\mathbb{N}}$ by

$$\Phi(f) = (f(q_1), f(q_2), f(q_3), \dots)$$

where $\{q_i\}$ enumerates D . If $\Phi(f) = \Phi(g)$, then $f(q) = g(q)$ for all $q \in D$. By continuity and density, $f = g$ on $[0, 1]$, so Φ is injective. Thus

$$\#C[0, 1] \leq \#(\mathbb{R}^{\mathbb{N}}) = \mathfrak{c}^{\aleph_0} = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0} = \mathfrak{c}.$$

Lower bound ($\#C[0, 1] \geq \mathfrak{c}$): The constant functions $\{f_r(x) = r : r \in \mathbb{R}\}$ form a subset of $C[0, 1]$ with cardinality $\mathfrak{c}$.

By Cantor-Bernstein theorem, $\#C[0, 1] = \mathfrak{c}$. □

We can now find bases for our two favorite subspaces.

- **Basis for Col(A):** The pivot columns of the *original* matrix A form a basis for Col(A). (Do not use the RREF columns, as EROs change the column space).
- **Basis for Nul(A):** The vectors found when writing the solution of $A\mathbf{x} = \mathbf{0}$ in parametric vector form form a basis for Nul(A).

Example 3.5.4

$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$ **Column Space:** Pivots are in columns 1 and 2. Basis for Col(A) = $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$. $\dim(\text{Col}(A)) = 2$.
Null Space: Solve $A\mathbf{x} = \mathbf{0}$. $x_1 + 2x_3 = 0 \implies x_1 = -2x_3$ $x_2 + 3x_3 = 0 \implies x_2 = -3x_3$ x_3 is free.
 Let $x_3 = t$. $\mathbf{x} = \begin{pmatrix} -2t \\ -3t \\ t \end{pmatrix} = t \begin{pmatrix} -2 \\ -3 \\ 1 \end{pmatrix}$. Basis for Nul(A) = $\left\{ \begin{pmatrix} -2 \\ -3 \\ 1 \end{pmatrix} \right\}$. $\dim(\text{Nul}(A)) = 1$.

Notice the connection to rank:

- $\dim(\text{Col}(A)) = (\text{Number of pivot columns}) = \text{rank}(A)$.
- $\dim(\text{Nul}(A)) = (\text{Number of free variables}) = n - \text{rank}(A)$.

This leads to one of the most important theorems in linear algebra.

Theorem 3.5.3: The Rank-Nullity Theorem

For an $m \times n$ matrix A ,

$$\dim(\text{Col}(A)) + \dim(\text{Nul}(A)) = n$$

or, equivalently,

$$\text{rank}(A) + \text{nullity}(A) = n$$

where n is the number of **columns** and $\text{nullity}(A) = \dim(\text{Nul}(A))$.

In our last example, $n = 3$. $\text{rank}(A) = 2$, $\text{nullity}(A) = 1$. $2 + 1 = 3$. The theorem holds. This theorem beautifully ties together the dimensions of the two fundamental subspaces associated with a matrix.

Coordinate Systems

A basis $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ for $\mathbb{R}^n$ acts like a new coordinate system. Because $\mathcal{B}$ spans $\mathbb{R}^n$ and is linearly independent, every $\mathbf{x} \in \mathbb{R}^n$ can be written *uniquely* as

$$\mathbf{x} = c_1\mathbf{b}_1 + \dots + c_n\mathbf{b}_n$$

Definition 3.5.6. The scalars $c_1, \dots, c_n$ are the **coordinates of $\mathbf{x}$ relative to the basis $\mathcal{B}$** . The **coordinate vector** of $\mathbf{x}$ (relative to $\mathcal{B}$) is

$$[\mathbf{x}]_{\mathcal{B}} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}$$

Let $P_{\mathcal{B}}$ be the **change-of-coordinates matrix** $P_{\mathcal{B}} = (\mathbf{b}_1 \ \dots \ \mathbf{b}_n)$. The equation $\mathbf{x} = c_1\mathbf{b}_1 + \dots + c_n\mathbf{b}_n$ is just the matrix equation

$$\mathbf{x} = P_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}$$

Since the columns of $P_{\mathcal{B}}$ are a basis, $P_{\mathcal{B}}$ is invertible (by the IMT).

$$[\mathbf{x}]_{\mathcal{B}} = P_{\mathcal{B}}^{-1} \mathbf{x}$$

This provides a way to "translate" between the standard coordinate system $\mathcal{E}$ and the new system $\mathcal{B}$.

3.5.6 Eigenvalues and Eigenvectors

Eigenvalues and eigenvectors are fundamental concepts in linear algebra that provide crucial insights into the structure of linear transformations. They play a central role in many applications, including vibration analysis, quantum mechanics, and data analysis.

The reason why we want to study eigenvalues and eigenvectors is that they help us understand how a linear transformation (represented by a matrix) acts on certain special directions in space. Specifically, an eigenvector is a direction that remains unchanged (up to scaling) when the transformation is applied, and the corresponding eigenvalue indicates how much the vector is stretched or compressed.

Definition 3.5.7 (Eigenvalues and Eigenvectors). Let A be an $n \times n$ square matrix. A scalar λ is called an **eigenvalue** of A if there exists a nonzero vector $\mathbf{v} \in \mathbb{R}^n$ such that

$$A\mathbf{v} = \lambda\mathbf{v}$$

The vector $\mathbf{v}$ is called an **eigenvector** corresponding to the eigenvalue λ .

Geometrically, an eigenvector $\mathbf{v}$ is a vector whose direction remains unchanged when transformed by A ; it is only scaled by the factor λ .

To find eigenvalues, we rewrite the equation $A\mathbf{v} = \lambda\mathbf{v}$ as

$$(A - \lambda I)\mathbf{v} = \mathbf{0}$$

This is a homogeneous system of linear equations. Since $\mathbf{v} \neq \mathbf{0}$, this system must have nontrivial solutions, which requires that the matrix $A - \lambda I$ be singular, i.e., its determinant must be zero.

Definition 3.5.8 (Characteristic Polynomial). The **characteristic polynomial** of a matrix A is defined as

$$p(\lambda) = \det(A - \lambda I)$$

This is an n th-degree polynomial in λ . The eigenvalues of A are the roots of the characteristic equation $p(\lambda) = 0$.

The roots of the characteristic polynomial may be real or complex. Repeated roots are called eigenvalues with algebraic multiplicity greater than 1. Each eigenvalue corresponds to an eigenspace.

Definition 3.5.9 (Eigenspace). For an eigenvalue λ , the corresponding **eigenspace** is the solution space of the homogeneous system $(A - \lambda I)\mathbf{v} = \mathbf{0}$, i.e., $\text{Nul}(A - \lambda I)$. The dimension of the eigenspace is called the **geometric multiplicity** of λ .

Example 3.5.5

For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, the characteristic polynomial is $p(\lambda) = (\lambda - 1)(\lambda - 3) = 0$. The eigenvalues are $\lambda_1 = 1, \lambda_2 = 3$. Solving $(A - \lambda I)\mathbf{v} = \mathbf{0}$ yields the corresponding eigenvectors: $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ for $\lambda_1 = 1$, and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for $\lambda_2 = 3$.

3.5.7 Diagonalization

Diagonalization is the process of transforming a matrix into diagonal form, which greatly simplifies computations involving matrix powers and exponentials. The geometric interpretation is that diagonalization aligns the coordinate system with the eigenvectors of the matrix, making the transformation represented by the matrix easier to understand.

Definition 3.5.10 (Diagonalizable Matrix). An $n \times n$ matrix A is said to be **diagonalizable** if there exists an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}$$

Equivalently, $P^{-1}AP = D$.

The diagonal entries of D are the eigenvalues of A , and the columns of P are the corresponding linearly independent eigenvectors.

Theorem 3.5.4

A matrix A is diagonalizable if and only if it has n linearly independent eigenvectors. This is equivalent to the condition that the geometric multiplicity of each eigenvalue equals its algebraic multiplicity (the multiplicity as a root of the characteristic polynomial).

Example 3.5.6

Continuing the previous example, $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 3$ with corresponding eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. These eigenvectors are linearly independent, so we can take

$$P = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$$

It's easy to verify that $A = PDP^{-1}$.

Once diagonalized, computing powers of A becomes straightforward:

$$A^k = (PDP^{-1})^k = PD^kP^{-1}$$

since D^k is simply obtained by raising each diagonal element to the k th power.

Inner Product Spaces

An inner product generalizes the dot product and provides a framework for defining lengths, angles, and orthogonality in vector spaces.

Definition 3.5.11 (Inner Product). Let V be a real vector space. An **inner product** is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ satisfying the following properties for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all scalars c :

1. $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$ (Symmetry)
2. $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$ (Linearity)
3. $\langle c\mathbf{u}, \mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle$
4. $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$, and $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ if and only if $\mathbf{u} = \mathbf{0}$ (Positive definiteness)

The most common example is the dot product (standard inner product) on $\mathbb{R}^n$:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n$$

Definition 3.5.12 (Norm and Distance). The **norm** (length) induced by an inner product is defined as

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$$

The **distance** between vectors $\mathbf{u}$ and $\mathbf{v}$ is $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.

Definition 3.5.13 (Orthogonality). Two vectors $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. A set of vectors is orthogonal if all pairs of distinct vectors in the set are orthogonal. If, in addition, each vector has unit norm, the set is **orthonormal**.

3.5.8 Orthogonal Bases and the Gram-Schmidt Process

In inner product spaces, orthogonal bases simplify many computations.

Theorem 3.5.5

If $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is an orthogonal basis for a subspace H , then for any $\mathbf{y} \in H$,

$$\mathbf{y} = c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k, \quad \text{where } c_i = \frac{\langle \mathbf{y}, \mathbf{v}_i \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle}$$

If the basis is orthonormal, then $c_i = \langle \mathbf{y}, \mathbf{v}_i \rangle$.

The Gram-Schmidt process converts any linearly independent set into an orthogonal basis.

Theorem 3.5.6: Gram-Schmidt Orthogonalization Process

Let $\{\mathbf{x}_1, \dots, \mathbf{x}_p\}$ be a basis for a subspace H . Define:

$$\begin{aligned} \mathbf{v}_1 &= \mathbf{x}_1 \\ \mathbf{v}_2 &= \mathbf{x}_2 - \frac{\langle \mathbf{x}_2, \mathbf{v}_1 \rangle}{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle} \mathbf{v}_1 \\ \mathbf{v}_3 &= \mathbf{x}_3 - \frac{\langle \mathbf{x}_3, \mathbf{v}_1 \rangle}{\langle \mathbf{v}_1, \mathbf{v}_1 \rangle} \mathbf{v}_1 - \frac{\langle \mathbf{x}_3, \mathbf{v}_2 \rangle}{\langle \mathbf{v}_2, \mathbf{v}_2 \rangle} \mathbf{v}_2 \\ &\vdots \\ \mathbf{v}_p &= \mathbf{x}_p - \sum_{i=1}^{p-1} \frac{\langle \mathbf{x}_p, \mathbf{v}_i \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle} \mathbf{v}_i \end{aligned}$$

Then $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is an orthogonal basis for H . Normalizing each vector yields an orthonormal basis.

3.5.9 Symmetric Matrices and Quadratic Forms

Real symmetric matrices have particularly nice properties that make them important in many applications.

Theorem 3.5.7: Spectral Theorem

Let A be an $n \times n$ real symmetric matrix. Then:

1. All eigenvalues of A are real.
2. A has n linearly independent eigenvectors, and eigenvectors corresponding to distinct eigenvalues are orthogonal.
3. A is orthogonally diagonalizable: there exists an orthogonal matrix Q (satisfying $Q^T = Q^{-1}$) and a diagonal matrix D such that

$$A = QDQ^T$$

Quadratic forms are homogeneous polynomials of degree 2 that can be represented in matrix form as $\mathbf{x}^T A \mathbf{x}$, where A is a symmetric matrix.

Definition 3.5.14 (Quadratic Form). A **quadratic form** is a function $Q : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j$$

where A is a symmetric matrix.

Through the orthogonal transformation $\mathbf{x} = Q\mathbf{y}$, a quadratic form can be reduced to its canonical form:

$$Q(\mathbf{x}) = \mathbf{y}^T D \mathbf{y} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2$$

where the λ_i are the eigenvalues of A .

Quadratic forms are classified based on the signs of their eigenvalues:

- **Positive definite:** All eigenvalues positive; $Q(\mathbf{x}) > 0$ for $\mathbf{x} \neq \mathbf{0}$.
- **Negative definite:** All eigenvalues negative.
- **Indefinite:** Eigenvalues have mixed signs.

This classification has important applications in optimization and the study of critical points in multivariable calculus.

Singular Value Decomposition (SVD) The Diagonalization Theorem ($A = PDP^{-1}$) applies only to square, diagonalizable matrices. The Spectral Theorem applies only to symmetric matrices. The SVD is the ultimate generalization: it applies to **any** $m \times n$ matrix.

Theorem 3.5.8: Singular Value Decomposition

Let A be an $m \times n$ matrix with rank r . Then there exists an $m \times n$ factorization of the form:

$$A = U \Sigma V^T$$

where:

- U is an $m \times m$ orthogonal matrix ($U^T U = I$). The columns of U are called the **left singular vectors**.
- V is an $n \times n$ orthogonal matrix ($V^T V = I$). The columns of V are called the **right singular vectors**.
- Σ is an $m \times n$ rectangular diagonal matrix with non-negative entries on the diagonal:

$$\Sigma = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}$$

where $D = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r)$ and $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$.

The scalars σ_i are called the **singular values** of A . They are the square roots of the non-zero eigenvalues of $A^T A$.

Geometric Interpretation: Any linear transformation $T(\mathbf{x}) = A\mathbf{x}$ maps the unit sphere in the domain to a hyperellipse in the codomain. The singular values are the lengths of the semi-axes of this hyperellipse.

Construction of SVD:

1. Compute the eigenvalues of the symmetric matrix $A^T A$. Let them be $\lambda_1 \geq \cdots \geq \lambda_n$.
2. The singular values are $\sigma_i = \sqrt{\lambda_i}$.

3. Find orthonormal eigenvectors of $A^T A$; these form the matrix V .
4. The first r columns of U are given by $\mathbf{u}_i = \frac{1}{\sigma_i} A \mathbf{v}_i$. Extend this set to an orthonormal basis for $\mathbb{R}^m$ to fill the rest of U .

3.6 Conclusions

In this chapter, we have learned a lot of concepts. From matrix to determinant, from rank to eigenvalue $\dots$. We can spot very beautiful symmetry between each part, we are actually using one single language to describe different things from the same perspective but have varying results. This is what makes mathematics attractive. Now we will focus on two concepts about Linear Algebra, to conclude what we have learned through out the journey:

3.6.1 Interpretations of Rank

Let A be an $m \times n$ matrix over a field $\mathbb{F}$ (e.g., $\mathbb{R}$ or $\mathbb{C}$). The rank of A , denoted as $\text{rank}(A)$ or $\rho(A)$, can be defined and interpreted in the following equivalent ways:

3.6.1.1 Vector Space Interpretations

- **Column Rank:** The dimension of the column space of A (the vector space spanned by its columns).

$$\text{rank}(A) = \dim(\text{Col}(A))$$

- **Row Rank:** The dimension of the row space of A (the vector space spanned by its rows). A fundamental property is that row rank equals column rank:

$$\dim(\text{Row}(A)) = \dim(\text{Col}(A))$$

- **Linear Independence:** The maximum number of linearly independent column vectors (or row vectors) in the matrix.

3.6.1.2 Computational/Algebraic Interpretations

- **Pivot Definition:** The number of pivots (leading 1s) in the Reduced Row Echelon Form (RREF) of A .
- **Determinantal Rank:** The order of the largest non-zero square minor of A . That is, r is the rank if there exists an $r \times r$ submatrix with a non-zero determinant, and every $(r+1) \times (r+1)$ minor is zero.
- **Decomposition Rank:** The smallest integer k such that A can be factored as $A = CR$, where C is $m \times k$ and R is $k \times n$.

3.6.1.3 Geometric and Mapping Interpretations

- **Image Dimension:** If we view A as a linear transformation $T : \mathbb{F}^n \rightarrow \mathbb{F}^m$ defined by $T(\mathbf{x}) = A\mathbf{x}$, the rank is the dimension of the image (range) of T :

$$\text{rank}(A) = \dim(\text{Im}(T))$$

- **Singular Value Decomposition (SVD):** The number of non-zero singular values of A .

3.6.2 The Rank-Nullity Theorem

The Rank-Nullity Theorem (often called the Fundamental Theorem of Linear Algebra) relates the dimensions of the domain, the image, and the kernel. Below are its expressions in different contexts.

3.6.2.1 1. Matrix Context

For an $m \times n$ matrix A :

Theorem 3.6.1: Matrix Rank-Nullity

The number of columns equals the sum of the rank and the nullity.

$$\text{rank}(A) + \text{nullity}(A) = n$$

- **rank(A):** The number of pivot columns (basic variables).
- **nullity(A):** The dimension of the null space ($\dim(\text{Null}(A))$), which corresponds to the number of free columns (free variables).
- **Interpretation:** Total Variables = Pivot Variables + Free Variables.

3.6.2.2 2. Linear Transformation Context

Let V and W be vector spaces, where V is finite-dimensional. Let $T : V \rightarrow W$ be a linear transformation.

Theorem 3.6.2: Linear Map Rank-Nullity

$$\dim(\text{Im}(T)) + \dim(\ker(T)) = \dim(V)$$

- $\dim(\text{Im}(T))$ is the rank of the transformation.
- $\dim(\ker(T))$ is the nullity (dimension of the kernel).
- Note that the sum equals the dimension of the *domain*, not the codomain.

3.6.2.3 3. Abstract Algebra Context (Isomorphism Theorems)

The theorem is a direct consequence of the **First Isomorphism Theorem** for vector spaces (or modules).

Theorem 3.6.3

$$V / \ker(T) \cong \text{Im}(T)$$

Taking dimensions of both sides:

$$\dim(V) - \dim(\ker(T)) = \dim(\text{Im}(T))$$

Rearranging this yields the standard Rank-Nullity equation.

3.6.2.4 4. Systems of Linear Equations

Consider the homogeneous system $A\mathbf{x} = \mathbf{0}$, where A is $m \times n$.

- The dimension of the solution space is $k = n - r$, where $r = \text{rank}(A)$.
- If $r = n$ (full column rank), the only solution is the trivial solution ($\mathbf{0}$), so nullity is 0.
- If $r < m$ (for the augmented system $A\mathbf{x} = \mathbf{b}$), existence of solutions depends on column space consistency.

3.6.3 A Unifying Viewpoint: The Vector-Space Axioms

There is no single universally accepted “axiom of linear algebra.” The subject can be organized computationally around elimination, geometrically around transformations, or abstractly around vector spaces and linear maps. Each organization highlights different questions.

The vector-space axioms are nevertheless one especially powerful unifying framework. They isolate two operations—addition and scalar multiplication—and state the rules needed for linear combination. Once these rules are available, concepts such as span, linear independence, basis, dimension, and linear transformation apply simultaneously to geometric vectors, matrices, polynomials, functions, and solution spaces of differential equations.

This should be read as a pedagogical viewpoint rather than a claim of exclusivity. For numerical work, matrix factorizations may be the most effective language. For geometry, invariant subspaces and orthogonality may be more natural. For functional analysis, the topology and norm on a vector space become as important as the algebraic axioms. The appropriate foundation depends on the problem being studied.

For reference, the abstract definition is repeated below.

Definition 3.6.1 (Vector Space). Let F be a field. A **vector space over** F is a set V equipped with vector addition and scalar multiplication such that, for all $u, v, w \in V$ and $a, b \in F$,

1. $u + v = v + u$ and $(u + v) + w = u + (v + w)$;
2. there is a zero vector 0 with $v + 0 = v$;
3. every v has an additive inverse $-v$;
4. $a(u + v) = au + av$ and $(a + b)v = av + bv$;
5. $(ab)v = a(bv)$ and $1v = v$.

Keywords: Eigenvalues, Eigenvectors, Diagonalization, Inner Product, Orthogonal Bases, Gram-Schmidt Process, Symmetric Matrices, Quadratic Forms

References:

Linear Algebra and Its Applications, 4th ed., Gilbert Strang, Cengage Learning, 2005.

Shanghai Jiao Tong University, School of Mathematical Sciences. Linear Algebra. China Machine Press.

3.7 Chapter Workshop

Worked example: Least squares as orthogonal projection

Fit a line through the origin, $y = ax$, to the data $(1, 1)$, $(2, 2)$, and $(3, 2)$. In matrix form,

$$A = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

The least-squares coefficient satisfies the normal equation

$$A^T A a = A^T b,$$

so

$$14a = 11, \quad a = \frac{11}{14}.$$

The residual $r = b - Aa$ is orthogonal to the column space of A because $A^T r = 0$. The computation is therefore a projection, not merely a formula for fitting data.

Application: Principal component analysis

After centering a data matrix X , the eigenvectors of $X^T X$ give directions of maximal sample variance. Equivalently, the right singular vectors of X provide orthogonal coordinates ordered by singular value. Truncating the singular-value decomposition produces a low-rank approximation and is used in compression, denoising, visualization, and latent-feature extraction.

3.8 Exercises

1. Row-reduce $\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 0 \\ 1 & 2 & 3 \end{pmatrix}$ and find its rank and nullity.
2. Find an orthonormal basis for the span of $(1, 1, 0)$ and $(1, 0, 1)$.
3. Determine the eigenvalues and eigenvectors of $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.
4. Prove that eigenvectors corresponding to distinct eigenvalues of a real symmetric matrix are orthogonal.
5. Find the least-squares solution of $x \approx 1$, $x \approx 2$, $x \approx 6$.
6. Explain why a matrix and the linear map it represents are not literally the same object.
7. Find a basis for the column space and null space of $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 1 & 3 \end{pmatrix}$.
8. Compute the standard matrix of the linear transformation $T(x, y) = (x + y, x - y)$.
9. Decide whether $\{(1, 0, 0), (0, 1, 0), (1, 1, 0)\}$ is linearly independent.

10. Diagonalize $\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$ and interpret its eigenvalues.
11. Use Gram-Schmidt to orthogonalize $(1, 1)$ and $(1, 0)$ in $\mathbb{R}^2$.
12. Verify directly that $\text{rank}(A) + \text{nullity}(A) = n$ for $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$.

Selected hints and answers

1. The second column is twice the first; elimination gives rank 2 and nullity 1.
2. Normalize the first vector, subtract its projection from the second, then normalize.
3. The eigenvalues are 3 and 1, with directions $(1, 1)$ and $(1, -1)$.
4. Compare $\langle Av, w \rangle$ with $\langle v, Aw \rangle$.
5. Minimize $(x - 1)^2 + (x - 2)^2 + (x - 6)^2$; the answer is the mean, $x = 3$.
6. A matrix depends on chosen bases; the abstract map does not. A change of basis changes the matrix by similarity or by left/right coordinate transformations.
7. The pivot columns form a basis for $\text{Col}(A)$; solve $A\mathbf{x} = \mathbf{0}$ for $\text{Nul}(A)$.
8. Apply T to the standard basis vectors $e_1 = (1, 0)$ and $e_2 = (0, 1)$.
9. Check whether one vector is a linear combination of the others, or row-reduce the matrix with these vectors as columns.
10. The matrix is already diagonal, so the eigenvalues are its diagonal entries and the eigenvectors are the coordinate axes.
11. Subtract the projection of the second vector onto the first, then normalize the resulting orthogonal pair.
12. This matrix has one pivot column, so its rank is 1 and its nullity is 1.

Chapter 4

Abstract Algebra

Learning objectives

By the end of this chapter, you should be able to:

- recognize groups, rings, fields, and modules in concrete examples;
- use homomorphisms, kernels, images, cosets, ideals, and quotient structures;
- apply the isomorphism theorems as classification tools;
- work with polynomial rings, irreducibility, and finite field extensions;
- connect algebraic structure with symmetry, coding, and cryptographic arithmetic.

Abstract algebra studies mathematical structures by extracting the rules that govern operations. In elementary algebra, one usually starts from numbers and equations. In abstract algebra, the order is reversed: we first specify a set, then specify operations on that set, and finally ask what can be deduced from the axioms alone.

This change of viewpoint is important. The same algebraic pattern appears in many places: integers under addition, non-zero real numbers under multiplication, rotations of a polygon, invertible matrices, permutations, polynomials, quotient arithmetic, and vector spaces. Abstract algebra is the language that recognizes these different objects as instances of the same structure.

The central idea of this chapter is therefore:

An algebraic structure is a set equipped with operations satisfying specified axioms. Once the axioms are fixed, every theorem proved from them is true for every model of those axioms.

We will introduce four major structures: groups, rings, fields, and modules. The route is cumulative. Groups describe one operation and invertibility; rings describe addition and multiplication together; fields describe arithmetic with division; modules describe linear algebra when the scalars are no longer required to form a field.

4.1 Operations and Algebraic Structures

At this point it is useful to slow down and separate two levels of thought. The first level is the concrete level: we compute with integers, matrices, permutations, or polynomials. The second level is the structural level: we ask which formal laws are actually being used. Abstract algebra becomes powerful precisely when we learn to move between these two levels. A calculation may take place in one familiar model, but the reason it works may depend only on associativity, identity, inverses, or distributivity. Once we identify the relevant laws, the same argument can be transported to every structure satisfying those laws.

A common beginner's mistake is to think that the operation symbol determines the structure. This is not true. The same symbol may hide very different operations, and the same operation may be written with different symbols. In group theory, for example, multiplication may mean ordinary multiplication, matrix multiplication, composition of functions, or addition written multiplicatively. Therefore one should always ask: what is the underlying set, what is the operation, and which axioms are valid?

Before defining groups or rings, we clarify what it means to have an operation on a set.

Definition 4.1.1 (Binary Operation). Let S be a set. A **binary operation** on S is a function

$$*: S \times S \rightarrow S.$$

For $a, b \in S$, we usually write $a * b$ instead of $*(a, b)$.

The condition $*: S \times S \rightarrow S$ already includes **closure**: whenever we combine two elements of S , the result is still in S .

Example 4.1.1: Operations and Non-operations

Addition is a binary operation on $\mathbb{Z}$, since $a + b \in \mathbb{Z}$ whenever $a, b \in \mathbb{Z}$. Subtraction is also a binary operation on $\mathbb{Z}$. However, division is not a binary operation on $\mathbb{Z}$, because $1/2 \notin \mathbb{Z}$. It is also not a binary operation on $\mathbb{R}$ unless division by 0 is excluded.

Definition 4.1.2 (Associativity and Commutativity). Let $*$ be a binary operation on S .

1. $*$ is **associative** if $(a * b) * c = a * (b * c)$ for all $a, b, c \in S$.
2. $*$ is **commutative** if $a * b = b * a$ for all $a, b \in S$.

Associativity is not merely a technical condition. Without it, expressions like $a * b * c * d$ are ambiguous, because different placements of parentheses may give different answers.

Example 4.1.2: A Non-associative Operation

On $\mathbb{R}$, define $a * b = a - b$. Then

$$(a * b) * c = (a - b) - c, \quad a * (b * c) = a - (b - c) = a - b + c.$$

These are usually different. Thus subtraction is not associative.

Definition 4.1.3 (Identity Element and Inverse). Let $*$ be a binary operation on S .

1. An element $e \in S$ is an **identity element** if $e * a = a * e = a$ for all $a \in S$.
2. If e is an identity element, an element $b \in S$ is an **inverse** of $a \in S$ if $a * b = b * a = e$.

Proposition 4.1.1: Uniqueness of Identity

If a binary operation has an identity element, then the identity element is unique.

Proof. Suppose e and e' are both identity elements. Since e is an identity, $e * e' = e'$. Since e' is an identity, $e * e' = e$. Hence $e = e'$. $\square$

4.2 Groups

The guiding intuition is that a group is a universe in which every permitted move can be reversed. The associativity axiom says that a sequence of moves can be performed without worrying about the placement of parentheses. The identity axiom says that there is a move which changes nothing. The inverse axiom

says that every move has an undo operation. These three requirements are exactly what is needed to treat algebraic manipulation as a reliable calculus of transformations.

This viewpoint also explains why many groups are non-commutative. If operations are thought of as actions performed in time, then order usually matters: rotating and then reflecting a figure is not necessarily the same as reflecting and then rotating it. Non-commutativity is therefore not an exotic defect but the normal behavior of transformations. Abelian groups are the special case in which the order of operations happens not to matter.

Groups are the algebraic structures that formalize the idea of reversible transformations. A rotation can be undone by a rotation in the opposite direction. A permutation can be undone by the inverse permutation. Adding an integer can be undone by adding its negative. These examples look different, but they obey the same axioms.

4.2.1 Definition and Basic Examples

Definition 4.2.1 (Group). A **group** is a pair $(G, *)$, where G is a set and $*$ is a binary operation on G , satisfying the following axioms:

1. **Associativity:** $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$.
2. **Identity:** There exists $e \in G$ such that $e * a = a * e = a$ for all $a \in G$.
3. **Inverses:** For every $a \in G$, there exists $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

If $a * b = b * a$ for all $a, b \in G$, the group is called **abelian**.

When the operation is clear, we write ab instead of $a * b$. For an additive group, we write the identity as 0 and the inverse of a as $-a$.

Example 4.2.1: Standard Groups

1. $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, and $(\mathbb{C}, +)$ are abelian groups.
2. $(\mathbb{R} \setminus \{0\}, \cdot)$ and $(\mathbb{C} \setminus \{0\}, \cdot)$ are abelian groups.
3. $(\mathbb{N}, +)$ is not a group, since positive natural numbers do not have additive inverses.
4. The set $GL_n(\mathbb{R})$ of invertible $n \times n$ real matrices is a group under matrix multiplication. It is non-abelian when $n \geq 2$.
5. The symmetric group S_n , consisting of all bijections from $\{1, \dots, n\}$ to itself, is a group under composition. It has $n!$ elements.

Example 4.2.2: Symmetry Group of a Triangle

An equilateral triangle has six rigid symmetries: three rotations and three reflections. These six transformations form a group under composition, usually denoted D_6 or D_3 depending on convention. The group is non-abelian because reflecting and then rotating need not equal rotating and then reflecting.

Remark 4.2.1

The operation in a group does not have to be numerical multiplication. It may be addition, composition of functions, matrix multiplication, or a geometric transformation. The word “multiplication” in group theory is often only notation.

4.2.2 Elementary Consequences of the Axioms

The definition of a group is short, but its consequences are strong.

Proposition 4.2.1: Uniqueness of Inverses

Let G be a group. For every $a \in G$, the inverse of a is unique.

Proof. Suppose b and c are both inverses of a . Then $ab = e$ and $ca = e$. Hence

$$b = eb = (ca)b = c(ab) = ce = c.$$

Therefore the inverse is unique. □

Proposition 4.2.2: Cancellation Laws

Let G be a group and $a, b, c \in G$.

1. If $ab = ac$, then $b = c$.
2. If $ba = ca$, then $b = c$.

Proof. If $ab = ac$, multiply both sides on the left by a^{-1} :

$$a^{-1}(ab) = a^{-1}(ac).$$

By associativity, $(a^{-1}a)b = (a^{-1}a)c$, so $eb = ec$, and hence $b = c$. The right cancellation law is similar. □

Proposition 4.2.3: Inverse of a Product

For $a, b \in G$,

$$(ab)^{-1} = b^{-1}a^{-1}.$$

Proof. We verify that $b^{-1}a^{-1}$ is the inverse of ab :

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aea^{-1} = e,$$

and similarly $(b^{-1}a^{-1})(ab) = e$. □

Definition 4.2.2 (Powers and Order of an Element). Let G be a group and $g \in G$.

1. For $n \in \mathbb{N}$, define $g^n = g \cdots g$ (n times), $g^0 = e$, and $g^{-n} = (g^{-1})^n$.
2. The **order** of g , denoted $|g|$ or $\text{ord}(g)$, is the smallest positive integer n such that $g^n = e$. If no such integer exists, g has infinite order.

Example 4.2.3

In $(\mathbb{Z}/6\mathbb{Z}, +)$, the element 2 has order 3, because

$$2 + 2 + 2 = 6 \equiv 0 \pmod{6},$$

and no smaller positive multiple of 2 is congruent to 0 modulo 6.

4.2.3 Subgroups and Generated Subgroups

Subgroups are the first mechanism by which we analyze a complicated group through smaller pieces. The important point is that a subgroup is not just any subset; it must be closed under the same algebraic life as the ambient group. Once we enter the subset, multiplication, identity, and inverses must keep us inside it. In this sense, a subgroup is a self-contained algebraic world embedded in a larger one.

The generated subgroup $\langle S \rangle$ should be understood as the algebraic closure of the chosen elements S . Starting with S , we are forced to include products, inverses, products of inverses, and so on. The result is the smallest environment in which all expressions built from elements of S make sense. This idea is analogous to the span of vectors in linear algebra: a small set of generators may determine a large structure.

A subgroup is a smaller group living inside a larger group.

Definition 4.2.3 (Subgroup). Let G be a group. A subset $H \subseteq G$ is a **subgroup**, written $H \leq G$, if H is itself a group under the operation inherited from G .

Theorem 4.2.1: Subgroup Test

A non-empty subset $H \subseteq G$ is a subgroup of G if and only if for all $x, y \in H$, we have $xy^{-1} \in H$.

Proof. If H is a subgroup, then $y^{-1} \in H$ and hence $xy^{-1} \in H$.

Conversely, assume H is non-empty and satisfies the stated condition. Choose $h \in H$. Taking $x = y = h$, we get $hh^{-1} = e \in H$. Taking $x = e$ and $y = h$, we get $eh^{-1} = h^{-1} \in H$. Finally, if $a, b \in H$, then $b^{-1} \in H$, and using the condition with $x = a$ and $y = b^{-1}$ gives $a(b^{-1})^{-1} = ab \in H$. Thus H contains the identity, inverses, and products, so it is a subgroup. $\square$

Definition 4.2.4 (Generated Subgroup). Let $S \subseteq G$. The **subgroup generated by S** , denoted $\langle S \rangle$, is the smallest subgroup of G containing S . Equivalently, it is the intersection of all subgroups of G containing S .

Definition 4.2.5 (Cyclic Group). A group G is **cyclic** if there exists $g \in G$ such that $G = \langle g \rangle$. Such an element g is called a **generator** of G .

Theorem 4.2.2: Classification of Cyclic Groups

Every cyclic group is isomorphic either to $(\mathbb{Z}, +)$ or to $(\mathbb{Z}/n\mathbb{Z}, +)$ for some positive integer n .

Proof. Let $G = \langle g \rangle$. Define $\varphi : \mathbb{Z} \rightarrow G$ by $\varphi(k) = g^k$. This map is a group homomorphism and is surjective. If g has infinite order, then $\ker \varphi = \{0\}$, so $\mathbb{Z} \cong G$. If g has order n , then $\ker \varphi = n\mathbb{Z}$, and the induced map gives $\mathbb{Z}/n\mathbb{Z} \cong G$. $\square$

4.2.4 Cosets and Lagrange's Theorem

Cosets are best understood as translated copies of a subgroup. If H records all moves that are considered internally invisible, then gH records all elements that differ from g by such an invisible move. This is why cosets naturally lead to equivalence classes. Two elements are in the same left coset precisely when one can be transformed into the other by multiplying by an element of H .

Lagrange's theorem is the first major counting theorem in group theory. Its content is not merely that $|H|$ divides $|G|$; it says that the group can be partitioned into disjoint blocks of equal size. This equal-size partition is the structural reason behind many divisibility results, including the fact that the order of an element in a finite group divides the order of the group.

Subgroups do not merely sit inside groups; they divide groups into equal-sized pieces.

Definition 4.2.6 (Cosets). Let $H \leq G$ and $g \in G$.

1. The **left coset** of H determined by g is $gH = \{gh : h \in H\}$.

2. The **right coset** of H determined by g is $Hg = \{hg : h \in H\}$.

Proposition 4.2.4: Cosets Partition the Group

The left cosets of H in G form a partition of G .

Proof. Every $g \in G$ lies in gH , since $g = ge$ and $e \in H$. Now suppose two cosets aH and bH intersect. Then there exist $h_1, h_2 \in H$ such that $ah_1 = bh_2$. Hence

$$b^{-1}a = h_2h_1^{-1} \in H.$$

For any $x \in aH$, write $x = ah$. Then

$$x = b(b^{-1}a)h \in bH.$$

Thus $aH \subseteq bH$. Symmetrically, $bH \subseteq aH$. Therefore two left cosets are either disjoint or equal. $\square$

Theorem 4.2.3: Lagrange's Theorem

Let G be a finite group and $H \leq G$. Then

$$|G| = [G : H]|H|,$$

where $[G : H]$ is the number of left cosets of H in G . In particular, $|H|$ divides $|G|$.

Proof. The left cosets partition G . For each $g \in G$, the map $H \rightarrow gH$ given by $h \mapsto gh$ is a bijection, so every coset has exactly $|H|$ elements. If there are $[G : H]$ cosets, then $|G| = [G : H]|H|$. $\square$

Corollary 4.2.1

If G is finite and $g \in G$, then the order of g divides $|G|$.

Proof. The subgroup generated by g has order $|g|$. Apply Lagrange's theorem to $\langle g \rangle \leq G$. $\square$

Corollary 4.2.2

Every group of prime order is cyclic.

Proof. Let $|G| = p$ be prime and choose $g \neq e$. The order of g divides p and is not 1, so it must be p . Thus $\langle g \rangle = G$. $\square$

4.2.5 Normal Subgroups and Quotient Groups

The need for normality appears when we try to make cosets into a group. For an arbitrary subgroup H , the product of two left cosets may depend on the chosen representatives, so the operation is not well-defined. Normality is exactly the condition that removes this ambiguity. It ensures that multiplying representatives and then passing to cosets gives a result independent of representative choices.

Thus quotient groups are not obtained by simply deleting elements. They are obtained by declaring all elements of H to be equivalent to the identity and then asking what algebraic structure remains. This is the group-theoretic version of reducing modulo an integer, where all multiples of n are treated as zero and the remaining arithmetic takes place among residue classes.

For a subgroup H , the set of cosets G/H is always a set, but it is not always a group. To multiply cosets by $(aH)(bH) = abH$, we need the answer to be independent of representatives. This is exactly what normality guarantees.

Definition 4.2.7 (Normal Subgroup). A subgroup $N \leq G$ is **normal**, written $N \trianglelefteq G$, if

$$gNg^{-1} = N$$

for all $g \in G$. Equivalently, $gng^{-1} \in N$ for all $g \in G$ and $n \in N$.

Proposition 4.2.5: Equivalent Forms of Normality

Let $N \leq G$. The following are equivalent:

1. $N \trianglelefteq G$.
2. $gN = Ng$ for every $g \in G$.
3. The operation $(aN)(bN) = abN$ is well-defined on the set of left cosets.

Definition 4.2.8 (Quotient Group). If $N \trianglelefteq G$, then the set of left cosets

$$G/N = \{gN : g \in G\}$$

forms a group under

$$(aN)(bN) = abN.$$

This group is called the **quotient group** of G by N .

Example 4.2.4: Integers Modulo n

The subgroup $n\mathbb{Z} \leq \mathbb{Z}$ is normal because $\mathbb{Z}$ is abelian. The quotient group

$$\mathbb{Z}/n\mathbb{Z}$$

is precisely the group of residue classes modulo n .

Remark 4.2.2

The quotient construction is one of the most important constructions in mathematics. It formalizes the idea that certain elements are to be regarded as equivalent. In $\mathbb{Z}/n\mathbb{Z}$, two integers are identified if their difference lies in $n\mathbb{Z}$.

4.2.6 Homomorphisms and Isomorphism Theorems

Homomorphisms are structure-preserving maps. They are more important than arbitrary functions because they respect the operation: applying the map after multiplying is the same as multiplying after applying the map. In this sense, a homomorphism translates one group language into another without breaking algebraic sentences.

The kernel measures the information lost by a homomorphism. Elements in the kernel become indistinguishable from the identity after mapping. The image measures the part of the target actually reached. The isomorphism theorems formalize the idea that every homomorphism factors into two conceptual steps: first collapse the kernel, then identify the resulting quotient with the image.

A homomorphism is a structure-preserving map. It does not merely send elements to elements; it sends algebraic relationships to algebraic relationships.

Definition 4.2.9 (Group Homomorphism). Let G and H be groups. A function $\varphi : G \rightarrow H$ is a **group homomorphism** if

$$\varphi(ab) = \varphi(a)\varphi(b)$$

for all $a, b \in G$.

Definition 4.2.10 (Kernel and Image). Let $\varphi : G \rightarrow H$ be a group homomorphism.

1. The **kernel** is $\ker \varphi = \{g \in G : \varphi(g) = e_H\}$.
2. The **image** is $\operatorname{im} \varphi = \{\varphi(g) : g \in G\}$.

Proposition 4.2.6

For a group homomorphism $\varphi : G \rightarrow H$, $\ker \varphi \trianglelefteq G$ and $\operatorname{im} \varphi \leq H$.

Proof. The image is closed under products and inverses because φ respects the operation. For the kernel, if $x, y \in \ker \varphi$, then

$$\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} = e_H e_H = e_H,$$

so $xy^{-1} \in \ker \varphi$. Thus the kernel is a subgroup. For normality, if $k \in \ker \varphi$ and $g \in G$, then

$$\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g)^{-1} = \varphi(g)e_H\varphi(g)^{-1} = e_H.$$

Thus $gkg^{-1} \in \ker \varphi$. □

Theorem 4.2.4: First Isomorphism Theorem for Groups

Let $\varphi : G \rightarrow H$ be a group homomorphism. Then

$$G/\ker \varphi \cong \operatorname{im} \varphi.$$

Proof. Define $\Phi : G/\ker \varphi \rightarrow \operatorname{im} \varphi$ by

$$\Phi(g \ker \varphi) = \varphi(g).$$

If $g \ker \varphi = g' \ker \varphi$, then $g'^{-1}g \in \ker \varphi$, so $\varphi(g'^{-1}g) = e_H$, and hence $\varphi(g) = \varphi(g')$. Thus Φ is well-defined. It is plainly surjective. Its kernel is the identity coset, so it is injective. Therefore it is an isomorphism. □

Remark 4.2.3

The first isomorphism theorem says that the only information lost by a homomorphism is exactly its kernel. Once the kernel is collapsed to the identity, the remaining quotient is the image.

4.2.7 Group Actions

A group action is a way for an abstract group to reveal itself through motion. Instead of studying group elements only as symbols, we let them act on a set and observe the resulting symmetries. This is why group actions connect algebra with geometry, combinatorics, and number theory. The same group may act on many different sets, and each action exposes different information.

The orbit-stabilizer principle is the basic bookkeeping law of group actions. The orbit counts where a point can go; the stabilizer counts the group elements that keep it fixed. Their sizes multiply to the size of the group in the finite case. This theorem is one of the clearest examples of a general theme: symmetry can be measured by decomposing movement into freedom and constraint.

Groups are not only objects; they also act on objects. This viewpoint explains why groups naturally describe symmetry.

Definition 4.2.11 (Group Action). Let G be a group and X a set. A **left action** of G on X is a map

$$G \times X \rightarrow X, \quad (g, x) \mapsto g \cdot x,$$

such that

1. $e \cdot x = x$ for all $x \in X$;
2. $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and $x \in X$.

Definition 4.2.12 (Orbit and Stabilizer). For a group action of G on X and $x \in X$, define

$$\text{Orb}(x) = \{g \cdot x : g \in G\},$$

and

$$\text{Stab}(x) = \{g \in G : g \cdot x = x\}.$$

Theorem 4.2.5: Orbit-Stabilizer Theorem

If G is finite and acts on X , then

$$|\text{Orb}(x)| = [G : \text{Stab}(x)]$$

for every $x \in X$. In particular,

$$|\text{Orb}(x)| |\text{Stab}(x)| = |G|.$$

Proof. Define a map from left cosets of $\text{Stab}(x)$ to $\text{Orb}(x)$ by

$$g\text{Stab}(x) \mapsto g \cdot x.$$

This map is well-defined because if $g^{-1}h \in \text{Stab}(x)$, then $h \cdot x = g \cdot x$. It is also bijective by definition of the orbit. Hence the number of orbit elements equals the number of cosets. $\square$

Definition 4.2.13 (Center and Centralizer). The **center** of a group G is

$$Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}.$$

The **centralizer** of $a \in G$ is

$$C_G(a) = \{g \in G : ga = ag\}.$$

Theorem 4.2.6: Class Equation

Let G be a finite group acting on itself by conjugation: $g \cdot x = gxg^{-1}$. Then

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)],$$

where x_i are representatives of the non-central conjugacy classes.

Remark 4.2.4

The class equation is a typical example of algebraic counting. It turns the abstract action of conjugation into a numerical equation. Many structure theorems about finite groups begin from this idea.

4.2.8 Sylow Theorems

Lagrange's theorem tells us that the order of a subgroup must divide the order of the group. The converse is false in general: if d divides $|G|$, there need not be a subgroup of order d . Sylow theory tells us that the converse is true for the largest powers of primes dividing $|G|$.

Definition 4.2.14 (Sylow p -subgroup). Let G be a finite group and suppose

$$|G| = p^k m,$$

where p is prime and $p \nmid m$. A subgroup of G of order p^k is called a **Sylow p -subgroup**.

Theorem 4.2.7: Sylow Theorems

Let G be a finite group with $|G| = p^k m$, where p is prime and $p \nmid m$. Then:

1. G has at least one Sylow p -subgroup.
2. Any two Sylow p -subgroups are conjugate.
3. If n_p is the number of Sylow p -subgroups, then

$$n_p \equiv 1 \pmod{p}, \quad n_p \mid m.$$

Example 4.2.5: A Group of Order 15

Let $|G| = 15 = 3 \cdot 5$. The number n_5 of Sylow 5-subgroups satisfies $n_5 \equiv 1 \pmod{5}$ and $n_5 \mid 3$. Hence $n_5 = 1$. Thus the Sylow 5-subgroup is normal. Similarly, $n_3 \equiv 1 \pmod{3}$ and $n_3 \mid 5$, so $n_3 = 1$ or 10 is impossible because $10 \nmid 5$; hence $n_3 = 1$. Both Sylow subgroups are normal. This strongly restricts the structure of G .

4.3 Rings

Rings are introduced when one operation is no longer enough. Addition gives an abelian group structure, while multiplication gives a second operation that interacts with addition through distributivity. This mirrors the arithmetic of integers and polynomials: addition organizes terms, multiplication combines them, and distributivity allows expansion and factorization.

The main conceptual difference between rings and fields is that division is no longer guaranteed. This absence of universal division is not a weakness; it is precisely what allows rings to model arithmetic phenomena such as divisibility, congruences, ideals, factorization, and zero divisors. Much of ring theory asks how much of familiar arithmetic survives when division is unavailable.

Groups describe one operation. Rings describe two operations, traditionally called addition and multiplication. The guiding example is $\mathbb{Z}$: addition forms an abelian group, multiplication is associative, and multiplication distributes over addition.

4.3.1 Definition and Examples

Definition 4.3.1 (Ring). A **ring** is a set R with two binary operations $+$ and $\cdot$ satisfying:

1. $(R, +)$ is an abelian group with identity 0.
2. Multiplication is associative: $(ab)c = a(bc)$ for all $a, b, c \in R$.
3. Distributive laws hold:

$$a(b + c) = ab + ac, \quad (a + b)c = ac + bc.$$

If there is an element $1 \in R$ with $1a = a1 = a$ for all $a \in R$, then R is a **ring with unity**. If $ab = ba$ for all $a, b \in R$, then R is **commutative**.

Example 4.3.1: Basic Rings

1. $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are commutative rings with unity.
2. $M_n(\mathbb{R})$, the set of all $n \times n$ real matrices, is a ring with unity. It is non-commutative for $n \geq 2$.
3. $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring with unity under modular addition and multiplication.
4. If R is a ring, then $R[x]$, the set of polynomials with coefficients in R , is a ring.

Proposition 4.3.1: Elementary Ring Identities

For any ring R and $a, b \in R$,

1. $a0 = 0a = 0$;
2. $a(-b) = -(ab)$ and $(-a)b = -(ab)$;
3. $(-a)(-b) = ab$.

Proof. We prove the first identity. Since $0 + 0 = 0$,

$$a0 = a(0 + 0) = a0 + a0.$$

Adding the additive inverse of $a0$ to both sides gives $0 = a0$. The identity $0a = 0$ is similar. The other statements follow from distributivity and uniqueness of additive inverses. $\square$

4.3.2 Units, Zero Divisors, and Domains

Definition 4.3.2 (Units and Zero Divisors). Let R be a ring with unity.

1. An element $u \in R$ is a **unit** if there exists $v \in R$ such that $uv = vu = 1$.
2. A non-zero element $a \in R$ is a **zero divisor** if there exists non-zero $b \in R$ such that $ab = 0$ or $ba = 0$.

Definition 4.3.3 (Integral Domain). An **integral domain** is a commutative ring with unity $1 \neq 0$ and no zero divisors.

Example 4.3.2

The ring $\mathbb{Z}/6\mathbb{Z}$ is not an integral domain because

$$\bar{2} \cdot \bar{3} = \bar{6} = \bar{0},$$

although $\bar{2}$ and $\bar{3}$ are non-zero. The ring $\mathbb{Z}/5\mathbb{Z}$ is a field, hence an integral domain.

Proposition 4.3.2: Cancellation in Domains

Let R be an integral domain. If $a \neq 0$ and $ab = ac$, then $b = c$.

Proof. From $ab = ac$, we get $a(b - c) = 0$. Since $a \neq 0$ and R has no zero divisors, $b - c = 0$. Hence $b = c$. $\square$

4.3.3 Ideals and Quotient Rings

Ideals play for rings the role that normal subgroups play for groups. They are exactly the subsets by which one may quotient while preserving the ring operations. The condition $ra \in I$ and $ar \in I$ says that once an element is declared to be zero modulo I , all of its ring multiples must also become zero. This is why ideals are the correct algebraic language for modular arithmetic.

For example, the ideal $n\mathbb{Z}$ records all integers that should vanish modulo n . The quotient ring $\mathbb{Z}/n\mathbb{Z}$ is obtained by identifying integers whose difference lies in $n\mathbb{Z}$. This example is the prototype for quotient rings in general: ideals encode the relations we impose, and quotients describe the arithmetic after those relations are imposed.

In group theory, normal subgroups are exactly the subgroups by which we can quotient. In ring theory, the corresponding objects are ideals.

Definition 4.3.4 (Ideal). Let R be a ring. A subset $I \subseteq R$ is a **left ideal** if:

1. $(I, +)$ is a subgroup of $(R, +)$;
2. for all $r \in R$ and $x \in I$, $rx \in I$.

Right ideals are defined similarly. If I is both a left and right ideal, it is called a **two-sided ideal**, or simply an **ideal** in the commutative case.

Example 4.3.3: Ideals in $\mathbb{Z}$

For every integer n , the set

$$n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$$

is an ideal of $\mathbb{Z}$. In fact, every ideal of $\mathbb{Z}$ has this form.

Definition 4.3.5 (Principal Ideal). Let R be a commutative ring and $a \in R$. The ideal generated by a is

$$(a) = \{ra : r \in R\}.$$

An ideal of this form is called a **principal ideal**.

Definition 4.3.6 (Quotient Ring). Let I be an ideal of a ring R . The quotient abelian group R/I becomes a ring under

$$(a + I) + (b + I) = (a + b) + I, \quad (a + I)(b + I) = ab + I.$$

This ring is called the **quotient ring** of R by I .

Example 4.3.4

The ring $\mathbb{Z}/n\mathbb{Z}$ is the quotient ring of $\mathbb{Z}$ by the ideal $n\mathbb{Z}$.

4.3.4 Ring Homomorphisms

Definition 4.3.7 (Ring Homomorphism). Let R and S be rings. A map $\varphi : R \rightarrow S$ is a **ring homomorphism** if

$$\varphi(a + b) = \varphi(a) + \varphi(b), \quad \varphi(ab) = \varphi(a)\varphi(b)$$

for all $a, b \in R$. If the rings have unity, one often also requires $\varphi(1_R) = 1_S$.

Definition 4.3.8 (Kernel and Image). For a ring homomorphism $\varphi : R \rightarrow S$,

$$\ker \varphi = \{r \in R : \varphi(r) = 0\}, \quad \text{im } \varphi = \{\varphi(r) : r \in R\}.$$

Proposition 4.3.3

The kernel of a ring homomorphism is an ideal, and the image is a subring.

Theorem 4.3.1: First Isomorphism Theorem for Rings

Let $\varphi : R \rightarrow S$ be a ring homomorphism. Then

$$R/\ker \varphi \cong \text{im } \varphi.$$

Example 4.3.5: Evaluation Homomorphism

Let F be a field and $a \in F$. Define

$$\text{ev}_a : F[x] \rightarrow F, \quad f(x) \mapsto f(a).$$

This is a ring homomorphism. Its kernel consists of all polynomials divisible by $x - a$. Hence

$$F[x]/(x - a) \cong F.$$

4.3.5 Prime and Maximal Ideals

Definition 4.3.9 (Prime and Maximal Ideals). Let R be a commutative ring with unity.

1. A proper ideal $P \subsetneq R$ is **prime** if $ab \in P$ implies $a \in P$ or $b \in P$.
2. A proper ideal $M \subsetneq R$ is **maximal** if there is no ideal I such that $M \subsetneq I \subsetneq R$.

Theorem 4.3.2: Quotients by Prime and Maximal Ideals

Let R be a commutative ring with unity.

1. R/P is an integral domain if and only if P is prime.
2. R/M is a field if and only if M is maximal.

Proof. For the first statement, in R/P we have

$$(a + P)(b + P) = P$$

if and only if $ab \in P$. Thus R/P has no zero divisors exactly when $ab \in P$ forces $a \in P$ or $b \in P$.

For the second statement, suppose R/M is a field. If $M \subsetneq I \subseteq R$, choose $a \in I \setminus M$. Then $a + M$ is non-zero in R/M , so it has an inverse $b + M$. Thus $ab - 1 \in M \subseteq I$, while $ab \in I$, hence $1 \in I$ and $I = R$. Therefore M is maximal. Conversely, if M is maximal and $a \notin M$, then the ideal generated by M and a is all of R , so there exist $m \in M$ and $r \in R$ such that $m + ra = 1$. Hence $(r + M)(a + M) = 1 + M$, so every non-zero element of R/M is invertible. $\square$

4.3.6 Polynomial Rings and Divisibility

Polynomial rings are the bridge between abstract algebra and equation solving.

Definition 4.3.10 (Divisibility). Let R be an integral domain and $a, b \in R$ with $a \neq 0$. We say a **divides** b , written $a \mid b$, if there exists $c \in R$ such that $b = ac$.

Definition 4.3.11 (Irreducible and Prime Elements). Let R be an integral domain and $p \in R$ be non-zero and not a unit.

1. p is **irreducible** if $p = ab$ implies that either a or b is a unit.
2. p is **prime** if $p \mid ab$ implies $p \mid a$ or $p \mid b$.

Proposition 4.3.4

In an integral domain, every prime element is irreducible.

Proof. Suppose p is prime and $p = ab$. Since $p \mid ab$, primality implies $p \mid a$ or $p \mid b$. Assume $p \mid a$. Then $a = pc$ for some c , so $p = ab = pcb$. Since $p \neq 0$ and the ring is a domain, cancellation gives $1 = cb$, so b is a unit. The other case is similar. $\square$

Definition 4.3.12 (Euclidean Domain, PID, and UFD). Let R be an integral domain.

1. R is a **Euclidean domain** if it has a Euclidean function allowing division with remainder.
2. R is a **principal ideal domain** (PID) if every ideal of R is principal.
3. R is a **unique factorization domain** (UFD) if every non-zero non-unit can be factored into irreducibles uniquely up to order and multiplication by units.

Theorem 4.3.3: Hierarchy of Domains

Fields $\subseteq$ Euclidean Domains $\subseteq$ PIDs $\subseteq$ UFDs $\subseteq$ Integral Domains.

Remark 4.3.1

The inclusions above are not merely terminology. Each stronger condition gives more tools. In a Euclidean domain, we can run the Euclidean algorithm. In a PID, ideals are controlled by single generators. In a UFD, factorization behaves like integer factorization.

Theorem 4.3.4: Division Algorithm in Polynomial Rings

Let F be a field. If $f(x), g(x) \in F[x]$ and $g(x) \neq 0$, then there exist unique $q(x), r(x) \in F[x]$ such that

$$f(x) = q(x)g(x) + r(x), \quad r(x) = 0 \text{ or } \deg r < \deg g.$$

Corollary 4.3.1

If F is a field, then $F[x]$ is a Euclidean domain, hence a PID and a UFD.

Theorem 4.3.5: Gauss's Lemma

If R is a UFD, then $R[x]$ is a UFD.

4.4 Fields

Fields are the algebraic setting in which the four elementary arithmetic operations behave as expected, except for division by zero. Because every non-zero element has a multiplicative inverse, linear equations over a field can be solved using the familiar methods of Gaussian elimination. This is why vector spaces are defined over fields in ordinary linear algebra.

Field extensions arise when a field is too small to contain the solutions we want. The passage from $\mathbb{Q}$ to $\mathbb{Q}(\sqrt{2})$ adds a root of $x^2 - 2$; the passage from $\mathbb{R}$ to $\mathbb{C}$ adds a root of $x^2 + 1$. The basic idea is to enlarge the number system just enough to make certain polynomial equations solvable while preserving field arithmetic.

Fields are rings in which every non-zero element is invertible. They are the natural algebraic setting for solving linear equations, building vector spaces, and studying roots of polynomials.

4.4.1 Definitions and First Examples

Definition 4.4.1 (Field). A **field** is a commutative ring F with unity $1 \neq 0$ such that every non-zero element of F is a unit.

Example 4.4.1

$\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are fields. The ring $\mathbb{Z}$ is not a field because 2 has no multiplicative inverse in $\mathbb{Z}$. The quotient ring $\mathbb{Z}/p\mathbb{Z}$ is a field if and only if p is prime.

Definition 4.4.2 (Characteristic). The **characteristic** of a field F is the smallest positive integer n such that

$$\underbrace{1 + 1 + \cdots + 1}_{n \text{ times}} = 0.$$

If no such n exists, the characteristic is 0.

Proposition 4.4.1

The characteristic of a field is either 0 or a prime number.

Proof. If $\text{char } F = n > 0$ and $n = ab$ with $1 < a, b < n$, then

$$0 = n \cdot 1 = (a \cdot 1)(b \cdot 1).$$

Since a field has no zero divisors, either $a \cdot 1 = 0$ or $b \cdot 1 = 0$, contradicting the minimality of n . Hence n is prime. $\square$

4.4.2 Field Extensions

Definition 4.4.3 (Field Extension). If $F \subseteq K$ are fields and the operations of F are the restrictions of the operations of K , then K is called a **field extension** of F , denoted K/F .

Every field extension K/F turns K into a vector space over F .

Definition 4.4.4 (Degree of an Extension). The **degree** of K/F is

$$[K : F] = \dim_F K.$$

If this dimension is finite, K/F is a **finite extension**.

Example 4.4.2

The field $\mathbb{C}$ is a degree 2 extension of $\mathbb{R}$, because every complex number can be uniquely written as $a + bi$ with $a, b \in \mathbb{R}$. Thus $\{1, i\}$ is a basis of $\mathbb{C}$ over $\mathbb{R}$.

Theorem 4.4.1: Tower Law

If $F \subseteq E \subseteq K$ are fields, then

$$[K : F] = [K : E][E : F]$$

whenever the degrees are finite.

Proof. Let $\{u_1, \dots, u_m\}$ be a basis of E over F , and let $\{v_1, \dots, v_n\}$ be a basis of K over E . Then the products $u_i v_j$ form a basis of K over F . Hence $[K : F] = mn$. $\square$

4.4.3 Algebraic Elements and Minimal Polynomials

Definition 4.4.5 (Algebraic and Transcendental Elements). Let K/F be a field extension and $\alpha \in K$.

1. α is **algebraic** over F if there exists a non-zero polynomial $f(x) \in F[x]$ such that $f(\alpha) = 0$.
2. α is **transcendental** over F if it is not algebraic over F .

Definition 4.4.6 (Minimal Polynomial). If α is algebraic over F , the **minimal polynomial** of α over F is the unique monic irreducible polynomial $m_\alpha(x) \in F[x]$ such that $m_\alpha(\alpha) = 0$.

Example 4.4.3

The element $\sqrt{2} \in \mathbb{R}$ is algebraic over $\mathbb{Q}$, and its minimal polynomial is $x^2 - 2$. The element $i \in \mathbb{C}$ is algebraic over $\mathbb{R}$, with minimal polynomial $x^2 + 1$.

Theorem 4.4.2: Simple Algebraic Extensions

Let α be algebraic over F with minimal polynomial $m_\alpha(x)$. Then

$$F(\alpha) \cong F[x]/(m_\alpha(x)).$$

In particular,

$$[F(\alpha) : F] = \deg m_\alpha.$$

Proof. Consider the evaluation homomorphism $\text{ev}_\alpha : F[x] \rightarrow F(\alpha)$ given by $f(x) \mapsto f(\alpha)$. Its kernel is the ideal generated by $m_\alpha(x)$. By the first isomorphism theorem for rings,

$$F[x]/(m_\alpha(x)) \cong \text{im}(\text{ev}_\alpha) = F[\alpha].$$

Since α is algebraic, $F[\alpha] = F(\alpha)$. The quotient has basis $1, x, \dots, x^{n-1}$, where $n = \deg m_\alpha$. □

4.4.4 Splitting Fields and Algebraic Closure

Definition 4.4.7 (Splitting Field). Let $f(x) \in F[x]$. A field extension K/F is a **splitting field** of f over F if:

1. f factors completely into linear factors in $K[x]$;
2. K is generated over F by the roots of f .

Example 4.4.4

The polynomial $x^2 - 2 \in \mathbb{Q}[x]$ has splitting field $\mathbb{Q}(\sqrt{2})$. The polynomial $x^3 - 2$ has one real root $\sqrt[3]{2}$ and two complex roots $\omega\sqrt[3]{2}$ and $\omega^2\sqrt[3]{2}$, where $\omega = e^{2\pi i/3}$. Its splitting field over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[3]{2}, \omega)$.

Definition 4.4.8 (Algebraically Closed Field). A field K is **algebraically closed** if every non-constant polynomial in $K[x]$ has a root in K .

Theorem 4.4.3: Fundamental Theorem of Algebra

The field $\mathbb{C}$ is algebraically closed.

4.4.5 Finite Fields

Theorem 4.4.4: Classification of Finite Fields

Let F be a finite field.

1. The characteristic of F is a prime p .
2. The number of elements of F is p^n for some $n \geq 1$.
3. For every prime p and every integer $n \geq 1$, there exists a finite field with p^n elements.
4. Any two finite fields with p^n elements are isomorphic.

Remark 4.4.1

A finite field with $q = p^n$ elements is denoted $\mathbb{F}_q$ or $GF(q)$. The multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $q - 1$.

4.5 Galois Theory

Galois theory is the point where the preceding structures meet. A polynomial gives a field extension by adjoining its roots, and the symmetries of that extension form a group of field automorphisms. The striking discovery is that algebraic information about intermediate fields corresponds to group-theoretic information about subgroups.

This correspondence turns questions about equations into questions about symmetry. To understand why an equation is solvable by radicals, one studies the structure of its Galois group. The philosophical lesson is that the difficulty of solving an equation is not located only in its coefficients; it is encoded in the way its roots can be permuted without violating algebraic relations.

Galois theory is the meeting point of groups and fields. It translates questions about roots of polynomials into questions about symmetry groups. The roots of a polynomial may be permuted by field automorphisms, and these permutations encode deep information about the original equation.

4.5.1 Field Automorphisms

Definition 4.5.1 (Field Automorphism). Let K/F be a field extension. An F -**automorphism** of K is a field isomorphism $\sigma : K \rightarrow K$ such that

$$\sigma(a) = a$$

for every $a \in F$.

Definition 4.5.2 (Galois Group). The set of all F -automorphisms of K is denoted

$$\text{Gal}(K/F).$$

Under composition, it forms a group called the **Galois group** of the extension K/F .

Example 4.5.1

For $K = \mathbb{Q}(\sqrt{2})$ and $F = \mathbb{Q}$, every $\mathbb{Q}$ -automorphism must send $\sqrt{2}$ to another root of $x^2 - 2$. Thus it either fixes $\sqrt{2}$ or sends it to $-\sqrt{2}$. Therefore

$$\text{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z}.$$

4.5.2 Normality and Separability

Definition 4.5.3 (Normal Extension). An algebraic extension K/F is **normal** if every irreducible polynomial in $F[x]$ having at least one root in K splits completely over K .

Definition 4.5.4 (Separable Polynomial and Extension). A polynomial is **separable** if it has no repeated roots in an algebraic closure. An algebraic extension is **separable** if every element of the extension has a separable minimal polynomial over the base field.

Definition 4.5.5 (Galois Extension). A finite extension K/F is called **Galois** if it is both normal and separable.

Remark 4.5.1

Over fields of characteristic 0, every irreducible polynomial is separable. Thus for extensions of $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$, separability usually causes no difficulty. Normality is often the more visible condition.

4.5.3 The Fundamental Theorem

Definition 4.5.6 (Fixed Field). Let K/F be a field extension and $H \leq \text{Gal}(K/F)$. The **fixed field** of H is

$$K^H = \{x \in K : \sigma(x) = x \text{ for all } \sigma \in H\}.$$

Theorem 4.5.1: Fundamental Theorem of Galois Theory

Let K/F be a finite Galois extension and let $G = \text{Gal}(K/F)$. There is a one-to-one inclusion-reversing correspondence

$$\{\text{subgroups } H \leq G\} \longleftrightarrow \{\text{intermediate fields } F \subseteq E \subseteq K\}$$

given by

$$H \mapsto K^H, \quad E \mapsto \text{Gal}(K/E).$$

Moreover, E/F is Galois if and only if $\text{Gal}(K/E)$ is a normal subgroup of G , and in that case

$$\text{Gal}(E/F) \cong G / \text{Gal}(K/E).$$

Remark 4.5.2

The correspondence reverses inclusion: larger subgroups fix more elements, so they correspond to smaller intermediate fields. This reversal is the conceptual heart of Galois theory.

4.5.4 Solvability by Radicals

Definition 4.5.7 (Solvable Group). A group G is **solvable** if there is a chain of subgroups

$$\{e\} = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_n = G$$

such that every quotient G_{i+1}/G_i is abelian.

Theorem 4.5.2: Galois Criterion for Solvability by Radicals

A polynomial over a field of characteristic 0 is solvable by radicals if its Galois group is solvable. Conversely, under the standard hypotheses of classical Galois theory, solvability by radicals forces the corresponding Galois group to be solvable.

Remark 4.5.3

The general polynomial of degree n has Galois group S_n . The group S_n is not solvable for $n \geq 5$. This is the structural reason why there is no general formula by radicals for polynomial equations of degree five and higher.

4.6 Modules

Modules generalize vector spaces by allowing the scalars to come from a ring rather than a field. This single change has large consequences. In a vector space, every non-zero scalar is invertible, so bases behave cleanly and dimension is well-defined. In a module, scalars may fail to be invertible or may even be zero divisors, so linear algebra becomes more delicate.

This is exactly why modules are useful. They retain the language of addition and scalar multiplication while being flexible enough to encode abelian groups, ideals, quotient structures, and solutions of linear equations over rings. One should read module theory as linear algebra after removing the simplifying assumption that scalars can always be divided.

Vector spaces are built over fields. Modules are what remain when the scalars are allowed to come from a ring. This generalization is natural, but it also removes some convenient properties of vector spaces. In particular, modules need not have bases.

4.6.1 Definitions and Examples

Definition 4.6.1 (*R*-Module). Let R be a ring with unity. A **left R -module** is an abelian group $(M, +)$ together with a scalar multiplication

$$R \times M \rightarrow M, \quad (r, m) \mapsto rm,$$

satisfying, for all $r, s \in R$ and $m, n \in M$,

1. $r(m + n) = rm + rn$;
2. $(r + s)m = rm + sm$;
3. $(rs)m = r(sm)$;
4. $1m = m$.

Example 4.6.1: Basic Modules

1. Every vector space over a field F is an F -module.
2. Every abelian group is a $\mathbb{Z}$ -module, where $n \cdot g$ means repeated addition.
3. A ring R is a module over itself.
4. Every ideal of a commutative ring R is an R -submodule of R .

Definition 4.6.2 (Submodule). Let M be an R -module. A subset $N \subseteq M$ is a **submodule** if N is an additive subgroup of M and $rn \in N$ for all $r \in R$, $n \in N$.

Definition 4.6.3 (Quotient Module). If N is a submodule of M , then the quotient abelian group M/N becomes an R -module by

$$r(m + N) = rm + N.$$

4.6.2 Module Homomorphisms and Exact Sequences

Definition 4.6.4 (Module Homomorphism). Let M and N be R -modules. A map $f : M \rightarrow N$ is an **R -module homomorphism** if

$$f(m_1 + m_2) = f(m_1) + f(m_2), \quad f(rm) = rf(m)$$

for all $m, m_1, m_2 \in M$ and $r \in R$.

Theorem 4.6.1: First Isomorphism Theorem for Modules

If $f : M \rightarrow N$ is an R -module homomorphism, then

$$M / \ker f \cong \operatorname{im} f.$$

Definition 4.6.5 (Exact Sequence). A sequence of R -modules and homomorphisms

$$\cdots \rightarrow M_{i-1} \xrightarrow{f_{i-1}} M_i \xrightarrow{f_i} M_{i+1} \rightarrow \cdots$$

is **exact at** M_i if

$$\operatorname{im} f_{i-1} = \ker f_i.$$

The sequence is **exact** if it is exact at every module.

Remark 4.6.1

Exactness is a compact way of recording algebraic information. For example, a sequence

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$$

is exact if and only if f is injective, g is surjective, and $C \cong B/f(A)$.

4.6.3 Free Modules and Failure of Bases

Definition 4.6.6 (Free Module). An R -module M is **free** if it has a basis, meaning there exists a subset $B \subseteq M$ such that every element of M can be written uniquely as a finite R -linear combination of elements of B .

Example 4.6.2

The module R^n is free with standard basis $e_1, \dots, e_n$. However, not every module is free. For instance, $\mathbb{Z}/n\mathbb{Z}$ is a $\mathbb{Z}$ -module, but it is not a free $\mathbb{Z}$ -module because every element has finite additive order.

Remark 4.6.2

This is a major difference between vector spaces and modules. Every vector space has a basis, but modules over general rings may have torsion, may fail to be free, and may have more complicated decompositions.

4.6.4 Finitely Generated Modules over PIDs

Definition 4.6.7 (Finitely Generated Module). An R -module M is **finitely generated** if there exist $m_1, \dots, m_n \in M$ such that every $m \in M$ can be written as

$$m = r_1 m_1 + \cdots + r_n m_n$$

for some $r_1, \dots, r_n \in R$.

Definition 4.6.8 (Torsion Element). Let M be an R -module over an integral domain R . An element $m \in M$ is a **torsion element** if there exists non-zero $r \in R$ such that $rm = 0$.

Theorem 4.6.2: Structure Theorem for Finitely Generated Modules over a PID

Let R be a PID and let M be a finitely generated R -module. Then

$$M \cong R^r \oplus R/(d_1) \oplus \cdots \oplus R/(d_k),$$

where $r \geq 0$, each d_i is a non-zero non-unit, and

$$d_1 \mid d_2 \mid \cdots \mid d_k.$$

The integer r and the invariant factors d_i are unique up to multiplication by units.

Example 4.6.3: Finite Abelian Groups

Taking $R = \mathbb{Z}$, the structure theorem implies that every finitely generated abelian group is isomorphic to

$$\mathbb{Z}^r \oplus \mathbb{Z}/d_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_k\mathbb{Z},$$

where $d_1 \mid d_2 \mid \cdots \mid d_k$.

Example 4.6.4: Linear Algebra Reinterpreted

Let V be a finite-dimensional vector space over F and let $T : V \rightarrow V$ be a linear transformation. We may turn V into an $F[x]$ -module by defining

$$f(x) \cdot v = f(T)(v).$$

The structure theorem for finitely generated modules over the PID $F[x]$ is one route to rational canonical form and Jordan canonical form.

4.7 Concrete Group Models and Classification Principles

The abstract definitions become useful only when they can be connected back to concrete groups. Cyclic groups model repeated application of one operation, permutation groups model finite symmetries, and direct products assemble independent systems. These constructions also illustrate a recurring goal of algebra: classify objects by reducing them to standard building blocks.

4.7.1 Cyclic Groups

Definition 4.7.1 (Cyclic Group). A group G is **cyclic** if there exists $g \in G$ such that

$$G = \langle g \rangle = \{g^n : n \in \mathbb{Z}\}.$$

The element g is called a **generator** of G .

Theorem 4.7.1: Classification of Cyclic Groups

Every cyclic group is isomorphic either to $(\mathbb{Z}, +)$ or to $(\mathbb{Z}/n\mathbb{Z}, +)$ for some positive integer n .

Proof. Let $G = \langle g \rangle$ and define $\varphi : \mathbb{Z} \rightarrow G$ by $\varphi(k) = g^k$. This is a surjective homomorphism. By the first isomorphism theorem,

$$G \cong \mathbb{Z} / \ker \varphi.$$

Every subgroup of $\mathbb{Z}$ has the form $n\mathbb{Z}$. If the kernel is $\{0\}$, then $G \cong \mathbb{Z}$. Otherwise $\ker \varphi = n\mathbb{Z}$, where n is the order of g , and $G \cong \mathbb{Z}/n\mathbb{Z}$. $\square$

Corollary 4.7.1

Every subgroup of a cyclic group is cyclic. If G has finite order n , then for every divisor $d \mid n$, there is exactly one subgroup of order d .

Proof. For $G = \langle g \rangle$, a subgroup corresponds under the quotient description to a subgroup of $\mathbb{Z}$. In the finite case, the subgroup of order d is generated by $g^{n/d}$. $\square$

Example 4.7.1: Generators of a Finite Cyclic Group

The additive group $\mathbb{Z}/n\mathbb{Z}$ is generated by the residue class $[k]$ exactly when $\gcd(k, n) = 1$. Therefore it has

$$\varphi(n)$$

generators, where φ is Euler's totient function.

4.7.2 Permutations and Cycle Decomposition**Theorem 4.7.2: Disjoint Cycle Decomposition**

Every permutation in S_n can be written as a product of disjoint cycles. This decomposition is unique up to the order of the cycles and cyclic rotation within each cycle.

Proof. Starting from any element a , repeatedly apply the permutation until the first repetition. Bijectivity forces the first repeated element to be a , producing a cycle. Remove those elements and repeat on the remaining set. Different orbits are disjoint, and every element belongs to exactly one orbit. $\square$

Definition 4.7.2 (Sign of a Permutation). A transposition exchanges two elements and fixes all others. Every permutation is a product of transpositions. Its **sign** is

$$\text{sgn}(\sigma) = (-1)^r,$$

where r is the number of transpositions in any such decomposition.

Theorem 4.7.3

The parity of r is independent of the chosen transposition decomposition. Hence

$$\text{sgn} : S_n \rightarrow \{1, -1\}$$

is a well-defined surjective homomorphism. Its kernel is the alternating group A_n .

Remark 4.7.1

The determinant formula from linear algebra contains the same sign homomorphism. The alternating signs in

$$\det A = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{1\sigma(1)} \cdots a_{n\sigma(n)}$$

are not an arbitrary convention; they encode the parity structure of the symmetric group.

4.7.3 Direct Products

Definition 4.7.3 (Direct Product of Groups). Let G and H be groups. Their **direct product** is the set $G \times H$ with operation

$$(g_1, h_1)(g_2, h_2) = (g_1g_2, h_1h_2).$$

Proposition 4.7.1

The order of $(g, h) \in G \times H$ is

$$\text{lcm}(|g|, |h|)$$

when both component orders are finite.

Example 4.7.2: Chinese Remainder Structure

If $\gcd(m, n) = 1$, then

$$\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$$

as additive groups and, more strongly, as rings. The isomorphism sends

$$[a]_{mn} \mapsto ([a]_m, [a]_n).$$

Thus arithmetic modulo a composite modulus splits into independent arithmetic modulo coprime factors.

4.8 Euclidean Domains, PIDs, and Unique Factorization

The integers and polynomial rings over fields share an important computational feature: division with remainder. Abstracting that feature leads to Euclidean domains, principal ideal domains, and unique factorization domains. The implications

$$\text{Euclidean domain} \implies \text{PID} \implies \text{UFD}$$

explain why gcd algorithms, Bezout identities, and prime factorizations work in these settings.

Definition 4.8.1 (Euclidean Domain). An integral domain R is a **Euclidean domain** if there exists a function

$$\delta : R \setminus \{0\} \rightarrow \mathbb{N}$$

such that for all $a, b \in R$ with $b \neq 0$, there exist $q, r \in R$ satisfying

$$a = bq + r,$$

where either $r = 0$ or $\delta(r) < \delta(b)$.

Example 4.8.1

The ring $\mathbb{Z}$ is Euclidean with $\delta(n) = |n|$. If F is a field, then $F[x]$ is Euclidean with $\delta(f) = \deg f$. The Gaussian integers $\mathbb{Z}[i]$ are Euclidean with norm

$$N(a + bi) = a^2 + b^2.$$

Theorem 4.8.1: Euclidean Domains Are PIDs

Every ideal in a Euclidean domain is principal.

Proof. Let I be a non-zero ideal and choose $d \in I \setminus \{0\}$ with minimal Euclidean value. For any $a \in I$, write

$$a = dq + r$$

with $r = 0$ or $\delta(r) < \delta(d)$. Since $r = a - dq \in I$, minimality forces $r = 0$. Thus d divides every element of I , so $I = (d)$. $\square$

Theorem 4.8.2: Bezout Identity in a PID

Let R be a PID and let $a, b \in R$. A greatest common divisor d of a and b can be written as

$$d = ax + by$$

for some $x, y \in R$.

Proof. The ideal (a, b) generated by a and b is principal, say $(a, b) = (d)$. Since $d \in (a, b)$, it has the form $ax + by$. Since both a and b belong to (d) , the element d divides both. Every common divisor of a and b divides every linear combination, hence divides d . $\square$

Definition 4.8.2 (Irreducible and Prime Elements). Let R be an integral domain and let $p \in R$ be a non-zero non-unit.

1. p is **irreducible** if $p = ab$ implies that a or b is a unit.
2. p is **prime** if $p \mid ab$ implies $p \mid a$ or $p \mid b$.

Proposition 4.8.1

Every prime element is irreducible. In a PID, every irreducible element is prime.

Proof. The first implication follows by applying primality to a factorization $p = ab$. For the second, suppose p is irreducible and $p \mid ab$. The ideal (p, a) is generated by some d dividing p . Since p is irreducible, either d is a unit or d is associated to p . In the second case $p \mid a$. In the first case Bezout gives $1 = px + ay$; multiplying by b gives

$$b = pbx + aby,$$

so $p \mid b$. $\square$

Theorem 4.8.3: Unique Factorization in a PID

Every PID is a unique factorization domain: every non-zero non-unit factors into irreducibles, and the factorization is unique up to order and multiplication by units.

Remark 4.8.1

The distinction between irreducible and prime matters outside UFDs. In $\mathbb{Z}[\sqrt{-5}]$, for example,

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}),$$

and these factorizations are not equivalent. Failure of unique factorization is one reason ideals become more fundamental than elements in algebraic number theory.

4.9 The Chinese Remainder Theorem and Interpolation

The Chinese remainder theorem is a structural decomposition theorem. It says that quotienting by an intersection of sufficiently independent ideals is the same as taking a product of quotients. In arithmetic, it reconstructs integers from residues. In polynomial algebra, it becomes interpolation.

Definition 4.9.1 (Comaximal Ideals). Ideals $I, J \leq R$ are **comaximal** if

$$I + J = R.$$

Equivalently, there exist $a \in I$ and $b \in J$ such that $a + b = 1$.

Theorem 4.9.1: Chinese Remainder Theorem

If $I_1, \dots, I_n$ are pairwise comaximal ideals in a commutative ring R , then the map

$$\Phi : R \rightarrow R/I_1 \times \cdots \times R/I_n, \quad r \mapsto (r + I_1, \dots, r + I_n)$$

is surjective and has kernel

$$I_1 \cap \cdots \cap I_n = I_1 \cdots I_n.$$

Therefore

$$R/(I_1 \cdots I_n) \cong R/I_1 \times \cdots \times R/I_n.$$

Two-Ideal Case. For comaximal ideals I and J , choose $a \in I$ and $b \in J$ with $a + b = 1$. Given residues $x + I$ and $y + J$, set

$$r = xb + ya.$$

Because $b \equiv 1 \pmod{I}$ and $a \equiv 1 \pmod{J}$, we have

$$r \equiv x \pmod{I}, \quad r \equiv y \pmod{J}.$$

Thus the map is surjective. Its kernel is $I \cap J$, and comaximality gives $I \cap J = IJ$. The general case follows by induction. $\square$

Example 4.9.1: Solving Simultaneous Congruences

Solve

$$x \equiv 2 \pmod{3}, \quad x \equiv 3 \pmod{5}.$$

Since $2 \cdot 3 + (-1) \cdot 5 = 1$, one convenient reconstruction is

$$x \equiv 2 \cdot (-5) + 3 \cdot 6 = 8 \pmod{15}.$$

Indeed, $8 \equiv 2 \pmod{3}$ and $8 \equiv 3 \pmod{5}$.

Theorem 4.9.2: Lagrange Interpolation

Let F be a field and let $a_1, \dots, a_n \in F$ be distinct. For arbitrary values $b_1, \dots, b_n \in F$, there exists a unique polynomial $p \in F[x]$ of degree less than n satisfying

$$p(a_i) = b_i \quad (1 \leq i \leq n).$$

It is given by

$$p(x) = \sum_{i=1}^n b_i \prod_{j \neq i} \frac{x - a_j}{a_i - a_j}.$$

Remark 4.9.1

This is the polynomial form of the Chinese remainder theorem applied to the pairwise comaximal ideals $(x - a_i)$. Evaluation at the points a_i is the quotient map

$$F[x] \longrightarrow \prod_{i=1}^n F[x]/(x - a_i) \cong F^n.$$

Thus interpolation is not an isolated formula; it is a quotient-ring decomposition.

4.10 Irreducibility and Constructing Field Extensions

A polynomial extension $F(\alpha)$ is controlled by the minimal polynomial of α . Consequently, the ability to recognize irreducible polynomials is central to field theory. Irreducibility plays the same role for polynomial rings that primality plays for the integers.

Theorem 4.10.1: Gauss's Lemma

A primitive polynomial in $\mathbb{Z}[x]$ is irreducible over $\mathbb{Z}$ if and only if it is irreducible over $\mathbb{Q}$. More generally, for a UFD R with fraction field K , a primitive polynomial is irreducible in $R[x]$ if and only if it is irreducible in $K[x]$.

Theorem 4.10.2: Eisenstein's Criterion

Let

$$f(x) = a_n x^n + \cdots + a_1 x + a_0 \in \mathbb{Z}[x].$$

If there exists a prime p such that

$$p \nmid a_n, \quad p \mid a_k \quad (0 \leq k < n), \quad p^2 \nmid a_0,$$

then f is irreducible over $\mathbb{Q}$.

Proof. Suppose $f = gh$ with non-constant primitive integer polynomials. Reducing modulo p gives

$$\bar{f}(x) = \bar{a}_n x^n.$$

Thus both $\bar{g}$ and $\bar{h}$ are monomials, so the constant terms of both g and h are divisible by p . Their product is the constant term a_0 , which would then be divisible by p^2 , a contradiction. $\square$

Example 4.10.1

For every prime p , the polynomial

$$x^{p-1} + x^{p-2} + \cdots + x + 1$$

is irreducible over $\mathbb{Q}$. Apply Eisenstein's criterion after substituting $x + 1$:

$$\frac{(x+1)^p - 1}{x}.$$

All non-leading coefficients are divisible by p , and the constant term is p , not divisible by p^2 .

Example 4.10.2: A Finite Field

The polynomial $x^2 + x + 1$ has no root in $\mathbb{F}_2$, so it is irreducible. Therefore

$$\mathbb{F}_2[x]/(x^2 + x + 1)$$

is a field with four elements. If α denotes the residue class of x , then

$$\alpha^2 + \alpha + 1 = 0, \quad \alpha^2 = \alpha + 1$$

in characteristic 2. Its elements are

$$0, \quad 1, \quad \alpha, \quad \alpha + 1.$$

4.11 Conclusion

Abstract algebra replaces computation with structure. Groups encode invertible operations and symmetry. Rings encode arithmetic with addition and multiplication. Fields provide the scalar systems needed for linear algebra and polynomial equations. Galois theory connects field extensions with groups of symmetries. Modules generalize vector spaces and reveal how linear algebra changes when the scalars are no longer a field.

The common method is the same throughout: define a structure by axioms, construct maps that preserve the structure, identify kernels and quotients, and then classify objects up to isomorphism. This pattern is one of the central organizing principles of modern mathematics.

4.12 Chapter Workshop

Worked example: Arithmetic in a quotient ring

In $\mathbb{F}_2[x]/(x^2 + x + 1)$, let $\alpha = [x]$. The defining relation gives

$$\alpha^2 = \alpha + 1.$$

Hence

$$\alpha(\alpha + 1) = \alpha^2 + \alpha = (\alpha + 1) + \alpha = 1.$$

Thus $\alpha + 1$ is the multiplicative inverse of α . The quotient contains four elements and every nonzero element is invertible, so it is a field. Computation in an extension field is polynomial reduction modulo an irreducible polynomial.

Application: Error-correcting codes

A binary linear code is a subspace $C \subseteq \mathbb{F}_2^n$. A generator matrix encodes messages, while a parity-check matrix tests whether a received vector lies in the code. The syndrome identifies a coset of C and can locate likely error patterns. Finite fields and quotient polynomials therefore provide the arithmetic used by practical communication and storage systems.

4.13 Exercises

1. List all subgroups of $\mathbb{Z}_{12}$.
2. Compute the units and zero divisors in $\mathbb{Z}_{10}$.

3. Prove that the kernel of a group homomorphism is normal.
4. Determine whether $x^3 + x + 1$ is irreducible over $\mathbb{F}_2$.
5. Use the Chinese remainder theorem to solve $x \equiv 2 \pmod{3}$ and $x \equiv 3 \pmod{5}$.
6. Explain why a module over a ring need not possess a basis.

Selected hints and answers

1. A cyclic group has one subgroup for each divisor of its order.
2. The units are classes coprime to 10: 1, 3, 7, 9. The nonzero zero divisors are 2, 4, 5, 6, 8.
3. If $k \in \ker \varphi$, then $\varphi(gkg^{-1}) = \varphi(g)e\varphi(g)^{-1} = e$.
4. A cubic over a field is reducible exactly when it has a root. Test 0 and 1.
5. The solution is 8 (mod 15).
6. Over a ring, torsion can prevent linear independence and spanning from behaving as they do over a field; $\mathbb{Z}/2\mathbb{Z}$ as a $\mathbb{Z}$ -module is a basic example.

Chapter 5

Mathematical Analysis II

Learning objectives

By the end of this chapter, you should be able to:

- distinguish pointwise, uniform, absolute, and conditional convergence;
- compute derivatives and Taylor approximations for multivariable functions;
- use the implicit-function theorem and constrained optimization;
- evaluate multiple, line, and surface integrals with appropriate coordinates;
- interpret Green, Gauss, and Stokes theorems and analyze Fourier expansions.

Now we will move on to another part of the analysis, which is Mathematical Analysis II. In this chapter, we will study the properties of functions of several variables, including limits, continuity, differentiability, and integrability. We will also introduce the concept of vector-valued functions and study their properties. This chapter will provide a solid foundation for further studies in multivariable calculus, differential equations, and mathematical physics.

5.1 Series

The key idea of this section is that an infinite series is not an infinite act of addition performed all at once. It is a limit process. We add finitely many terms, obtain a sequence of partial sums, and then ask whether that sequence stabilizes. This distinction is essential: the behavior of the terms a_n alone is not enough; what matters is the accumulated behavior of the partial sums.

Many convergence tests can be understood as methods for comparing partial sums with simpler objects. Positive series are controlled by monotonicity and boundedness. Alternating series are controlled by cancellation. Dirichlet and Abel tests are controlled by bounded oscillation multiplied by monotone decay. Uniform convergence of function series adds one more layer: the stabilization must occur at the same speed for every point in the domain.

Series are an essential concept in mathematical analysis, providing a way to represent functions as sums of simpler components. They are widely used in various fields, including physics, engineering, and economics, to model complex phenomena and solve problems.

5.1.1 Numerical Series

The study of infinite series is the discrete analogue of improper integrals. It allows us to define sums of infinitely many terms, providing the foundation for representing functions as power series (Taylor series) in later analysis.

Definition 5.1.1 (Infinite Series). Given a sequence of real numbers $\{a_n\}_{n=1}^{\infty}$, the formal expression

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$$

is called an **infinite series**. The number a_n is called the general term.

To assign a value to this infinite sum, we look at finite sums.

5.1.2 Upper and Lower Limits

Upper and lower limits are especially useful because they preserve information about eventual upper and lower behavior even when an ordinary limit does not exist. Instead of asking where the whole sequence goes, we ask where its subsequences can accumulate. This is why $\limsup$ and $\liminf$ are natural tools in convergence tests involving roots, ratios, and oscillatory expressions.

A practical way to remember them is this: $\limsup x_n$ is the limiting ceiling seen from far out in the sequence, while $\liminf x_n$ is the limiting floor. If these two quantities coincide, the ceiling and the floor squeeze the sequence into a genuine limit. If they differ, the sequence has persistent oscillation or multiple asymptotic behaviors.

To study the convergence of series, it is often helpful to use the concept of upper and lower limits of sequences, especially when the sequence does not converge.

Definition 5.1.2 (Limit Point). Let $\{x_n\}$ be a sequence. A value $\xi \in \mathbb{R} \cup \{+\infty, -\infty\}$ is called a **limit point** of the sequence if there exists a subsequence $\{x_{n_k}\}$ such that $\lim_{k \rightarrow \infty} x_{n_k} = \xi$.

Definition 5.1.3 (Upper and Lower Limits). Let E be the set of all limit points of a sequence $\{x_n\}$. The **upper limit** (limit superior) and **lower limit** (limit inferior) are defined as:

$$\limsup_{n \rightarrow \infty} x_n = \sup E, \quad \liminf_{n \rightarrow \infty} x_n = \inf E$$

Theorem 5.1.1

A sequence $\{x_n\}$ converges to L (possibly $\pm\infty$) if and only if $\limsup x_n = \liminf x_n = L$.

This theorem is really obvious: if the sequence converges, then all its subsequences converge to the same limit, so the set of limit points is just $\{L\}$, and both $\limsup$ and $\liminf$ equal L . Conversely, if $\limsup$ and $\liminf$ coincide at L , then all limit points are L , which means every subsequence has a further subsequence converging to L , forcing the whole sequence to converge to L .

Property 5.1.1

For any two sequences $\{x_n\}$ and $\{y_n\}$:

1. $\limsup(x_n + y_n) \leq \limsup x_n + \limsup y_n$
2. $\liminf(x_n + y_n) \geq \liminf x_n + \liminf y_n$

Equality holds if at least one of the sequences converges.

Definition 5.1.4 (Convergence). Let S_n denote the n -th **partial sum** of the series:

$$S_n = \sum_{k=1}^n a_k = a_1 + \cdots + a_n$$

If the sequence of partial sums $\{S_n\}$ converges to a limit S (i.e., $\lim_{n \rightarrow \infty} S_n = S$), we say the series **converges** to S , and write $\sum_{n=1}^{\infty} a_n = S$. If $\{S_n\}$ does not converge, the series **diverges**.

Theorem 5.1.2: Cauchy Criterion for Series

The series $\sum a_n$ converges if and only if for every $\epsilon > 0$, there exists an integer N such that for all $n > N$ and any $p \geq 1$:

$$|S_{n+p} - S_n| = |a_{n+1} + a_{n+2} + \cdots + a_{n+p}| < \epsilon$$

Theorem 5.1.3: Necessary Condition for Convergence

If the series $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

The converse is **false**. For example, the harmonic series $\sum \frac{1}{n}$ diverges even though $\lim \frac{1}{n} = 0$. This is the most fundamental trap in series analysis.

Property 5.1.2: Linearity

If $\sum a_n = A$ and $\sum b_n = B$, then for any constants α, β :

$$\sum (\alpha a_n + \beta b_n) = \alpha A + \beta B$$

5.1.2.1 Tests for Positive Series

A series $\sum a_n$ is called a **positive series** if $a_n \geq 0$ for all n . For such series, the partial sum sequence $\{S_n\}$ is monotonically increasing.

Theorem 5.1.4

A positive series converges if and only if its sequence of partial sums is bounded above.

Comparison Tests**Theorem 5.1.5: Direct Comparison Test**

Let $0 \leq a_n \leq b_n$ for all n .

1. If $\sum b_n$ converges, then $\sum a_n$ converges.
2. If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Theorem 5.1.6: Limit Comparison Test

Let $a_n > 0, b_n > 0$. Consider the limit $L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$.

1. If $0 < L < +\infty$, then $\sum a_n$ and $\sum b_n$ converge or diverge together.
2. If $L = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.
3. If $L = +\infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Integral Test and p -Series**Theorem 5.1.7: Integral Test**

Let $f(x)$ be a continuous, positive, and decreasing function on $[1, +\infty)$ such that $f(n) = a_n$. Then the series $\sum_{n=1}^{\infty} a_n$ converges if and only if the improper integral $\int_1^{+\infty} f(x)dx$ converges.

From this, we derive the convergence of the famous p -series:

Corollary 5.1.1: p -Series

The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$:

- Converges if $p > 1$.
- Diverges if $p \leq 1$.

More rigorous expressions of this theorem can be described as:

Theorem 5.1.8

Assume $f(x)$ is continuous, positive and integrable on $[a, A]$ for any $A > a > 0$. Pick an increasing sequence $\{x_n\}$ such that $x_n \rightarrow +\infty$ as $n \rightarrow +\infty$ and $a_1 = a$. Then the series $\sum_{n=1}^{\infty} \int_{a_n}^{a_{n+1}} f(x)dx$ and the improper integral $\int_a^{+\infty} f(x)dx$ either both converge or both diverge. More specifically, if the function is decreasing, then we have:

$$\sum_{n=1}^{\infty} \int_{a_n}^{a_{n+1}} f(x)dx = \int_a^{+\infty} f(x)dx$$

Remark 5.1.1

1. The choice of the sequence $\{a_n\}$ is not unique. A common choice is $a_n = n$, which recovers the standard integral test. However, other choices can be made depending on the specific function $f(x)$ and the context of the problem.
2. If the function $f(x)$ is not positive, and $\int_a^{+\infty} f(x)dx$ converges, we can judge the convergence of $\sum_{n=1}^{\infty} \int_{a_n}^{a_{n+1}} f(x)dx$ by considering the absolute values of the integrals. But if $\int_a^{+\infty} f(x)dx$ diverges, we cannot conclude anything about the series without additional information.

Here is some applications of the integral test:

Example 5.1.1

Determine the convergence of the series $\sum_{n=2}^{\infty} \frac{1}{n \ln^q n}$.

Solution: Consider the function $f(x) = \frac{1}{x \ln^q x}$ for $x \geq 2$. This function is continuous, positive, and decreasing for $x > e$ when $q > 0$. We apply the integral test:

$$\int_2^{+\infty} \frac{1}{x \ln^q x} dx$$

Let $t = \ln x$, then $dx = e^t dt$ and when $x = 2$, $t = \ln 2$. The integral becomes:

$$\int_{\ln 2}^{+\infty} \frac{1}{t^q} dt$$

This integral converges if and only if $q > 1$. Therefore, by the integral test, the series $\sum_{n=2}^{\infty} \frac{1}{n \ln^q n}$ converges if and only if $q > 1$ and diverges otherwise.

Also, we can use the integral test to determine the convergence of improper integrals:

Example 5.1.2

Determine the convergence of the integral $\int_0^{+\infty} \frac{dx}{1+x^2 \sin^2 x}$.

Solution: The key process is to find a suitable sequence $\{a_n\}$ to apply the integral test. We can choose $a_n = n\pi$ for $n = 0, 1, 2, \dots$. Then we consider the series:

$$\sum_{n=0}^{\infty} \int_{n\pi}^{(n+1)\pi} \frac{dx}{1+x^2 \sin^2 x}$$

Let $a_n = \int_{n\pi}^{(n+1)\pi} \frac{dx}{1+x^2 \sin^2 x}$. We have:

$$\int_{n\pi}^{(n+1)\pi} \frac{dx}{1+x^2 \sin^2 x} = \int_0^{\pi} \frac{dt}{1+(n\pi+t)^2 \sin^2 t} > \int_0^{\frac{1}{(n+1)\pi}} \frac{dt}{1+(n\pi+t)^2 \sin^2 t}$$

As we can observe that:

$$(n\pi+t)^2 \sin^2 t < (n+1)^2 \pi^2 t^2 < (n+1)^2 \pi^2 \cdot \frac{1}{(n+1)^2 \pi^2} = 1$$

Thus, we have:

$$\int_0^{\frac{1}{(n+1)\pi}} \frac{dt}{1+(n\pi+t)^2 \sin^2 t} > \int_0^{\frac{1}{(n+1)\pi}} \frac{dt}{2} = \frac{1}{2(n+1)\pi}$$

Therefore, $a_n > \frac{1}{2(n+1)\pi}$. Since the series $\sum \frac{1}{n}$ diverges, by the comparison test, the series $\sum a_n$ diverges. Hence, by the integral test, the integral $\int_0^{+\infty} \frac{dx}{1+x^2 \sin^2 x}$ diverges.

Example 5.1.3

Determine the convergence of the integral $\int_0^{+\infty} \frac{dx}{1+x^4 \sin^2 x}$.

Solution: We again choose the sequence $\{a_n\}$ as $a_n = n\pi$ for $n = 0, 1, 2, \dots$. Then we consider the series:

$$\sum_{n=0}^{\infty} \int_{n\pi}^{(n+1)\pi} \frac{dx}{1+x^4 \sin^2 x}$$

Let $a_n = \int_{n\pi}^{(n+1)\pi} \frac{dx}{1+x^4 \sin^2 x}$. We have:

$$\int_{n\pi}^{(n+1)\pi} \frac{dx}{1+x^4 \sin^2 x} = \int_0^{\frac{\pi}{2}} \frac{dt}{1+(n\pi+t)^4 \sin^2 t} + \int_{\frac{\pi}{2}}^{\pi} \frac{dt}{1+(n\pi+t)^4 \sin^2 t} = I_1 + I_2$$

For I_1 , we have:

$$I_1 > \int_0^{\frac{\pi}{2}} \frac{dt}{1+(n\pi+\frac{\pi}{2})^4} = \frac{\pi/2}{1+(n\pi+\frac{\pi}{2})^4} > \frac{\pi/2}{2(n+1)^4 \pi^4} = \frac{1}{4(n+1)^4 \pi^3}$$

For I_2 , we have:

$$I_2 > \int_{\frac{\pi}{2}}^{\pi} \frac{dt}{1+(n\pi+\pi)^4} = \frac{\pi/2}{1+(n\pi+\pi)^4} > \frac{\pi/2}{2(n+1)^4 \pi^4} = \frac{1}{4(n+1)^4 \pi^3}$$

Therefore, $a_n = I_1 + I_2 > \frac{1}{2(n+1)^4 \pi^3}$. Since the series $\sum \frac{1}{n^4}$ converges, by the comparison test, the series $\sum a_n$ converges. Hence, by the integral test, the integral $\int_0^{+\infty} \frac{dx}{1+x^4 \sin^2 x}$ converges.

A Quick Detour: Upper and Lower Limits ($\limsup$ and $\liminf$) Sometimes sequences refuse to settle down and bounce around like a hyperactive kangaroo. When a classic limit doesn't exist, we track their extreme accumulation points instead. These are the **Limit Superior** ($\limsup_{n \rightarrow \infty} x_n$) and **Limit Inferior** ($\liminf_{n \rightarrow \infty} x_n$). A quirky feature of these is their arithmetic—it's not your standard algebra! For example, $\limsup(x_n + y_n) \leq \limsup x_n + \limsup y_n$. They only strictly obey the equals sign when one of the sequences actually converges. Keep this in mind when Cauchy drops a $\limsup$ on you!

Ratio and Root Tests These are the most commonly used tests for computation.

Theorem 5.1.9: D'Alembert's Ratio Test

Let $\sum a_n$ be a positive series and let

$$\rho = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$$

- If $\rho < 1$, the series converges.
- If $\rho > 1$, the series diverges.
- If $\rho = 1$, the test is **inconclusive** (e.g., $1/n$ diverges but $1/n^2$ converges, both have $\rho = 1$).

The proof of this theorem relies on:

$$\liminf_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} \leq \liminf_{n \rightarrow \infty} \sqrt[n]{x_n} \leq \limsup_{n \rightarrow \infty} \sqrt[n]{x_n} \leq \limsup_{n \rightarrow \infty} \frac{x_{n+1}}{x_n}$$

Theorem 5.1.10: Cauchy's Root Test

Let $\sum a_n$ be a positive series and let

$$\rho = \limsup_{n \rightarrow \infty} \sqrt[n]{a_n}$$

- If $\rho < 1$, the series converges.
- If $\rho > 1$, the series diverges.
- If $\rho = 1$, the test is inconclusive.

Raabe's Test When the Ratio Test yields $\rho = 1$, Raabe's test provides a finer criterion.

Theorem 5.1.11: Raabe's Test

Let $a_n > 0$. Consider the limit:

$$R = \lim_{n \rightarrow \infty} n \left(\frac{a_n}{a_{n+1}} - 1 \right)$$

- If $R > 1$, the series converges.
- If $R < 1$, the series diverges.
- If $R = 1$, the test is inconclusive.

5.1.2.2 Alternating Series

An alternating series has the form $\sum_{n=1}^{\infty} (-1)^{n-1} a_n$ where $a_n > 0$.

Theorem 5.1.12: Leibniz Test

If the sequence $\{a_n\}$ satisfies:

1. Monotonicity: $a_{n+1} \leq a_n$ for all n ;
2. Limit zero: $\lim_{n \rightarrow \infty} a_n = 0$;

then the alternating series $\sum (-1)^{n-1} a_n$ converges. Furthermore, the sum S satisfies $|S - S_n| \leq a_{n+1}$.

5.1.2.3 Absolute and Conditional Convergence

For series with arbitrary signs, we distinguish two types of convergence.

Definition 5.1.5. A series $\sum a_n$ is called **absolutely convergent** if the series of absolute values $\sum |a_n|$ converges. If $\sum a_n$ converges but $\sum |a_n|$ diverges, the series is called **conditionally convergent**.

Theorem 5.1.13

Absolute convergence implies convergence.

$$\sum |a_n| < \infty \implies \sum a_n \text{ converges.}$$

Example 5.1.4

The series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \dots$ converges by Leibniz Test, but $\sum \frac{1}{n}$ diverges. Thus, it is **conditionally convergent**.

5.1.2.4 Dirichlet's and Abel's Tests

For general series of the form $\sum_{n=1}^{\infty} a_n b_n$, where standard tests fail, these tests are powerful.

Theorem 5.1.14: Dirichlet's Test

The series $\sum_{n=1}^{\infty} a_n b_n$ converges if:

1. The partial sums of $\sum a_n$ are bounded (i.e., $|\sum_{k=1}^n a_k| \leq M$).
2. The sequence $\{b_n\}$ is monotonic and $\lim_{n \rightarrow \infty} b_n = 0$.

Example Application: Proving $\sum \frac{\sin n}{n}$ converges. Here $a_n = \sin n$ (bounded sums) and $b_n = 1/n$ (goes to 0).

Theorem 5.1.15: Abel's Test

The series $\sum_{n=1}^{\infty} a_n b_n$ converges if:

1. The series $\sum a_n$ converges.
2. The sequence $\{b_n\}$ is monotonic and bounded.

5.1.2.5 Properties of Series**Theorem 5.1.16: Associativity**

If a series converges, we can group terms (insert parentheses) without changing the sum. However, removing parentheses from a grouped series may alter convergence (e.g., $(1-1)+(1-1)+\dots$ converges to 0, but $1-1+1-1\dots$ diverges).

Theorem 5.1.17: Riemann Rearrangement Theorem

- If a series is **absolutely convergent**, any rearrangement of its terms converges to the same sum.
- If a series is **conditionally convergent**, for any real number L (or $\pm\infty$), there exists a rearrangement of terms such that the series converges to L .

This highlights the dangerous nature of conditional convergence: the order of summation matters!

5.1.2.6 Product of Series

Now we shall dive into the topic of multiplying two series together. We have following definition:

Definition 5.1.6. Assume we have two series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$. Their **Cauchy Product** is defined as the sum of this following elements:

$$\begin{pmatrix} a_1b_1 & a_1b_2 & a_1b_3 & a_1b_4 & \cdots \\ a_2b_1 & a_2b_2 & a_2b_3 & a_2b_4 & \cdots \\ a_3b_1 & a_3b_2 & a_3b_3 & a_3b_4 & \cdots \\ a_4b_1 & a_4b_2 & a_4b_3 & a_4b_4 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

The reason we define the product of two series in this way is that, due to the properties of rearranged sequences, changing the order of summation may alter the sum of the series. For example, we have following two arrangements of the Cauchy product:

Cauchy arrangement

$$\sum_{i=1}^{\infty} a_i \cdot \sum_{j=1}^{\infty} b_j = \sum_{n=2}^{\infty} \sum_{k=1}^{n-1} a_k b_{n-k} = a_1b_1 + (a_1b_2 + a_2b_1) + (a_1b_3 + a_2b_2 + a_3b_1) + \cdots$$

Square arrangement

$$\sum_{i=1}^{\infty} a_i \cdot \sum_{j=1}^{\infty} b_j = \sum_{n=1}^{\infty} \left(\sum_{i=1}^n (a_i b_n + a_n b_i) - a_n b_n \right)$$

Under certain conditions, both arrangements converge to the same sum. This is formalized in the following theorem:

Theorem 5.1.18

Assume we have two series $\sum_{n=1}^{\infty} a_n = A$ and $\sum_{n=1}^{\infty} b_n = B$. If all of them converges absolutely, then their Cauchy product converges to AB , regardless of the arrangement used.

Proof. Let

$$c_n = \sum_{k=1}^{n-1} a_k b_{n-k}, \quad n \geq 2.$$

Absolute convergence gives

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |a_i b_j| = \left(\sum_{i=1}^{\infty} |a_i| \right) \left(\sum_{j=1}^{\infty} |b_j| \right) < \infty.$$

Therefore the double series may be summed in any order. Summing by diagonals gives

$$\sum_{n=2}^{\infty} c_n = \sum_{n=2}^{\infty} \sum_{k=1}^{n-1} a_k b_{n-k} = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_i b_j.$$

Summing first in j and then in i gives

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_i b_j = \sum_{i=1}^{\infty} a_i \left(\sum_{j=1}^{\infty} b_j \right) = BA = AB.$$

Thus the Cauchy product converges absolutely to AB , and every rearrangement has the same sum. $\square$

5.1.2.7 Infinite Product

Definition 5.1.7. Assume $\{a_n\}$ is a sequence such that $a_n \neq 0$ for all n . The **infinite product** $\prod_{n=1}^{\infty} a_n$ is defined as the limit of the partial products:

$$P_N = \prod_{n=1}^N a_n$$

If the limit $\lim_{N \rightarrow \infty} P_N$ exists and is non-zero, we say the infinite product converges to that limit. Else, the infinite product diverges.

Just like series, we call P_N the N -th partial product of the infinite product.

Mind that if the infinite product "converges to zero", we still say it diverges. We will see why in the following theorem.

Here are some properties of infinite products:

Theorem 5.1.19

If the infinite product $\prod_{n=1}^{\infty} a_n$ converges to a non-zero limit, then $\lim_{n \rightarrow \infty} a_n = 1$.
If the infinite product $\prod_{n=1}^{\infty} a_n$ converges to a non-zero limit, then $\lim_{m \rightarrow \infty} \prod_{n=m+1}^{\infty} a_n = 1$.

Sometimes, it is easier to study the convergence of infinite products by expressing it in terms of $p_n = 1 + a_n$:

$$\prod_{n=1}^{\infty} p_n = \prod_{n=1}^{\infty} (1 + a_n)$$

Let's go back to the question of why we say the infinite product diverges if it converges to zero. This is because of the following theorem:

Theorem 5.1.20

Assume $a_n > -1$ for all n . The infinite product $\prod_{n=1}^{\infty} (1 + a_n)$ converges to a non-zero limit if and only if the series $\sum_{n=1}^{\infty} \ln(1 + a_n)$ converges.
Furthermore, if $a_n > -1$ does not change sign, then the infinite product $\prod_{n=1}^{\infty} (1 + a_n)$ converges to a non-zero limit if and only if the series $\sum_{n=1}^{\infty} a_n$ converges.

Using this theorem, we can see that if the infinite product converges to zero, then the series $\sum \ln(1 + a_n)$ diverges to $-\infty$. This means that the terms a_n do not approach zero fast enough, leading to divergence. We can apply the properties of series to study the convergence of infinite products.

Corollary 5.1.2

Assume $p_n = 1 + a_n$, and a_n does not change sign, then $\prod_{n=1}^{\infty} (1 + a_n)$ converges in the absolute sense if and only if $\sum_{n=1}^{\infty} a_n$ converges.

Corollary 5.1.3

Assume $p_n = 1 + a_n$, if $\sum a_n$ converges, then the infinite product $\prod (1 + a_n)$ converges if and only if $\sum a_n^2$ converges.

Also, mathematicians use the infinite product deduce some famous formulas, for example:

Wallis Law:

$$\frac{\pi}{2} = \prod_{n=1}^{\infty} \frac{4n^2}{4n^2 - 1} = \left(\frac{2 \cdot 2}{1 \cdot 3} \right) \left(\frac{4 \cdot 4}{3 \cdot 5} \right) \left(\frac{6 \cdot 6}{5 \cdot 7} \right) \cdots$$

Viète Law:

$$\frac{2}{\pi} = \prod_{n=1}^{\infty} \cos \frac{\pi}{2^{n+1}} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2 + \sqrt{2}}}{2} \cdot \frac{\sqrt{2 + \sqrt{2 + \sqrt{2}}}}{2} \cdots$$

Stirling Formula: When $n \rightarrow \infty$, we have the famous approximation:

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e} \right)^n$$

5.1.3 Function Series

Function series generalize numerical series by summing functions instead of numbers. They are crucial for representing functions as power series, Fourier series, etc.

Now we will broaden our discussion to series of functions. Consider a sequence of functions $\{f_n(x)\}$ defined on a set $E \subseteq \mathbb{R}$. We define the series of functions as:

Definition 5.1.8 (Function Series). A **function series** is an expression of the form:

$$\sum_{n=1}^{\infty} u_n(x) = u_1(x) + u_2(x) + u_3(x) + \cdots$$

where each $u_n(x)$ is a function defined on E .

How can we determine whether such a series converges? We need to use the concept of convergence of series:

Definition 5.1.9. Assume $u_n(x)$ is a function defined on a set E . For each fixed $x \in E$, we can consider the numerical series $\sum_{n=1}^{\infty} u_n(x)$. If this series converges to a limit $S(x)$, we say that the function series converges at the point x , and we call the point the **point of convergence**.

For all the points of convergence, we call the set of them the **domain of convergence**. Then we can define the sum function of the series:

Definition 5.1.10 (Sum Function). If the function series $\sum_{n=1}^{\infty} u_n(x)$ converges at every point $x \in D$, we define the **sum function** $S(x)$ as:

$$S(x) = \sum_{n=1}^{\infty} u_n(x), \quad x \in D$$

Likewise, we can define the sequence of partial sums:

Definition 5.1.11 (Partial Sums). The N -th **partial sum** of the function series $\sum_{n=1}^{\infty} u_n(x)$ is defined as:

$$S_N(x) = \sum_{n=1}^N u_n(x)$$

for each $x \in E$.

Which means, the function series and $\{S_n(x)\}$ are actually equivalent. To make things simpler, we will use the sequence of partial sums to study the convergence of function series.

After all, how can we define the convergence of function series? There are two main types: pointwise convergence and uniform convergence. First, we have pointwise convergence:

Definition 5.1.12 (Pointwise Convergence). Let $\{f_n\}$ be a sequence of functions defined on a set $E \subseteq \mathbb{R}$. We say that $\{f_n\}$ **converges pointwise** to a function f on E if for every $x \in E$:

$$\lim_{n \rightarrow \infty} f_n(x) = f(x)$$

Similarly, a series $\sum_{n=1}^{\infty} f_n(x)$ converges pointwise to $S(x)$ if the sequence of partial sums converges pointwise to $S(x)$.

Pointwise convergence means that for each fixed x , the sequence $\{f_n(x)\}$ converges to $f(x)$.

However, pointwise convergence is often too weak to preserve important properties of functions, such as continuity, differentiability, and integrability.

Example 5.1.5: Failure of Pointwise Convergence

Consider $f_n(x) = x^n$ on the interval $[0, 1]$. For any $x \in [0, 1)$, $\lim_{n \rightarrow \infty} x^n = 0$. For $x = 1$, $\lim_{n \rightarrow \infty} 1^n = 1$. The limit function is:

$$f(x) = \begin{cases} 0 & 0 \leq x < 1 \\ 1 & x = 1 \end{cases}$$

Although each f_n is continuous, the limit function f is **discontinuous**. Pointwise convergence fails to preserve continuity.

But wait, there's more! Let's look at more instances where pointwise convergence absolutely ruins our calculus dreams:

Example 5.1.6: Calculus Goes Bonkers

Consider the sequence $S_n(x) = nx(1 - x^2)^n$ on $[0, 1]$. For any $x \in [0, 1]$, the pointwise limit is clearly $S(x) = 0$. You would logically expect the integral to be 0, right? But let's actually integrate the sequence:

$$\int_0^1 S_n(x) dx = \int_0^1 nx(1 - x^2)^n dx = -\frac{n}{2} \int_0^1 (1 - x^2)^n d(1 - x^2) = \frac{n}{2(n+1)} \rightarrow \frac{1}{2}$$

Pointwise convergence just watched the integral become 1/2 and basically said "I'll pretend I didn't see that."

Example 5.1.7: Derivatives Jumping Out the Window

Take the series $\sum_{n=1}^{\infty} \frac{\sin(n^2 x)}{n^2}$. It converges pointwise (and even absolutely) to some nice continuous function. But if we try to naively take the derivative term-by-term, we get $\sum_{n=1}^{\infty} \cos(n^2 x)$, which diverges horribly.

To fix this catastrophic failure of properties, we need to bring in the heavy artillery: a stronger form of convergence.

5.1.3.1 Uniform Convergence

Uniform convergence is a strengthening of pointwise convergence. In pointwise convergence, for each fixed x we may choose a different threshold N after which the approximation is good. In uniform convergence, one threshold N must work for all x in the domain. This change is small in notation but large in meaning: it prevents convergence from becoming slower and slower as x moves through the domain.

This is the reason uniform convergence preserves many analytic properties. Continuity, integration, and sometimes differentiation involve neighborhoods or limiting processes that depend on x . If convergence is only pointwise, these operations may fail to commute with the limit. Uniform convergence supplies the global control needed to exchange limits with other operations.

Definition 5.1.13 (Uniform Convergence). A sequence of functions $\{f_n\}$ converges **uniformly** to f on E if for every $\epsilon > 0$, there exists an integer $N(\epsilon)$ (dependent on ϵ but **independent of** x) such that for all $n > N(\epsilon)$ and all $x \in E$:

$$|f_n(x) - f(x)| < \epsilon$$

We write $f_n \xrightarrow{D} f$ on E .

Geometrically, this means that for $n > N$, the graph of $f_n(x)$ lies entirely within a "tube" of width 2ϵ centered around $f(x)$.

Theorem 5.1.21: Uniform Convergence via Sup-norm

Let $\{f_n\}$ be a sequence of functions defined on E . $\{f_n\}$ converges uniformly to f on E if and only if

$$\lim_{n \rightarrow \infty} \left(\sup_{x \in E} |f_n(x) - f(x)| \right) = 0$$

Example 5.1.8

Consider $f_n(x) = (1-x)x^n$ on $[0, 1]$. For any $x \in [0, 1]$, $f(x) = \lim f_n(x) = 0$. The maximum of $f_n(x)$ occurs at $x = \frac{n}{n+1}$, where

$$f_n\left(\frac{n}{n+1}\right) = \frac{1}{n+1} \left(\frac{n}{n+1}\right)^n \rightarrow 0 \cdot e^{-1} = 0$$

Since the maximum approach zero, f_n converges uniformly to 0 on $[0, 1]$.

What if we want to boldly destroy the illusion of uniform convergence? Instead of finding the exact supremum, we can evaluate the function at a *moving target*!

Theorem 5.1.22: Sequence Criterion / Moving Target Trick

Let $\{f_n(x)\}$ converge pointwise to $f(x)$ on E . Then $\{f_n(x)\}$ **converges uniformly** to $f(x)$ on E if and only if for **any** sequence $\{x_n\} \subset E$, we have:

$$\lim_{n \rightarrow \infty} |f_n(x_n) - f(x_n)| = 0$$

If you can find even one sequence $\{x_n\}$ where the limit is not zero, uniform convergence is dead in the water.

Example 5.1.9

Consider $f_n(x) = \left(1 + \frac{x}{n}\right)^n$ on $[0, +\infty)$. The pointwise limit is famously $f(x) = e^x$. Does it converge uniformly? Let's pick a moving target sequence $x_n = n$. The distance becomes:

$$|f_n(x_n) - f(x_n)| = \left| \left(1 + \frac{n}{n}\right)^n - e^n \right| = |2^n - e^n| \rightarrow +\infty$$

Uniform convergence practically left the chat. (However, note that it does converge uniformly on any bounded interval $[0, t]$!)

Definition 5.1.14 (Compact Convergence / Uniform Convergence on Compact Sets). Let $\{f_n\}$ be a sequence of functions defined on a set $E \subseteq \mathbb{R}$. We say that $\{f_n\}$ converges **compactly** (or uniformly on compact subsets) to a function f on E if for every compact subset $K \subseteq E$, the sequence $\{f_n\}$ converges uniformly to f on K .

Compact convergence is weaker than uniform convergence on the entire set E , but stronger than pointwise convergence. For open sets or unbounded domains where global uniform convergence often fails, compact convergence serves as a natural and highly useful substitute. Importantly, since continuity is a local property, compact convergence is sufficient to ensure that the limit function remains continuous.

Now let's return to the topic of uniform convergence. We have the following criterion for uniform convergence:

Theorem 5.1.23: Cauchy Criterion for Uniform Convergence

The sequence $\{f_n\}$ converges uniformly on E if and only if for every $\epsilon > 0$, there exists N such that for all $m, n > N$ and all $x \in E$:

$$|f_n(x) - f_m(x)| < \epsilon$$

Seriously, is there any limit process Cauchy hasn't put his name on? Cauchy, you really won big time. It is a fundamental tool for proving convergence without explicitly finding the limit. Its core idea is to show that the sequence becomes "eventually close" to itself, which implies convergence.

For series $\sum u_n(x)$, the most practical tool for proving uniform convergence is the Weierstrass M-Test.

Theorem 5.1.24: Weierstrass M-Test

Let $\{u_n(x)\}$ be a sequence of functions defined on E . Suppose there exists a sequence of positive constants $\{M_n\}$ such that:

1. $|u_n(x)| \leq M_n$ for all $x \in E$ and all $n \geq 1$.
2. The numerical series $\sum_{n=1}^{\infty} M_n$ converges.

Then the series $\sum_{n=1}^{\infty} u_n(x)$ converges **uniformly** and absolutely on E .

Just like numerical series, when absolute convergence (M-Test) isn't an option, we have to rely on the dynamic duo: Abel and Dirichlet.

Theorem 5.1.25: A-D Tests for Uniform Convergence

Consider the function series $\sum_{n=1}^{\infty} a_n(x)b_n(x)$ defined on E . It converges uniformly on E if either of the following conditions hold:

1. **Dirichlet's Test:** The partial sums of $\sum b_n(x)$ are **uniformly bounded** on E (i.e., bounded evenly everywhere, all at once, by a single M), and the sequence $a_n(x)$ is monotonic with respect to n for each x and converges **uniformly** to 0 on E .
2. **Abel's Test:** The series $\sum b_n(x)$ converges uniformly on E , and $a_n(x)$ is monotonic with respect to n for each x and is **uniformly bounded** across all n and $x \in E$.

5.1.3.2 Properties of Uniformly Convergent Series

Uniform convergence allows us to interchange the order of limit operations, which is rigorous justification for "term-by-term" calculus. In short: **The operation on the infinite sum equals the infinite sum of the operations.**

Theorem 5.1.26: Continuity

If a sequence of continuous functions $\{f_n\}$ converges uniformly to f on an interval E , then the limit function f is continuous on E . Consequently, for a series of continuous functions $\sum u_n(x)$, if the series converges uniformly to $S(x)$, then $S(x)$ is continuous:

$$\lim_{x \rightarrow x_0} \sum_{n=1}^{\infty} u_n(x) = \sum_{n=1}^{\infty} \lim_{x \rightarrow x_0} u_n(x)$$

Theorem 5.1.27: Term-by-Term Integration

Let each $\{u_n(x)\}$ be Riemann integrable on $[a, b]$, and suppose the series $\sum_{n=1}^{\infty} u_n(x)$ converges uniformly to $S(x)$ on $[a, b]$. Then the limit function $S(x)$ is also Riemann integrable on $[a, b]$, and the integration and summation can be safely interchanged:

$$\int_a^b S(x) dx = \sum_{n=1}^{\infty} \int_a^b u_n(x) dx$$

Theorem 5.1.28: Term-by-Term Differentiation

Let $\{u_n(x)\}$ be a sequence of continuously differentiable functions on $[a, b]$ (meaning each u'_n is continuous). Suppose the series $\sum_{n=1}^{\infty} u_n(x)$ converges at some point $x_0 \in [a, b]$. If the series of derivatives $\sum_{n=1}^{\infty} u'_n(x)$ converges **uniformly** on $[a, b]$, then the original series converges uniformly on $[a, b]$ to a continuously differentiable function $S(x)$, and:

$$S'(x) = \left(\sum_{n=1}^{\infty} u_n(x) \right)' = \sum_{n=1}^{\infty} u'_n(x)$$

There is also a powerful theorem that guarantees uniform convergence under specific conditions:

Theorem 5.1.29: Dini's Theorem

Let $\{S_n(x)\}$ be a sequence of continuous functions defined on a compact interval $[a, b]$. If:

1. $S_n(x)$ converges pointwise to a continuous function $S(x)$ on $[a, b]$,
2. For each fixed $x \in [a, b]$, the sequence $\{S_n(x)\}$ is monotonic with respect to n ,

then the convergence is **uniform** on $[a, b]$.

5.1.4 Power Series

Power series form the most well-behaved class of function series. Inside their interval or disk of convergence, they behave almost like polynomials: they may be added, multiplied, integrated, and differentiated term by term. The radius of convergence is the boundary between this stable polynomial-like behavior and possible divergence.

However, the boundary itself requires separate attention. The root or ratio test usually determines convergence strictly inside and strictly outside the radius, but it says nothing decisive at endpoints. Endpoint analysis must return to ordinary series tests. This is why power series problems naturally split into three parts: find the radius, test the interior, and examine the boundary.

A power series is a specific type of function series of the form:

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots$$

where a_n are the coefficients of the series, and x_0 is the center of convergence. For simplicity in notation and proofs, we often set $x_0 = 0$ by a simple change of variables $y = x - x_0$, transforming the series into $\sum_{n=0}^{\infty} a_n x^n$.

5.1.4.1 Radius of Convergence

Unlike general function series, which might converge on complicated sets, power series possess a highly structured domain of convergence. The set of points where a power series converges is always an interval centered at x_0 .

Theorem 5.1.30: Cauchy-Hadamard Theorem

Given a power series $\sum a_n x^n$, let

$$\rho = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}$$

The **radius of convergence** R is defined as:

$$R = \begin{cases} 0 & \text{if } \rho = +\infty \\ 1/\rho & \text{if } 0 < \rho < +\infty \\ +\infty & \text{if } \rho = 0 \end{cases}$$

The series converges absolutely for every x such that $|x| < R$ and diverges for every x such that $|x| > R$. The interval $(-R, R)$ is called the interval of convergence. The behavior at the endpoints $x = \pm R$ must be investigated individually depending on the specific series.

In practice, if the sequence of coefficients is well-behaved, the ratio test provides a simpler way to compute the radius of convergence:

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

provided this limit exists or is $+\infty$.

Property 5.1.3: Uniform Convergence and Smoothness

Inside the interval of convergence $(-R, R)$, the power series converges **uniformly** on any closed and bounded sub-interval $[-r, r]$ where $0 < r < R$. Because of this uniform convergence, the limit function $S(x) = \sum_{n=0}^{\infty} a_n x^n$ is continuous on $(-R, R)$. Even more impressively, a power series can be differentiated and integrated term-by-term inside its interval of convergence. The resulting series will have the exact same radius of convergence R , which implies that a power series defines an infinitely differentiable (C^∞) function inside its radius of convergence.

5.1.4.2 Abel's Theorem

We know that uniform convergence guarantees continuity. Since a power series converges uniformly on any compact subset of $(-R, R)$, the sum function is continuous on $(-R, R)$. But what happens if the series converges at an endpoint $x = R$? Does the continuity extend to the boundary?

Theorem 5.1.31: Abel's Continuity Theorem

If the power series $\sum_{n=0}^{\infty} a_n x^n$ converges at $x = R$ (respectively, $x = -R$), then the sum function $f(x) = \sum_{n=0}^{\infty} a_n x^n$ is continuous at $x = R$ from the left (respectively, at $x = -R$ from the right). That is to say:

$$\lim_{x \rightarrow R^-} \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_n R^n$$

This theorem provides a powerful tool for formally evaluating sums of numerical series by defining an auxiliary power series and taking the limit. For example, consider the expansion:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

By the ratio test, this series has $R = 1$. At $x = 1$, the series is the alternating harmonic series $1 - 1/2 + 1/3 - \dots$, which converges by the Alternating Series Test. Therefore, by Abel's Theorem, we can take the limit as $x \rightarrow 1^-$ on the function $\ln(1+x)$ to deduce that $\ln 2 = 1 - 1/2 + 1/3 - \dots$.

Here are some standard examples for calculating the sum function of a power series using term-by-term calculus:

Example 5.1.10

Find the sum function $S(x)$ of the power series $\sum_{n=1}^{\infty} \frac{x^n}{n!}$.

Solution: Using the ratio test, the radius of convergence is $R = +\infty$. For $x \in (-\infty, +\infty)$, we differentiate term-by-term:

$$S'(x) = \sum_{n=1}^{\infty} \frac{nx^{n-1}}{n!} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!} = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + \sum_{k=1}^{\infty} \frac{x^k}{k!} = 1 + S(x)$$

Thus, we obtain a first-order linear differential equation: $S'(x) - S(x) = 1$. The solution is $S(x) = Ce^x - 1$. Since $S(0) = 0$, we have $C - 1 = 0 \implies C = 1$. Therefore, $S(x) = e^x - 1$. Consequently, the entire series including $n = 0$ is $\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + S(x) = e^x$.

Example 5.1.11

Find the sum function $S(x)$ of the power series $\sum_{n=0}^{\infty} \frac{x^n}{n+1}$.

Solution: The radius of convergence is $R = 1$. The domain of convergence is $[-1, 1)$. For $x \neq 0$:

$$S(x) = \sum_{n=0}^{\infty} \frac{x^n}{n+1} = \frac{1}{x} \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} = \frac{1}{x} \sum_{n=0}^{\infty} \int_0^x t^n dt$$

Since the power series converges uniformly within its interval of convergence, we can interchange the integral and the sum:

$$S(x) = \frac{1}{x} \int_0^x \left(\sum_{n=0}^{\infty} t^n \right) dt = \frac{1}{x} \int_0^x \frac{1}{1-t} dt = -\frac{1}{x} \ln(1-x)$$

For $x = 0$, the series yields $S(0) = 1$. Notice that $\lim_{x \rightarrow 0} -\frac{\ln(1-x)}{x} = 1$, so the function is continuous at $x = 0$.

Taylor and Maclaurin Series

We have seen that every power series represents a C^∞ function inside its interval of convergence. We now ask the inverse question: given an infinitely differentiable function $f(x)$, can it be represented by a power series? And if so, what must the coefficients a_n be?

Theorem 5.1.32: Taylor Series

If a function $f(x)$ can be expanded as a power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ with a non-zero radius of convergence R , then by repeatedly differentiating and evaluating at $x = x_0$, we find that the coefficients are uniquely determined as:

$$a_n = \frac{f^{(n)}(x_0)}{n!}$$

The resulting series $\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$ is called the **Taylor Series** of f centered at x_0 . If $x_0 = 0$, it is specifically called the **Maclaurin Series**.

Remark 5.1.2: Analyticity

It is crucial to note that simply being C^∞ (having derivatives of all orders) is not enough to guarantee that the Taylor series of $f(x)$ actually converges to $f(x)$. The series might only converge at x_0 , or it might converge to a completely different function. Functions that are equal to their Taylor series on an open interval are called **real analytic functions**. A famous example of a smooth but non-analytic function is $f(x) = e^{-1/x^2}$ for $x \neq 0$ and $f(0) = 0$, where all derivatives at 0 vanish, making its Maclaurin series identically zero.

Thus, a function can be expanded as a Taylor series converging to the function itself if and only if the remainder $R_n(x)$ in Taylor's formula approaches zero as $n \rightarrow \infty$. Here, we present the integral form of the remainder, which can be explicitly derived using integration by parts.

Theorem 5.1.33: Taylor's Theorem with Integral Remainder

If $f(x)$ has continuous derivatives up to order $n + 1$ in a neighborhood $U(x_0, r)$, then for any $x \in U(x_0, r)$, we have:

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + r_n(x)$$

where the remainder $r_n(x)$ is given by:

$$r_n(x) = \frac{1}{n!} \int_{x_0}^x f^{(n+1)}(t) (x - t)^n dt$$

Proof. Let $r_n(x) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$. By differentiating $r_n(x)$ with respect to x , most terms cancel out:

$$r'_n(x) = f'(x) - f'(x_0) - \sum_{k=2}^n \frac{f^{(k)}(x_0)}{(k-1)!} (x - x_0)^{k-1} = \frac{f^{(n+1)}(x)}{n!} (x - x)^n$$

Wait, let's take a simpler approach by differentiating with respect to the continuous variable. Observe that $r_n(x_0) = r'_n(x_0) = \dots = r_n^{(n)}(x_0) = 0$, and we know $r_n^{(n+1)}(x) = f^{(n+1)}(x)$. We can express $r_n(x)$ through repeated integration from x_0 to x . Let's start with a single layer:

$$r_n(x) = r_n(x) - r_n(x_0) = \int_{x_0}^x r'_n(t) dt = \int_{x_0}^x r'_n(t) d(t - x)$$

Applying integration by parts:

$$= r'_n(t)(t - x) \Big|_{x_0}^x - \int_{x_0}^x r''_n(t)(t - x) dt = \int_{x_0}^x r''_n(t)(x - t) dt$$

Repeating this process $n - 1$ more times gives:

$$r_n(x) = \frac{1}{n!} \int_{x_0}^x r_n^{(n+1)}(t)(x - t)^n dt = \frac{1}{n!} \int_{x_0}^x f^{(n+1)}(t)(x - t)^n dt$$

This completes the proof. □

Example 5.1.12: Standard Maclaurin Expansions

The following cornerstone expansions appear frequently throughout analysis and physics. Notice the intervals of convergence:

$$\begin{aligned}
 e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots & (R = \infty) \\
 \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots & (R = \infty) \\
 \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots & (R = \infty) \\
 \frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots & (R = 1) \\
 \ln(1+x) &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots & (R = 1) \\
 (1+x)^\alpha &= 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{n!} x^n & (R = 1)
 \end{aligned}$$

5.2 Limits and Continuity in Euclidean Space

The passage from one-variable analysis to several-variable analysis is not merely a change of notation. In one variable, a point can approach another point only from the left or from the right. In $\mathbb{R}^n$, there are infinitely many paths of approach. As a result, limits and continuity become more geometric: we must control behavior in every direction simultaneously.

The norm, inner product, open balls, closed sets, compactness, and connectedness provide the language for this control. They allow us to state multivariable versions of familiar theorems without relying on pictures. For example, compactness replaces the closed bounded interval, and connectedness replaces the intuitive idea of an interval having no gaps.

Moving from single-variable calculus to multivariable calculus requires a generalized setting. We replace the real line $\mathbb{R}$ with the n -dimensional Euclidean space $\mathbb{R}^n$. While many concepts generalize naturally, the geometry of $\mathbb{R}^n$ introduces new complexities, particularly regarding directions of approach for limits.

5.2.1 The Structure of Euclidean Space $\mathbb{R}^n$

The space $\mathbb{R}^n$ is the set of all ordered n -tuples of real numbers. An element $\mathbf{x} \in \mathbb{R}^n$ is written as $\mathbf{x} = (x_1, x_2, \dots, x_n)$, where x_i are the components.

Definition 5.2.1 (Inner Product and Norm). For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the **Euclidean inner product** (or dot product) is defined as:

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^n x_i y_i$$

The **Euclidean norm** (or length) of $\mathbf{x}$ is defined as:

$$\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{\sum_{i=1}^n x_i^2}$$

The **distance** between two points $\mathbf{x}$ and $\mathbf{y}$ is $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$.

The geometry of $\mathbb{R}^n$ is governed by two fundamental inequalities.

Theorem 5.2.1: Cauchy-Schwarz Inequality

For any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$$

Equality holds if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent.

Proof. For any real number λ , consider the non-negative quantity:

$$\|\lambda \mathbf{x} + \mathbf{y}\|^2 = (\lambda \mathbf{x} + \mathbf{y}) \cdot (\lambda \mathbf{x} + \mathbf{y}) = \lambda^2 \|\mathbf{x}\|^2 + 2\lambda(\mathbf{x} \cdot \mathbf{y}) + \|\mathbf{y}\|^2 \geq 0$$

Since this quadratic in λ is always non-negative, its discriminant must be less than or equal to zero:

$$\Delta = 4(\mathbf{x} \cdot \mathbf{y})^2 - 4\|\mathbf{x}\|^2 \|\mathbf{y}\|^2 \leq 0 \implies |\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$$

This completes the proof. $\square$

Theorem 5.2.2: Triangle Inequality

For any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$$

Proof. By the Cauchy-Schwarz inequality:

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) = \|\mathbf{x}\|^2 + 2(\mathbf{x} \cdot \mathbf{y}) + \|\mathbf{y}\|^2 \\ &\leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\| \|\mathbf{y}\| + \|\mathbf{y}\|^2 = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2 \end{aligned}$$

Taking the square root of both sides gives the desired inequality. $\square$

5.2.2 Basic Topology of $\mathbb{R}^n$

To define limits rigorously, we need the language of point set topology. The concept of an "open interval" in $\mathbb{R}$ generalizes to an "open ball" (or neighborhood) in $\mathbb{R}^n$.

Definition 5.2.2 (Neighborhood and Punctured Neighborhood). Let $\mathbf{a} \in \mathbb{R}^n$ and $\delta > 0$. The **neighborhood** (open ball) of radius δ centered at $\mathbf{a}$ is the set:

$$U(\mathbf{a}, \delta) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{a}\| < \delta\}$$

The **punctured neighborhood** (deleted neighborhood) is $\mathring{U}(\mathbf{a}, \delta) = U(\mathbf{a}, \delta) \setminus \{\mathbf{a}\}$.

Given a point set $E \subseteq \mathbb{R}^n$ and a point $P \in \mathbb{R}^n$, we can classify P relative to E :

- **Interior point:** P is an interior point of E if there exists a neighborhood $U(P)$ such that $U(P) \subseteq E$.
- **Exterior point:** P is an exterior point of E if there exists a neighborhood $U(P)$ such that $U(P) \cap E = \emptyset$.
- **Boundary point:** P is a boundary point of E if every neighborhood $U(P)$ contains at least one point in E and at least one point not in E .
- **Accumulation point (Limit point):** P is an accumulation point of E if every punctured neighborhood $\mathring{U}(P, \delta)$ contains at least one point of E (implying it contains infinitely many points of E). The set of all accumulation points of E is called the derived set, denoted as E' .
- **Isolated point:** P is an isolated point of E if $P \in E$ but P is not an accumulation point of E .

Definition 5.2.3 (Open and Closed Sets). A set $S \subseteq \mathbb{R}^n$ is called **open** if every point of S is an interior point. A set $S \subseteq \mathbb{R}^n$ is called **closed** if it contains all of its accumulation points (i.e., $S' \subseteq S$). The **closure** of a set E , denoted $\overline{E}$, is defined as $\overline{E} = E \cup E'$. A set is closed if and only if $E = \overline{E}$.

Theorem 5.2.3

A set $S \subseteq \mathbb{R}^n$ is closed if and only if its complement $S^c = \mathbb{R}^n \setminus S$ is open.

Proof. ($\implies$) Assume S is closed. Let $\mathbf{x} \in S^c$. Since $\mathbf{x} \notin S$ and S contains all its accumulation points, $\mathbf{x}$ is not an accumulation point of S . Thus, there exists a neighborhood $U(\mathbf{x}, \delta)$ such that $U(\mathbf{x}, \delta) \cap S = \emptyset$, meaning $U(\mathbf{x}, \delta) \subseteq S^c$. Hence, S^c is open.

($\impliedby$) Assume S^c is open. Let $\mathbf{x}$ be an accumulation point of S . If $\mathbf{x} \notin S$, then $\mathbf{x} \in S^c$. Since S^c is open, there is a neighborhood $U(\mathbf{x}, \delta) \subseteq S^c$, which means $U(\mathbf{x}, \delta) \cap S = \emptyset$. This contradicts that $\mathbf{x}$ is an accumulation point of S . Thus, $\mathbf{x} \in S$, and S is closed. $\square$

Theorem 5.2.4: De Morgan's Laws

For any family of sets $\{S_\alpha\}$ in $\mathbb{R}^n$:

$$\left(\bigcup_{\alpha} S_{\alpha} \right)^c = \bigcap_{\alpha} S_{\alpha}^c, \quad \left(\bigcap_{\alpha} S_{\alpha} \right)^c = \bigcup_{\alpha} S_{\alpha}^c$$

Proof. $\mathbf{x} \in \left(\bigcup_{\alpha} S_{\alpha} \right)^c \iff \forall \alpha, \mathbf{x} \notin S_{\alpha} \iff \forall \alpha, \mathbf{x} \in S_{\alpha}^c \iff \mathbf{x} \in \bigcap_{\alpha} S_{\alpha}^c$. This proves the first identity. The second is proved similarly. $\square$

Property 5.2.1

Any union of open sets is open. Any finite intersection of open sets is open. By De Morgan's laws, any intersection of closed sets is closed, and any finite union of closed sets is closed. **Counterexample for infinite intersection of open sets:** The sets $S_i = (-1 - 1/i, 1 + 1/i)$ are open, but their infinite intersection $\bigcap_{i=1}^{\infty} S_i = [-1, 1]$ is closed, not open.

5.2.2.1 Fundamental Theorems of $\mathbb{R}^n$

The real numbers $\mathbb{R}$ rely on the completeness axiom. In $\mathbb{R}^n$, the equivalent completeness properties are formulated through the following equivalent theorems:

Theorem 5.2.5: Nested Closed Rectangle Theorem

Let $\Delta_k = \prod_{i=1}^n [a_{ki}, b_{ki}]$ be a sequence of closed rectangles in $\mathbb{R}^n$ such that $\Delta_1 \supseteq \Delta_2 \supseteq \dots \supseteq \Delta_k \supseteq \dots$. If the diameter of Δ_k approaches 0 as $k \rightarrow \infty$ (i.e., $\lim_{k \rightarrow \infty} \sqrt{\sum (b_{ki} - a_{ki})^2} = 0$), then there exists a unique point $\eta \in \bigcap_{k=1}^{\infty} \Delta_k$.

Theorem 5.2.6: Cantor's Intersection Theorem

Let $\{S_k\}$ be a sequence of non-empty, closed, and bounded subsets of $\mathbb{R}^n$ such that $S_1 \supseteq S_2 \supseteq \dots \supseteq S_k \supseteq \dots$. If the diameter $\text{diam}(S_k) = \sup_{\mathbf{x}, \mathbf{y} \in S_k} \|\mathbf{x} - \mathbf{y}\| \rightarrow 0$ as $k \rightarrow \infty$, then there is a unique point belonging to $\bigcap_{k=1}^{\infty} S_k$.

Theorem 5.2.7: Bolzano-Weierstrass Theorem

Every bounded infinite sequence in $\mathbb{R}^n$ has a convergent subsequence.

Theorem 5.2.8: Cauchy Convergence Principle

A sequence $\{\mathbf{x}_k\}$ in $\mathbb{R}^n$ converges if and only if it is a Cauchy sequence, meaning that for any $\epsilon > 0$, there exists N such that for all $j, k > N$, $\|\mathbf{x}_j - \mathbf{x}_k\| < \epsilon$.

5.2.3 Limits of Functions of Several Variables

When testing a multivariable limit, path checking is a diagnostic tool rather than a complete proof method. If two paths lead to different limiting values, the limit cannot exist. But if several convenient paths give the same answer, this only gives evidence, not proof. A proof must bound $|f(x) - L|$ in terms of the distance from x to the limiting point, or otherwise reduce the expression to a known theorem.

This distinction explains many standard examples. Along straight lines a function may appear to have one limit, while along a curve such as $y = x^2$ it may behave differently. The hidden problem is that multivariable functions can concentrate their irregular behavior along thin curves that are invisible to a small finite list of test paths.

Let $f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a function defined on a domain D . We are interested in the behavior of $f(\mathbf{x})$ as $\mathbf{x}$ approaches a point $\mathbf{a}$.

Definition 5.2.4 (Limit). Let $\mathbf{a}$ be an accumulation point of D . We say that the limit of $f(\mathbf{x})$ as $\mathbf{x}$ approaches $\mathbf{a}$ is $\mathbf{L}$, written as:

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = \mathbf{L}$$

if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $\mathbf{x} \in D$:

$$0 < \|\mathbf{x} - \mathbf{a}\| < \delta \implies \|f(\mathbf{x}) - \mathbf{L}\| < \epsilon$$

5.2.3.1 Path Dependence and Non-existence of Limits

In single-variable calculus ($n = 1$), x can approach a from only two directions (left or right). In $\mathbb{R}^n$ ($n \geq 2$), $\mathbf{x}$ can approach $\mathbf{a}$ from **infinitely many directions** along infinitely many different paths (lines, parabolas, spirals, etc.).

Theorem 5.2.9

If $\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = L$, then for any continuous curve $\gamma(t)$ ending at $\mathbf{a}$, the limit along the curve must be L . Consequently, if $f(\mathbf{x})$ approaches different values along two different paths ending at $\mathbf{a}$, the limit **does not exist**.

Example 5.2.1

Consider the function $f(x, y) = \frac{xy}{x^2 + y^2}$. We investigate the limit at $(0, 0)$.

1. Approach along the x -axis ($y = 0$):

$$\lim_{x \rightarrow 0} f(x, 0) = \lim_{x \rightarrow 0} \frac{0}{x^2} = 0$$

2. Approach along the line $y = x$:

$$\lim_{x \rightarrow 0} f(x, x) = \lim_{x \rightarrow 0} \frac{x^2}{x^2 + x^2} = \frac{1}{2}$$

Since $0 \neq 1/2$, the limit $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ **does not exist**.

Remark 5.2.1

It is not enough to check all straight lines passing through the origin. Consider the famous counterexample:

$$f(x, y) = \frac{xy^2}{x^2 + y^4}$$

Along any line $y = mx$ (and $x = 0$), the limit is 0. However, along the parabola $x = y^2$, the limit is $1/2$. Thus, the limit does not exist.

5.2.3.2 Iterated Limits vs. Simultaneous Limits

For a function of two variables $f(x, y)$, we can define **iterated limits**:

$$L_{12} = \lim_{x \rightarrow a} \left(\lim_{y \rightarrow b} f(x, y) \right) \quad \text{and} \quad L_{21} = \lim_{y \rightarrow b} \left(\lim_{x \rightarrow a} f(x, y) \right)$$

Proposition 5.2.1

The existence of iterated limits does not guarantee that the simultaneous limit $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ exists. Even if $L_{12} = L_{21}$, the simultaneous limit may not exist. Conversely, if the simultaneous limit exists, the iterated limits may not exist (because the inner limits might not be defined). However, if the simultaneous limit exists AND the inner limits exist, then they must all be equal.

5.2.4 Continuity

The definition of continuity in $\mathbb{R}^n$ is formally identical to that in $\mathbb{R}$.

Definition 5.2.5 (Continuity). A function $f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **continuous** at a point $\mathbf{a} \in D$ if:

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = f(\mathbf{a})$$

If f is continuous at every point in D , we say f is continuous on D .

Elementary operations preserve continuity:

- The sum, difference, product, and quotient (denominator non-zero) of continuous functions are continuous.
- The composition of continuous functions is continuous. That is, if f is continuous at $\mathbf{a}$ and g is continuous at $f(\mathbf{a})$, then $g \circ f$ is continuous at $\mathbf{a}$.

5.2.5 Properties of Continuous Functions on Compact Sets

The notions of boundedness and extreme values rely on the domain being "compact".

Definition 5.2.6 (Open Cover and Compact Set). Let S be a set in $\mathbb{R}^n$. A collection of open sets $\{U_\alpha\}$ is called an **open cover** of S if $S \subseteq \bigcup_\alpha U_\alpha$. A set S is **compact** if every open cover of S has a finite subcover. That is, there exist $U_{\alpha_1}, \dots, U_{\alpha_m}$ such that $S \subseteq \bigcup_{i=1}^m U_{\alpha_i}$.

Theorem 5.2.10: Heine-Borel Theorem

A subset $S \subseteq \mathbb{R}^n$ is compact if and only if it is closed and bounded.

Remark 5.2.2

Note that this theorem is uniquely true for $\mathbb{R}^n$ (more generally, for finite-dimensional normed spaces). In infinite-dimensional normed spaces, a closed and bounded set is not guaranteed to be compact (e.g., a closed unit ball).

In addition, there is an equivalent definition based on accumulation points: S is compact if and only if every infinite subset of S has at least one accumulation point in S .

Outline of Proof for $\mathbb{R}^2$. ($\implies$) If S is compact, we can cover S with open disks $U(0, k)$ for $k = 1, 2, \dots$. A finite subcover implies S is bounded. To show S is closed, assume $a \notin S$ is an accumulation point. We can construct an open cover of S using sets $V_n = \{\mathbf{x} : \|\mathbf{x} - a\| > 1/n\}$. Since $a \notin S$, $\bigcup_{n=1}^{\infty} V_n \supseteq S$. However, since a is an accumulation point, any finite number of V_n 's will fail to cover some points in S that are very close to a . This contradicts compactness, so S must be closed.

($\impliedby$) If S is closed and bounded, we can enclose it in a large closed rectangle. Using the Nested Closed Rectangle Theorem or Bolzano-Weierstrass, one can systematically show that any open cover must have a finite subcover by dividing the rectangle into smaller pieces repeatedly. $\square$

Theorem 5.2.11: Weierstrass Extreme Value Theorem

Let $K \subseteq \mathbb{R}^n$ be a compact set and let $f : K \rightarrow \mathbb{R}$ be a continuous function. Then:

1. f is bounded on K .
2. f attains its maximum and minimum values on K . That is, there exist points $\mathbf{x}_{\min}, \mathbf{x}_{\max} \in K$ such that for all $\mathbf{x} \in K$:

$$f(\mathbf{x}_{\min}) \leq f(\mathbf{x}) \leq f(\mathbf{x}_{\max})$$

Theorem 5.2.12: Compactness and Continuity

Continuous mappings map compact sets to compact sets. If $K \subseteq \mathbb{R}^n$ is compact and $f : K \rightarrow \mathbb{R}^m$ is continuous, then $f(K)$ is compact in $\mathbb{R}^m$.

Theorem 5.2.13: Uniform Continuity

Let $K \subseteq \mathbb{R}^n$ be a compact set. If $f : K \rightarrow \mathbb{R}^m$ is continuous on K , then f is **uniformly continuous** on K .

Definition 5.2.7 (Uniform Continuity). f is uniformly continuous on D if for every $\epsilon > 0$, there exists $\delta > 0$ such that for all $\mathbf{x}, \mathbf{y} \in D$:

$$\|\mathbf{x} - \mathbf{y}\| < \delta \implies \|f(\mathbf{x}) - f(\mathbf{y})\| < \epsilon$$

Note that δ depends only on ϵ , not on the location $\mathbf{x}$.

Remark 5.2.3

If the domain is not compact (e.g., open or unbounded), a continuous function need not be bounded or uniformly continuous. For example, $f(x) = 1/x$ on $(0, 1)$ is continuous but unbounded and not uniformly continuous.

5.2.6 Connectedness and the Intermediate Value Theorem

To generalize the Intermediate Value Theorem, we need the concept of connectedness.

Definition 5.2.8 (Path-Connected Set). A set $D \subseteq \mathbb{R}^n$ is **path-connected** if for any two points $\mathbf{a}, \mathbf{b} \in D$, there exists a continuous curve $\gamma : [0, 1] \rightarrow D$ such that $\gamma(0) = \mathbf{a}$ and $\gamma(1) = \mathbf{b}$.

Theorem 5.2.14: Generalized Intermediate Value Theorem

Let $D \subseteq \mathbb{R}^n$ be a path-connected set and let $f : D \rightarrow \mathbb{R}$ be continuous. If $\mathbf{a}, \mathbf{b} \in D$ and $f(\mathbf{a}) < c < f(\mathbf{b})$, then there exists a point $\mathbf{c} \in D$ such that $f(\mathbf{c}) = c$.

This ensures that the image of a path-connected set under a continuous real-valued function is an interval.

5.3 Differential Calculus of Multivariable Functions

Partial derivatives measure change along coordinate axes, but differentiability asks for something stronger: the function must be well approximated by one linear map in all directions at once. Thus differentiability is a statement about the existence of a good tangent plane, not merely about the existence of directional slopes. This is why partial derivatives alone do not guarantee differentiability.

The total differential is the linear part of the best local approximation. When f is differentiable at a point, the error after subtracting this linear part is small compared with the distance to the point. In symbols, the remainder is $o(\|h\|)$. This little- o condition is the precise analytic form of the phrase “locally linear.”

From here, we formally transition into the differential calculus of multivariable functions. We will first introduce partial derivatives and then introduce the idea of directional derivatives.

5.3.1 Partial Derivatives

Definition 5.3.1 (Partial Derivative). Let $f(x_1, x_2, \dots, x_n)$ be a function defined in a neighborhood of a point $\mathbf{x}_0 = (x_{0,1}, x_{0,2}, \dots, x_{0,n})$. If the following limit exists:

$$\lim_{x_i \rightarrow x_{0,i}} \frac{f(x_{0,1}, \dots, x_i, \dots, x_{0,n}) - f(x_{0,1}, \dots, x_{0,i}, \dots, x_{0,n})}{x_i - x_{0,i}} = A$$

then we say that f has a **partial derivative** with respect to its i -th variable at $\mathbf{x}_0$, and we denote it by $\frac{\partial f}{\partial x_i}(\mathbf{x}_0) = A$.

The definition of a partial derivative is very similar to that of a single-variable derivative, except we restrict the change to only one specific coordinate direction, keeping all other variables constant.

However, studying the rate of change only along the coordinate axes is often insufficient for a complete multidimensional analysis. This motivates the definition of the directional derivative.

Definition 5.3.2 (Directional Derivative). Let $f(\mathbf{x})$ be a function defined in a neighborhood of $\mathbf{x}_0$. For any given non-zero vector $\mathbf{u}$ (often taken as a unit vector), if the following limit exists:

$$\lim_{t \rightarrow 0} \frac{f(\mathbf{x}_0 + t\mathbf{u}) - f(\mathbf{x}_0)}{t} = A$$

then we say that f has a **directional derivative** at $\mathbf{x}_0$ along the direction $\mathbf{u}$, denoted by $D_{\mathbf{u}}f(\mathbf{x}_0) = A$.

5.3.1.1 Higher Partial Derivatives

Once first-order partial derivatives exist, they may themselves be differentiated. For a function $z = f(x, y)$, the second-order partial derivatives are denoted by

$$\begin{aligned} f_{xx} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right), & f_{xy} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right), \\ f_{yx} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right), & f_{yy} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right). \end{aligned}$$

The mixed derivatives f_{xy} and f_{yx} need not be equal merely because they exist. A reliable sufficient condition is continuity.

Theorem 5.3.1: Schwarz–Clairaut Theorem

Let f be defined in a neighborhood of (x_0, y_0) . If the mixed second-order partial derivatives f_{xy} and f_{yx} are continuous in a neighborhood of (x_0, y_0) , then

$$f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0).$$

Proof. Consider the rectangular second difference

$$I = \frac{f(x_0 + h, y_0 + k) - f(x_0 + h, y_0) - f(x_0, y_0 + k) + f(x_0, y_0)}{hk}.$$

Define

$$\phi(x) = f(x, y_0 + k) - f(x, y_0).$$

Applying the one-variable mean value theorem first in x and then in y gives

$$I = f_{xy}(\xi_1, \eta_1),$$

where ξ_1 lies between x_0 and $x_0 + h$, and η_1 lies between y_0 and $y_0 + k$. Similarly, defining

$$\psi(y) = f(x_0 + h, y) - f(x_0, y)$$

and applying the mean value theorem first in y and then in x gives

$$I = f_{yx}(\xi_2, \eta_2).$$

Hence $f_{xy}(\xi_1, \eta_1) = f_{yx}(\xi_2, \eta_2)$. Letting $(h, k) \rightarrow (0, 0)$ and using the continuity of the mixed derivatives yields the result. $\square$

Note

The theorem is often remembered as: continuous mixed second partial derivatives may be interchanged. Existence alone is not enough.

5.3.2 Differentiability and the Total Differential

Partial derivatives measure change along coordinate axes. Differentiability is stronger: it requires the whole increment of the function to be approximated by one linear map, uniformly with respect to all directions.

Definition 5.3.3 (Differentiability). Let $f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be defined in a neighborhood of $\mathbf{x}_0$. We say that f is **differentiable** at $\mathbf{x}_0$ if there exists a linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$f(\mathbf{x}_0 + \Delta \mathbf{x}) - f(\mathbf{x}_0) = L(\Delta \mathbf{x}) + o(\|\Delta \mathbf{x}\|), \quad \Delta \mathbf{x} \rightarrow \mathbf{0}.$$

Equivalently, there exist constants $A_1, \dots, A_n$ such that

$$\Delta f = A_1 \Delta x_1 + \dots + A_n \Delta x_n + o\left(\sqrt{\Delta x_1^2 + \dots + \Delta x_n^2}\right).$$

Theorem 5.3.2: Coefficients of the Differential

If f is differentiable at $\mathbf{x}_0$, then all partial derivatives $\partial f / \partial x_i$ exist at $\mathbf{x}_0$, and

$$A_i = \frac{\partial f}{\partial x_i}(\mathbf{x}_0).$$

Thus the total differential is

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \cdots + \frac{\partial f}{\partial x_n} dx_n.$$

Proof. In the definition of differentiability, set all increments equal to 0 except Δx_i . Then

$$f(\mathbf{x}_0 + \Delta x_i \mathbf{e}_i) - f(\mathbf{x}_0) = A_i \Delta x_i + o(|\Delta x_i|).$$

Dividing by Δx_i and letting $\Delta x_i \rightarrow 0$ gives the i -th partial derivative and shows that it equals A_i . $\square$

Theorem 5.3.3: Differentiability Implies Continuity

If f is differentiable at $\mathbf{x}_0$, then f is continuous at $\mathbf{x}_0$.

Proof. Since

$$\Delta f = L(\Delta \mathbf{x}) + o(\|\Delta \mathbf{x}\|),$$

and every linear map in finite-dimensional Euclidean space is bounded, $L(\Delta \mathbf{x}) \rightarrow 0$ as $\Delta \mathbf{x} \rightarrow 0$. The remainder also tends to 0. Hence $\Delta f \rightarrow 0$. $\square$

Theorem 5.3.4: A Practical Sufficient Condition

If all first-order partial derivatives of f exist in a neighborhood of $\mathbf{x}_0$ and are continuous at $\mathbf{x}_0$, then f is differentiable at $\mathbf{x}_0$.

Proof for two variables. Let $\Delta x = h$ and $\Delta y = k$. Write

$$\begin{aligned} \Delta f &= f(x_0 + h, y_0 + k) - f(x_0, y_0) \\ &= [f(x_0 + h, y_0 + k) - f(x_0, y_0 + k)] + [f(x_0, y_0 + k) - f(x_0, y_0)]. \end{aligned}$$

By the one-variable mean value theorem,

$$\Delta f = f_x(x_0 + \theta_1 h, y_0 + k)h + f_y(x_0, y_0 + \theta_2 k)k,$$

where $\theta_1, \theta_2 \in (0, 1)$. Subtracting $f_x(x_0, y_0)h + f_y(x_0, y_0)k$ and using the continuity of f_x and f_y shows that the remainder is $o(\sqrt{h^2 + k^2})$. $\square$

Note

The logical implications worth remembering are

$$C^1 \implies \text{differentiable} \implies \text{continuous and all partial derivatives exist.}$$

The reverse implications are generally false.

Example 5.3.1: Partial Derivatives Do Not Guarantee Differentiability

Define

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Along the coordinate axes, the function is identically 0, so $f_x(0, 0) = f_y(0, 0) = 0$. However, along $y = x^2$,

$$f(x, x^2) = \frac{x^4}{2x^4} = \frac{1}{2},$$

so f is not continuous at the origin. Therefore it is not differentiable there.

5.3.2.1 Higher Differentials

For a C^2 function $f(x, y)$, the second differential is

$$d^2 f = f_{xx} dx^2 + 2f_{xy} dx dy + f_{yy} dy^2.$$

More generally, one may formally write

$$d^k f = \left(dx \frac{\partial}{\partial x} + dy \frac{\partial}{\partial y} \right)^k f$$

whenever the required partial derivatives exist and are sufficiently regular. The binomial form is especially useful in Taylor expansions.

5.3.3 Vector-Valued Functions and Jacobian Matrices

Although scalar-valued functions are the main object in elementary multivariable calculus, the cleanest formulation of the chain rule uses vector-valued functions.

Definition 5.3.4 (Vector-Valued Function). A function $\mathbf{f} : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be written componentwise as

$$\mathbf{f}(\mathbf{x}) = \begin{pmatrix} f_1(x_1, \dots, x_n) \\ f_2(x_1, \dots, x_n) \\ \vdots \\ f_m(x_1, \dots, x_n) \end{pmatrix}.$$

Definition 5.3.5 (Jacobian Matrix). If the component functions have first-order partial derivatives, the Jacobian matrix of $\mathbf{f}$ at $\mathbf{x}$ is

$$J_{\mathbf{f}}(\mathbf{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}.$$

Definition 5.3.6 (Differentiability for Vector-Valued Functions). The map $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at $\mathbf{x}_0$ if there exists an $m \times n$ matrix A such that

$$\mathbf{f}(\mathbf{x}_0 + \Delta \mathbf{x}) - \mathbf{f}(\mathbf{x}_0) = A \Delta \mathbf{x} + o(\|\Delta \mathbf{x}\|).$$

When differentiability holds, $A = J_{\mathbf{f}}(\mathbf{x}_0)$.

Theorem 5.3.5: Matrix Chain Rule

Let $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at $\mathbf{x}_0$, and let $\mathbf{g} : \mathbb{R}^m \rightarrow \mathbb{R}^p$ be differentiable at $\mathbf{y}_0 = \mathbf{f}(\mathbf{x}_0)$. Then $\mathbf{g} \circ \mathbf{f}$ is differentiable at $\mathbf{x}_0$, and

$$J_{\mathbf{g} \circ \mathbf{f}}(\mathbf{x}_0) = J_{\mathbf{g}}(\mathbf{y}_0)J_{\mathbf{f}}(\mathbf{x}_0).$$

5.3.4 The Gradient and Directional Derivatives

For a scalar-valued function $f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, the gradient collects all first-order partial derivatives into one vector:

$$\nabla f(\mathbf{x}) = \left(\frac{\partial f}{\partial x_1}(\mathbf{x}), \dots, \frac{\partial f}{\partial x_n}(\mathbf{x}) \right).$$

In two variables,

$$\nabla f(x, y) = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}.$$

Theorem 5.3.6: Directional Derivative Formula

If f is differentiable at $\mathbf{x}_0$ and $\mathbf{u}$ is a unit vector, then

$$D_{\mathbf{u}}f(\mathbf{x}_0) = \nabla f(\mathbf{x}_0) \cdot \mathbf{u}.$$

Consequently, the largest directional derivative is $\|\nabla f(\mathbf{x}_0)\|$, attained in the direction of $\nabla f(\mathbf{x}_0)$ if the gradient is nonzero.

Proof. Using differentiability,

$$f(\mathbf{x}_0 + t\mathbf{u}) - f(\mathbf{x}_0) = \nabla f(\mathbf{x}_0) \cdot (t\mathbf{u}) + o(|t|).$$

Divide by t and let $t \rightarrow 0$. □

The gradient obeys the familiar differentiation rules:

$$\begin{aligned} \nabla(af + bg) &= a\nabla f + b\nabla g, \\ \nabla(fg) &= f\nabla g + g\nabla f, \\ \nabla\left(\frac{f}{g}\right) &= \frac{g\nabla f - f\nabla g}{g^2}, \quad g \neq 0. \end{aligned}$$

5.3.5 Chain Rules for Multivariable Functions

Let

$$z = f(x, y), \quad x = x(u, v), \quad y = y(u, v).$$

If f is differentiable at $(x(u_0, v_0), y(u_0, v_0))$ and x, y are differentiable at (u_0, v_0) , then

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u}, \quad \frac{\partial z}{\partial v} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v}.$$

This is simply the matrix chain rule written component by component.

5.3.5.1 Invariance of the First Differential

The first differential has the same formal expression under a change of variables:

$$\begin{aligned} dz &= f_x dx + f_y dy \\ &= f_x(x_u du + x_v dv) + f_y(y_u du + y_v dv) \\ &= z_u du + z_v dv. \end{aligned}$$

This is called the **invariance of the first differential**. It is a compact way to remember the multivariable chain rule.

5.3.6 Taylor Formula for Functions of Two Variables

Taylor's formula is the local polynomial approximation produced by repeatedly applying the differential operator

$$h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}.$$

Theorem 5.3.7: Taylor Formula

Suppose f has continuous partial derivatives up to order $r + 1$ in a neighborhood of (x_0, y_0) . For $(x, y) = (x_0 + h, y_0 + k)$,

$$f(x_0 + h, y_0 + k) = f(x_0, y_0) + \sum_{j=1}^r \frac{1}{j!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^j f(x_0, y_0) + R_r,$$

where

$$R_r = o\left((h^2 + k^2)^{r/2}\right)$$

for the Peano form of the remainder. A Lagrange-type form is

$$R_r = \frac{1}{(r+1)!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^{r+1} f(x_0 + \theta h, y_0 + \theta k), \quad 0 < \theta < 1.$$

The second-order expansion is especially important:

$$\begin{aligned} f(x_0 + h, y_0 + k) &= f(x_0, y_0) + f_x(x_0, y_0)h + f_y(x_0, y_0)k \\ &\quad + \frac{1}{2} (f_{xx}(x_0, y_0)h^2 + 2f_{xy}(x_0, y_0)hk + f_{yy}(x_0, y_0)k^2) \\ &\quad + o(h^2 + k^2). \end{aligned}$$

Theorem 5.3.8: Mean Value Formula

If f has continuous first-order partial derivatives on a convex domain containing (x_0, y_0) and $(x_0 + h, y_0 + k)$, then for some $\theta \in (0, 1)$,

$$\begin{aligned} f(x_0 + h, y_0 + k) - f(x_0, y_0) \\ = f_x(x_0 + \theta h, y_0 + \theta k)h + f_y(x_0 + \theta h, y_0 + \theta k)k. \end{aligned}$$

5.3.7 Implicit Functions

The implicit function theorem explains when an equation can be solved locally for some variables in terms of others. Its core idea is linearization. If the relevant derivative with respect to the variables to be solved

for is invertible, then near the point the nonlinear equation behaves like an invertible linear equation. The invertibility condition prevents the solution set from folding vertically or losing dimension.

Geometrically, an equation $F(x, y) = 0$ often defines a curve. If $F_y \neq 0$, the curve is locally the graph of a function $y = f(x)$. If $F_y = 0$, this graph representation may fail: the curve may have a vertical tangent, a crossing, or a singularity. Thus the derivative condition is not a technical decoration but the mechanism that selects a stable local coordinate system.

Many relations are naturally written as equations rather than explicit formulas. The basic form is

$$F(x, y) = 0.$$

The central question is when this equation locally determines y as a function of x .

Theorem 5.3.9: Implicit Function Theorem: One Equation

Let $F(x, y)$ have continuous first-order partial derivatives in a neighborhood of (x_0, y_0) . If

$$F(x_0, y_0) = 0, \quad F_y(x_0, y_0) \neq 0,$$

then there exists a neighborhood of x_0 on which the equation $F(x, y) = 0$ uniquely determines a C^1 function $y = f(x)$ with $f(x_0) = y_0$. Moreover,

$$\frac{dy}{dx} = -\frac{F_x(x, y)}{F_y(x, y)}.$$

Theorem 5.3.10: Implicit Function Theorem: One Dependent Variable

Let $F(x_1, \dots, x_n, y)$ have continuous first-order partial derivatives near $(x_1^0, \dots, x_n^0, y_0)$. If

$$F(x_1^0, \dots, x_n^0, y_0) = 0, \quad F_y(x_1^0, \dots, x_n^0, y_0) \neq 0,$$

then the equation $F(x_1, \dots, x_n, y) = 0$ locally determines a unique C^1 function

$$y = f(x_1, \dots, x_n),$$

and

$$\frac{\partial y}{\partial x_i} = -\frac{F_{x_i}}{F_y}, \quad i = 1, \dots, n.$$

5.3.7.1 Systems of Implicit Equations

For two equations

$$\begin{cases} F(x, y, u, v) = 0, \\ G(x, y, u, v) = 0, \end{cases}$$

one wants to solve locally for u and v as functions of x and y .

Theorem 5.3.11: Implicit Function Theorem for a System

Suppose F and G have continuous first-order partial derivatives near (x_0, y_0, u_0, v_0) and

$$F(x_0, y_0, u_0, v_0) = G(x_0, y_0, u_0, v_0) = 0.$$

If

$$\frac{\partial(F, G)}{\partial(u, v)} = \begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}_{(x_0, y_0, u_0, v_0)} \neq 0,$$

then the system locally determines unique C^1 functions

$$u = u(x, y), \quad v = v(x, y).$$

Their derivatives satisfy

$$\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = - \begin{pmatrix} F_u & F_v \\ G_u & G_v \end{pmatrix}^{-1} \begin{pmatrix} F_x & F_y \\ G_x & G_y \end{pmatrix}.$$

Note

In computations, it is often simpler to differentiate the two equations directly with respect to x or y , then solve the resulting linear system for u_x, v_x or u_y, v_y .

5.3.8 Geometric Applications**5.3.8.1 Space Curves: Tangent Lines and Normal Planes**

Let a space curve be given by

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}.$$

If $\mathbf{r}'(t_0) \neq \mathbf{0}$, then $\mathbf{r}'(t_0)$ is a tangent vector to the curve at $\mathbf{r}(t_0)$.

The tangent line is

$$\frac{x - x(t_0)}{x'(t_0)} = \frac{y - y(t_0)}{y'(t_0)} = \frac{z - z(t_0)}{z'(t_0)},$$

with the usual convention that a zero denominator means the corresponding coordinate is constant. The normal plane is

$$x'(t_0)(x - x(t_0)) + y'(t_0)(y - y(t_0)) + z'(t_0)(z - z(t_0)) = 0.$$

If the curve is instead given as the intersection of two surfaces

$$\begin{cases} F(x, y, z) = 0, \\ G(x, y, z) = 0, \end{cases}$$

and ∇F and ∇G are not parallel, then a tangent vector is

$$\nabla F \times \nabla G = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ F_x & F_y & F_z \\ G_x & G_y & G_z \end{vmatrix}.$$

Equivalently,

$$\nabla F \times \nabla G = \frac{\partial(F, G)}{\partial(y, z)}\mathbf{i} + \frac{\partial(F, G)}{\partial(z, x)}\mathbf{j} + \frac{\partial(F, G)}{\partial(x, y)}\mathbf{k}.$$

5.3.8.2 Surfaces: Tangent Planes and Normal Lines

For a surface given implicitly by

$$F(x, y, z) = 0,$$

the gradient is normal to the surface. If $P_0 = (x_0, y_0, z_0)$ is on the surface and $\nabla F(P_0) \neq \mathbf{0}$, then the tangent plane is

$$F_x(P_0)(x - x_0) + F_y(P_0)(y - y_0) + F_z(P_0)(z - z_0) = 0.$$

The normal line is

$$\frac{x - x_0}{F_x(P_0)} = \frac{y - y_0}{F_y(P_0)} = \frac{z - z_0}{F_z(P_0)}.$$

For an explicit surface $z = f(x, y)$, the same formula follows from $F(x, y, z) = f(x, y) - z$:

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

5.3.9 Extrema of Multivariable Functions

5.3.9.1 Unconstrained Extrema

Definition 5.3.7 (Critical Point). Let $f(x, y)$ have first-order partial derivatives. A point (x_0, y_0) is called a critical point if

$$f_x(x_0, y_0) = 0, \quad f_y(x_0, y_0) = 0.$$

Theorem 5.3.12: Necessary Condition for an Interior Extremum

If f has first-order partial derivatives at an interior point (x_0, y_0) and has a local maximum or local minimum there, then

$$f_x(x_0, y_0) = f_y(x_0, y_0) = 0.$$

Theorem 5.3.13: Second Derivative Test

Suppose f has continuous second-order partial derivatives near a critical point (x_0, y_0) . Put

$$A = f_{xx}(x_0, y_0), \quad B = f_{xy}(x_0, y_0), \quad C = f_{yy}(x_0, y_0),$$

and

$$\Delta = AC - B^2.$$

Then:

1. If $\Delta > 0$ and $A > 0$, f has a local minimum at (x_0, y_0) .
2. If $\Delta > 0$ and $A < 0$, f has a local maximum at (x_0, y_0) .
3. If $\Delta < 0$, f has no local extremum at (x_0, y_0) ; the point is a saddle point.
4. If $\Delta = 0$, the test is inconclusive.

Idea. At a critical point, the first-order terms in Taylor's formula vanish. The local behavior is governed by the quadratic form

$$\frac{1}{2} (Ah^2 + 2Bhk + Ck^2).$$

The sign definiteness of this quadratic form is determined by $\Delta = AC - B^2$ and A . □

5.3.9.2 Constrained Extrema and Lagrange Multipliers

The geometric idea behind Lagrange multipliers is tangency. On the constraint curve or surface, an extremum occurs where the level set of the objective can no longer be moved to a higher or lower value without leaving the constraint. Therefore the relevant normal vectors must be linearly dependent.

Theorem 5.3.14: Lagrange Multiplier Method: One Constraint

To find extrema of $f(x, y, z)$ subject to

$$g(x, y, z) = 0,$$

look for solutions of

$$\nabla f + \lambda \nabla g = \mathbf{0}, \quad g = 0.$$

Equivalently, define

$$L(x, y, z, \lambda) = f(x, y, z) + \lambda g(x, y, z)$$

and solve

$$\begin{cases} L_x = f_x + \lambda g_x = 0, \\ L_y = f_y + \lambda g_y = 0, \\ L_z = f_z + \lambda g_z = 0, \\ L_\lambda = g(x, y, z) = 0. \end{cases}$$

Theorem 5.3.15: Multiple Constraints

For constraints

$$g(x, y, z) = 0, \quad h(x, y, z) = 0,$$

possible constrained extrema satisfy

$$\nabla f + \lambda \nabla g + \mu \nabla h = \mathbf{0}, \quad g = 0, \quad h = 0.$$

Equivalently, use

$$L = f + \lambda g + \mu h.$$

Note

Lagrange multiplier equations give candidate points. Boundary points, singular points of the constraint, and points where the rank condition fails must still be checked separately when they occur.

5.4 Multiple Integrals

Multiple integrals should be read in two complementary ways. Analytically, they are limits of sums over small pieces of a region. Geometrically, they compute accumulated quantities: area, volume, mass, charge, probability, or average value. The integrand gives local density, and the integral adds these densities over the region.

The main difficulty is often not the integration itself but the description of the domain. Changing the order of integration or changing variables is essentially a way of choosing better coordinates for the same region. A good coordinate system makes the boundary simple and the integrand manageable; a poor coordinate system can turn a simple problem into a long computation.

The construction of multiple integrals begins with the geometric question of area. For double integrals, the domain must have a well-defined area in the Jordan sense.

5.4.1 Jordan Area and Double Integrals

Let $D \subseteq \mathbb{R}^2$ be bounded and contained in a rectangle

$$U = [a, b] \times [c, d].$$

Partition U into small rectangles

$$R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j].$$

The inner area is obtained from rectangles completely contained in D , while the outer area is obtained from rectangles that intersect D . If the supremum of all inner sums equals the infimum of all outer sums, then D has area, denoted $m(D)$.

Theorem 5.4.1: Jordan Measurability Criterion

A bounded plane region D has Jordan area if and only if its boundary ∂D has area zero.

Definition 5.4.1 (Double Integral). Let f be defined on a bounded Jordan measurable region D . Divide D into subregions ΔD_i with areas $\Delta \sigma_i$, choose a sample point $P_i \in \Delta D_i$, and let

$$\lambda = \max_i \text{diam}(\Delta D_i).$$

If the limit

$$\lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(P_i) \Delta \sigma_i$$

exists and is independent of the subdivision and the sample points, then f is integrable on D , and

$$\iint_D f(x, y) d\sigma = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(P_i) \Delta \sigma_i.$$

5.4.1.1 Darboux Sums and Integrability

For each subregion ΔD_i , set

$$M_i = \sup_{(x,y) \in \Delta D_i} f(x, y), \quad m_i = \inf_{(x,y) \in \Delta D_i} f(x, y).$$

The upper and lower Darboux sums are

$$U = \sum_{i=1}^n M_i \Delta \sigma_i, \quad L = \sum_{i=1}^n m_i \Delta \sigma_i.$$

The function f is integrable if and only if

$$\lim_{\lambda \rightarrow 0} (U - L) = 0.$$

Equivalently, if $\omega_i = M_i - m_i$ is the oscillation of f on ΔD_i , then

$$\lim_{\lambda \rightarrow 0} \sum_{i=1}^n \omega_i \Delta \sigma_i = 0.$$

Theorem 5.4.2: Continuity Implies Integrability

If D is a bounded closed Jordan measurable region and f is continuous on D , then f is integrable on D .

5.4.2 Multiple Integrals

The definition extends directly to $\mathbb{R}^n$: area elements $\Delta\sigma_i$ are replaced by volume elements ΔV_i , and double integrals are replaced by n -fold integrals.

For a three-dimensional body D with density $\rho(x, y, z)$, the mass is

$$M = \iiint_D \rho(x, y, z) dV,$$

and the center of mass is

$$\bar{x} = \frac{1}{M} \iiint_D x\rho(x, y, z) dV, \quad \bar{y} = \frac{1}{M} \iiint_D y\rho(x, y, z) dV, \quad \bar{z} = \frac{1}{M} \iiint_D z\rho(x, y, z) dV.$$

For a plane lamina with density $\rho(x, y)$,

$$M = \iint_D \rho(x, y) d\sigma,$$

$$\bar{x} = \frac{1}{M} \iint_D x\rho(x, y) d\sigma, \quad \bar{y} = \frac{1}{M} \iint_D y\rho(x, y) d\sigma.$$

5.4.3 Basic Properties of Double Integrals

Assume all functions involved are integrable on the relevant Jordan measurable domains.

Property 5.4.1

1. For constants a, b ,

$$\iint_D (af + bg) d\sigma = a \iint_D f d\sigma + b \iint_D g d\sigma.$$

2. If $D = D_1 \cup D_2$ and the interiors of D_1 and D_2 do not overlap, then

$$\iint_D f d\sigma = \iint_{D_1} f d\sigma + \iint_{D_2} f d\sigma.$$

3. If $f(x, y) \leq g(x, y)$ for all $(x, y) \in D$, then

$$\iint_D f d\sigma \leq \iint_D g d\sigma.$$

4. If f is integrable on D , then $|f|$ is integrable on D , and

$$\left| \iint_D f d\sigma \right| \leq \iint_D |f| d\sigma.$$

Theorem 5.4.3: Mean Value Theorem for Double Integrals

If f and g are integrable on D , and g does not change sign, then there exists

$$\xi \in \left[\inf_D f, \sup_D f \right]$$

such that

$$\iint_D fg \, d\sigma = \xi \iint_D g \, d\sigma.$$

If f is continuous on a connected compact region D , then there exists $(x_0, y_0) \in D$ such that

$$\iint_D fg \, d\sigma = f(x_0, y_0) \iint_D g \, d\sigma.$$

5.5 Evaluation of Multiple Integrals

The definition of a multiple integral is conceptually important, but in computation we almost never return to the definition directly. The practical problem is to reduce a multiple integral to repeated one-dimensional integrals. This is the role of Fubini's theorem and the reduction formulas for different types of regions.

5.5.1 Reduction of Double Integrals

A plane region is called x -simple if it can be written as

$$D = \{(x, y) : a \leq x \leq b, \varphi_1(x) \leq y \leq \varphi_2(x)\}.$$

For such a region,

$$\iint_D f(x, y) \, d\sigma = \int_a^b \left(\int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) \, dy \right) dx.$$

Similarly, if

$$D = \{(x, y) : c \leq y \leq d, \psi_1(y) \leq x \leq \psi_2(y)\},$$

then

$$\iint_D f(x, y) \, d\sigma = \int_c^d \left(\int_{\psi_1(y)}^{\psi_2(y)} f(x, y) \, dx \right) dy.$$

The two formulas are not two different integrals. They are two descriptions of the same geometric accumulation. Changing the order of integration means redrawing the same region with the other variable as the outer variable.

Theorem 5.5.1: Fubini-Type Reduction

If f is continuous on a bounded closed Jordan measurable region D , then the double integral over D can be computed by the corresponding iterated integral whenever D is represented as a simple region.

Note

In applications, most mistakes come from the domain, not from the antiderivative. A reliable workflow is: first draw D , then decide which variable is outer, then write the projection interval, and only then write the inner boundary functions.

5.5.2 Changing the Order of Integration

Suppose an integral is given as

$$\int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) dy dx.$$

It represents the region

$$D = \{(x, y) : a \leq x \leq b, \varphi_1(x) \leq y \leq \varphi_2(x)\}.$$

To reverse the order, we must describe the same set as

$$D = \{(x, y) : c \leq y \leq d, \psi_1(y) \leq x \leq \psi_2(y)\}.$$

Then

$$\int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) dy dx = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y) dx dy.$$

If a horizontal or vertical line intersects the region in several disjoint intervals, the region must be split. This splitting is not a technical annoyance; it is the price of forcing a complicated domain into a simple-region form.

Example 5.5.1: A Typical Order Change

Let

$$D = \{(x, y) : 0 \leq x \leq 1, x^2 \leq y \leq x\}.$$

Since $x^2 \leq y \leq x$ implies $0 \leq y \leq 1$, and for a fixed y the inequalities become

$$y \leq x \leq \sqrt{y},$$

we have

$$\int_0^1 \int_{x^2}^x f(x, y) dy dx = \int_0^1 \int_y^{\sqrt{y}} f(x, y) dx dy.$$

5.5.3 Change of Variables

For a one-variable integral, substitution changes the coordinate on an interval. For a double integral, a change of variables changes the coordinate grid on a plane region. The correction factor is the absolute value of the Jacobian determinant.

Let

$$x = x(u, v), \quad y = y(u, v),$$

and assume that this transformation maps a region G in the uv -plane one-to-one onto a region D in the xy -plane, with continuous first partial derivatives and nonzero Jacobian except possibly on a negligible set. Then

$$\iint_D f(x, y) dx dy = \iint_G f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv,$$

where

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}.$$

Example 5.5.2: Polar Coordinates

The polar transformation is

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Its Jacobian is

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r.$$

Hence

$$dxdy = r drd\theta.$$

This is why polar coordinates are effective when the region contains circles, sectors, disks, annuli, or expressions involving $x^2 + y^2$.

5.5.4 Triple Integrals

For a solid region $\Omega \subseteq \mathbb{R}^3$, the triple integral is defined similarly:

$$\iiint_{\Omega} f(x, y, z) dV.$$

If

$$\Omega = \{(x, y, z) : (x, y) \in D, z_1(x, y) \leq z \leq z_2(x, y)\},$$

then

$$\iiint_{\Omega} f(x, y, z) dV = \iint_D \left(\int_{z_1(x, y)}^{z_2(x, y)} f(x, y, z) dz \right) dxdy.$$

The same idea applies if we project onto the yz -plane or xz -plane. Again, the essential step is to understand the geometry of the solid.

5.5.5 Cylindrical and Spherical Coordinates

Cylindrical coordinates are

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z,$$

with volume element

$$dV = r drd\theta dz.$$

They are useful for cylinders, cones, rotational solids, and integrands involving $x^2 + y^2$.

Spherical coordinates may be written as

$$x = \rho \sin \varphi \cos \theta, \quad y = \rho \sin \varphi \sin \theta, \quad z = \rho \cos \varphi,$$

where $\rho \geq 0$, $0 \leq \varphi \leq \pi$, and $0 \leq \theta < 2\pi$. The volume element is

$$dV = \rho^2 \sin \varphi d\rho d\varphi d\theta.$$

Note

The symbols θ and φ are not universal. Some books interchange their roles. The formula is safe only after checking how the angles are defined.

5.6 Line and Surface Integrals

Multiple integrals add quantities over regions with area or volume. Line integrals and surface integrals add quantities over lower-dimensional geometric objects: curves and surfaces. The main new issue is that curves and surfaces may carry orientation.

5.6.1 Arc Length and First-Type Line Integrals

Let a smooth curve Γ be parametrized by

$$\mathbf{r}(t) = (x(t), y(t), z(t)), \quad \alpha \leq t \leq \beta.$$

Its arc length element is

$$ds = \|\mathbf{r}'(t)\| dt = \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2} dt.$$

Thus the first-type line integral is

$$\int_{\Gamma} f(x, y, z) ds = \int_{\alpha}^{\beta} f(x(t), y(t), z(t)) \|\mathbf{r}'(t)\| dt.$$

This integral is independent of the orientation of the curve. It measures a scalar density accumulated along a curve, such as mass of a wire.

5.6.2 Second-Type Line Integrals

Let

$$\mathbf{F} = (P, Q, R)$$

be a vector field. The second-type line integral along an oriented curve Γ is

$$\int_{\Gamma} P dx + Q dy + R dz = \int_{\alpha}^{\beta} (Px'(t) + Qy'(t) + Rz'(t)) dt,$$

where P, Q, R are evaluated at $\mathbf{r}(t)$. Equivalently,

$$\int_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = \int_{\alpha}^{\beta} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt.$$

Reversing the orientation changes the sign:

$$\int_{-\Gamma} \mathbf{F} \cdot d\mathbf{r} = - \int_{\Gamma} \mathbf{F} \cdot d\mathbf{r}.$$

Note

The first-type integral uses ds and has no orientation. The second-type integral uses dx, dy, dz and depends on orientation. This distinction is one of the most common sources of errors.

5.6.3 Path Independence and Potential Functions

A vector field $\mathbf{F} = (P, Q, R)$ is called conservative on a region D if there exists a scalar function U such that

$$\nabla U = \mathbf{F}.$$

Then

$$\int_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = U(B) - U(A)$$

for any curve Γ from A to B . In this case, the line integral is path independent.

In a simply connected region in the plane, if P, Q have continuous first partial derivatives, then

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

is the standard test for path independence of

$$\int P dx + Q dy.$$

In three dimensions, the corresponding condition is

$$\nabla \times \mathbf{F} = \mathbf{0},$$

again under suitable simply connected assumptions.

5.6.4 Surface Area

Let a smooth surface Σ be parametrized by

$$\mathbf{r}(u, v) = (x(u, v), y(u, v), z(u, v)), \quad (u, v) \in D.$$

Set

$$E = \mathbf{r}_u \cdot \mathbf{r}_u, \quad F = \mathbf{r}_u \cdot \mathbf{r}_v, \quad G = \mathbf{r}_v \cdot \mathbf{r}_v.$$

Then

$$dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du dv = \sqrt{EG - F^2} du dv,$$

and the surface area is

$$S = \iint_D \sqrt{EG - F^2} du dv.$$

If the surface is given by $z = g(x, y)$ over a plane region D , then

$$dS = \sqrt{1 + g_x^2 + g_y^2} dx dy,$$

so

$$S = \iint_D \sqrt{1 + g_x^2 + g_y^2} dx dy.$$

If the surface is given implicitly by $H(x, y, z) = 0$ and $H_z \neq 0$, then locally

$$dS = \frac{\|\nabla H\|}{|H_z|} dx dy.$$

5.6.5 First-Type Surface Integrals

For a scalar field f on Σ ,

$$\iint_{\Sigma} f(x, y, z) dS = \iint_D f(\mathbf{r}(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| du dv.$$

If Σ is given by $z = g(x, y)$, then

$$\iint_{\Sigma} f(x, y, z) dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} dx dy.$$

This integral is the surface analogue of a first-type line integral: it accumulates scalar density over a surface.

5.6.6 Second-Type Surface Integrals

For a vector field $\mathbf{F} = (P, Q, R)$, the second-type surface integral is the flux through an oriented surface:

$$\iint_{\Sigma} \mathbf{F} \cdot \mathbf{n} \, dS.$$

Using a parametrization $\mathbf{r}(u, v)$, choosing the orientation given by $\mathbf{r}_u \times \mathbf{r}_v$, we have

$$\iint_{\Sigma} \mathbf{F} \cdot \mathbf{n} \, dS = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, dudv.$$

Equivalently,

$$\mathbf{r}_u \times \mathbf{r}_v = \left(\frac{\partial(y, z)}{\partial(u, v)}, \frac{\partial(z, x)}{\partial(u, v)}, \frac{\partial(x, y)}{\partial(u, v)} \right),$$

so

$$\iint_{\Sigma} \mathbf{F} \cdot \mathbf{n} \, dS = \iint_D \left[P \frac{\partial(y, z)}{\partial(u, v)} + Q \frac{\partial(z, x)}{\partial(u, v)} + R \frac{\partial(x, y)}{\partial(u, v)} \right] dudv.$$

It is common to write the same flux integral as

$$\iint_{\Sigma} P \, dydz + Q \, dzdx + R \, dx dy.$$

This is not a new object. It is a compact notation for $\mathbf{F} \cdot \mathbf{n} \, dS$.

Note

Changing the chosen side of the surface changes the sign of the second-type surface integral. It does not change a first-type surface integral.

5.7 Green, Gauss, and Stokes Formulas

Green's formula, Gauss' formula, and Stokes' formula are the structural core of vector calculus. They all express the same idea in different dimensions: an integral over a boundary can be converted into an integral over the interior, or conversely.

5.7.1 Green's Formula

Let $D \subseteq \mathbb{R}^2$ be a bounded region with positively oriented boundary ∂D . Positive orientation means that the region stays on the left as one moves along the boundary, equivalently counterclockwise for the usual orientation of the plane.

Theorem 5.7.1: Green's Formula

If P and Q have continuous first partial derivatives on a region containing D , then

$$\int_{\partial D} P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy.$$

For a rectangle, this follows directly from the one-dimensional fundamental theorem of calculus. For a general region, one decomposes the region into small pieces. The integrals along internal edges cancel because the two neighboring pieces traverse each shared edge in opposite directions.

Corollary 5.7.1: Area Formulas

Taking suitable P, Q in Green's formula gives

$$A(D) = \iint_D 1 \, dxdy = \int_{\partial D} x \, dy = - \int_{\partial D} y \, dx = \frac{1}{2} \int_{\partial D} x \, dy - y \, dx.$$

5.7.2 Gauss' Formula

Let $\Omega \subseteq \mathbb{R}^3$ be a solid region with outward oriented boundary surface $\partial\Omega$.

Theorem 5.7.2: Gauss' Formula

If $\mathbf{F} = (P, Q, R)$ has continuous first partial derivatives on a region containing Ω , then

$$\iint_{\partial\Omega} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_{\Omega} \nabla \cdot \mathbf{F} \, dV,$$

that is,

$$\iint_{\partial\Omega} P \, dydz + Q \, dzdx + R \, dxdy = \iiint_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dxdydz.$$

Gauss' formula says that total outward flux equals total source strength inside the region. In physical language, divergence measures local source density.

5.7.3 Stokes' Formula

Let Σ be an oriented smooth surface with positively oriented boundary $\partial\Sigma$. The positive orientation is determined by the right-hand rule: if the thumb points in the chosen normal direction, the fingers curl in the positive boundary direction.

Theorem 5.7.3: Stokes' Formula

If $\mathbf{F} = (P, Q, R)$ has continuous first partial derivatives on a region containing Σ , then

$$\int_{\partial\Sigma} P \, dx + Q \, dy + R \, dz = \iint_{\Sigma} (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS.$$

Equivalently,

$$\begin{aligned} \int_{\partial\Sigma} P \, dx + Q \, dy + R \, dz &= \iint_{\Sigma} \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dydz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) dzdx \\ &\quad + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy. \end{aligned}$$

Stokes' formula converts circulation along a boundary curve into the flux of curl through a spanning surface. Green's formula is the planar special case of Stokes' formula.

5.7.4 Gradient, Divergence, Curl, and Conservative Fields

For a scalar field $u(x, y, z)$,

$$\nabla u = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right)$$

is the gradient. It points in the direction of fastest increase.

For a vector field $\mathbf{F} = (P, Q, R)$,

$$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

is the divergence, measuring local source or sink strength.

The curl is

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}.$$

It measures local rotational tendency.

A field is conservative if $\mathbf{F} = \nabla U$ for some potential U . In a simply connected region, the condition $\nabla \times \mathbf{F} = \mathbf{0}$ is the usual criterion for conservativeness. In that case, line integrals depend only on endpoints.

5.8 Parameterized Integrals

A parameterized integral is an integral in which one variable is integrated out while the other variables are treated as parameters. For example,

$$I(y) = \int_a^b f(x, y) dx.$$

The main question is whether the good properties of f transfer to I : continuity, differentiability, and integrability with respect to the parameter.

5.8.1 Proper Parameterized Integrals

Assume $f(x, y)$ is defined on the rectangle

$$[a, b] \times [c, d].$$

If f is continuous on this rectangle, then

$$I(y) = \int_a^b f(x, y) dx$$

is continuous on $[c, d]$.

Theorem 5.8.1: Differentiation under the Integral Sign

If f and f_y are continuous on $[a, b] \times [c, d]$, then

$$I(y) = \int_a^b f(x, y) dx$$

is differentiable on $[c, d]$, and

$$I'(y) = \int_a^b \frac{\partial f}{\partial y}(x, y) dx.$$

Theorem 5.8.2: Integration with Respect to the Parameter

If f is continuous on $[a, b] \times [c, d]$, then

$$\int_c^d \left(\int_a^b f(x, y) dx \right) dy = \int_a^b \left(\int_c^d f(x, y) dy \right) dx.$$

The conceptual reason is simple: on a compact rectangle, continuity gives uniform control. This uniformity is exactly what allows the limiting operation hidden inside the integral to commute with changes in the parameter.

5.8.2 Improper Parameterized Integrals

Now consider

$$I(y) = \int_a^{+\infty} f(x, y) dx.$$

For each fixed y , this is an improper integral. To obtain continuity or differentiability of $I(y)$, pointwise convergence is not enough. We need uniform convergence with respect to the parameter.

Definition 5.8.1 (Uniform Convergence of an Improper Parameterized Integral). The improper integral

$$\int_a^{+\infty} f(x, y) dx$$

converges uniformly for $y \in E$ if for every $\varepsilon > 0$, there exists $A > a$ such that for all $A_2 > A_1 > A$ and all $y \in E$,

$$\left| \int_{A_1}^{A_2} f(x, y) dx \right| < \varepsilon.$$

Theorem 5.8.3: Weierstrass Test for Improper Parameterized Integrals

If

$$|f(x, y)| \leq g(x), \quad y \in E,$$

and

$$\int_a^{+\infty} g(x) dx$$

converges, then

$$\int_a^{+\infty} f(x, y) dx$$

converges uniformly on E .

Theorem 5.8.4: Continuity and Differentiation

If $f(x, y)$ is continuous and the improper integral

$$\int_a^{+\infty} f(x, y) dx$$

converges uniformly on $[c, d]$, then $I(y)$ is continuous on $[c, d]$. If, in addition, f_y is continuous and

$$\int_a^{+\infty} f_y(x, y) dx$$

converges uniformly on $[c, d]$, then

$$I'(y) = \int_a^{+\infty} f_y(x, y) dx.$$

Note

The recurring theme is the same as for function series: pointwise convergence only tells us what happens at each fixed parameter, while uniform convergence lets us preserve analytic structure.

5.9 Euler Integrals

The classical Euler integrals are the Beta function and the Gamma function. They are important not only as special functions, but also as examples of how improper integrals, parameterized integrals, and changes of variables interact.

5.9.1 The Beta Function

For $p > 0$ and $q > 0$, define

$$B(p, q) = \int_0^1 x^{p-1} (1-x)^{q-1} dx.$$

The integral may have singularities at both endpoints. Near 0, the convergence is controlled by x^{p-1} ; near 1, it is controlled by $(1-x)^{q-1}$. Hence the natural domain is

$$p > 0, \quad q > 0.$$

The Beta function is symmetric:

$$B(p, q) = B(q, p),$$

which follows from the substitution $x = 1 - t$.

A useful recursion is

$$B(p, q) = \frac{q-1}{p+q-1} B(p, q-1), \quad q > 1.$$

It is obtained by integration by parts.

Another useful form is obtained by the substitution $x = \sin^2 \theta$:

$$B(p, q) = 2 \int_0^{\pi/2} \sin^{2p-1} \theta \cos^{2q-1} \theta d\theta.$$

Equivalently,

$$\int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta = \frac{1}{2} B\left(\frac{m+1}{2}, \frac{n+1}{2}\right),$$

whenever $m > -1$ and $n > -1$.

5.9.2 The Gamma Function

For $p > 0$, define

$$\Gamma(p) = \int_0^{+\infty} x^{p-1} e^{-x} dx.$$

Near 0, the convergence is controlled by x^{p-1} ; near $+\infty$, the exponential factor dominates any power. Hence $\Gamma(p)$ is defined for $p > 0$.

The fundamental recursion is

$$\Gamma(p+1) = p\Gamma(p),$$

which follows from integration by parts:

$$\Gamma(p+1) = \int_0^{+\infty} x^p e^{-x} dx = p \int_0^{+\infty} x^{p-1} e^{-x} dx.$$

Since

$$\Gamma(1) = 1,$$

we get

$$\Gamma(n+1) = n!, \quad n \in \mathbb{N}.$$

Thus the Gamma function extends the factorial from positive integers to positive real numbers.

A central value is

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi},$$

which follows from the Gaussian integral.

5.9.3 Relation Between Beta and Gamma

The key identity is

$$B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}, \quad p > 0, \quad q > 0.$$

One standard proof starts from

$$\Gamma(p)\Gamma(q) = \int_0^{+\infty} \int_0^{+\infty} t^{p-1} s^{q-1} e^{-(t+s)} dt ds,$$

then changes variables by

$$r = t + s, \quad x = \frac{t}{t+s}.$$

The Jacobian is r , and the integral separates into

$$\left(\int_0^1 x^{p-1} (1-x)^{q-1} dx \right) \left(\int_0^{+\infty} r^{p+q-1} e^{-r} dr \right).$$

Therefore

$$\Gamma(p)\Gamma(q) = B(p, q)\Gamma(p+q).$$

There are several deeper identities, such as the reflection formula and Stirling's formula, but for ordinary analysis computations the defining integrals, recursion formulas, and the Beta–Gamma relation are the essential tools.

5.10 Fourier Series

Taylor series approximate a function near one point by powers of $x - x_0$. Fourier series approximate a function on an interval by trigonometric waves. The key idea is not local polynomial approximation, but orthogonal decomposition.

5.10.1 Trigonometric Orthogonality

On $[-\pi, \pi]$, the basic orthogonality relations are

$$\int_{-\pi}^{\pi} \cos nx \cos mx \, dx = 0 \quad (n \neq m),$$

$$\int_{-\pi}^{\pi} \sin nx \sin mx \, dx = 0 \quad (n \neq m),$$

$$\int_{-\pi}^{\pi} \sin nx \cos mx \, dx = 0.$$

Also,

$$\int_{-\pi}^{\pi} \cos^2 nx \, dx = \pi, \quad \int_{-\pi}^{\pi} \sin^2 nx \, dx = \pi$$

for $n \geq 1$, and

$$\int_{-\pi}^{\pi} 1^2 \, dx = 2\pi.$$

This is exactly the same idea as orthogonal coordinates in linear algebra. Fourier coefficients are coordinates of a function along trigonometric basis directions.

5.10.2 Fourier Coefficients and Fourier Series

For an integrable 2π -periodic function f , define

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, dx,$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx.$$

The Fourier series of f is

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx).$$

Its m -th partial sum is

$$S_m(x) = \frac{a_0}{2} + \sum_{n=1}^m (a_n \cos nx + b_n \sin nx).$$

For a function on $[-l, l]$, the corresponding expansion is

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right),$$

with

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} \, dx, \quad b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} \, dx.$$

5.10.3 Complex Form

Using Euler's formula, the Fourier series can also be written as

$$\sum_{n=-\infty}^{+\infty} c_n e^{inx},$$

where

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} \, dx.$$

The real and complex forms are equivalent. The complex form makes orthogonality especially transparent:

$$\int_{-\pi}^{\pi} e^{inx} e^{-ikx} \, dx = \int_{-\pi}^{\pi} e^{i(n-k)x} \, dx = \begin{cases} 2\pi, & n = k, \\ 0, & n \neq k. \end{cases}$$

5.10.4 Bessel Inequality and the Least-Squares View

Let

$$\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)g(x) dx.$$

Then S_m is the orthogonal projection of f onto the finite-dimensional space

$$\text{span}\{1, \cos x, \sin x, \dots, \cos mx, \sin mx\}.$$

Therefore S_m is the best approximation to f in that subspace in the mean-square sense:

$$\int_{-\pi}^{\pi} |f(x) - S_m(x)|^2 dx$$

is minimal among all trigonometric polynomials of degree at most m .

This immediately gives Bessel's inequality:

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} f^2(x) dx.$$

5.10.5 Riemann–Lebesgue Lemma

If f is integrable on $[a, b]$, then

$$\lim_{n \rightarrow \infty} \int_a^b f(x) \sin nx dx = 0, \quad \lim_{n \rightarrow \infty} \int_a^b f(x) \cos nx dx = 0.$$

The intuition is high-frequency cancellation. The function f changes relatively slowly compared with the rapid oscillation of $\sin nx$ and $\cos nx$, so positive and negative contributions cancel in the limit.

5.10.6 Dirichlet Kernel and Pointwise Convergence

The partial sum can be written as an integral operator:

$$S_m(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x+u) D_m(u) du,$$

where

$$D_m(u) = \frac{1}{2} + \sum_{n=1}^m \cos nu = \frac{\sin(m + \frac{1}{2})u}{2 \sin \frac{u}{2}}.$$

This is the Dirichlet kernel. It is the continuous analogue of the projection matrix from finite-dimensional linear algebra.

The convergence of $S_m(x)$ is therefore controlled by the behavior of f near x . Far from x , the oscillation in $D_m(u)$ tends to cancel by the Riemann–Lebesgue lemma. Near $u = 0$, the kernel has a singular shape, so local regularity of f matters.

Theorem 5.10.1: Dirichlet–Jordan Criterion

If f is 2π -periodic, piecewise smooth or of bounded variation in a neighborhood of x , and the one-sided limits $f(x+0)$ and $f(x-0)$ exist, then the Fourier series converges at x to

$$\frac{f(x+0) + f(x-0)}{2}.$$

In particular, if f is continuous at x , then the Fourier series converges to $f(x)$.

Theorem 5.10.2: Dini-Type Local Criterion

Let

$$\sigma(x) = \frac{f(x+0) + f(x-0)}{2}$$

and

$$E_x(u) = f(x+u) + f(x-u) - 2\sigma(x).$$

If for some $\delta > 0$,

$$\int_0^\delta \left| \frac{E_x(u)}{u} \right| du < +\infty,$$

then

$$S_m(x) \rightarrow \sigma(x).$$

A Lipschitz or Hölder condition near x implies the Dini condition. Thus these tests are all ways of ensuring that the local error is controlled strongly enough to survive the singular behavior of the Dirichlet kernel.

5.10.7 What Fourier Series Really Adds

Fourier series should be understood at two levels.

First, algebraically, it is an orthogonal expansion. The coefficients are coordinates, and the partial sums are projections.

Second, analytically, pointwise convergence is not automatic from projection. The projection viewpoint controls global mean-square error, while pointwise convergence depends on the local interaction between f and the Dirichlet kernel.

This distinction explains why Fourier theory is both natural and subtle: the construction is linear-algebraic, but the convergence question is genuinely analytic.

5.11 Conclusion

Mathematical Analysis II extends one-variable analysis in two directions. The first direction is multivariable analysis: Euclidean space, multivariable limits, differentiation, implicit functions, extrema, multiple integrals, and vector calculus. The second direction is limiting processes of functions: function series, power series, parameterized integrals, Euler integrals, and Fourier series.

The central idea remains unchanged: analysis is the study of when limiting operations are valid. Can we interchange a limit and an integral? Can we differentiate a series term by term? Can we compute an integral by changing coordinates? Can a boundary integral be replaced by an interior integral? Each theorem in this chapter gives one precise situation in which the answer is yes.

5.12 Chapter Workshop

Worked example: Constrained optimization with a geometric check

Maximize $f(x, y) = xy$ subject to $x^2 + y^2 = 1$. The Lagrange equations are

$$y = 2\lambda x, \quad x = 2\lambda y, \quad x^2 + y^2 = 1.$$

At a nonzero solution, $4\lambda^2 = 1$, so $y = \pm x$. The constraint gives $x^2 = y^2 = 1/2$. Therefore the maximum is $1/2$ and the minimum is $-1/2$. Geometrically, the level curves $xy = c$ are tangent to the unit circle at the extrema, which is exactly the condition $\nabla f = \lambda \nabla g$.

Application: Optimization and automatic differentiation

For a loss function $L(\theta)$, gradient-based algorithms use the first-order model

$$L(\theta + h) \approx L(\theta) + \nabla L(\theta)^T h.$$

The chain rule on Jacobian matrices is the mathematical core of reverse-mode automatic differentiation. Hessians and quadratic Taylor models describe curvature and motivate Newton and quasi-Newton methods.

Application: Fourier representation and compression

A periodic signal can be approximated by a small number of Fourier modes. Keeping the largest coefficients replaces time-domain samples by coordinates in an orthogonal frequency basis. The same projection principle appears in audio compression, spectral numerical methods, and the analysis of linear partial differential equations.

5.13 Exercises

1. Determine whether $\sum_{n=1}^{\infty} x^n/n$ converges uniformly on $[0, 1/2]$ and on $[0, 1]$.
2. Find the gradient and Hessian of $f(x, y) = x^2 + xy + 2y^2$ and classify its critical point.
3. Evaluate $\iint_D (x + y) dA$ over the triangle $D = \{x \geq 0, y \geq 0, x + y \leq 1\}$.
4. Use polar coordinates to evaluate $\iint_{x^2+y^2 \leq 1} (x^2 + y^2) dA$.
5. Determine whether $F(x, y) = (-y/(x^2 + y^2), x/(x^2 + y^2))$ is conservative on $\mathbb{R}^2 \setminus \{0\}$.
6. Find the Fourier coefficients of the 2π -periodic function $f(x) = x$ on $(-\pi, \pi)$.
7. Give an example of pointwise convergence that is not uniform convergence.

Selected hints and answers

1. The Weierstrass test gives uniform convergence on $[0, 1/2]$. It is not uniform on $[0, 1]$ because the sum $-\log(1 - x)$ is unbounded there.
2. $\nabla f = (2x + y, x + 4y)$; the Hessian is positive definite, so the origin is a strict global minimum.
3. Integrate $0 \leq y \leq 1 - x, 0 \leq x \leq 1$; the value is $1/3$.
4. The integral is $\int_0^{2\pi} \int_0^1 r^2 r dr d\theta = \pi/2$.
5. Its curl vanishes away from the origin, but the integral around the unit circle is 2π , so it is not conservative on this non-simply-connected domain.
6. The function is odd: $a_n = 0$, and $b_n = 2(-1)^{n+1}/n$.
7. On $[0, 1]$, $f_n(x) = x^n$ converges pointwise to a discontinuous limit and therefore not uniformly.

Part III

Spaces and Complex Structure

Chapter 6

Topology

Learning objectives

By the end of this chapter, you should be able to:

- move between metric definitions and open-set definitions;
- compute closure, interior, boundary, and limit points;
- test continuity using inverse images and compare spaces up to homeomorphism;
- use connectedness, compactness, and separation axioms in proofs;
- construct and reason about product and quotient topologies.

Topology is often described as geometry without measurement. This slogan is useful but incomplete. A better description is that topology studies which properties of spaces are preserved by continuous deformation. Lengths, angles, and areas may change, but nearness, separation, connectedness, compactness, and continuity remain meaningful.

The transition from metric spaces to topological spaces is motivated by abstraction. In a metric space, open sets are defined using distances. But many arguments in analysis use only the open sets themselves, not the numerical values of the distances. Topology isolates this open-set structure and asks what can still be proved once distance is removed from the foundations.

Topology is the study of the qualitative structure of spaces. In Euclidean geometry, we measure length, angle, and area. In metric spaces, we measure distance. In topology, we keep only the information needed to speak about nearness, convergence, continuity, connectedness, and compactness.

This may seem like a loss of information, but it is in fact a powerful abstraction. Many theorems in analysis do not depend on the exact numerical value of a distance. The Intermediate Value Theorem depends on connectedness. The Extreme Value Theorem depends on compactness. Continuity can be described without mentioning a formula for distance. Topology isolates these structural ideas.

The basic object is a set together with a chosen collection of subsets called open sets. Once the open sets are specified, the language of limits and continuity can be rebuilt.

6.1 From Metric Spaces to Topological Spaces

6.1.1 Open Sets in Metric Spaces

In a metric space, the open ball is the primitive local object. Saying that a set is open means that every point of the set has a small protective neighborhood still contained in the set. This captures the intuitive

idea that points inside an open set are not sitting on the edge. Around each point there is room to move slightly without leaving the set.

Notice that openness is relative to the ambient space. The interval $(0, 1)$ is open in $\mathbb{R}$, while the same set viewed inside $[0, 1]$ is also open in the subspace topology because it equals $[0, 1] \cap (0, 1)$. This dependence on context is not a defect; it is exactly what allows topology to describe spaces internally.

Let us recall the metric-space intuition. If (X, d) is a metric space, then an open ball centered at $x \in X$ with radius $r > 0$ is

$$B_r(x) = \{y \in X : d(x, y) < r\}.$$

A subset $U \subseteq X$ is open if for every $x \in U$, there exists $r > 0$ such that $B_r(x) \subseteq U$.

Example 6.1.1

In $\mathbb{R}$ with the usual metric $d(x, y) = |x - y|$, the interval $(0, 1)$ is open. For every $x \in (0, 1)$, we may choose

$$r < \min\{x, 1 - x\},$$

so that $(x - r, x + r) \subseteq (0, 1)$.

The important observation is that once we know which subsets are open, we can define continuity without explicitly using the metric.

Proposition 6.1.1: Topological Form of Continuity in Metric Spaces

Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \rightarrow Y$ is continuous if and only if for every open set $V \subseteq Y$, the preimage $f^{-1}(V)$ is open in X .

Proof. Assume first that f is ε - δ continuous. Let $V \subseteq Y$ be open and let $x \in f^{-1}(V)$. Then $f(x) \in V$, so there exists $\varepsilon > 0$ such that $B_\varepsilon(f(x)) \subseteq V$. By continuity, there exists $\delta > 0$ such that $d_X(x, x') < \delta$ implies $d_Y(f(x), f(x')) < \varepsilon$. Thus $B_\delta(x) \subseteq f^{-1}(V)$, so $f^{-1}(V)$ is open.

Conversely, assume preimages of open sets are open. For $x \in X$ and $\varepsilon > 0$, the ball $B_\varepsilon(f(x))$ is open in Y , so $f^{-1}(B_\varepsilon(f(x)))$ is open in X and contains x . Therefore it contains some ball $B_\delta(x)$, giving the usual ε - δ condition. $\square$

This proposition motivates the axiomatic definition of topology.

6.2 Topological Spaces

6.2.1 Definition of a Topology

A topology keeps only the information needed to speak about openness. The axioms say that the whole space and the empty set are open, that arbitrary unions of open sets are open, and that finite intersections of open sets are open. These rules match the behavior of open sets in metric spaces, but they no longer require a distance function.

The asymmetry between arbitrary unions and finite intersections is important. Arbitrary unions of open sets remain open because a point belongs to at least one open set and can use that set's local neighborhood. Infinite intersections may fail to be open because the available neighborhoods may shrink to zero size. For instance, $\bigcap_{n=1}^{\infty} (-1/n, 1/n) = \{0\}$ is not open in $\mathbb{R}$.

Definition 6.2.1 (Topology). Let X be a set. A **topology** on X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ satisfying:

1. $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$;
2. arbitrary unions of elements of $\mathcal{T}$ belong to $\mathcal{T}$;

3. finite intersections of elements of $\mathcal{T}$ belong to $\mathcal{T}$.

The pair $(X, \mathcal{T})$ is called a **topological space**. Elements of $\mathcal{T}$ are called **open sets**.

The asymmetry between arbitrary unions and finite intersections is essential. In $\mathbb{R}$, arbitrary unions of open intervals are open, but arbitrary intersections of open sets need not be open.

Example 6.2.1

For each $n \in \mathbb{N}$, the interval $(-1/n, 1/n)$ is open in $\mathbb{R}$. However,

$$\bigcap_{n=1}^{\infty} (-1/n, 1/n) = \{0\},$$

and $\{0\}$ is not open in the usual topology of $\mathbb{R}$.

6.2.2 Fundamental Examples

The examples in this subsection should not be memorized as a disconnected list. They show that a topology can be coarse or fine depending on how many open sets are allowed. A coarse topology has fewer open sets, so it distinguishes fewer local behaviors. A fine topology has more open sets, so it can distinguish more local behaviors. The indiscrete topology is maximally coarse, while the discrete topology is maximally fine.

This comparison also explains why continuity depends on the chosen topology. If the domain topology is very fine, many preimages are open and continuity becomes easier. If the codomain topology is very fine, there are more open sets whose preimages must be checked and continuity becomes harder. Therefore topology is not just a property of a set; it controls which functions out of and into that set are considered continuous.

Example 6.2.2: Discrete Topology

For any set X , the collection $\mathcal{T} = \mathcal{P}(X)$ is a topology. Every subset is open. This is called the **discrete topology**.

Example 6.2.3: Indiscrete Topology

For any set X , the collection $\mathcal{T} = \{\emptyset, X\}$ is a topology. This is called the **indiscrete topology** or **trivial topology**.

Example 6.2.4: Usual Topology on $\mathbb{R}^n$

The topology generated by Euclidean open balls in $\mathbb{R}^n$ is called the usual topology. Unless otherwise stated, $\mathbb{R}^n$ is equipped with this topology.

Example 6.2.5: Cofinite Topology

Let X be a set. Define

$$\mathcal{T} = \{\emptyset\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}.$$

Then $\mathcal{T}$ is a topology on X , called the **cofinite topology**.

Proof. The empty set belongs to $\mathcal{T}$, and $X \setminus X = \emptyset$ is finite, so $X \in \mathcal{T}$. If $\{U_i\}_{i \in I}$ are non-empty open sets, then

$$X \setminus \bigcup_{i \in I} U_i = \bigcap_{i \in I} (X \setminus U_i),$$

which is a subset of one finite set $X \setminus U_j$ if at least one U_j is non-empty; hence it is finite. Thus arbitrary unions are open. If $U_1, \dots, U_n$ are open, then

$$X \setminus \bigcap_{k=1}^n U_k = \bigcup_{k=1}^n (X \setminus U_k),$$

a finite union of finite sets, hence finite. Thus finite intersections are open. $\square$

Example 6.2.6: Particular Point Topology

Fix $p \in X$. Define

$$\mathcal{T} = \{\emptyset\} \cup \{U \subseteq X : p \in U\}.$$

Then $\mathcal{T}$ is a topology. In this topology, every non-empty open set must contain the special point p .

6.2.3 Comparing Topologies

Given two topologies on the same set, saying that one is finer than the other means that it contains more open sets. This order relation is a way of comparing how much local information the topologies remember. The finer topology can make distinctions that the coarser topology cannot.

One should be careful with the direction of continuity under this comparison. The identity map from a finer topology to a coarser topology is continuous, because every open set in the coarser topology is already open in the finer topology. The reverse identity map may fail to be continuous. This small observation is often the quickest way to check whether two topologies are genuinely different.

Definition 6.2.2 (Finer and Coarser Topologies). Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be topologies on the same set X . We say $\mathcal{T}_1$ is **finer** than $\mathcal{T}_2$ if

$$\mathcal{T}_2 \subseteq \mathcal{T}_1.$$

Equivalently, $\mathcal{T}_2$ is **coarser** than $\mathcal{T}_1$.

A finer topology has more open sets. More open sets make it easier for subsets to be open, but harder for functions into the space to be continuous.

Example 6.2.7

On any set X , the discrete topology is the finest topology, while the indiscrete topology is the coarsest topology.

Remark 6.2.1

Topology is not only about sets; it is about sets together with a chosen standard of local visibility. A subset is open if the topology allows us to see it as a neighborhood-like object.

6.3 Bases and Subbases

A basis is a compressed description of a topology. Instead of listing all open sets, we list a collection of basic open sets and then allow arbitrary unions. This mirrors the way intervals generate the usual topology on $\mathbb{R}$. Most open sets are too numerous and irregular to list directly, but they can be assembled from simple pieces.

The value of a basis is both conceptual and practical. Conceptually, it tells us what local neighborhoods look like. Practically, it allows us to prove continuity and openness by checking only basic open sets. In many spaces, especially product spaces, the topology is easiest to understand through its basis rather than through all open sets at once.

It is often inconvenient to list every open set. Instead, we describe a smaller collection of basic open sets from which all open sets can be built.

6.3.1 Basis for a Topology

Definition 6.3.1 (Basis). Let X be a set. A collection $\mathcal{B} \subseteq \mathcal{P}(X)$ is a **basis** for a topology on X if:

1. for every $x \in X$, there exists $B \in \mathcal{B}$ such that $x \in B$;
2. if $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B}$, then there exists $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

The topology generated by $\mathcal{B}$ consists of all unions of elements of $\mathcal{B}$.

Example 6.3.1

Open intervals (a, b) form a basis for the usual topology on $\mathbb{R}$. Open balls form a basis for the usual topology on $\mathbb{R}^n$.

Proposition 6.3.1

If $\mathcal{B}$ is a basis on X , then the collection of all unions of basis elements is a topology on X .

Proof. The empty union gives $\emptyset$, and the first basis condition ensures that X is the union of all basis elements. Arbitrary unions of unions of basis elements are still unions of basis elements. For finite intersections, it is enough to consider two open sets U and V . If $x \in U \cap V$, then $x \in B_1 \subseteq U$ and $x \in B_2 \subseteq V$ for some basis elements. By the second basis condition, there exists B_3 such that $x \in B_3 \subseteq B_1 \cap B_2 \subseteq U \cap V$. Thus $U \cap V$ is a union of basis elements. $\square$

6.3.2 Subbasis

A subbasis is an even more economical generating device than a basis. Starting from a subbasis, we first take finite intersections to form a basis, and then arbitrary unions to form the topology. The two-step procedure is important: finite intersections create local compatibility among the chosen subbasic conditions, while arbitrary unions allow open sets to be assembled from those local pieces.

Subbases are especially useful when a topology is defined by requiring many maps to be continuous. For example, product topologies can be described using inverse images of open sets under coordinate projection maps. The subbasis captures the minimum open-set information needed to make those projections continuous.

Definition 6.3.2 (Subbasis). A collection $\mathcal{S} \subseteq \mathcal{P}(X)$ is a **subbasis** for a topology on X if the set of all finite intersections of elements of $\mathcal{S}$ forms a basis. The topology generated by $\mathcal{S}$ is the collection of arbitrary unions of such finite intersections.

Example 6.3.2

In $\mathbb{R}$, the collection of all rays $(-\infty, a)$ and (a, ∞) is a subbasis for the usual topology.

6.3.3 Countability Axioms

Definition 6.3.3 (First and Second Countability). A topological space X is **first countable** if every point has a countable local basis. It is **second countable** if the topology has a countable basis.

Example 6.3.3

The usual topology on $\mathbb{R}^n$ is second countable. A countable basis is given by all open balls whose centers have rational coordinates and whose radii are positive rational numbers.

Proposition 6.3.2

Every second countable space is first countable.

Proof. Let $\mathcal{B}$ be a countable basis for X . For a fixed point $x \in X$, the collection of all basis elements containing x is countable. It is a local basis at x . $\square$

6.4 Closed Sets, Closure, Interior, and Boundary

Closure, interior, and boundary are three ways of describing how a set sits inside a space. The interior consists of points safely inside the set. The closure consists of points that cannot be separated from the set by an open neighborhood. The boundary consists of points where every neighborhood sees both the set and its complement.

These notions are topological because they depend only on open sets. A point is in the closure of A if every open neighborhood of the point intersects A . This definition does not mention distance, although in metric spaces it agrees with the more familiar sequential intuition: points in the closure can often be approached by sequences from the set.

Open sets are primary in the axioms, but closed sets are equally important.

Definition 6.4.1 (Closed Set). Let X be a topological space. A subset $C \subseteq X$ is **closed** if $X \setminus C$ is open.

Proposition 6.4.1: Properties of Closed Sets

In a topological space X :

1. $\emptyset$ and X are closed;
2. arbitrary intersections of closed sets are closed;
3. finite unions of closed sets are closed.

Proof. These properties follow from the axioms for open sets and De Morgan's laws:

$$X \setminus \bigcap_i C_i = \bigcup_i (X \setminus C_i), \quad X \setminus \bigcup_{k=1}^n C_k = \bigcap_{k=1}^n (X \setminus C_k).$$

$\square$

6.4.1 Closure and Interior

Definition 6.4.2 (Closure). Let $A \subseteq X$. The **closure** of A , denoted $\overline{A}$, is the intersection of all closed sets containing A .

Definition 6.4.3 (Interior). Let $A \subseteq X$. The **interior** of A , denoted A° or $\text{Int}(A)$, is the union of all open sets contained in A .

Definition 6.4.4 (Boundary). The **boundary** of A is

$$\partial A = \overline{A} \setminus A^\circ.$$

Equivalently,

$$\partial A = \overline{A} \cap \overline{X \setminus A}.$$

Example 6.4.1

In $\mathbb{R}$ with the usual topology, for $A = (0, 1)$,

$$\overline{A} = [0, 1], \quad A^\circ = (0, 1), \quad \partial A = \{0, 1\}.$$

For $A = \mathbb{Q} \subseteq \mathbb{R}$,

$$\overline{\mathbb{Q}} = \mathbb{R}, \quad \text{Int}(\mathbb{Q}) = \emptyset.$$

Proposition 6.4.2: Closure via Open Neighborhoods

Let $A \subseteq X$ and $x \in X$. Then $x \in \overline{A}$ if and only if every open set U containing x intersects A .

Proof. If some open neighborhood U of x is disjoint from A , then $X \setminus U$ is a closed set containing A but not containing x , so $x \notin \overline{A}$. Conversely, if $x \notin \overline{A}$, then $X \setminus \overline{A}$ is an open neighborhood of x disjoint from A . $\square$

6.4.2 Limit Points and Density

Definition 6.4.5 (Limit Point). Let $A \subseteq X$. A point $x \in X$ is a **limit point** of A if every open neighborhood of x contains a point of A different from x .

Definition 6.4.6 (Dense Subset). A subset $A \subseteq X$ is **dense** in X if $\overline{A} = X$.

Example 6.4.2

Both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$ with the usual topology. Every open interval contains both rational and irrational numbers.

6.5 Constructing New Spaces

New spaces are often built from old ones. The subspace topology describes what openness means inside a subset. The product topology describes what openness means when several coordinates vary independently. The quotient topology describes what openness means after gluing points together. These three constructions are the basic grammar of topology.

It is helpful to associate each construction with a geometric operation. Subspaces correspond to cutting out a part of a space. Products correspond to forming coordinate systems or configuration spaces. Quotients correspond to identification, such as gluing the ends of an interval to form a circle. Many important spaces are most naturally defined by quotient constructions.

Topology becomes powerful when we can construct new spaces from old spaces.

6.5.1 Subspace Topology

Definition 6.5.1 (Subspace Topology). Let $(X, \mathcal{T})$ be a topological space and let $Y \subseteq X$. The **subspace topology** on Y is

$$\mathcal{T}_Y = \{Y \cap U : U \in \mathcal{T}\}.$$

Example 6.5.1

The interval $[0, 1]$ as a subspace of $\mathbb{R}$ has open sets of the form $[0, 1] \cap U$, where U is open in $\mathbb{R}$. For example, $[0, 1/2)$ is open in $[0, 1]$, although it is not open in $\mathbb{R}$.

6.5.2 Product Topology

The product topology is designed so that coordinate projections are continuous and so that open neighborhoods describe independent variation in finitely many coordinates. In a finite product such as $X \times Y$, basic open sets look like $U \times V$. In infinite products, however, basic open sets restrict only finitely many coordinates and leave all other coordinates unrestricted.

This finite restriction rule is not arbitrary. It is what makes compactness behave well under products and what prevents the topology from becoming too fine. The product topology therefore balances two needs: it remembers the topology of each coordinate while still allowing infinite products to retain useful global properties.

Definition 6.5.2 (Product Topology). Let X and Y be topological spaces. The **product topology** on $X \times Y$ is generated by the basis

$$\mathcal{B} = \{U \times V : U \subseteq X \text{ open}, V \subseteq Y \text{ open}\}.$$

Example 6.5.2

The usual topology on $\mathbb{R}^2$ is the same as the product topology on $\mathbb{R} \times \mathbb{R}$, where each copy of $\mathbb{R}$ has the usual topology.

Proposition 6.5.1: Continuity into a Product

A map $f : Z \rightarrow X \times Y$ is continuous if and only if its coordinate maps $\pi_X \circ f : Z \rightarrow X$ and $\pi_Y \circ f : Z \rightarrow Y$ are continuous.

6.5.3 Quotient Topology

The quotient topology is usually the least intuitive construction at first because open sets are defined indirectly. A set in the quotient is declared open exactly when its full preimage upstairs is open. This is the correct definition because the quotient map is supposed to be continuous by construction, and the topology is chosen to be the strongest topology with that property.

Geometrically, quotient spaces should be imagined as spaces obtained by identifying points. A circle can be formed by identifying the two endpoints of an interval. A torus can be formed by identifying opposite edges of a square. The quotient topology is the formal rule that makes these gluing pictures mathematically precise.

Definition 6.5.3 (Quotient Topology). Let X be a topological space, let Y be a set, and let $q : X \rightarrow Y$ be a surjective map. The **quotient topology** on Y is defined by declaring $U \subseteq Y$ open if and only if $q^{-1}(U)$ is open in X .

Example 6.5.3: Circle as a Quotient

The circle S^1 can be obtained by identifying the endpoints of the interval $[0, 1]$. Formally, define an equivalence relation on $[0, 1]$ by $0 \sim 1$ and $x \sim x$ otherwise. The quotient space $[0, 1]/\sim$ is homeomorphic to S^1 .

Remark 6.5.1

Quotient topology is the rigorous language of gluing. Many geometric spaces are built by taking a simple space and identifying selected points or subspaces.

6.6 Separation Axioms

Separation axioms measure how well a topology can distinguish points and closed sets. In a Hausdorff space, two distinct points can be placed into disjoint neighborhoods. This property is so common in analysis that it is easy to overlook, but it is essential for many familiar results. For example, limits of sequences are unique in Hausdorff spaces.

Weaker topologies may fail to separate points, which means the space intentionally forgets some distinctions. Stronger separation properties make the topology behave more like metric spaces. Thus separation axioms form a scale: as we move up the scale, the topology contains enough open sets to support more refined arguments.

Topological spaces can be very strange. Separation axioms measure how well a topology distinguishes points.

Definition 6.6.1 (T_0 , T_1 , and Hausdorff Spaces). Let X be a topological space.

1. X is T_0 if for any distinct $x, y \in X$, there exists an open set containing one of them but not the other.
2. X is T_1 if for any distinct $x, y \in X$, there exists an open set containing x but not y , and an open set containing y but not x .
3. X is **Hausdorff** or T_2 if for any distinct $x, y \in X$, there exist disjoint open sets U, V such that $x \in U$ and $y \in V$.

Proposition 6.6.1

Every Hausdorff space is T_1 , and every T_1 space is T_0 .

Example 6.6.1

Every metric space is Hausdorff. If $x \neq y$, let $r = d(x, y)/3$. Then $B_r(x)$ and $B_r(y)$ are disjoint open neighborhoods.

Proposition 6.6.2

A space X is T_1 if and only if every singleton set $\{x\}$ is closed.

Proof. If X is T_1 , then for every $y \neq x$, there is an open set U_y containing y but not x . Thus

$$X \setminus \{x\} = \bigcup_{y \neq x} U_y$$

is open, so $\{x\}$ is closed. Conversely, if every singleton is closed, then $X \setminus \{y\}$ is an open set containing x but not y whenever $x \neq y$. $\square$

Theorem 6.6.1: Uniqueness of Limits in Hausdorff Spaces

If X is Hausdorff, then a sequence in X has at most one limit.

Proof. Suppose $x_n \rightarrow x$ and $x_n \rightarrow y$ with $x \neq y$. Since X is Hausdorff, choose disjoint open sets U and V with $x \in U$ and $y \in V$. Eventually $x_n \in U$, and eventually $x_n \in V$. Hence eventually $x_n \in U \cap V$, impossible because $U \cap V = \emptyset$. $\square$

6.7 Continuity and Homeomorphism

The topological definition of continuity is inverse-image based: $f : X \rightarrow Y$ is continuous if the preimage of every open set in Y is open in X . This definition may look less intuitive than the ε - δ definition, but it captures the same idea without numerical distances. It says that whenever the target imposes a local test of nearness, the source passes that test through preimages.

Homeomorphisms are the isomorphisms of topology. If two spaces are homeomorphic, they may look geometrically different, but they have the same topological structure. A coffee mug and a torus are often used as a visual metaphor because each has one hole; topology regards them as equivalent under continuous deformation, provided tearing and gluing are not allowed.

6.7.1 Continuous Functions

Definition 6.7.1 (Continuity). Let X and Y be topological spaces. A function $f : X \rightarrow Y$ is **continuous** if for every open set $V \subseteq Y$, the preimage $f^{-1}(V)$ is open in X .

Proposition 6.7.1: Equivalent Form Using Closed Sets

A function $f : X \rightarrow Y$ is continuous if and only if for every closed set $C \subseteq Y$, the preimage $f^{-1}(C)$ is closed in X .

Proof. This follows from

$$f^{-1}(Y \setminus C) = X \setminus f^{-1}(C).$$

□

Proposition 6.7.2: Composition of Continuous Functions

If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then $g \circ f : X \rightarrow Z$ is continuous.

Proof. For any open set $W \subseteq Z$,

$$(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W)).$$

Since g is continuous, $g^{-1}(W)$ is open in Y . Since f is continuous, $f^{-1}(g^{-1}(W))$ is open in X . □

6.7.2 Homeomorphisms

Definition 6.7.2 (Homeomorphism). A function $f : X \rightarrow Y$ is a **homeomorphism** if:

1. f is bijective;
2. f is continuous;
3. f^{-1} is continuous.

If such a map exists, X and Y are called **homeomorphic**.

Homeomorphic spaces are considered the same from the topological viewpoint. They may look geometrically different, but their open-set structures are equivalent.

Example 6.7.1

The open intervals $(0, 1)$ and $\mathbb{R}$ are homeomorphic. One possible homeomorphism is

$$f(x) = \tan\left(\pi x - \frac{\pi}{2}\right).$$

Remark 6.7.1

A bijective continuous function need not be a homeomorphism. The inverse must also be continuous. However, a continuous bijection from a compact space to a Hausdorff space is always a homeomorphism.

6.8 Connectedness

Connectedness formalizes the idea of being in one piece. A space is disconnected if it can be separated into two non-empty open parts that do not interact. The definition is deliberately phrased in terms of open sets rather than pictures, because topological spaces need not come with an obvious visual representation.

The intermediate value theorem is a fundamental example of connectedness in action. The interval $[a, b]$ is connected, and a continuous image of a connected set is connected. Since connected subsets of $\mathbb{R}$ are intervals, a continuous real-valued function on $[a, b]$ cannot jump over intermediate values. Thus a classical theorem of calculus becomes a theorem about preservation of connectedness.

Connectedness formalizes the idea that a space consists of one piece.

6.8.1 Definitions

Definition 6.8.1 (Separation). A **separation** of a topological space X is a pair of non-empty disjoint open sets $U, V \subseteq X$ such that

$$X = U \cup V.$$

Definition 6.8.2 (Connected Space). A topological space X is **connected** if it has no separation.

Proposition 6.8.1: Equivalent Characterization

A space X is connected if and only if the only subsets of X that are both open and closed are $\emptyset$ and X .

Proof. If $A \subseteq X$ is both open and closed and is neither $\emptyset$ nor X , then A and $X \setminus A$ form a separation. Conversely, if $X = U \cup V$ is a separation, then U is open and also closed because $U = X \setminus V$. $\square$

Theorem 6.8.1: Intervals Are Connected

Every interval in $\mathbb{R}$ is connected.

Proof. We prove the idea for an interval I . Suppose $I = A \cup B$ is a separation, with $a \in A$ and $b \in B$, say $a < b$. Let

$$S = A \cap [a, b].$$

Using the least upper bound property of $\mathbb{R}$, one shows that $c = \sup S$ must belong neither consistently to A nor to B without contradicting the openness of A and B in the subspace I . Hence no such separation exists. $\square$

6.8.2 Continuous Images of Connected Spaces

Theorem 6.8.2

If $f : X \rightarrow Y$ is continuous and X is connected, then $f(X)$ is connected.

Proof. Suppose $f(X)$ were disconnected. Then $f(X) = U \cup V$ for non-empty disjoint open sets in the subspace topology of $f(X)$. The preimages $f^{-1}(U)$ and $f^{-1}(V)$ are non-empty disjoint open subsets of X whose union is X , contradicting connectedness. $\square$

Theorem 6.8.3: Intermediate Value Theorem

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. If $f(a) < y < f(b)$ or $f(b) < y < f(a)$, then there exists $c \in (a, b)$ such that $f(c) = y$.

Proof. The interval $[a, b]$ is connected, and the continuous image $f([a, b])$ is connected in $\mathbb{R}$. Connected subsets of $\mathbb{R}$ are intervals. Hence $f([a, b])$ contains every value between $f(a)$ and $f(b)$. $\square$

6.8.3 Path Connectedness

Definition 6.8.3 (Path Connectedness). A space X is **path connected** if for any $x, y \in X$, there exists a continuous map $\gamma : [0, 1] \rightarrow X$ such that $\gamma(0) = x$ and $\gamma(1) = y$.

Proposition 6.8.2

Every path connected space is connected.

Proof. Suppose X is path connected but disconnected, with separation $X = U \cup V$. Choose $x \in U$ and $y \in V$. Let $\gamma : [0, 1] \rightarrow X$ be a path from x to y . Then $\gamma([0, 1])$ is connected as a continuous image of a connected interval, but it intersects both U and V , giving a separation of $\gamma([0, 1])$. This is impossible. $\square$

Remark 6.8.1

The converse is false. There exist connected spaces that are not path connected, such as the topologist's sine curve. Thus path connectedness is stronger than connectedness.

6.9 Compactness

Compactness is one of the most important replacement principles in analysis. In $\mathbb{R}^n$, compactness is equivalent to being closed and bounded, but the open-cover definition is more fundamental. It says that whenever a space is covered by open sets, finitely many of those sets already suffice. Compactness is therefore a finiteness principle hidden inside an infinite setting.

Many theorems in analysis rely on this finite extraction property. Continuous functions on compact spaces attain maxima and minima because open neighborhoods that control local behavior can be reduced to finitely many neighborhoods. Uniform continuity on compact metric spaces follows for the same reason: local control can be upgraded to global control by compactness.

Compactness is one of the central ideas of topology and analysis. In finite sets, many arguments work because a finite cover has finitely many pieces. Compactness generalizes this finite behavior to infinite spaces.

6.9.1 Open Covers

Definition 6.9.1 (Open Cover). Let X be a topological space and $A \subseteq X$. An **open cover** of A is a collection of open sets $\{U_i\}_{i \in I}$ such that

$$A \subseteq \bigcup_{i \in I} U_i.$$

A **subcover** is a subcollection that still covers A .

Definition 6.9.2 (Compactness). A topological space X is **compact** if every open cover of X has a finite subcover.

Example 6.9.1

Every finite topological space is compact. Given any open cover, choose one open set containing each point. Since there are finitely many points, this gives a finite subcover.

Example 6.9.2

The open interval $(0, 1)$ is not compact in the usual topology. The collection

$$\left\{ \left(\frac{1}{n}, 1 \right) : n \geq 2 \right\}$$

is an open cover of $(0, 1)$, but no finite subcollection covers points sufficiently close to 0.

6.9.2 Compactness in $\mathbb{R}^n$

Theorem 6.9.1: Heine-Borel Theorem

A subset $K \subseteq \mathbb{R}^n$ is compact if and only if it is closed and bounded.

Remark 6.9.1

The Heine-Borel theorem is special to finite-dimensional Euclidean spaces. In general metric spaces, closed and bounded does not necessarily imply compact.

Theorem 6.9.2: Bolzano-Weierstrass Theorem

Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Remark 6.9.2

In $\mathbb{R}^n$, compactness, sequential compactness, and closed boundedness are equivalent. In arbitrary topological spaces, these equivalences may fail.

6.9.3 Continuous Images of Compact Spaces

Theorem 6.9.3

If $f : X \rightarrow Y$ is continuous and X is compact, then $f(X)$ is compact.

Proof. Let $\{V_i\}_{i \in I}$ be an open cover of $f(X)$. Then $\{f^{-1}(V_i)\}_{i \in I}$ is an open cover of X . Since X is compact, finitely many preimages cover X , say $f^{-1}(V_{i_1}), \dots, f^{-1}(V_{i_n})$. Then $V_{i_1}, \dots, V_{i_n}$ cover $f(X)$. $\square$

Theorem 6.9.4: Extreme Value Theorem

Let X be compact and let $f : X \rightarrow \mathbb{R}$ be continuous. Then f attains a maximum and a minimum on X .

Proof. The image $f(X)$ is compact in $\mathbb{R}$. By Heine-Borel, it is closed and bounded. Since it is bounded, it has a supremum and an infimum. Since it is closed, these values belong to $f(X)$. Therefore f attains its maximum and minimum. $\square$

Theorem 6.9.5: Compact to Hausdorff

Let X be compact, Y Hausdorff, and $f : X \rightarrow Y$ a continuous bijection. Then f is a homeomorphism.

Proof. It suffices to show that f is a closed map. Let $C \subseteq X$ be closed. Since X is compact, C is compact. The continuous image $f(C)$ is compact in Y . In a Hausdorff space, compact subsets are closed. Hence $f(C)$ is closed, so f^{-1} is continuous. $\square$

6.10 Metric Spaces Revisited

Returning to metric spaces after developing topology is useful because it clarifies what extra structure a metric provides. A metric gives numerical distances, sequences, Cauchy conditions, completeness, and quantitative estimates. A topology gives open sets and continuity. Every metric determines a topology, but not every topology comes from a metric.

This distinction matters in advanced mathematics. Some spaces are naturally topological but not naturally metric; others admit many different metrics producing the same topology. When a proof uses only open sets, it is topological. When it uses distances, Cauchy sequences, or explicit bounds, it is metric. Recognizing this difference helps identify the true assumptions behind a theorem.

Many topological spaces come from metrics, but not all topologies are metrizable.

Definition 6.10.1 (Metrizable Space). A topological space $(X, \mathcal{T})$ is **metrizable** if there exists a metric d on X such that $\mathcal{T}$ is exactly the topology generated by the open balls of d .

Example 6.10.1

Every discrete topological space is metrizable. Define

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Then every singleton is open, so every subset is open.

Remark 6.10.1

Metrizability is a strong property. Metric spaces are always Hausdorff and first countable. Therefore any topology failing these properties cannot come from a metric.

6.11 Separation Axioms Beyond the Hausdorff Condition

The axioms T_0 , T_1 , and T_2 compare neighborhoods of points. The next separation axioms compare points with closed sets, and then compare two closed sets with each other. This progression is not cosmetic. Many constructions in analysis require continuous functions that distinguish sets, and such functions cannot be produced from the Hausdorff condition alone.

There are several conventions in the literature. In these notes, a *regular space* and a *normal space* are not assumed to be T_1 unless this is stated explicitly. Thus T_3 means regular plus T_1 , and T_4 means normal plus T_1 .

Definition 6.11.1 (Regularity and Normality). Let X be a topological space.

1. X is **regular** if whenever $x \in X$ and $F \subseteq X$ is closed with $x \notin F$, there exist disjoint open sets U, V such that $x \in U$ and $F \subseteq V$.
2. X is **completely regular** if whenever $x \notin F$ and F is closed, there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(x) = 0$ and $f|_F = 1$.
3. X is **normal** if whenever $A, B \subseteq X$ are disjoint closed sets, there exist disjoint open sets U, V such that $A \subseteq U$ and $B \subseteq V$.

Definition 6.11.2 (The Separation Hierarchy). A space is called

T_0	Kolmogorov,
T_1	all singleton sets are closed,
T_2	Hausdorff,
T_3	regular and T_1 ,
$T_{3\frac{1}{2}}$	completely regular and T_1 ,
T_4	normal and T_1 .

A $T_{3\frac{1}{2}}$ space is also called a **Tychonoff space**.

Proposition 6.11.1: Closure Form of Regularity

A space X is regular if and only if for every point x and every open set U containing x , there exists an open set V such that

$$x \in V \subseteq \overline{V} \subseteq U.$$

Proof. Assume first that X is regular. The set $F = X \setminus U$ is closed and does not contain x . Choose disjoint open sets V, W with $x \in V$ and $F \subseteq W$. Since $V \cap W = \emptyset$, no point of W lies in $\overline{V}$. Therefore $\overline{V} \subseteq X \setminus W \subseteq U$.

Conversely, suppose the closure condition holds. Let F be closed and $x \notin F$. Apply the condition to $U = X \setminus F$ and choose V with $x \in V \subseteq \overline{V} \subseteq U$. Then V and $X \setminus \overline{V}$ are disjoint open sets containing x and F , respectively. $\square$

Theorem 6.11.1: Metric Spaces Are Normal

Every metric space is normal. Consequently, every metric space is T_4 and hence also $T_{3\frac{1}{2}}$, T_3 , T_2 , T_1 , and T_0 .

Proof. Let A and B be disjoint closed subsets of a metric space (X, d) . Define

$$d(x, A) = \inf_{a \in A} d(x, a), \quad d(x, B) = \inf_{b \in B} d(x, b).$$

Both functions are continuous. Set

$$U = \{x : d(x, A) < d(x, B)\}, \quad V = \{x : d(x, B) < d(x, A)\}.$$

These sets are open and disjoint. If $a \in A$, then $d(a, A) = 0$, while $d(a, B) > 0$ because B is closed and $a \notin B$. Thus $A \subseteq U$, and similarly $B \subseteq V$. $\square$

Example 6.11.1: Why the Hypotheses Matter

The cofinite topology on an infinite set is T_1 but not Hausdorff: any two non-empty open sets intersect. Thus the implication $T_1 \Rightarrow T_2$ fails. The Sierpinski space $X = \{0, 1\}$ with topology $\{\emptyset, \{1\}, X\}$ is T_0 but not T_1 . These examples show that the lower separation axioms are genuinely distinct.

Theorem 6.11.2: Urysohn's Lemma

Let X be a normal space and let $A, B \subseteq X$ be disjoint closed sets. Then there exists a continuous function $f : X \rightarrow [0, 1]$ such that

$$f|_A = 0, \quad f|_B = 1.$$

Construction Outline. The proof repeatedly uses normality to insert closures between open sets. For each dyadic rational $r \in [0, 1]$, construct an open set U_r so that

$$A \subseteq U_0, \quad \overline{U_r} \subseteq U_s \quad (r < s), \quad X \setminus U_1 \supseteq B.$$

Define

$$f(x) = \inf\{r : x \in U_r\},$$

with the convention that the infimum of the empty set is 1. The nesting relation implies

$$f^{-1}([0, a)) = \bigcup_{r < a} U_r, \quad f^{-1}((a, 1]) = \bigcup_{r > a} (X \setminus \overline{U_r}),$$

so both relevant inverse images are open. Hence f is continuous and has the required boundary values. $\square$

Theorem 6.11.3: Tietze Extension Theorem

Let X be a normal space, let $A \subseteq X$ be closed, and let $f : A \rightarrow [-1, 1]$ be continuous. Then there exists a continuous function $F : X \rightarrow [-1, 1]$ such that $F|_A = f$.

Remark 6.11.1

Urysohn's lemma separates two closed sets by a continuous real-valued function. Tietze's theorem extends an already defined continuous function from a closed subspace. The second theorem is deeper, but its proof repeatedly applies the first theorem to approximate the remaining error. These results explain why normality is the natural topological condition for constructing real-valued functions.

6.12 Components, Local Connectedness, and the Topologist's Sine Curve

Connectedness is a global condition: the whole space cannot be separated into two non-empty open pieces. Path connectedness is more constructive: every pair of points can be joined by a continuous path. Local connectedness asks for connected neighborhoods near each point. None of these conditions should be confused with the others.

Definition 6.12.1 (Connected and Path Components). For $x \in X$, the **connected component** of x is the union of all connected subsets of X containing x . The **path component** of x is the union of all path connected subsets containing x .

Proposition 6.12.1

Connected components are closed, pairwise disjoint, and maximal connected subsets. Path components are pairwise disjoint and maximal path connected subsets, but they need not be closed in an arbitrary topological space.

Proof. If two connected subsets intersect, their union is connected. Hence the union used to define a connected component is connected, and two different components cannot intersect. The closure of a connected set is connected, so maximality forces each connected component to contain its closure and therefore to be

closed. The path-component statement follows from the analogous fact that two intersecting path connected sets have path connected union. $\square$

Definition 6.12.2 (Local Connectedness). A space X is **locally connected at x** if every neighborhood of x contains a connected open neighborhood of x . It is **locally path connected at x** if every neighborhood of x contains a path connected open neighborhood of x . The space is locally connected or locally path connected if the condition holds at every point.

Theorem 6.12.1

If X is locally path connected, then its connected components and path components coincide, and each component is open.

Proof. Fix $x \in X$ and let P be its path component. For each $y \in P$, choose a path connected open neighborhood U_y of y . Every point of U_y can be joined to y , and y can be joined to x , so $U_y \subseteq P$. Hence P is open. The path components form a disjoint open partition. A connected subset cannot meet two different members of such a partition, so the connected component of x is contained in P . The reverse containment is automatic because path connected sets are connected. $\square$

Example 6.12.1: The Topologist's Sine Curve

Let

$$S = \left\{ \left(x, \sin \frac{1}{x} \right) : 0 < x \leq 1 \right\} \cup (\{0\} \times [-1, 1]) \subseteq \mathbb{R}^2.$$

The graph part

$$G = \left\{ \left(x, \sin \frac{1}{x} \right) : 0 < x \leq 1 \right\}$$

is the continuous image of the connected interval $(0, 1]$, so G is connected. Its closure is exactly S , and the closure of a connected set is connected. Therefore S is connected.

However, no path in S joins a point of the vertical segment to a point of G . Suppose such a path existed. As the first coordinate approaches 0 through positive values, the second coordinate would have to follow $\sin(1/x)$, which oscillates infinitely often between values near -1 and 1 . Continuity at the first moment when the path leaves the vertical segment would be impossible. Hence S is connected but not path connected. It is also not locally connected at points of the vertical segment.

Remark 6.12.1

This example is valuable because the failure is not caused by disconnected pieces placed far apart. The graph accumulates on the entire vertical segment, so closure creates connectedness without creating paths. It is a model example of why limit behavior in topology can be subtler than geometric intuition suggests.

6.13 Compactness Beyond Closed and Bounded Sets

The slogan “compact means closed and bounded” is valid in finite-dimensional Euclidean space, but it is not the definition and it fails in many other spaces. The open-cover definition is robust because it is preserved by continuous maps and makes sense without a metric. Several equivalent formulations reveal different ways in which compactness converts infinitely many local conditions into finitely many global conditions.

6.13.1 The Finite Intersection Property

Definition 6.13.1 (Finite Intersection Property). A family $\mathcal{F}$ of subsets of X has the **finite intersection property** if every finite subfamily has non-empty intersection.

Theorem 6.13.1: Closed-Set Characterization of Compactness

A topological space X is compact if and only if every family of closed sets with the finite intersection property has non-empty total intersection.

Proof. Suppose X is compact and $\{F_i\}_{i \in I}$ is a family of closed sets with empty total intersection. Then $\{X \setminus F_i\}_{i \in I}$ is an open cover of X . A finite subcover would imply that the corresponding finite intersection of the F_i is empty, contradicting the finite intersection property.

Conversely, suppose every closed family with the finite intersection property has non-empty intersection. If an open cover $\{U_i\}_{i \in I}$ had no finite subcover, then every finite intersection of the closed sets $X \setminus U_i$ would be non-empty. Their total intersection would therefore be non-empty, contradicting the assumption that the U_i cover X . $\square$

6.13.2 Sequential Compactness in Metric Spaces

Definition 6.13.2 (Sequential Compactness). A space is **sequentially compact** if every sequence has a convergent subsequence whose limit belongs to the space.

Theorem 6.13.2

For metric spaces, compactness and sequential compactness are equivalent.

Idea of Proof. If a metric space is compact, cover it by finitely many balls of radius 1. One of these balls contains infinitely many terms of the sequence. Cover that ball by finitely many balls of radius $1/2$ and choose one containing infinitely many of the remaining terms. Continuing inductively gives a nested selection of terms whose mutual distances tend to zero. Compactness supplies a cluster point, and the selected subsequence converges to it.

For the converse, assume sequential compactness. First, the space is totally bounded. Otherwise there would exist $\varepsilon > 0$ and a sequence (x_n) with

$$d(x_m, x_n) \geq \varepsilon \quad (m \neq n),$$

which has no convergent subsequence.

Now let $\mathcal{U}$ be an open cover. We claim that $\mathcal{U}$ has a Lebesgue number. If not, for each n choose x_n such that $B_{1/n}(x_n)$ is contained in no member of $\mathcal{U}$. Sequential compactness gives a subsequence $x_{n_k} \rightarrow x$. Choose $U \in \mathcal{U}$ and $r > 0$ with $B_r(x) \subseteq U$. For large k , both $d(x_{n_k}, x) < r/2$ and $1/n_k < r/2$, so

$$B_{1/n_k}(x_{n_k}) \subseteq B_r(x) \subseteq U,$$

a contradiction. Finally, total boundedness supplies finitely many balls of radius smaller than the Lebesgue number, and one member of $\mathcal{U}$ contains each ball. These finitely many members form a finite subcover. $\square$

Theorem 6.13.3: Lebesgue Number Lemma

Let (X, d) be a compact metric space and let $\mathcal{U}$ be an open cover of X . Then there exists $\delta > 0$ such that every subset of diameter less than δ is contained in some member of $\mathcal{U}$.

Proof. For each $x \in X$, choose $U_x \in \mathcal{U}$ containing x , and choose $r_x > 0$ such that $B_{2r_x}(x) \subseteq U_x$. The balls $B_{r_x}(x)$ cover X , so finitely many suffice: $B_{r_1}(x_1), \dots, B_{r_n}(x_n)$. Let $\delta = \min r_i$. If a set A has diameter less than δ and contains a point a , choose i with $a \in B_{r_i}(x_i)$. Then every $y \in A$ satisfies

$$d(y, x_i) \leq d(y, a) + d(a, x_i) < \delta + r_i \leq 2r_i,$$

so $A \subseteq B_{2r_i}(x_i) \subseteq U_{x_i}$. □

Corollary 6.13.1: Uniform Continuity on Compact Spaces

Every continuous map from a compact metric space to a metric space is uniformly continuous.

Proof. Let $f : X \rightarrow Y$ be continuous and fix $\varepsilon > 0$. For each $x \in X$, choose an open neighborhood U_x such that $f(U_x)$ lies in the ball of radius $\varepsilon/2$ around $f(x)$. A Lebesgue number δ for the cover $\{U_x\}$ ensures that whenever $d(p, q) < \delta$, the two-point set $\{p, q\}$ lies in one U_x . Hence

$$d_Y(f(p), f(q)) < \varepsilon.$$

□

6.13.3 Finite Products and the First Form of Tychonoff's Theorem

Theorem 6.13.4: Finite Product Theorem

If X and Y are compact, then $X \times Y$ is compact in the product topology. Consequently, every finite product of compact spaces is compact.

Tube-Lemma Proof. Let $\mathcal{U}$ be an open cover of $X \times Y$. Fix $x \in X$. The slice $\{x\} \times Y$ is homeomorphic to Y and hence compact, so finitely many members of $\mathcal{U}$ cover it. Their union is an open set containing $\{x\} \times Y$. By the tube lemma, there exists an open neighborhood N_x of x such that

$$N_x \times Y$$

is contained in that finite union. The sets N_x cover X . Compactness of X gives finitely many of them, and collecting the corresponding finite subfamilies of $\mathcal{U}$ yields a finite cover of $X \times Y$. □

Theorem 6.13.5: Tube Lemma

Let Y be compact and let $W \subseteq X \times Y$ be open with $\{x_0\} \times Y \subseteq W$. Then there exists an open neighborhood N of x_0 such that $N \times Y \subseteq W$.

Proof. For each $y \in Y$, choose basic open sets $N_y \subseteq X$ and $V_y \subseteq Y$ such that

$$(x_0, y) \in N_y \times V_y \subseteq W.$$

The sets V_y cover Y , so finitely many $V_{y_1}, \dots, V_{y_n}$ suffice. Then

$$N = N_{y_1} \cap \dots \cap N_{y_n}$$

is an open neighborhood of x_0 , and $N \times Y \subseteq W$. □

Remark 6.13.1

The full Tychonoff theorem states that an arbitrary product of compact spaces is compact. Its standard proof uses the axiom of choice, often through ultrafilters or maximal families with the finite intersection property. The finite theorem above contains the geometric core and is sufficient for most elementary applications.

6.14 Products and Quotients as Universal Constructions

The definitions of product and quotient topologies can appear asymmetric. The product topology is generated by inverse images under projections, while the quotient topology is tested by inverse images under one surjective map. Their common feature is a universal property: each topology is exactly the one that makes a certain continuity problem as easy as possible.

Theorem 6.14.1: Universal Property of Products

Let Z be a topological space and let $f : Z \rightarrow X \times Y$. Write

$$f(z) = (f_1(z), f_2(z)).$$

Then f is continuous if and only if both coordinate maps $f_1 : Z \rightarrow X$ and $f_2 : Z \rightarrow Y$ are continuous.

Proof. If f is continuous, then $f_1 = \pi_X \circ f$ and $f_2 = \pi_Y \circ f$ are continuous. Conversely, for every basic open set $U \times V$,

$$f^{-1}(U \times V) = f_1^{-1}(U) \cap f_2^{-1}(V),$$

which is open when f_1 and f_2 are continuous. □

Theorem 6.14.2: Universal Property of Quotients

Let $q : X \rightarrow Y$ be a quotient map and let $g : Y \rightarrow Z$ be a function. Then g is continuous if and only if $g \circ q : X \rightarrow Z$ is continuous.

Proof. One direction follows from composition. Conversely, assume $g \circ q$ is continuous. For every open set $W \subseteq Z$,

$$q^{-1}(g^{-1}(W)) = (g \circ q)^{-1}(W)$$

is open in X . By the definition of the quotient topology, $g^{-1}(W)$ is open in Y . □

Example 6.14.1: The Circle from an Interval

Define $q : [0, 1] \rightarrow S^1$ by

$$q(t) = e^{2\pi it}.$$

The map is continuous, surjective, and identifies exactly the endpoints 0 and 1. Since $[0, 1]$ is compact and S^1 is Hausdorff, the induced bijection

$$[0, 1]/(0 \sim 1) \longrightarrow S^1$$

is a homeomorphism. This argument is stronger than a picture: it proves that the quotient topology is precisely the usual topology on the circle.

Example 6.14.2: A Non-Hausdorff Quotient

Start with two copies of the real line and identify corresponding points except at the origins. The resulting “line with two origins” is locally indistinguishable from $\mathbb{R}$, but the two origins cannot be separated by disjoint open sets. Quotients can therefore destroy the Hausdorff property. Whenever a space is built by gluing, separation axioms must be checked rather than assumed.

6.15 Conclusion

Topology begins by replacing distance with open sets. From this simple abstraction, we recover the language of continuity, convergence, compactness, and connectedness. The power of topology is that it reveals which theorems of analysis depend on metric quantities and which depend only on the qualitative structure of open sets.

The essential pattern is the same as in previous chapters: identify the primitive data, state axioms, define structure-preserving maps, and then study properties invariant under those maps. In topology, the primitive data are open sets, the structure-preserving maps are continuous functions, and the notion of sameness is homeomorphism.

6.16 Chapter Workshop

Worked example: Why quotient topology constructs a circle

Let $q : [0, 1] \rightarrow S^1$ be $q(t) = e^{2\pi it}$. It is continuous and identifies exactly the endpoints 0 and 1. Because $[0, 1]$ is compact and S^1 is Hausdorff, the induced continuous bijection

$$[0, 1]/(0 \sim 1) \longrightarrow S^1$$

is a homeomorphism. The quotient topology is therefore the precise operation of gluing endpoints, not merely a suggestive picture.

Application: Topology of data representations

A data representation is often useful only up to a transformation: phase may be periodic, rotations may be equivalent, or permutations may not change the object. Quotient spaces formalize these identifications. In manifold learning and topological data analysis, connected components, loops, and higher-dimensional holes are used as features that are stable under continuous deformation rather than under exact coordinates.

6.17 Exercises

1. In the usual topology on $\mathbb{R}$, find the interior, closure, boundary, and set of limit points of $\{0\} \cup \{1/n : n \in \mathbb{N}\}$.
2. Prove that the continuous image of a connected space is connected.
3. Give an open cover of $(0, 1)$ with no finite subcover.
4. Show that a finite subset of a Hausdorff space is closed.
5. Compare the product topology and box topology on $\mathbb{R}^{\mathbb{N}}$ by giving a box-open set that is not product-open.

6. Explain why compactness, rather than closed-and-bounded language, is the topological property preserved by homeomorphisms.

Selected hints and answers

1. The interior is empty; the closure and set itself coincide; the boundary is the whole set; 0 is the only limit point.
2. If the image were separated into two disjoint nonempty open subsets, their inverse images would separate the domain.
3. Use $U_n = (1/n, 1)$ together with a suitable indexing shift, or $U_n = (0, 1 - 1/n)$ for $n \geq 2$.
4. Singletons are closed in a Hausdorff space; use a finite union.
5. The set $\prod_{n=1}^{\infty} (-1/n, 1/n)$ is box-open but contains no basic product neighborhood of the zero sequence.
6. “Closed and bounded” depends on a metric and is special to Euclidean spaces; compactness is defined purely by open covers and is preserved by continuous maps and homeomorphisms.

Chapter 7

Complex Analysis

Learning objectives

By the end of this chapter, you should be able to:

- compute with complex numbers and elementary holomorphic functions;
- use the Cauchy–Riemann equations without confusing necessity and sufficiency;
- evaluate contour integrals using antiderivatives, Cauchy’s formula, and residues;
- classify isolated singularities through Laurent series;
- apply the argument principle, maximum principles, and conformal viewpoints.

Complex analysis studies functions of a complex variable. At first glance, it may look like ordinary calculus performed on the complex plane. However, this impression is misleading. Complex differentiability is far more rigid than real differentiability. A function that is differentiable once in the complex sense is automatically infinitely differentiable and locally represented by a power series. This is one of the most beautiful phenomena in mathematics.

The central reason is that complex numbers contain both algebraic and geometric information. Multiplication by a complex number combines scaling and rotation. Therefore, a complex derivative is not merely a rate of change; it is a local similarity transformation. This geometric constraint forces complex differentiable functions to have strong structure.

This chapter develops the basic route of complex analysis:

complex numbers $\rightarrow$ complex differentiability $\rightarrow$ Cauchy theory $\rightarrow$ power series $\rightarrow$ residues.

7.1 Complex Numbers and the Complex Plane

The complex plane should be read simultaneously as an algebraic field and as a geometric plane. Addition corresponds to vector translation, while multiplication corresponds to scaling and rotation. This dual interpretation is the reason complex numbers are so efficient in geometry, differential equations, Fourier analysis, and signal processing. They package two real coordinates into one object whose arithmetic remembers planar geometry.

Whenever possible, it is useful to translate a complex formula into a geometric statement. The modulus $|z|$ is distance from the origin, $|z - w|$ is distance between points, conjugation reflects across the real axis, and multiplication by $e^{i\theta}$ rotates through angle θ . These interpretations make later ideas such as conformality and residues much less mysterious.

7.1.1 Algebraic Definition

Definition 7.1.1 (Complex Numbers). The set of complex numbers is

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}, i^2 = -1\}.$$

For $z = a + bi$, the real and imaginary parts are

$$\operatorname{Re} z = a, \quad \operatorname{Im} z = b.$$

Addition and multiplication are defined by the rules

$$(a + bi) + (c + di) = (a + c) + (b + d)i,$$

and

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

Definition 7.1.2 (Complex Conjugate and Modulus). For $z = a + bi \in \mathbb{C}$, define its **complex conjugate** by

$$\bar{z} = a - bi,$$

and its **modulus** by

$$|z| = \sqrt{a^2 + b^2}.$$

Proposition 7.1.1: Basic Identities

For $z, w \in \mathbb{C}$,

1. $z\bar{z} = |z|^2$;
2. $\overline{z + w} = \bar{z} + \bar{w}$;
3. $\overline{z\bar{w}} = \bar{z}w$;
4. $|zw| = |z||w|$;
5. if $z \neq 0$, then $z^{-1} = \bar{z}/|z|^2$.

Proof. Each identity follows directly from writing $z = a + bi$ and $w = c + di$. For example,

$$z\bar{z} = (a + bi)(a - bi) = a^2 + b^2 = |z|^2.$$

If $z \neq 0$, then

$$z \frac{\bar{z}}{|z|^2} = \frac{z\bar{z}}{|z|^2} = 1.$$

□

7.1.2 Geometric Interpretation

A complex number $z = a + bi$ can be viewed as the point (a, b) in the plane. Under this identification, addition of complex numbers is vector addition.

Multiplication has a richer meaning. If z has modulus r and argument θ , then multiplication by z scales lengths by r and rotates angles by θ .

Definition 7.1.3 (Argument). For a non-zero complex number $z = a + bi$, an **argument** of z is any real number θ such that

$$a = |z| \cos \theta, \quad b = |z| \sin \theta.$$

The set of all arguments differs by integer multiples of 2π .

Definition 7.1.4 (Polar Form). Every non-zero complex number can be written as

$$z = r(\cos \theta + i \sin \theta),$$

where $r = |z| > 0$ and θ is an argument of z .

7.1.3 Euler's Formula

Euler's formula is more than a memorable identity; it is the bridge between exponential growth and circular motion. In real analysis, the exponential function and trigonometric functions appear to describe different phenomena. In complex analysis they become different shadows of the same function. This unification is one reason complex methods simplify oscillatory problems.

The formula also explains why polar form is natural. Writing $z = re^{i\theta}$ separates size from direction. Multiplication then becomes especially transparent: sizes multiply and angles add. This is often the simplest way to compute powers, roots, and rotations.

Theorem 7.1.1: Euler's Formula

For every real number θ ,

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

Proof via Taylor Series. Using the Taylor series expansions,

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, \quad \sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!},$$

we substitute $x = i\theta$:

$$e^{i\theta} = 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + \cdots.$$

Grouping real and imaginary parts gives

$$e^{i\theta} = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) + i \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right),$$

which equals $\cos \theta + i \sin \theta$. □

Corollary 7.1.1: Exponential Form

Every non-zero complex number can be written as

$$z = re^{i\theta},$$

where $r = |z|$ and θ is an argument of z .

Proposition 7.1.2: Multiplication in Polar Form

If

$$z_1 = r_1 e^{i\theta_1}, \quad z_2 = r_2 e^{i\theta_2},$$

then

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}.$$

Proof. Using laws of exponentials,

$$z_1 z_2 = r_1 r_2 e^{i\theta_1} e^{i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)}.$$

□

Theorem 7.1.2: De Moivre's FormulaFor $n \in \mathbb{Z}$,

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

Proof. Since $\cos \theta + i \sin \theta = e^{i\theta}$,

$$(\cos \theta + i \sin \theta)^n = (e^{i\theta})^n = e^{in\theta} = \cos(n\theta) + i \sin(n\theta).$$

□

Example 7.1.1: Roots of UnityThe solutions of $z^n = 1$ are

$$z_k = e^{2\pi i k/n}, \quad k = 0, 1, \dots, n-1.$$

Geometrically, these are the vertices of a regular n -gon on the unit circle.

7.2 Topology of the Complex Plane

The topology of the complex plane is the stage on which complex functions live. Open sets matter because complex differentiability is defined only at points with room to move in every nearby direction. If the domain were not open, the difference quotient might be forced to approach along only part of the plane, which would not capture genuine complex differentiability.

Domains are required to be connected because many central theorems depend on the ability to move from one point to another without leaving the region. The identity theorem, for example, uses connectedness to propagate equality from a small neighborhood to the whole domain. Without connectedness, a holomorphic function could behave independently on different components.

Since $\mathbb{C}$ is identified with $\mathbb{R}^2$, it inherits the usual topology from the Euclidean plane.

Definition 7.2.1 (Open Disk). For $z_0 \in \mathbb{C}$ and $r > 0$, the open disk centered at z_0 with radius r is

$$D(z_0, r) = \{z \in \mathbb{C} : |z - z_0| < r\}.$$

Definition 7.2.2 (Domain). A subset $\Omega \subseteq \mathbb{C}$ is called a **domain** if it is open and connected.

Definition 7.2.3 (Limit and Continuity). Let $f : \Omega \rightarrow \mathbb{C}$ and let z_0 be a limit point of Ω . We say

$$\lim_{z \rightarrow z_0} f(z) = L$$

if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < |z - z_0| < \delta \implies |f(z) - L| < \varepsilon.$$

The function f is continuous at z_0 if $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

The definition is formally identical to the real-variable definition, but there is a crucial difference: z can approach z_0 from infinitely many directions in the plane.

7.3 Complex Differentiability

The decisive point in complex differentiability is that the difference quotient must have a single limit no matter how h approaches zero. In real one-variable calculus there are only two basic directions of approach, left and right. In the complex plane there are infinitely many directions, and they must all agree. This requirement forces the derivative to behave like multiplication by one complex number.

Thus complex differentiability is much stronger than differentiability of the real and imaginary parts as functions of two real variables. A general real differentiable map from $\mathbb{R}^2$ to $\mathbb{R}^2$ can stretch in different directions, shear, or distort angles. A complex derivative, when non-zero, can only scale and rotate infinitesimally. This geometric rigidity is the source of most powerful theorems in the subject.

7.3.1 Definition of Complex Derivative

Definition 7.3.1 (Complex Derivative). Let $\Omega \subseteq \mathbb{C}$ be open and let $f : \Omega \rightarrow \mathbb{C}$. We say f is **complex differentiable** at $z_0 \in \Omega$ if the limit

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

exists as a complex number.

This looks like the usual derivative, but h approaches 0 through the complex plane. The quotient must approach the same limit along every direction.

Example 7.3.1

The function $f(z) = z^2$ is complex differentiable everywhere, and

$$f'(z) = 2z.$$

Indeed,

$$\frac{(z+h)^2 - z^2}{h} = 2z + h \rightarrow 2z.$$

Example 7.3.2: Complex Conjugation

The function $f(z) = \bar{z}$ is not complex differentiable at any point. At z_0 ,

$$\frac{\overline{z_0 + h} - \bar{z}_0}{h} = \frac{\bar{h}}{h}.$$

If h is real, this quotient is 1. If $h = it$ with $t \in \mathbb{R}$, then $\bar{h}/h = (-it)/(it) = -1$. The limit depends on direction, so it does not exist.

7.3.2 Holomorphic Functions

The word “holomorphic” should be read as a local condition. A function is holomorphic on a region if every point of the region has a neighborhood where the complex derivative exists. This local definition is deceptively modest: after Cauchy’s theorem is established, holomorphicity implies integral formulas, power series expansions, and strong global restrictions.

It is also useful to distinguish being complex differentiable at a single point from being holomorphic on an open set. A function may satisfy the Cauchy-Riemann equations at one isolated point for accidental reasons. Holomorphicity requires stable differentiability throughout a neighborhood, and that stability is what unlocks the full theory.

Definition 7.3.2 (Holomorphic Function). Let $\Omega \subseteq \mathbb{C}$ be open. A function $f : \Omega \rightarrow \mathbb{C}$ is **holomorphic** on Ω if it is complex differentiable at every point of Ω .

Some authors use the word **analytic** to mean locally representable by a power series. A major theorem of complex analysis states that holomorphic and analytic are equivalent.

Proposition 7.3.1: Rules of Differentiation

If f and g are complex differentiable at z_0 , then:

1. $(f + g)'(z_0) = f'(z_0) + g'(z_0)$;
2. $(fg)'(z_0) = f'(z_0)g(z_0) + f(z_0)g'(z_0)$;
3. if $g(z_0) \neq 0$, then

$$\left(\frac{f}{g}\right)'(z_0) = \frac{f'(z_0)g(z_0) - f(z_0)g'(z_0)}{g(z_0)^2};$$

4. the chain rule holds: $(g \circ f)'(z_0) = g'(f(z_0))f'(z_0)$.

7.4 Cauchy-Riemann Equations

The Cauchy-Riemann equations express the compatibility between real two-dimensional differentiability and complex multiplication. If the real Jacobian matrix of $f = u + iv$ is to represent multiplication by a complex number $a + bi$, it must have the special form

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

The equations $u_x = v_y$ and $u_y = -v_x$ are exactly the statement that the Jacobian has this form.

This matrix viewpoint is often the fastest way to remember the equations. The derivative of a holomorphic function is not an arbitrary linear transformation of the plane. It is a similarity transformation: a uniform scaling combined with a rotation. The Cauchy-Riemann equations are the coordinate expression of that fact.

Let

$$f(z) = u(x, y) + iv(x, y), \quad z = x + iy,$$

where $u, v : \Omega \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$.

If f is complex differentiable at $z_0 = x_0 + iy_0$, then the limit defining $f'(z_0)$ must be the same along the real and imaginary directions.

Along the real direction, $h \in \mathbb{R}$:

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

Along the imaginary direction, $h = it$:

$$f'(z_0) = \frac{u(x_0, y_0 + t) - u(x_0, y_0) + i(v(x_0, y_0 + t) - v(x_0, y_0))}{it}.$$

Since division by i equals multiplication by $-i$, this gives

$$f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0).$$

Equating real and imaginary parts yields the Cauchy-Riemann equations.

Theorem 7.4.1: Cauchy-Riemann Equations

If $f = u + iv$ is complex differentiable at $z_0 = x_0 + iy_0$ and the relevant partial derivatives exist, then

$$u_x = v_y, \quad u_y = -v_x$$

at (x_0, y_0) .

Theorem 7.4.2: Sufficient Condition

Let $f = u + iv$ be defined on an open set Ω . If u and v have continuous first partial derivatives and satisfy the Cauchy-Riemann equations on Ω , then f is holomorphic on Ω .

Example 7.4.1

For $f(z) = z^2$, we have

$$z^2 = (x + iy)^2 = x^2 - y^2 + i(2xy).$$

Thus $u = x^2 - y^2$ and $v = 2xy$. Then

$$u_x = 2x = v_y, \quad u_y = 2y = -v_x.$$

The Cauchy-Riemann equations hold everywhere.

Example 7.4.2

For $f(z) = \bar{z} = x - iy$, we have $u = x$ and $v = -y$. Then

$$u_x = 1, \quad v_y = -1.$$

The first Cauchy-Riemann equation fails everywhere, so f is not holomorphic.

7.4.1 Geometric Meaning

If $f'(z_0) \neq 0$, then near z_0 ,

$$f(z_0 + h) \approx f(z_0) + f'(z_0)h.$$

Multiplication by $f'(z_0)$ scales and rotates small vectors. Therefore holomorphic functions with non-zero derivative preserve angles locally. Such maps are called conformal.

Definition 7.4.1 (Conformal Map). A holomorphic function f is **conformal** at z_0 if $f'(z_0) \neq 0$. It is conformal on a domain if it is conformal at every point of that domain.

7.5 Elementary Holomorphic Functions

Elementary functions in complex analysis are not merely copied from real analysis. They are usually defined by power series or by algebraic combinations of the complex exponential, because these definitions automatically produce holomorphic functions and preserve the identities that matter most. Once defined this way, the functions often reveal behavior impossible in the real setting.

For instance, complex sine and cosine are no longer bounded. The exponential function is periodic in the imaginary direction because $e^{z+2\pi i} = e^z$. These facts are not exceptions; they are consequences of extending real functions into a two-dimensional algebraic setting.

7.5.1 Exponential and Trigonometric Functions

Definition 7.5.1 (Complex Exponential). For $z \in \mathbb{C}$, define

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

This series converges for all $z \in \mathbb{C}$ and defines an entire function.

Proposition 7.5.1

The complex exponential satisfies

$$(e^z)' = e^z, \quad e^{z+w} = e^z e^w.$$

Definition 7.5.2 (Complex Sine and Cosine). Define

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

Proposition 7.5.2

For all $z \in \mathbb{C}$,

$$\frac{d}{dz} \sin z = \cos z, \quad \frac{d}{dz} \cos z = -\sin z.$$

7.5.2 Complex Logarithm

The logarithm is the first place where topology visibly enters complex analysis. Locally, away from the origin and away from a chosen branch cut, one can choose a continuous argument and define a single-valued logarithm. Globally on $\mathbb{C} \setminus \{0\}$ this is impossible because a loop around the origin changes the argument by 2π .

This failure is not an algebraic inconvenience but a topological obstruction. The punctured plane has a hole, and the argument records how many times a path winds around that hole. Later, residues and contour integrals will turn this winding behavior into exact numerical invariants.

The logarithm is more subtle in complex analysis because the argument is multi-valued.

If $z = re^{i\theta}$, then a logarithm of z should satisfy

$$e^w = z.$$

Writing $w = u + iv$, this gives $e^u = r$ and $v = \theta + 2\pi k$.

Definition 7.5.3 (Multi-valued Logarithm). For $z \neq 0$, the complex logarithm is the multi-valued expression

$$\log z = \ln |z| + i(\arg z + 2\pi k), \quad k \in \mathbb{Z}.$$

Definition 7.5.4 (Principal Branch). The **principal argument** $\text{Arg } z$ is chosen in $(-\pi, \pi]$. The corresponding principal logarithm is

$$\text{Log } z = \ln |z| + i \text{Arg } z.$$

It is defined on $\mathbb{C} \setminus (-\infty, 0]$ if one wants a continuous single-valued branch.

Remark 7.5.1

The impossibility of defining a continuous single-valued logarithm on all of $\mathbb{C} \setminus \{0\}$ is topological. Going once around the origin changes the argument by 2π .

7.6 Complex Integration

A contour integral is a line integral written in complex notation. The path supplies the direction of motion through the factor $\gamma'(t)$, while the function supplies the complex value being accumulated. Because complex multiplication contains rotation, the integral is sensitive not only to the values of f but also to the orientation and winding of the path.

The example $\int_{|z|=1} 1/z \, dz = 2\pi i$ should be treated as a warning sign. In ordinary real calculus, a derivative with an antiderivative integrates to zero around closed loops. Here $1/z$ is locally the derivative of a logarithm, but no global single-valued logarithm exists on the punctured plane. The non-zero integral detects this global obstruction.

7.6.1 Curves and Contour Integrals

Definition 7.6.1 (Path). A **path** in $\mathbb{C}$ is a continuous function

$$\gamma : [a, b] \rightarrow \mathbb{C}.$$

If γ is continuously differentiable except possibly at finitely many points, it is called a **piecewise smooth path**.

Definition 7.6.2 (Contour Integral). Let f be continuous on a set containing the image of a piecewise smooth path $\gamma : [a, b] \rightarrow \mathbb{C}$. The **contour integral** of f along γ is

$$\int_{\gamma} f(z) \, dz = \int_a^b f(\gamma(t)) \gamma'(t) \, dt.$$

Example 7.6.1

Let $\gamma(t) = e^{it}$, $0 \leq t \leq 2\pi$, be the unit circle traversed counterclockwise. Then

$$\int_{\gamma} \frac{1}{z} \, dz = \int_0^{2\pi} \frac{1}{e^{it}} i e^{it} \, dt = \int_0^{2\pi} i \, dt = 2\pi i.$$

This example is fundamental: $1/z$ has no global primitive on $\mathbb{C} \setminus \{0\}$.

Proposition 7.6.1: Estimation Lemma

If $|f(z)| \leq M$ on a path γ of length L , then

$$\left| \int_{\gamma} f(z) \, dz \right| \leq ML.$$

Proof. By the definition of contour integral,

$$\left| \int_{\gamma} f(z) \, dz \right| \leq \int_a^b |f(\gamma(t))| |\gamma'(t)| \, dt \leq M \int_a^b |\gamma'(t)| \, dt = ML.$$

□

7.6.2 Primitives

Definition 7.6.3 (Primitive). Let f be holomorphic on a domain Ω . A function F is a **primitive** of f on Ω if $F' = f$ on Ω .

Theorem 7.6.1: Fundamental Theorem for Complex Line Integrals

If $F' = f$ on a domain containing a path γ from a to b , then

$$\int_{\gamma} f(z) \, dz = F(b) - F(a).$$

In particular, the integral of f around any closed path is 0.

Proof. Using the definition of contour integral and the chain rule,

$$\int_{\gamma} f(z) dz = \int_{\alpha}^{\beta} f(\gamma(t))\gamma'(t) dt = \int_{\alpha}^{\beta} (F \circ \gamma)'(t) dt = F(\gamma(\beta)) - F(\gamma(\alpha)).$$

□

7.7 Cauchy's Theorem and Cauchy's Integral Formula

Cauchy's theorem says that holomorphic functions have path-independent integrals on simply connected domains. Conceptually, holomorphicity is a local condition, while simple connectedness removes global holes. When both are present, closed contour integrals vanish. If a hole is present, the local condition may still hold but a contour can wind around the hole and produce a non-zero integral.

Cauchy's integral formula is even stronger. It says that the value of a holomorphic function at an interior point is determined by its values on a surrounding circle. This is a profound difference from real analysis: holomorphic functions cannot be freely modified at one point without affecting their behavior elsewhere. Boundary data controls interior data.

The Cauchy theory is the core of elementary complex analysis. It explains why holomorphic functions are rigid.

Theorem 7.7.1: Cauchy's Theorem, Simple Form

Let Ω be a simply connected domain and let f be holomorphic on Ω . Then for every closed piecewise smooth path γ in Ω ,

$$\int_{\gamma} f(z) dz = 0.$$

Remark 7.7.1

The condition that Ω is simply connected cannot be omitted. The function $1/z$ is holomorphic on $\mathbb{C} \setminus \{0\}$, but its integral around the unit circle is $2\pi i$.

Theorem 7.7.2: Cauchy's Integral Formula

Let f be holomorphic on an open set containing the closed disk $\overline{D}(z_0, R)$. If $0 < r < R$, then

$$f(z_0) = \frac{1}{2\pi i} \int_{|z-z_0|=r} \frac{f(z)}{z-z_0} dz,$$

where the circle is oriented counterclockwise.

Idea of Proof. The function

$$g(z) = \frac{f(z) - f(z_0)}{z - z_0}$$

for $z \neq z_0$ has a removable singularity at z_0 because f is complex differentiable. Therefore its integral around the circle is 0 by Cauchy's theorem. Hence

$$\int \frac{f(z)}{z - z_0} dz = f(z_0) \int \frac{1}{z - z_0} dz = 2\pi i f(z_0).$$

□

Theorem 7.7.3: Cauchy's Integral Formula for Derivatives

Under the same assumptions,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{|z-z_0|=r} \frac{f(z)}{(z-z_0)^{n+1}} dz.$$

Corollary 7.7.1

If f is holomorphic on Ω , then f is infinitely complex differentiable on Ω .

Remark 7.7.2

This is a sharp contrast with real analysis. A real function may be differentiable once but not twice. In complex analysis, one derivative automatically gives infinitely many derivatives.

7.8 Power Series and Analyticity

Power series are the local coordinate system of complex analysis. Around any point inside its domain, a holomorphic function can be reconstructed from its derivatives at that point. This means that local derivative data determine the function in a whole neighborhood, and by analytic continuation often much farther.

The theorem “holomorphic implies analytic” is one of the deepest surprises of a first course. In real analysis, smooth functions need not equal their Taylor series. In complex analysis, the existence of one complex derivative on an open set is enough to force a convergent Taylor expansion. Cauchy's integral formula is the mechanism behind this dramatic strengthening.

7.8.1 Power Series

Definition 7.8.1 (Power Series). A **power series** centered at z_0 is a series of the form

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n.$$

Theorem 7.8.1: Radius of Convergence

For every power series $\sum a_n (z - z_0)^n$, there exists $R \in [0, \infty]$ such that the series converges absolutely for $|z - z_0| < R$ and diverges for $|z - z_0| > R$.

Theorem 7.8.2: Termwise Differentiation

A power series is holomorphic inside its disk of convergence, and it may be differentiated term by term:

$$\left(\sum_{n=0}^{\infty} a_n (z - z_0)^n \right)' = \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}.$$

The derivative has the same radius of convergence.

7.8.2 Holomorphic Implies Analytic

Theorem 7.8.3: Taylor Expansion of Holomorphic Functions

Let f be holomorphic on a disk $D(z_0, R)$. Then for $|z - z_0| < R$,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n.$$

Idea of Proof. Choose a circle $|\zeta - z_0| = r$ with $|z - z_0| < r < R$. By Cauchy's integral formula,

$$f(z) = \frac{1}{2\pi i} \int_{|\zeta - z_0| = r} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

We rewrite

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0} \cdot \frac{1}{1 - \frac{z - z_0}{\zeta - z_0}} = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\zeta - z_0)^{n+1}}.$$

Substituting this into the integral and using Cauchy's formula for derivatives gives the Taylor series. $\square$

Theorem 7.8.4: Identity Theorem

Let f and g be holomorphic on a domain Ω . If the set

$$\{z \in \Omega : f(z) = g(z)\}$$

has a limit point in Ω , then $f = g$ on all of Ω .

Remark 7.8.1

The identity theorem is another sign of rigidity. A holomorphic function on a domain is determined by its values on any subset with a limit point.

7.9 Zeros and Singularities

Singularities are not merely points where a formula is undefined. They are local fingerprints of how a holomorphic function fails to extend. A removable singularity means the failure is artificial; a pole means the function blows up like a finite negative power; an essential singularity means the behavior is genuinely wild.

The classification is powerful because it converts analytic behavior near a point into algebraic information in a Laurent series. Instead of trying to inspect the function directly, we inspect the negative powers. The principal part tells us exactly what kind of obstruction sits at the singularity.

7.9.1 Zeros of Holomorphic Functions

Definition 7.9.1 (Zero of Order m). Let f be holomorphic near z_0 . We say z_0 is a **zero of order m** if

$$f(z) = (z - z_0)^m g(z),$$

where g is holomorphic near z_0 and $g(z_0) \neq 0$.

Proposition 7.9.1

If f is holomorphic and not identically zero on a domain, then the zeros of f are isolated.

7.9.2 Isolated Singularities

Definition 7.9.2 (Isolated Singularity). Let f be holomorphic on $0 < |z - z_0| < R$. Then z_0 is called an **isolated singularity** of f .

Definition 7.9.3 (Types of Isolated Singularities). An isolated singularity z_0 is:

1. **removable** if f can be redefined at z_0 to become holomorphic;
2. a **pole** if $|f(z)| \rightarrow \infty$ as $z \rightarrow z_0$;
3. **essential** if it is neither removable nor a pole.

Example 7.9.1

The function $\sin z/z$ has a removable singularity at 0, because

$$\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots.$$

The function $1/z^m$ has a pole of order m at 0. The function $e^{1/z}$ has an essential singularity at 0.

7.10 Laurent Series and Residues

Residues isolate the one Laurent coefficient that contour integration can see. Around a small circle centered at the singularity, every power $(z - z_0)^n$ integrates to zero except the power $n = -1$. Therefore the coefficient of $(z - z_0)^{-1}$ controls the entire local contribution of that singularity to a contour integral.

This is why the residue theorem is computationally so efficient. A global integral around a complicated contour can be reduced to local algebra near finitely many singularities. The theorem is also conceptually important: it shows that analytic, algebraic, and topological data are measuring the same phenomenon from different angles.

7.10.1 Laurent Series**Theorem 7.10.1: Laurent Expansion**

Let f be holomorphic on an annulus

$$r < |z - z_0| < R.$$

Then f has a unique Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

valid in the annulus.

A Laurent expansion is like a Taylor expansion, but it allows negative powers. The terms with negative powers capture the behavior of f near the singularity.

Definition 7.10.1 (Principal Part). In the Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n,$$

the terms with $n < 0$ form the **principal part**.

Proposition 7.10.1: Classification by Laurent Series

Let z_0 be an isolated singularity of f .

1. z_0 is removable if the principal part is zero.
2. z_0 is a pole of order m if the principal part has finitely many terms and the lowest power is $(z - z_0)^{-m}$.
3. z_0 is essential if the principal part has infinitely many non-zero terms.

7.10.2 Residues

Definition 7.10.2 (Residue). Let f have an isolated singularity at z_0 and Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n.$$

The coefficient a_{-1} is called the **residue** of f at z_0 , denoted

$$\text{Res}(f, z_0) = a_{-1}.$$

Theorem 7.10.2: Residue Theorem

Let f be holomorphic on a domain Ω except for finitely many isolated singularities $z_1, \dots, z_n$. Let γ be a positively oriented simple closed contour in Ω enclosing these singularities and no others. Then

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f, z_k).$$

Idea of Proof. Around each singularity, write the Laurent expansion. All terms integrate to 0 around a small circle except the $(z - z_k)^{-1}$ term, whose integral is $2\pi i$. Cauchy's theorem then deforms the original contour into the sum of small circles around the singularities. $\square$

Proposition 7.10.2: Residue at a Simple Pole

If $f(z) = g(z)/h(z)$, where g and h are holomorphic near z_0 , $g(z_0) \neq 0$, $h(z_0) = 0$, and $h'(z_0) \neq 0$, then z_0 is a simple pole and

$$\text{Res}(f, z_0) = \frac{g(z_0)}{h'(z_0)}.$$

Example 7.10.1

Compute

$$\int_{|z|=2} \frac{1}{z^2 + 1} dz.$$

The singularities are $z = i$ and $z = -i$, both inside the circle. Since

$$z^2 + 1 = (z - i)(z + i),$$

we have

$$\operatorname{Res}\left(\frac{1}{z^2 + 1}, i\right) = \frac{1}{2i}, \quad \operatorname{Res}\left(\frac{1}{z^2 + 1}, -i\right) = \frac{1}{-2i}.$$

The sum of residues is 0, so the integral is 0.

Example 7.10.2

For the unit circle,

$$\int_{|z|=1} \frac{e^z}{z^3} dz$$

can be read from the Taylor expansion

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots.$$

Thus

$$\frac{e^z}{z^3} = z^{-3} + z^{-2} + \frac{1}{2}z^{-1} + \frac{1}{6} + \cdots.$$

The residue at 0 is $1/2$, so the integral equals

$$2\pi i \cdot \frac{1}{2} = \pi i.$$

7.11 Major Theorems and Consequences

The major theorems at the end of the chapter should be read as consequences of rigidity. Liouville's theorem says that an entire function cannot be both globally bounded and non-constant. The fundamental theorem of algebra follows by applying this analytic rigidity to polynomials. The maximum modulus principle says that non-constant holomorphic functions cannot have interior peaks in absolute value.

These results show that holomorphic functions are controlled by global constraints. Unlike arbitrary differentiable functions, they cannot be patched together freely. Their local behavior, boundary behavior, and global domain structure are tightly linked.

Theorem 7.11.1: Liouville's Theorem

If f is entire and bounded, then f is constant.

Sketch. By Cauchy's estimate,

$$|f'(z_0)| \leq \frac{M}{R}$$

for every $R > 0$, where M bounds $|f|$. Letting $R \rightarrow \infty$ gives $f'(z_0) = 0$ for every z_0 . Therefore f is constant. $\square$

Theorem 7.11.2: Fundamental Theorem of Algebra

Every non-constant polynomial with complex coefficients has at least one complex root.

Proof via Liouville. Suppose $p(z)$ is a non-constant polynomial with no zeros. Then $1/p(z)$ is entire. Since $|p(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$, the function $1/p(z)$ is bounded outside a large disk. It is also bounded on the closed disk by continuity. Hence $1/p$ is bounded entire, so it is constant by Liouville's theorem. Thus p is constant, a contradiction. $\square$

Theorem 7.11.3: Maximum Modulus Principle

Let f be holomorphic on a domain Ω . If $|f|$ attains a local maximum at an interior point and f is not constant, then this is impossible. Therefore a non-constant holomorphic function cannot attain its maximum modulus in the interior of a domain.

Remark 7.11.1

The maximum modulus principle is another expression of rigidity. Holomorphic functions cannot have arbitrary local behavior; their values inside a region are controlled by their boundary behavior.

7.12 A More Complete Form of Cauchy's Theory

The simple form of Cauchy's theorem is enough for many computations, but it hides two important points. First, the theorem does not require the derivative f' to be continuous; complex differentiability alone is sufficient. Second, the value of a contour integral depends not merely on which singularities lie "inside" a curve, but on how many times the curve winds around each singularity. These refinements lead naturally to Goursat's theorem and the winding number.

7.12.1 Goursat's Theorem

Theorem 7.12.1: Cauchy–Goursat Theorem for a Triangle

Let f be holomorphic on an open set containing a triangle T and its interior. Then

$$\int_{\partial T} f(z) dz = 0.$$

No continuity assumption on f' is required.

Nested-Triangle Argument. Divide T into four congruent triangles. The integrals over the interior edges cancel, so the integral over ∂T is the sum of the four boundary integrals. At least one small triangle, call it T_1 , satisfies

$$\left| \int_{\partial T_1} f(z) dz \right| \geq \frac{1}{4} \left| \int_{\partial T} f(z) dz \right|.$$

Repeat the construction to obtain nested triangles

$$T \supseteq T_1 \supseteq T_2 \supseteq \cdots$$

whose diameters tend to zero and whose boundary integrals retain at least the corresponding fraction of the original integral. Compactness gives a unique point z_0 in all the triangles.

Since f is differentiable at z_0 ,

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \varepsilon(z)(z - z_0),$$

where $\varepsilon(z) \rightarrow 0$ as $z \rightarrow z_0$. The constant and linear terms have primitives, so their integrals around ∂T_n vanish. The remaining integral is bounded by

$$\sup_{z \in T_n} |\varepsilon(z)| \cdot \text{diam}(T_n) \cdot \text{length}(\partial T_n),$$

which tends to zero faster than the scaling inherited from the first triangle. Hence the original integral must be zero. $\square$

Theorem 7.12.2: Cauchy's Theorem on Simply Connected Domains

Let Ω be simply connected and let f be holomorphic on Ω . Then every closed piecewise smooth contour γ in Ω satisfies

$$\int_{\gamma} f(z) dz = 0.$$

Equivalently, f has a primitive on Ω .

Remark 7.12.1

The equivalence with the existence of a primitive is worth remembering. Vanishing integrals around closed contours imply path independence: if γ_1 and γ_2 join z_0 to z , then following γ_1 and returning along γ_2 gives a closed contour. One may therefore define

$$F(z) = \int_{z_0}^z f(\zeta) d\zeta,$$

and path independence makes F well-defined. A local difference-quotient argument then gives $F' = f$.

7.12.2 Winding Number and the General Integral Formula

Definition 7.12.1 (Winding Number). Let γ be a closed piecewise smooth contour and let a not lie on its image. The **winding number** of γ about a is

$$\text{Ind}(\gamma, a) = \frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z - a}.$$

Proposition 7.12.1

The winding number is an integer. It is constant as a moves within any connected component of $\mathbb{C} \setminus \gamma$, and it is 0 on the unbounded component.

Theorem 7.12.3: General Cauchy Integral Formula

Let f be holomorphic on a region containing a closed contour γ and every point around which γ has non-zero winding number. If a does not lie on γ , then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - a} dz = \text{Ind}(\gamma, a) f(a).$$

Example 7.12.1

If $\gamma(t) = e^{3it}$ for $0 \leq t \leq 2\pi$, then γ traverses the unit circle three times counterclockwise. Hence

$$\text{Ind}(\gamma, 0) = 3, \quad \int_{\gamma} \frac{e^z}{z} dz = 6\pi i.$$

The geometric region enclosed by the curve has not changed, but the integral records multiplicity.

Theorem 7.12.4: Morera's Theorem

Let f be continuous on a domain Ω . If

$$\int_{\partial T} f(z) dz = 0$$

for every triangle T whose closure lies in Ω , then f is holomorphic on Ω .

Proof. On a disk contained in Ω , fix z_0 and define

$$F(z) = \int_{[z_0, z]} f(\zeta) d\zeta.$$

The triangular integral condition makes this definition path independent within the disk. Continuity of f implies $F' = f$, so f is locally a derivative of a holomorphic function and is therefore holomorphic. $\square$

7.13 Singularities: Criteria and Local Behavior

The Laurent-series classification is exact, but in applications one often recognizes a singularity without computing the entire series. Boundedness detects removable singularities, reciprocal behavior detects poles, and failure of both indicates an essential singularity. These tests connect algebraic expansions with geometric behavior near the exceptional point.

Theorem 7.13.1: Riemann's Removable Singularity Theorem

Let f be holomorphic on $0 < |z - z_0| < R$. The following conditions are equivalent:

1. f extends holomorphically to z_0 ;
2. f is bounded in some punctured neighborhood of z_0 ;
3. $\lim_{z \rightarrow z_0} (z - z_0)f(z) = 0$.

Main Idea. If f is bounded near z_0 , define

$$g(z) = \begin{cases} (z - z_0)^2 f(z), & z \neq z_0, \\ 0, & z = z_0. \end{cases}$$

Then g is complex differentiable at z_0 with $g'(z_0) = 0$. Since g has a zero of order at least two, $g(z)/(z - z_0)^2$ extends holomorphically, giving the desired extension of f . The other implications follow directly from the extension or from Cauchy's formula. $\square$

Theorem 7.13.2: Characterizations of a Pole

For a function holomorphic on a punctured neighborhood of z_0 , the following are equivalent:

1. z_0 is a pole of f ;
2. $|f(z)| \rightarrow \infty$ as $z \rightarrow z_0$;
3. $1/f$ extends holomorphically to z_0 and has a zero there.

If the zero of $1/f$ has order m , then the pole of f has order m .

Proposition 7.13.1: Residue at a Pole of Order m

If z_0 is a pole of order m , then

$$\operatorname{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)].$$

Example 7.13.1

For

$$f(z) = \frac{e^z}{(z-1)^3},$$

the point $z = 1$ is a pole of order 3. Thus

$$\operatorname{Res}(f, 1) = \frac{1}{2!} \left. \frac{d^2}{dz^2} e^z \right|_{z=1} = \frac{e}{2}.$$

Theorem 7.13.3: Casorati–Weierstrass

If z_0 is an essential singularity of f , then the image of every punctured neighborhood of z_0 is dense in $\mathbb{C}$.

Proof. Suppose some disk $D(w_0, \varepsilon)$ is omitted near z_0 . Then

$$g(z) = \frac{1}{f(z) - w_0}$$

is holomorphic and bounded near z_0 , so g has a removable singularity. If the extension satisfies $g(z_0) \neq 0$, then $f = w_0 + 1/g$ also extends, so z_0 is removable. If $g(z_0) = 0$, then $1/g$ has a pole, so f has a pole. Both conclusions contradict essentiality. $\square$

Remark 7.13.1

The function $e^{1/z}$ illustrates the theorem. Along sequences for which $1/z$ has large positive real part, the modulus becomes arbitrarily large; along sequences with large negative real part, it approaches 0; and by varying the imaginary part, the argument winds rapidly. An essential singularity is therefore not merely “more divergent” than a pole. It is qualitatively less ordered.

7.14 Residue Calculus for Real Integrals

The residue theorem becomes especially useful when a real integral can be embedded into a contour integral. The contour is chosen so that one part of it reproduces the desired real integral, while the remaining parts either vanish or can be related to the same unknown quantity. The main work is therefore not the residue computation but the design of the contour.

7.14.1 Rational Functions on the Real Line

Example 7.14.1: The Basic Cauchy Integral

Let $a > 0$. Consider

$$I = \int_{-\infty}^{\infty} \frac{dx}{x^2 + a^2}.$$

Integrate $1/(z^2 + a^2)$ over the upper semicircle of radius R . The only pole in the upper half-plane is $z = ia$, with residue

$$\text{Res}\left(\frac{1}{z^2 + a^2}, ia\right) = \frac{1}{2ia}.$$

The integral over the semicircular arc is bounded by

$$\frac{\pi R}{R^2 - a^2} \rightarrow 0.$$

Therefore

$$I = 2\pi i \cdot \frac{1}{2ia} = \frac{\pi}{a}.$$

Example 7.14.2: A Fourth-Degree Denominator

Consider

$$I = \int_{-\infty}^{\infty} \frac{dx}{x^4 + 1}.$$

The poles in the upper half-plane are

$$z_1 = e^{i\pi/4}, \quad z_2 = e^{3i\pi/4}.$$

Since the poles are simple,

$$\text{Res}\left(\frac{1}{z^4 + 1}, z_k\right) = \frac{1}{4z_k^3}.$$

Their sum is

$$\frac{1}{4e^{3i\pi/4}} + \frac{1}{4e^{9i\pi/4}} = -\frac{i}{2\sqrt{2}}.$$

The upper semicircular contribution tends to zero, and hence

$$I = 2\pi i \left(-\frac{i}{2\sqrt{2}}\right) = \frac{\pi}{\sqrt{2}}.$$

7.14.2 Trigonometric Integrals and the Unit Circle

For integrals over $[0, 2\pi]$, the substitution

$$z = e^{i\theta}, \quad d\theta = \frac{dz}{iz}, \quad \cos \theta = \frac{1}{2} \left(z + \frac{1}{z}\right)$$

turns a trigonometric integral into a contour integral around $|z| = 1$.

Example 7.14.3

Assume $a > |b| > 0$. Then

$$I = \int_0^{2\pi} \frac{d\theta}{a + b \cos \theta}$$

becomes

$$I = \frac{2}{i} \int_{|z|=1} \frac{dz}{bz^2 + 2az + b}.$$

The roots are

$$z_{\pm} = \frac{-a \pm \sqrt{a^2 - b^2}}{b},$$

and exactly one of them lies inside the unit circle. Evaluating the residue at that root gives

$$I = \frac{2\pi}{\sqrt{a^2 - b^2}}.$$

Note

A contour argument must include three checks: which poles lie inside the contour, why the auxiliary contour pieces vanish or cancel, and whether the orientation is positive. Most sign errors in residue calculations come from skipping one of these checks.

7.15 The Argument Principle and Rouché's Theorem

A holomorphic function changes both magnitude and argument along a contour. When the contour closes, the net change in argument counts how many times the image curve winds around the origin. Zeros and poles are exactly the points where this winding can be created or destroyed. The argument principle makes this relation precise.

Theorem 7.15.1: Argument Principle

Let f be meromorphic on a region containing a positively oriented simple closed contour γ and its interior. Assume that f has no zeros or poles on γ . If N and P denote the numbers of zeros and poles inside γ , counted with multiplicity, then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = N - P.$$

Proof. Near a zero a of order m , write $f(z) = (z - a)^m g(z)$ with $g(a) \neq 0$. Then

$$\frac{f'}{f} = \frac{m}{z - a} + \frac{g'}{g},$$

so the residue is m . Near a pole of order m , the same calculation gives residue $-m$. The residue theorem now yields the formula. $\square$

Corollary 7.15.1: Change of Argument

Under the same hypotheses, the total change in a continuous choice of $\arg f(z)$ as z traverses γ is

$$2\pi(N - P).$$

Theorem 7.15.2: Rouché's Theorem

Let f and g be holomorphic on and inside a simple closed contour γ . If

$$|g(z)| < |f(z)|$$

for every $z \in \gamma$, then f and $f + g$ have the same number of zeros inside γ , counted with multiplicity.

Proof. For $0 \leq t \leq 1$, define $F_t = f + tg$. On γ ,

$$|tg| \leq |g| < |f|,$$

so F_t never vanishes there. The integer

$$\frac{1}{2\pi i} \int_{\gamma} \frac{F'_t}{F_t} dz$$

depends continuously on t , hence is constant. At $t = 0$ it counts the zeros of f , and at $t = 1$ it counts the zeros of $f + g$. $\square$

Example 7.15.1

Determine the number of zeros of

$$p(z) = z^7 + 4z^3 + 1$$

inside $|z| = 2$. On the circle,

$$|z^7| = 128, \quad |4z^3 + 1| \leq 33 < 128.$$

By Rouché's theorem, p and z^7 have the same number of zeros in the disk. Therefore all seven zeros of p , counted with multiplicity, lie inside $|z| < 2$.

Example 7.15.2

Determine the number of zeros of

$$q(z) = z^5 + 10z + 3$$

inside $|z| = 1$. On the unit circle,

$$|z^5 + 3| \leq 4 < 10 = |10z|.$$

Thus q and $10z$ have the same number of zeros in the unit disk, namely one.

7.16 Maximum Principles, Open Mapping, and Schwarz's Lemma

The maximum modulus principle is not an isolated fact. It belongs to a family of results expressing the same rigidity: non-constant holomorphic functions send open sets to open sets, cannot attain interior extrema of modulus, and are severely constrained when mapping the unit disk into itself.

Theorem 7.16.1: Mean Value Property

If f is holomorphic on a neighborhood of the closed disk $\overline{D}(z_0, r)$, then

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt.$$

Proof. Parametrize the circle in Cauchy's integral formula by $z = z_0 + re^{it}$. Since $dz = ire^{it}dt$, the factors re^{it} cancel. $\square$

Theorem 7.16.2: Minimum Modulus Principle

Let f be holomorphic and non-vanishing on a domain. If $|f|$ attains a local minimum at an interior point, then f is constant.

Proof. The reciprocal $1/f$ is holomorphic. A local minimum of $|f|$ is a local maximum of $|1/f|$, so the maximum modulus principle forces $1/f$, and hence f , to be constant. $\square$

Theorem 7.16.3: Open Mapping Theorem

A non-constant holomorphic function maps open sets to open sets.

Idea of Proof. Let $w_0 = f(z_0)$. Choose a small circle around z_0 on which $f(z) - w_0$ has no zeros. Its modulus has a positive minimum m on the circle. If $|w - w_0| < m$, then on that circle

$$|w - w_0| < |f(z) - w_0|.$$

Rouche's theorem shows that $f(z) - w$ and $f(z) - w_0$ have the same number of zeros inside. Since the second function vanishes at z_0 , so does the first. Thus every w sufficiently close to w_0 lies in the image. $\square$

Theorem 7.16.4: Schwarz's Lemma

Let f be holomorphic on the unit disk $\mathbb{D}$, suppose $f(0) = 0$, and assume $|f(z)| \leq 1$ for all $z \in \mathbb{D}$. Then

$$|f(z)| \leq |z|, \quad |f'(0)| \leq 1.$$

If equality holds at some non-zero point or if $|f'(0)| = 1$, then

$$f(z) = e^{i\theta} z$$

for some real θ .

Proof. Define $g(z) = f(z)/z$ for $z \neq 0$ and $g(0) = f'(0)$. The singularity at 0 is removable, so g is holomorphic. On $|z| = r < 1$,

$$|g(z)| \leq \frac{1}{r}.$$

The maximum modulus principle gives $|g(z)| \leq 1/r$ for $|z| \leq r$. Letting $r \rightarrow 1$ yields $|g(z)| \leq 1$, which proves both inequalities. Equality forces g to have an interior maximum modulus, hence to be constant of modulus 1. $\square$

7.17 Geometric Visualizations

Complex formulas are often easier to understand when one tracks how curves and regions move. A holomorphic map is locally a rotation and dilation wherever its derivative is non-zero. Singularities and residues, by contrast, are detected by how contours wind around exceptional points.

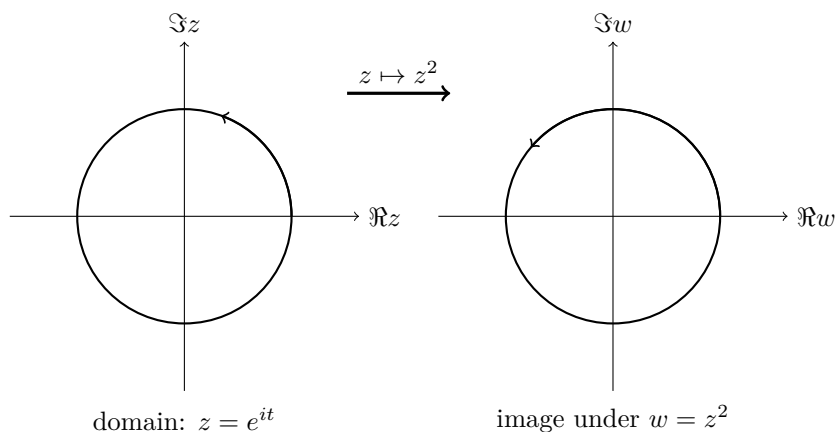


Figure 7.1: The map $z \mapsto z^2$ doubles arguments. One traversal of an arc in the z -plane produces twice the angular change in the w -plane.

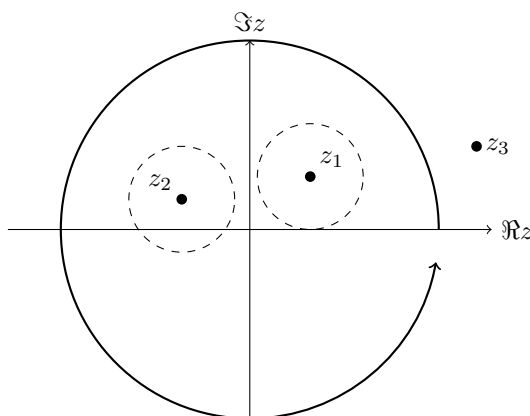


Figure 7.2: A residue contour encloses z_1 and z_2 but not z_3 . The contour integral is $2\pi i$ times the sum of the enclosed residues, with winding numbers included.

Remark 7.17.1

The first picture encodes conformality: near a point where $f'(z_0) \neq 0$, small directions are rotated by $\arg f'(z_0)$ and scaled by $|f'(z_0)|$. The second picture encodes locality of residues: the global contour integral is assembled from data in arbitrarily small neighborhoods of the enclosed singularities.

7.18 Conclusion

Complex analysis begins with a small change in the definition of derivative: the increment h is allowed to approach 0 from every direction in the complex plane. This small change creates an extremely rigid theory. Holomorphic functions satisfy the Cauchy-Riemann equations, admit contour integral formulas, possess power series expansions, and have isolated singularities whose behavior is encoded by Laurent series.

The residue theorem is the computational climax of the elementary theory. It converts difficult contour integrals into algebraic data: the coefficients of $(z - z_0)^{-1}$ in local Laurent expansions. Conceptually, complex analysis shows how algebra, geometry, topology, and analysis become inseparable once the real line is replaced by the complex plane.

7.19 Chapter Workshop

Worked example: A real integral from residues

Evaluate

$$I = \int_{-\infty}^{\infty} \frac{dx}{x^2 + 1}.$$

Integrate $1/(z^2 + 1)$ over the upper semicircle. The only enclosed pole is $z = i$, with residue

$$\text{Res}_{z=i} \frac{1}{(z-i)(z+i)} = \frac{1}{2i}.$$

The semicircular contribution tends to zero, so the residue theorem gives

$$I = 2\pi i \cdot \frac{1}{2i} = \pi.$$

The method succeeds because the complex extension converts a real integral into local information at an isolated singularity.

Application: Potential flow and two-dimensional fields

If $f(z) = \phi(x, y) + i\psi(x, y)$ is holomorphic, the Cauchy–Riemann equations imply that ϕ and ψ are harmonic and that their level curves meet orthogonally. In two-dimensional electrostatics and ideal fluid flow, ϕ may represent a potential and ψ a stream function. Conformal maps transport simple geometries to more complicated domains while preserving local angles.

Application: Frequency response

Rational transfer functions are studied through their poles and zeros in the complex plane. Pole locations govern growth, decay, resonance, and stability; contour methods recover inverse transforms and asymptotic behavior. Complex analysis therefore provides the natural language for linear systems and signal processing.

7.20 Exercises

1. Verify the Cauchy–Riemann equations for $f(z) = z^2$ and identify its real and imaginary parts.
2. Evaluate $\int_{|z|=2} \frac{e^z}{z-1} dz$.
3. Find the Laurent series of $1/[z(z-1)]$ valid in $0 < |z| < 1$.
4. Classify the singularity of $\sin z/z^3$ at 0 and compute its residue.
5. Use the argument principle to count the zeros of $z^3 - 2$ inside $|z| = 2$.
6. Explain why a nonconstant holomorphic function cannot attain an interior maximum of its modulus.

Selected hints and answers

1. $u = x^2 - y^2$ and $v = 2xy$; check $u_x = v_y$ and $u_y = -v_x$.
2. Cauchy's integral formula gives $2\pi ie$.

3. Write $1/[z(z-1)] = -z^{-1}(1+z+z^2+\cdots)$.
4. The expansion is $z^{-2} - 1/6 + \cdots$; the singularity is a pole of order 2 and the residue is 0.
5. All three zeros have modulus $2^{1/3} < 2$.
6. The maximum-modulus principle follows from Cauchy's integral formula or the mean-value property; equality forces local constancy and hence global constancy on a connected domain.

Part IV

Discrete and Probabilistic Models

Chapter 8

Applied Mathematics: Graph Theory

Learning objectives

By the end of this chapter, you should be able to:

- model a discrete system as a graph and choose an appropriate graph representation;
- analyze degrees, connectivity, paths, cycles, trees, planarity, and coloring;
- execute basic traversal and shortest-path algorithms;
- distinguish deterministic graph theorems from probabilistic threshold statements;
- connect graph models with routing, networks, scheduling, and graph-based learning.

8.1 Introduction to Graph Theory

Graph theory is the study of mathematical structures used to model pairwise relations between objects. It provides the theoretical foundation for analyzing networks, data structures, scheduling, optimization, and computational complexity. This chapter establishes rigorous definitions of graph-theoretic structures and provides detailed proofs for central theorems concerning connectivity, planarity, and traversability.

8.1.1 Basic Concepts and Classification

This section covers the fundamental building blocks of graph theory, including definitions, representations, connectivity, and specific classifications of graphs.

8.1.1.1 Fundamental Definitions and Theorems

We begin by distinguishing between different types of graph structures based on the nature of their edges.

Definition 8.1.1 (Simple Graph). A **simple graph** $G = (V, E)$ consists of a non-empty finite set of vertices V and a set E of 2-element subsets of V (unordered pairs of distinct vertices). In a simple graph, there are no loops (edges connecting a vertex to itself) and no multiple edges (more than one edge between the same pair of vertices).

Definition 8.1.2 (Multigraph). A **multigraph** $G = (V, E)$ is a graph where multiple edges (also called parallel edges) are allowed between the same pair of vertices. Formally, E is a multiset of unordered pairs of vertices. Loops are typically not allowed in a strict multigraph definition, though conventions vary.

Definition 8.1.3 (Pseudograph). A **pseudograph** is a generalization of a multigraph that allows both multiple edges between distinct vertices and **loops** (edges connecting a vertex to itself).

Definition 8.1.4 (Directed Graph). A **directed graph** (or digraph) $G = (V, E)$ consists of a set of vertices V and a set of **ordered** pairs $E \subseteq V \times V$, called arcs or directed edges. An edge (u, v) is directed from u

to v .

Definition 8.1.5 (Degree). The **degree** of a vertex v , denoted $\deg(v)$, is the number of edges incident to it. In a graph with loops, each loop contributes 2 to the degree. In a directed graph, we distinguish between in-degree ($\deg^-(v)$) and out-degree ($\deg^+(v)$).

According to these definitions, we can now state and prove a fundamental theorem regarding vertex degrees.

Theorem 8.1.1: The Handshaking Lemma

For any undirected graph $G = (V, E)$, $\sum_{v \in V} \deg(v) = 2|E|$.

Proof. We count the number of incident pairs (v, e) where $v \in V, e \in E$ and $v \in e$. Summing over vertices, each vertex v contributes $\deg(v)$ pairs, yielding $\sum \deg(v)$. Summing over edges, each edge $\{u, v\}$ has exactly two endpoints, contributing exactly 2 to the sum. Thus, the sum of degrees equals twice the number of edges. $\square$

8.1.1.2 Graph Operations and Isomorphism

Definition 8.1.6 (Graph Isomorphism). Two graphs $G = (V, E)$ and $H = (W, F)$ are **isomorphic** if there exists a bijection $f : V \rightarrow W$ such that $\{u, v\} \in E$ if and only if $\{f(u), f(v)\} \in F$.

Just like isotope in chemistry, isomorphic graphs are structurally identical, differing only in the labeling of vertices.

Theorem 8.1.2: Isomorphism Invariants

If two graphs are isomorphic, they must have:

- Same number of vertices and edges
- Same degree sequence
- Same number of cycles of each length
- Same connectivity properties

This invariants remain unchanged under isomorphism, providing necessary (but not sufficient) conditions for two graphs to be isomorphic.

8.1.1.3 Degree Sequences and Realizability

A sequence of integers is **graphic** if there exists a simple graph with that degree sequence.

Theorem 8.1.3: Havel-Hakimi Theorem

Let $S = (d_1, d_2, \dots, d_n)$ be a finite list of nonnegative integers that is non-increasing (i.e., $d_1 \geq d_2 \geq \dots \geq d_n$). Let S' be the list obtained by deleting d_1 and subtracting 1 from the next d_1 elements of S . Then, S is graphic if and only if S' is graphic.

Proof. Intuition: This theorem provides a recursive algorithm to check if a degree sequence is valid.

($\Leftarrow$) If S' is graphic, there exists a graph G' with degree sequence S' . We can construct a graph G with sequence S by adding a new vertex v_1 and connecting it to the d_1 vertices whose degrees were reduced by 1 in S' . Since we connect v_1 to distinct vertices, the result is a simple graph.

($\Rightarrow$) If S is graphic, there exists a realization G . We must show that there exists a *specific* realization where the vertex with degree d_1 connects to the d_1 vertices with the highest degrees. Let v_1 be a vertex with degree d_1 . Let $N(v_1)$ be its neighbors. If $N(v_1)$ consists of the d_1 vertices with the highest degrees (excluding v_1), we are done (removing v_1 yields S'). If not, there exist vertices x, y such that $x \in N(v_1)$, $y \notin N(v_1)$, but

$\deg(y) > \deg(x)$ (or $\deg(y) \geq \deg(x)$ where y is a preferred high-degree target). Since $\deg(y) \geq \deg(x)$, there must be a vertex z such that z is adjacent to y but not to x . We perform an **edge swap**: remove edges $\{v_1, x\}$ and $\{y, z\}$, and add edges $\{v_1, y\}$ and $\{x, z\}$. This preserves all degrees but moves a neighbor of v_1 to a higher-degree vertex. Repeating this process eventually ensures v_1 is connected to the d_1 highest-degree vertices. Removing v_1 then yields a graph with sequence S' . $\square$

The key point of the proof is to **SWAP** edges to ensure the highest-degree vertices are connected to the vertex with larger degree, thus preserving the degree sequence while constructing a valid graph.

8.1.1.4 Graph Representations

Definition 8.1.7 (Adjacency Matrix). For a graph $G = (V, E)$ with n vertices, the **adjacency matrix** A is an $n \times n$ matrix where $A_{ij} = 1$ if vertices i and j are adjacent, and 0 otherwise.

Adjacency matrices can be applied to different types of graphs.

Theorem 8.1.4

For an adjacency matrix A :

- A_{ij}^k counts the number of walks of length k from vertex i to j
- The eigenvalues of A provide information about graph structure
- For regular graphs, the largest eigenvalue equals the degree

Adjacency matrices are particularly useful for algorithmic applications and spectral graph theory. In the adjacency matrix, the element A_{ij}^k of the k -th power A^k represents the number of paths of length k from vertex i to vertex j .

Definition 8.1.8 (Incidence Matrix). The **incidence matrix** M of a graph with n vertices and m edges is an $n \times m$ matrix where $M_{ve} = 1$ if vertex v is incident to edge e , and 0 otherwise.

8.1.1.5 Connectivity

Connectivity measures the resilience of a graph.

Definition 8.1.9 (Paths of undirected Graphs). A path of length n , ($n \geq 0$) from u to v in an undirected graph $G = (V, E)$ is a finite sequence of vertices $u = v_0, v_1, v_2, \dots, v_n = v$ such that each set pair $\{v_{i-1}, v_i\}$ is an edge in E .

With the definition of paths, we can define connectivity now.

Definition 8.1.10. A graph G is **connected** if there exists a path between every pair of vertices in G . Otherwise, it is **disconnected**.

The definitions of directed graphs is similar, except that the edges are ordered pairs.

Definition 8.1.11 (Paths of directed Graphs). A path of length n , ($n \geq 0$) from u to v in a directed graph $G = (V, E)$ is a finite sequence of vertices $u = v_0, v_1, v_2, \dots, v_n = v$ such that each ordered pair (v_{i-1}, v_i) is an edge in E .

But the connectivity definition is a bit different. We have two types of connectivity in directed graphs.

Definition 8.1.12. A directed graph G is **strongly connected** if there exists a directed path from every vertex to every other vertex. It is **weakly connected** if the underlying undirected graph (obtained by replacing all directed edges with undirected edges) is connected.

However, simply understanding the concept of connectivity is not enough. We need to quantify how connected a graph is. This leads us to the concepts of vertex and edge connectivity.

Definition 8.1.13. 1. A **cut vertex** is a vertex $v \in V$ such that the removal of the vertex v results in more connected components than G .

2. A subset V' in V is called a vertex cut if the subgraph $G - V'$ is disconnected.

3. The **vertex connectivity** $\kappa(G)$ is the minimum number of vertices whose removal disconnects G or results in a single vertex.

Likewise, we can define edge connectivity using the same format.

Definition 8.1.14. 1. A **bridge** (or cut edge) is an edge $e \in E$ such that the removal of the edge e results in more connected components than G .

2. A subset E' in E is called an edge cut if the subgraph $G - E'$ is disconnected.
3. The **edge connectivity** $\lambda(G)$ is the minimum number of edges whose removal disconnects G .

Theorem 8.1.5: Whitney's Inequality

For any graph G , the vertex connectivity $\kappa(G)$, edge connectivity $\lambda(G)$, and minimum degree $\delta(G)$ satisfy:

$$\kappa(G) \leq \lambda(G) \leq \delta(G)$$

Proof. Part 1: $\lambda(G) \leq \delta(G)$ Let v be a vertex with minimum degree $\delta(G)$. The set of edges incident to v forms an edge cut separating v from the rest of the graph (assuming $n > 1$). This cut has size $\delta(G)$. Since $\lambda(G)$ is the size of the *minimum* edge cut, $\lambda(G) \leq \delta(G)$.

Part 2: $\kappa(G) \leq \lambda(G)$ Consider a minimum edge cut $[S, \bar{S}]$ consisting of $\lambda(G)$ edges.

- **Case A:** If every vertex in S is adjacent to every vertex in $\bar{S}$, then $\lambda(G) = |S||\bar{S}|$. In this case, removing all vertices in S (or $\bar{S}$, whichever is smaller) disconnects the graph (or leaves a trivial graph). Since $|S| \leq |S||\bar{S}|$ (assuming non-trivial sets), the inequality holds.
- **Case B:** If not all edges exist between S and $\bar{S}$, there exist $x \in S$ and $y \in \bar{S}$ such that $\{x, y\} \notin E$. We construct a vertex cut. For every edge $e = \{u, v\}$ in the edge cut, choose the endpoint that belongs to S (or consistently $\bar{S}$). Let T be this set of vertices. Removing T breaks all paths between $S \setminus T$ and $\bar{S}$. We can optimize this selection: strictly speaking, Menger's Theorem or a direct construction shows that a vertex cut of size $\leq \lambda(G)$ exists.

Thus, $\kappa(G) \leq \lambda(G)$. □

8.1.1.6 Special Graph Classes

In this part we will introduce some special classes of graphs along with their properties.

Complete Graphs

Definition 8.1.15. A **complete graph** K_n is a simple graph where every pair of distinct vertices is connected by an edge.

Complete graphs are maximally connected and serve as important examples in graph theory. Here are some of the properties of complete graphs.

Theorem 8.1.6

For K_n :

- Number of edges: $\frac{n(n-1)}{2}$
- Regular of degree $n - 1$
- Hamiltonian and Eulerian for $n \geq 3$

Bipartite Graphs Another important class of graphs is bipartite graphs. They are widely used in modeling relationships between two distinct sets as well as data structures like matchings. Databases, recommendation systems, and scheduling problems often utilize bipartite graphs.

Definition 8.1.16. A **bipartite graph** is a graph $G = (V, E)$ where the vertex set V can be partitioned into two disjoint sets X and Y such that every edge in E connects a vertex in X to a vertex in Y . There are no edges between vertices within the same set.

A wonderful property of bipartite graphs is defined by the concept of matchings. A matching is a set of edges without common vertices.

Definition 8.1.17. A **matching** in a graph $G = (V, E)$ is a subset of edges $M \subseteq E$ such that no two edges in M share a common vertex. A maximal matching is a matching with the largest number of edges.

Definition 8.1.18. A **complete matching** is a matching that covers every vertex of the graph. In a bipartite graph $G = (X, Y, E)$, a complete matching pairs every vertex in X with a unique vertex in Y .

Theorem 8.1.7: Characterization of Bipartite Graphs

A graph G is bipartite if and only if it contains no odd cycles.

Proof. ($\Rightarrow$) Let G be bipartite with partitions X and Y . Any path must alternate between sets: $x_1 \in X, y_1 \in Y, x_2 \in X, \dots$. To return to the starting set X to close a cycle, one must traverse an even number of steps. Thus, any cycle must have even length.

($\Leftarrow$) Assume G has no odd cycles. We assume G is connected (otherwise apply to components). Pick a start vertex v_0 . Define sets $V_i = \{v \in V \mid d(v_0, v) = i\}$. Let $X = \bigcup_{k \text{ even}} V_k$ and $Y = \bigcup_{k \text{ odd}} V_k$. We claim X and Y are independent sets. Suppose there is an edge between $u, w \in X$. Then $u \in V_i, w \in V_j$ with i, j even. There is a path from v_0 to u of length i and to w of length j . Combining these with edge $\{u, w\}$ creates a closed walk of length $i + j + 1$ (odd). While a closed walk is not a simple cycle, a closed odd walk *must* contain a simple odd cycle (standard lemma). This contradicts the hypothesis. Thus, no edges exist within X (similarly for Y). G is bipartite. $\square$

There is a famous theorem regarding matchings in bipartite graphs, known as Hall's Marriage Theorem. The theorem provides a necessary and sufficient condition for the existence of a perfect matching that saturates one part of the bipartition.

Theorem 8.1.8: Hall's Marriage Theorem

Let $G = (X, Y, E)$ be a bipartite graph. There exists a matching that saturates X if and only if for every subset $S \subseteq X$, $|N(S)| \geq |S|$, where $N(S)$ denotes the set of neighbors of vertices in S .

Proof. ($\Rightarrow$) If a matching saturates X , every vertex in S is matched to a distinct vertex in $N(S)$ (its partner). Thus, the number of available neighbors must be at least the number of elements in S .

($\Leftarrow$) *Proof by induction on $|X|$.* Base case $|X| = 1$: If $|N(S)| \geq |S| \implies \deg(x) \geq 1$, edge exists, matching exists. Assume true for $|X| < k$. Consider $|X| = k$.

- **Case 1 (Strong Condition):** If for all proper subsets $\emptyset \subset S \subset X$, we have the strict inequality $|N(S)| \geq |S| + 1$. Pick any edge $\{u, v\}$ with $u \in X$. Match them. Consider $G' = G - \{u, v\}$. For any subset $S' \subseteq X \setminus \{u\}$, its neighborhood in G' is $N_{G'}(S') = N_G(S') \setminus \{v\}$. Since $|N_G(S')| \geq |S'| + 1$, we have $|N_{G'}(S')| \geq |S'|$. By induction, the rest of X can be matched.
- **Case 2 (Tight Condition):** There exists a "critical" proper subset $A \subset X$ where $|N(A)| = |A|$. By the induction hypothesis (since $|A| < k$), A can be matched to $N(A)$. Let $G' = G - (A \cup N(A))$. We must show G' satisfies Hall's condition for the remaining set $X' = X \setminus A$. Let $S' \subseteq X'$. Note that $N_G(S' \cup A) = N_{G'}(S') \cup N(A)$. By Hall's condition on G : $|N_G(S' \cup A)| \geq |S' \cup A| = |S'| + |A|$. Since $N(A)$ and $N_{G'}(S')$ are disjoint (by definition of removing $N(A)$): $|N_{G'}(S')| + |N(A)| \geq |S'| + |A|$.

Substituting $|N(A)| = |A|$, we get $|N_{G'}(S')| \geq |S'|$. Thus, X' can also be matched. Combining matchings yields the result.

□

Note on Matchings

- **König's Theorem:** In any bipartite graph, the size of the maximum matching equals the size of the minimum vertex cover.
- **Regular Bipartite Graphs:** Every k -regular bipartite graph has a perfect matching (for $k \geq 1$).

Planar Graphs Another important class of graphs is planar graphs. They are widely used in geographic mapping, circuit design, and network visualization.

Definition 8.1.19 (Planar Graph). A graph is **planar** if it can be drawn on a plane without any edges crossing.

One of the most famous results in planar graph theory is Euler's formula, which relates the number of vertices, edges, and faces in a connected planar graph.

Theorem 8.1.9: Euler's Formula

For any connected planar graph with v vertices, e edges, and f faces (regions), the following holds:

$$v - e + f = 2$$

Proof. Proof by induction on the number of edges e .

- **Base Case:** If $e = 0$, then $v = 1$ (since connected). There is 1 face (the outer infinite region).

$$1 - 0 + 1 = 2$$

The formula holds.

- **Inductive Step:** Assume the formula holds for all connected planar graphs with fewer than k edges. Let G have k edges.

- **Subcase 1: G is a tree.** Then G has no cycles. $e = v - 1$ and $f = 1$.

$$v - (v - 1) + 1 = 2$$

Formula holds.

- **Subcase 2: G contains a cycle.** Choose an edge E_{cycle} that lies on a cycle. Since it is on a cycle, it separates two distinct faces. Removing E_{cycle} merges these two faces into one, decreasing f by 1. The number of edges e decreases by 1. The number of vertices v remains constant. The graph remains connected. Let the parameters for $G - E_{cycle}$ be v', e', f' . By induction: $v' - e' + f' = 2$. Substitute $v' = v$, $e' = e - 1$, $f' = f - 1$:

$$v - (e - 1) + (f - 1) = 2 \implies v - e + 1 + f - 1 = 2 \implies v - e + f = 2$$

The formula holds for all connected planar graphs.

□

We see that Euler's formula in graph theory shares a similar structure to Euler's characteristic in topology, highlighting the deep connections between these fields. Actually, Euler's formula can be viewed as a special case of the Euler characteristic for surfaces, where planar graphs correspond to graphs embedded on a sphere (which has Euler characteristic 2). We can also use the formula to derive the Euler's formula in topology

by considering graphs drawn on surfaces of different genus. Readers interested in topology may explore this connection further.

So how can we define whether a graph is planar or not? Kuratowski's Theorem provides a complete characterization of planar graphs in terms of forbidden subgraphs. But before stating the theorem, we need to define the concept of **homeomorphism**.

Definition 8.1.20. A graph H is a **homeomorphism** of a graph G if H can be obtained from G by subdividing edges (replacing each edge with a path of length 2).

Theorem 8.1.10: Kuratowski's Theorem

A finite graph is planar if and only if it contains no subgraph that is a subdivision of K_5 or $K_{3,3}$.

After all, why do we care about planar graphs? One of the most famous problems in graph theory is the Four Color Theorem, which states that any planar graph can be colored using at most four colors such that no two adjacent vertices share the same color. This theorem has important applications in map coloring, scheduling, and network design.

Theorem 8.1.11: Four Color Theorem

Every planar graph is 4-colorable.

8.2 Paths and Circuits

In graph theory, paths and circuits are fundamental concepts that describe ways to traverse a graph. A path is a sequence of vertices connected by edges, while a circuit is a closed path that starts and ends at the same vertex. This section explores key theorems related to Eulerian circuits and Hamiltonian cycles, providing rigorous proofs for each. We shall pay special attention to two special types of traversals: Eulerian circuits, which cover every edge exactly once, and Hamiltonian cycles, which visit every vertex exactly once.

8.2.1 Eulerian Circuits

Definition 8.2.1. An **Eulerian circuit** in a graph is a closed trail that visits every edge exactly once.

To define whether a graph has an Eulerian circuit, we can use the following theorem.

Theorem 8.2.1

A connected graph G has an Eulerian circuit (a closed trail covering every edge exactly once) if and only if every vertex in G has an even degree.

Proof. ($\Rightarrow$) If there is an Eulerian circuit, consider the traversal of the circuit. Every time the trail enters a vertex v , it must subsequently leave v via a different edge. Thus, the edges incident to v are used in pairs (entry and exit). Consequently, the total degree of every vertex must be even.

($\Leftarrow$) Assume all vertices have even degree. We construct a trail T of maximum length.

1. **T must be a closed cycle.** Suppose T starts at u and ends at v ($u \neq v$). The number of edges in T incident to v is odd (one entry for every exit, plus one final entry). However, $\deg(v)$ is even in G . Thus, there is at least one unused edge incident to v . We could extend T through this edge, contradicting the assumption that T is a maximal trail. Thus, T must end at u .
2. **T covers all edges.** Suppose T does not include all edges. Since G is connected, there must be an unused edge $e = \{x, y\}$ where x lies on T . Consider the subgraph $G' = G - E(T)$. Since T is a cycle, every vertex in T has even degree in T . Since vertices in G have even degree, vertices in G' must also

have even degree (even - even = even). We can start a new trail T' from x in G' . Because degrees are even, this trail can be extended until it returns to x . We can then splice T' into T to form a longer closed trail, contradicting the maximality of T .

Therefore, T must contain all edges. □

Also, there is another concept similar to Eulerian circuits, called Euler paths.

Definition 8.2.2. An **Euler path** in a graph is a trail that visits every edge exactly once but does not necessarily start and end at the same vertex.

Similarly, we can define whether a graph has an Euler path or not.

Theorem 8.2.2: Euler Trail Criterion

A connected graph G has an Euler trail if and only if it has either zero or two vertices of odd degree. The trail is closed precisely when there are no odd-degree vertices; when there are exactly two, every Euler trail starts at one odd vertex and ends at the other.

Proof. In any trail, every internal visit to a vertex uses one entering and one leaving edge. Hence all vertices other than the two possible endpoints are incident with an even number of used edges. If the trail uses every edge, the graph can therefore have only zero or two odd-degree vertices.

Conversely, if every degree is even, the preceding Euler-circuit theorem applies. If exactly two vertices u and v have odd degree, add one temporary edge joining u to v . Every degree in the enlarged connected graph is now even, so it has an Euler circuit. Removing the temporary edge breaks that circuit into an Euler trail in the original graph, beginning at one of u, v and ending at the other. □

8.2.2 Hamiltonian Cycles

Another important concept in graph theory is the existence of Hamiltonian cycles.

Definition 8.2.3. A **Hamiltonian cycle** in a graph is a closed loop that visits every vertex exactly once.

Unfortunately, there is no simple necessary and sufficient condition for the existence of Hamiltonian cycles like there is for Eulerian circuits. However, we do have some sufficient conditions, such as Ore's Theorem.

Theorem 8.2.3: Ore's Theorem

Let G be a simple graph with $n \geq 3$ vertices. If for every pair of non-adjacent vertices u and v ,

$$\deg(u) + \deg(v) \geq n$$

then G is Hamiltonian.

Proof. Proof by Contradiction. Assume the condition holds but G is not Hamiltonian. We add edges to G as long as the graph remains non-Hamiltonian. Let the resulting maximal non-Hamiltonian graph be G^* . In G^* , adding any single edge $\{u, v\}$ creates a Hamiltonian cycle. This implies there is a Hamiltonian path in G^* from u to v : $u = v_1 \rightarrow v_2 \rightarrow \cdots \rightarrow v_n = v$. Since edge $\{u, v\}$ does not exist in G^* , the degree condition applies: $\deg(u) + \deg(v) \geq n$.

We look for a "crossover" index. Let $S = \{i \mid \{u, v_{i+1}\} \in E(G^*)\}$ be the indices where u connects to the node after v_i . Let $T = \{i \mid \{v_i, v\} \in E(G^*)\}$ be the indices where v connects to v_i . Note that indices are taken from $\{1, \dots, n-1\}$. The size $|S| = \deg(u)$ and $|T| = \deg(v)$. Sum of sizes: $|S| + |T| \geq n$. The available indices are $1, \dots, n-1$. By the Pigeonhole Principle, $S \cap T \neq \emptyset$. Let $k \in S \cap T$. Then $\{u, v_{k+1}\} \in E$ and $\{v_k, v\} \in E$. We can construct a cycle:

$$u \rightarrow v_{k+1} \rightarrow v_{k+2} \cdots \rightarrow v \rightarrow v_k \rightarrow v_{k-1} \cdots \rightarrow v_2 \rightarrow u$$

This cycle visits every vertex exactly once, making G^* Hamiltonian, which is a contradiction. □

8.3 Trees and Forests

Trees and forests are fundamental structures in graph theory with numerous applications in computer science, biology, and network design. A tree is a connected acyclic graph, while a forest is a disjoint union of trees. This section explores the properties and characterizations of trees and forests, providing rigorous definitions and theorems.

8.3.1 Definitions and Characterization

Definition 8.3.1 (Trees). A tree is a connected undirected simple graph without simple circuits (acyclic).

Forests are closely related to trees. You can think of a forest as a collection of trees. Remember that a tree is a special case of a forest with only one connected component.

Definition 8.3.2 (Forests). A forest is a undirected simple graph without simple circuits (acyclic).

There is a relatively abstract characterization of trees.

Theorem 8.3.1

An undirected graph G is a tree iff any two vertices in G is connected by a unique simple path.

Theorem 8.3.2: Characterization of Trees

A graph G with $|V| = n$ vertices is a tree if and only if it is connected and $|E| = n - 1$.

Theorem 8.3.3: Corollary

A connected simple graph $G = (V, E)$ satisfies that $|E| \geq |V| - 1$.

Proof. If a connected simple graph G has $|E| < |V| - 1$, then we can add edges to G until it has $|V| - 1$ edges. The resulting graph is still connected, but by the previous theorem, it must be a tree. However, a tree with n vertices has exactly $n - 1$ edges, so adding edges to reach $n - 1$ edges contradicts the assumption that we started with fewer than $n - 1$ edges. Therefore, any connected simple graph must have at least $|V| - 1$ edges. $\square$

Proof. $(\Rightarrow)$ **Tree $\Rightarrow$ Connected and $n - 1$ edges.** We proceed by induction on n . *Base Case:* For $n = 1$, $e = 0$. The condition holds ($0 = 1 - 1$). *Inductive Step:* Assume all trees with k vertices have $k - 1$ edges. Let T be a tree with $k + 1$ vertices. Since T is acyclic and finite ($n \geq 2$), it must contain at least two vertices of degree 1 (leaves). Let v be a leaf and e be the incident edge. Consider $T' = T - \{v\}$. T' is still connected (removing a leaf does not break connectivity) and acyclic. Thus T' is a tree with k vertices. By the inductive hypothesis, T' has $k - 1$ edges. Since T has exactly one more edge than T' , $|E(T)| = (k - 1) + 1 = k = (k + 1) - 1$.

A more rigorous proof is done by randomly selecting a vertex in the tree, and discussing all the vertices in the neighborhood of the vertex. Once we remove the vertex, all the neighbors become roots of subtrees. We can apply the inductive hypothesis on each subtree because they all satisfy the inductive hypothesis, and sum up the number of edges. Mark that according to the definition of tree, there is no cycle in the graph, so there is no edge between any two subtrees, and the intersections between subtrees are empty. Thus the total number of edges is the sum of edges in each subtree plus the number of edges connecting the root vertex to each subtree (which is exactly the number of subtrees). This gives us the desired result.

$(\Leftarrow)$ **Connected and $n - 1$ edges $\Rightarrow$ Tree.** Assume G is connected with $n - 1$ edges. We must show G is acyclic. Suppose G contains a cycle. Remove an edge from this cycle. The graph remains connected. Repeat this process until the graph is acyclic. Let the resulting graph be T . T is a tree (connected and acyclic) on n vertices. By the first part of this proof, T must have $n - 1$ edges. However, we started with

$n - 1$ edges and removed at least one edge to break the cycle. This implies G initially had $> n - 1$ edges, a contradiction. Thus, G contains no cycles. $\square$

8.3.2 Rooted Trees

Definition 8.3.3 (Rooted Tree). A rooted tree is a tree with a designated vertex called the root, and the designated vertex is the root of the tree.

Definition 8.3.4. Assume we have a rooted tree (V, E) with the root r . Then we define that:

1. The **level** of a vertex v is the length of the unique simple path from r to v .
2. The **height** of the tree is the maximum level of any vertex in the tree.
3. The partial order $\leq$ on V is defined as: for any two vertices u, v in V , $u \leq v$ if and only if u lies on the unique simple path from r to v .
4. The **parent** of a vertex $v \neq r$ is the unique vertex u such that $\{u, v\} \in E$ and u is on the unique simple path from r to v .
5. The **children** of a vertex v is the set of vertices $\{u \in V \mid \{u, v\} \in E \text{ and } v \text{ is the parent of } u\}$.
6. A **leaf** is a vertex with no children (degree 1 if not the root, degree 0 if it is the root).

Definition 8.3.5. If G is a rooted tree, then G is m -ary if every internal vertex has at most m children. If every internal vertex has exactly m children, then G is a **full m -ary tree**.

Definition 8.3.6. A rooted tree G is binary if it is 2-ary.

Theorem 8.3.4

In a full m -ary tree with n internal vertices, the number of leaves L is given by:

$$L = (m - 1)n + 1$$

Theorem 8.3.5

An m -ary tree of height h has at most m^h leaves.

The proofs of the above two theorems are left as exercises for the reader. Hints: use induction on the number of internal vertices for the first theorem, and induction on the height for the second theorem.

Definition 8.3.7. A rooted m -ary tree of height h is balanced if all its leaves are at level h or $h - 1$.

8.3.3 Spanning Trees

Definition 8.3.8 (Spanning Tree). If $G = (V, E)$ is a connected graph, then a **spanning tree** of G is a subgraph $T = (V, E')$ that is a tree. T contains all the vertices of G .

Theorem 8.3.6

A simple graph G is connected if and only if it has a spanning tree.

Proof. (**If**) If G has a spanning tree T , then T is connected (by definition of tree). Since T is a subgraph of G and contains all vertices of G , G must also be connected. (**Only if**) Assume G is connected. We can construct a spanning tree by starting with an arbitrary vertex and performing a depth-first search (DFS) or breadth-first search (BFS) to explore all vertices. As we explore, we add edges to our spanning tree whenever we encounter a new vertex. Since G is connected, this process will eventually visit all vertices, resulting in a spanning tree that includes all vertices of G . Actually, a spanning tree can be obtained by

removing edges from G until no cycles remain while ensuring the graph remains connected. This process will yield a spanning tree. $\square$

Definition 8.3.9 (Weighted Graph). A **weighted graph** is a graph $G = (V, E)$ together with a weight function $w : E \rightarrow \mathbb{R}$ that assigns a real number (weight) to each edge.

Definition 8.3.10 (Minimum Spanning Tree). In a weighted connected graph G , a **minimum spanning tree** is a spanning tree with the smallest total edge weight.

To find a minimum spanning tree, we can use algorithms such as Kruskal's or Prim's algorithm. **Prim's Algorithm:**

1. Start with an arbitrary vertex and initialize the tree with this vertex.
2. At each step, add the minimum-weight edge that connects a vertex in the tree to a vertex outside the tree.
3. Repeat until all vertices are included in the tree.

Theorem 8.3.7: Prim's Algorithm Correctness

Prim's algorithm correctly finds a minimum spanning tree in a connected weighted graph.

Proof. We will prove that the spanning tree T produced by Prim's algorithm is a minimum spanning tree (MST) of the given connected, undirected graph $G = (V, E)$ with weight function $w : E \rightarrow \mathbb{R}$.

The core of the proof relies on the **MST property** (also called the **cut property**):

Let $G = (V, E)$ be a connected, undirected graph with edge weights. For any nonempty proper subset $S \subset V$ (i.e., S is a set containing some, but not all, vertices of the graph), let e be a minimum-weight edge with one endpoint in S and the other in $V \setminus S$. Then e is contained in **some** minimum spanning tree of G .

Prim's algorithm, at each step, chooses exactly such a minimum-weight edge (a *light edge*) crossing the cut $(S, V \setminus S)$, where S is the set of vertices already included in the partially constructed tree.

We will now prove that every edge added by Prim's algorithm belongs to at least one MST. The proof proceeds by **contradiction**.

Assume, for the sake of contradiction, that the tree T produced by Prim's algorithm is *not* a minimum spanning tree. Then there exists at least one MST, say T' , of G such that $w(T') < w(T)$. Since both T and T' span all vertices of G but have different edge sets, there must be an edge e that is selected by Prim's algorithm at some step but is *not* in T' .

Let us consider the moment when Prim's algorithm adds edge $e = (u, v)$ to T . At that moment, the algorithm's vertex set S contains u (without loss of generality) but not v . Since e is not in T' , if we add e to T' , it creates a unique cycle C in $T' \cup \{e\}$.

Within this cycle C , there must be at least one other edge $e' = (x, y)$ that also crosses the cut $(S, V \setminus S)$ (with $x \in S$ and $y \in V \setminus S$). This is because the cycle starts and ends in S and must leave and re-enter S ; edge e provides one crossing, so there must be another crossing to complete the cycle.

By Prim's algorithm's greedy choice, e is a minimum-weight edge crossing the cut $(S, V \setminus S)$. Therefore, we have

$$w(e) \leq w(e').$$

Now, consider the new tree $T'' = T' \setminus \{e'\} \cup \{e\}$. T'' is also a spanning tree of G because it is connected and has exactly $|V| - 1$ edges. The weight of T'' is

$$w(T'') = w(T') - w(e') + w(e).$$

Since $w(e) \leq w(e')$, we have $w(T'') \leq w(T')$.

- If $w(e) < w(e')$, then $w(T'') < w(T')$, which contradicts the assumption that T' is an MST (since an MST must have minimum possible weight).
- If $w(e) = w(e')$, then $w(T'') = w(T')$, meaning T'' is also an MST. However, T'' now contains the edge e selected by Prim's algorithm.

Therefore, in either case, the edge e chosen by Prim's algorithm is contained in at least one MST (namely, either T' or T''). This argument can be applied iteratively to every edge added by Prim's algorithm. Consequently, the final tree T produced by Prim's algorithm must be a minimum spanning tree itself.

Thus, Prim's algorithm correctly computes a minimum spanning tree of the given graph G .

□

Another popular algorithm for finding a minimum spanning tree is Kruskal's algorithm. **Kruskal's Algorithm**

1. Choose the initial graph G_1 with all the vertices included and no edges.
2. Construct G_{i+1} by adding the smallest weight edge that forms a forest.
3. Finally, G_{m-1} is a minimum spanning tree, where m is the number of vertices.

Mind that we view each step as adding an edge to the existing forest, connecting two isolated components, ensuring that no cycles are formed. Every isolated vertex is considered a component.

Proof. Kruskal's Algorithm Correctness We will prove that the spanning tree T produced by Kruskal's algorithm is a minimum spanning tree (MST) of the given connected, undirected graph $G = (V, E)$ with weight function $w : E \rightarrow \mathbb{R}$. The core of the proof relies on the **MST property** (also called the **cut property**):

Let $S \subset V$ be a subset of vertices, and let e be a minimum-weight edge crossing the cut $(S, V \setminus S)$. Then, e belongs to every MST of G .

Kruskal's algorithm, at each step, chooses exactly such a minimum-weight edge (a *light edge*) that does not form a cycle with the edges already selected.

We will now prove that every edge added by Kruskal's algorithm belongs to at least one MST. The proof proceeds by **contradiction**.

Assume, for the sake of contradiction, that the tree T produced by Kruskal's algorithm is *not* a minimum spanning tree. Then there exists at least one MST, say T' , of G such that $w(T') < w(T)$. Since both T and T' span all vertices of G but have different edge sets, there must be an edge e that is selected by Kruskal's algorithm at some step but is *not* in T' . We choose the first such edge in the order selected by Kruskal's algorithm.

Let us consider the moment when Kruskal's algorithm adds edge $e = (u, v)$ to T . At that moment, the algorithm's selected edges form a forest, and adding e connects two distinct components of this forest, or adding e connects two vertices in a connected component, which is contradicted to the rule of the algorithm, we shall ignore this case. Let S be the set of vertices in one of these components (say, the component containing u). Since e is not in T' , if we add e to T' , it creates a unique cycle C in $T' \cup \{e\}$.

Within this cycle C , there must be at least one other edge $e' = (x, y)$ that also crosses the cut $(S, V \setminus S)$ (with $x \in S$ and $y \in V \setminus S$). This is because the cycle starts and ends in S and must leave and re-enter S ; edge e provides one crossing, so there must be another crossing to complete the cycle.

By Kruskal's algorithm's greedy choice, e is a minimum-weight edge crossing the cut $(S, V \setminus S)$. Therefore, we have

$$w(e) \leq w(e').$$

Now, consider the new tree $T'' = T' \setminus \{e'\} \cup \{e\}$. T'' is also a spanning tree of G because it is connected and has exactly $|V| - 1$ edges. The weight of T'' is

$$w(T'') = w(T') - w(e') + w(e).$$

Since $w(e) \leq w(e')$, we have $w(T'') \leq w(T')$. But this contradicts the assumption that $w(T') < w(T)$, since T'' is a spanning tree of G and $w(T'') \leq w(T')$. Therefore, our assumption must be false, and Kruskal's algorithm produces a minimum spanning tree.

□

Another possible application of spanning trees is the Huffman coding tree, which is widely used in data compression algorithms.

The Huffman coding tree is a binary tree used for lossless data compression. It assigns variable-length codes to input characters based on their frequencies, with more frequent characters receiving shorter codes. The tree is constructed using a greedy algorithm that combines the two least frequent nodes iteratively until a single tree is formed.

Here are the steps to construct a Huffman coding tree:

1. Construct the forest consisting of single-node trees for each character, with the weight of each node equal to the frequency of the corresponding character.
2. Sort the weights of the trees in the forest in non-decreasing order.
3. While there is more than one tree in the forest:
 - (a) Select the two trees with the smallest weights.
 - (b) Create a new internal node with these two trees as children, and assign it a weight equal to the sum of their weights.
 - (c) Insert the new tree back into the forest, maintaining the sorted order of weights.
4. The remaining tree in the forest is the Huffman coding tree.

Each leaf node in the Huffman coding tree represents a character, and the path from the root to the leaf determines the binary code for that character (left edge = 0, right edge = 1).

For example, consider the characters A, B, C, and D with frequencies 5, 9, 12, and 13, respectively. The Huffman coding tree would be constructed as follows:

1. Create single-node trees: A(5), B(9), C(12), D(13).
2. Sort: A(5), B(9), C(12), D(13).
3. Combine A and B: New node AB(14) with children A and B.
4. Sort: C(12), D(13), AB(14).
5. Combine C and D: New node CD(25) with children C and D.
6. Sort: AB(14), CD(25).
7. Combine AB and CD: New root node ABCD(39) with children AB and CD.

8. Then the final Huffman coding tree is formed.

Why the Huffman coding tree is optimal can be proved using a greedy choice argument and an exchange argument, showing that any other prefix-free code would result in a longer average code length.

Proof. Huffman's Algorithm Correctness

We will prove that the binary tree T produced by Huffman's algorithm is an optimal prefix-free code for the given set of characters with their frequencies.

Notation:

- Let C be the set of characters.
- Let $f(c)$ be the frequency of character $c \in C$.
- Let $d_T(c)$ be the depth of character c in the tree T .
- The cost of the code represented by tree T is given by:

$$\text{Cost}(T) = \sum_{c \in C} f(c) \cdot d_T(c).$$

We will prove the optimality of Huffman's algorithm using induction on the number of characters $|C|$.

Base case: For $|C| = 2$, the optimal code is trivial: assign one character to '0' and the other to '1'. Huffman's algorithm produces this code, which is optimal. **Inductive step:** Assume Huffman's algorithm produces an optimal code for any set of k characters. We will show it also produces an optimal code for $k + 1$ characters. Let $C = \{c_1, c_2, \dots, c_{k+1}\}$ be the set of characters with frequencies $f(c_1) \leq f(c_2) \leq \dots \leq f(c_{k+1})$. Huffman's algorithm combines the two characters with the smallest frequencies, say c_1 and c_2 , into a new character c_{12} with frequency $f(c_{12}) = f(c_1) + f(c_2)$. By the inductive hypothesis, Huffman's algorithm produces an optimal code for the reduced set $C' = \{c_{12}, c_3, \dots, c_{k+1}\}$. Let T' be the optimal tree for C' . We can construct the tree T for C by replacing the leaf node for c_{12} in T' with an internal node having c_1 and c_2 as its children. The cost of the tree T can be expressed as:

$$\text{Cost}(T) = \text{Cost}(T') + f(c_1) + f(c_2).$$

Now, consider any other prefix-free code tree T^* for the original set C . We can construct a tree T'^* for the reduced set C' by merging the leaves for c_1 and c_2 into a single leaf for c_{12} . The cost of T^* can be expressed as:

$$\text{Cost}(T^*) = \text{Cost}(T'^*) + f(c_1) + f(c_2).$$

By the inductive hypothesis, we have:

$$\text{Cost}(T') \leq \text{Cost}(T'^*).$$

Thus,

$$\text{Cost}(T) = \text{Cost}(T') + f(c_1) + f(c_2) \leq \text{Cost}(T'^*) + f(c_1) + f(c_2) = \text{Cost}(T^*).$$

Therefore, Huffman's algorithm produces an optimal prefix-free code for any set of $k + 1$ characters. By induction, Huffman's algorithm produces an optimal prefix-free code for any set of characters. □

We have to claim that the construction process of the Huffman coding tree itself is an implementation of Kruskal's algorithm, since at each step we are combining two trees with the smallest weights, which is equivalent to adding the smallest weight edge that connects two components in Kruskal's algorithm.

Also, as we dive deeper into the processes, the construction itself is a kind of mathematical induction, since at each step we are reducing the problem size by one (combining two characters into one), and we can prove the optimality using induction as shown in the proof above.

Graph Algorithms Implementation

The following Python code implements BFS, DFS, Prim's Algorithm, and Kruskal's Algorithm using adjacency matrices represented as nested lists.

BFS Algorithm

```

1 def find_shortest_path_BFS(matrix, start, end):
2     """
3     Find shortest path using Breadth-First Search (unweighted).
4     """
5     vertices_count = len(matrix)
6     if not (isinstance(start, int) and isinstance(end, int)):
7         raise TypeError("Start and End vertices must be integers")
8     if not (0 <= start < vertices_count and 0 <= end < vertices_count):
9         raise ValueError("Vertex index out of range")
10
11     queue = [start]
12     memory = {start: None}
13     found = False
14     idx = 0
15
16     while idx < len(queue):
17         curr = queue[idx]
18         if curr == end:
19             found = True
20             break
21
22         for i in range(vertices_count):
23             if matrix[curr][i] != 0 and i not in memory:
24                 memory[i] = curr
25                 queue.append(i)
26                 if i == end:
27                     found = True
28                     break
29         if found:
30             break
31         idx += 1
32
33     if not found:
34         return None
35
36     path = []
37     index = end
38     while index is not None:
39         path.append(index)
40         index = memory[index]
41     return path[::-1]

```

DFS Algorithm

```

1 def find_path_DFS(matrix, start, end):
2     """
3     Find a path using Depth-First Search.
4     """
5     vertices_count = len(matrix)
6     if not (isinstance(start, int) and isinstance(end, int)):
7         raise TypeError("Start and End vertices must be integers")
8     if not (0 <= start < vertices_count and 0 <= end < vertices_count):
9         raise ValueError("Vertex index out of range")
10
11     stack = [start]
12     memory = {start: None}
13     found = False
14
15     while len(stack) > 0:
16         curr = stack.pop()
17         if curr == end:

```

```

18         found = True
19         break
20
21     for i in range(vertices_count):
22         if matrix[curr][i] != 0 and i not in memory:
23             memory[i] = curr
24             stack.append(i)
25
26 if not found:
27     return None
28
29 path = []
30 curr_node = end
31 while curr_node is not None:
32     path.append(curr_node)
33     curr_node = memory[curr_node]
34 return path[::-1]

```

Prim's Algorithm

```

1 def minimum_spanning_tree_prim(weights):
2     """
3     Implementation of Prim's algorithm for MST.
4     Returns the MST as an adjacency matrix.
5     """
6     if not isinstance(weights, list):
7         raise TypeError("Weights must be a nested list (matrix)")
8     if len(weights) == 0:
9         raise ValueError("Weights matrix cannot be empty")
10    if len(weights) != len(weights[0]):
11        raise ValueError("Weights matrix must be square")
12
13    n = len(weights)
14    INF = float('inf')
15    key = [INF] * n
16    parent = [None] * n
17    mst_set = [False] * n
18    key[0] = 0
19    parent[0] = -1
20
21    for _ in range(n):
22        min_val = INF
23        u = -1
24        for v in range(n):
25            if not mst_set[v] and key[v] < min_val:
26                min_val = key[v]
27                u = v
28
29        if u == -1:
30            break
31        mst_set[u] = True
32        for v in range(n):
33            w = weights[u][v]
34            if w > 0 and not mst_set[v] and w < key[v]:
35                key[v] = w
36                parent[v] = u
37
38    mst_matrix = [[0] * n for _ in range(n)]
39    for i in range(1, n):
40        if parent[i] is not None:
41            u, v = parent[i], i
42            weight = weights[u][v]
43            mst_matrix[u][v] = weight
44            mst_matrix[v][u] = weight
45    return mst_matrix

```

Kruskal's Algorithm


```

1 def minimum_spanning_tree_kruskal(weights):
2     """
3     Implementation of Kruskal's algorithm for MST.
4     Returns the MST as an adjacency matrix.
5     """
6     if not isinstance(weights, list):
7         raise TypeError("Weights must be a nested list (matrix)")
8
9     n = len(weights)
10    edges = []
11    for i in range(n):
12        for j in range(i + 1, n):
13            if weights[i][j] > 0:
14                edges.append((weights[i][j], i, j))
15    edges.sort()
16
17    parent = list(range(n))
18
19    def find(i):
20        if parent[i] == i:
21            return i
22        parent[i] = find(parent[i])
23        return parent[i]
24
25    def union(i, j):
26        root_i = find(i)
27        root_j = find(j)
28        if root_i != root_j:
29            parent[root_i] = root_j
30            return True
31        return False
32
33    mst_matrix = [[0] * n for _ in range(n)]
34
35    for w, u, v in edges:
36        if union(u, v):
37            mst_matrix[u][v] = w
38            mst_matrix[v][u] = w
39    return mst_matrix

```

8.4 Random Graphs

8.4.1 Basic Properties of Random Graphs

Degree Distribution

Theorem 8.4.1: Degree Distribution of $G(n, p)$

In $G(n, p)$, the degree of each vertex follows a binomial distribution $\text{Bin}(n - 1, p)$. For large n and small p , it approximates a Poisson distribution $\text{Po}(\lambda)$ with $\lambda = p(n - 1)$.

Threshold Functions and Phase Transitions Many properties in random graphs exhibit threshold phenomena: as the parameter p crosses a critical threshold p_c , the property changes from almost surely not holding to almost surely holding.

Theorem 8.4.2: Connectivity Threshold

For $G(n, p)$, the threshold function for connectivity is $p_c = \frac{\ln n}{n}$. More precisely:

- If $p = \frac{\ln n + c}{n}$, then $\mathbb{P}(G(n, p) \text{ is connected}) \rightarrow e^{-e^{-c}}$ as $n \rightarrow \infty$.
- If $p \ll \frac{\ln n}{n}$, then $G(n, p)$ is almost surely disconnected.
- If $p \gg \frac{\ln n}{n}$, then $G(n, p)$ is almost surely connected.

Remark 8.4.1: With high probability

For a sequence of random graphs, an event occurs **with high probability** if its probability tends to 1 as $n \rightarrow \infty$.

Theorem 8.4.3: Giant-Component Phase Transition

Fix $c > 0$ and let $G \sim G(n, c/n)$.

- If $0 < c < 1$, then with high probability every connected component has order $O(\log n)$.
- If $c = 1$, the largest components have order $n^{2/3}$ in the critical scale.
- If $c > 1$, then with high probability there is a unique giant component of order

$$\rho n + o(n),$$

where $\rho \in (0, 1)$ is the nonzero solution of

$$\rho = 1 - e^{-c\rho}.$$

All remaining components have order $O(\log n)$.

Distances When the expected degree np grows, a breadth-first search initially behaves like a branching process with approximately np children per discovered vertex. This gives the useful heuristic

$$\text{typical distance} \approx \frac{\log n}{\log(np)}.$$

Exact diameter statements depend on the regime of p and require additional correction terms, so the heuristic should not be quoted as a universal theorem.

Clustering Coefficient

Definition 8.4.1 (Local Clustering Coefficient). For a vertex i of degree $k_i \geq 2$, the local clustering coefficient is

$$C_i = \frac{2e_i}{k_i(k_i - 1)},$$

where e_i is the number of edges between neighbors of i .

Proposition 8.4.1: Clustering in $G(n, p)$

Conditional on the neighbor set of a vertex with degree at least two, every possible edge between two of its neighbors is present independently with probability p . Consequently,

$$\mathbb{E}[C_i \mid k_i \geq 2] = p.$$

This is one reason the basic Erdős–Rényi model often underestimates the high clustering observed in empirical networks.

8.4.2 Important Theorems in Random Graph Theory

Thresholds for Monotone Properties

Theorem 8.4.4: Existence of a Threshold Function

Let $\mathcal{P}$ be a nontrivial monotone increasing graph property. There is a function $p^*(n)$ such that

$$\mathbb{P}(G(n, p) \in \mathcal{P}) \longrightarrow \begin{cases} 0, & p/p^*(n) \rightarrow 0, \\ 1, & p/p^*(n) \rightarrow \infty. \end{cases}$$

The theorem asserts the existence of a transition scale; determining its location and width is a separate problem.

Subgraph Appearance

Theorem 8.4.5: Threshold for a Fixed Subgraph

For a fixed graph H , define

$$m(H) = \max_{H' \subseteq H, v(H') \geq 1} \frac{e(H')}{v(H')}.$$

Then the threshold for $G(n, p)$ to contain a copy of H is

$$p^*(n) = n^{-1/m(H)}.$$

Thus the probability tends to 0 when $p \ll p^*(n)$ and to 1 when $p \gg p^*(n)$. If H is strictly balanced, then $m(H) = e(H)/v(H)$.

Hamiltonicity

Theorem 8.4.6: Hamiltonicity Threshold

Let

$$p = \frac{\log n + \log \log n + c_n}{n}.$$

Then

$$\mathbb{P}(G(n, p) \text{ is Hamiltonian}) \longrightarrow \begin{cases} 0, & c_n \rightarrow -\infty, \\ 1, & c_n \rightarrow +\infty. \end{cases}$$

The threshold coincides asymptotically with the disappearance of vertices of degree 0 or 1.

8.4.3 Algorithmic Generation of Random Graphs

The $G(n, p)$ model can be generated using a simple algorithm that considers each potential edge independently with probability p .

8.4.4 Applications of Random Graphs

Complex Network Modeling Random graph models provide foundational frameworks for modeling social networks, the Internet, biological networks, and other complex systems. While classical models like $G(n, p)$ differ from real networks in certain statistical properties, they serve as the basis for developing more accurate models.

Network Robustness Connectivity thresholds in random graphs provide baseline models for robustness under independent vertex or edge failures. Real networks may behave differently because degree distributions, clustering, and dependencies are not captured by $G(n, p)$.

Epidemic Spread Random graph models are used to study disease spread in populations, where vertices represent individuals and edges represent contacts.

8.4.5 Conclusion and Future Directions

Random graph theory offers a powerful mathematical framework for studying complex networks. Although classical models like $G(n, p)$ differ from real-world networks in certain statistical properties (e.g., degree distribution, clustering), they remain foundational. Current research directions include:

- Developing more realistic random graph models
- Studying dynamic random graph processes
- Exploring algorithms and computational problems on random graphs
- Applying random graph theory to machine learning and data science

8.5 Chapter Workshop

Worked example: Breadth-first search and shortest unweighted paths

Suppose an unweighted graph is explored from a source s by breadth-first search. Vertices are placed in layers

$$L_k = \{v : d(s, v) = k\}.$$

When a vertex in L_k is processed, every previously undiscovered neighbor is assigned to L_{k+1} . No shorter path can be discovered later, because all vertices at distance at most k have already been processed. Thus breadth-first search computes shortest-path distance in $O(|V| + |E|)$ time when adjacency lists are used.

Application: Graph neural networks

A message-passing graph neural network updates a vertex representation by aggregating features from its neighbors. After k layers, the representation depends on a k -hop neighborhood, paralleling repeated multiplication by an adjacency-derived operator. Graph symmetry matters: aggregation should be invariant to the ordering of neighbors, while the overall model should transform consistently under relabeling of vertices.

Application: Network design

Spanning trees model minimum-cost connection without cycles; shortest paths model routing; cuts measure vulnerability and capacity; matchings model assignment; and colorings model conflict-free scheduling. The graph abstraction is useful because the same theorem or algorithm applies to many systems once vertices and edges have been chosen correctly.

8.6 Exercises

1. Prove the handshaking identity $\sum_{v \in V} \deg(v) = 2|E|$.
2. Run breadth-first search on a graph of your choice with at least six vertices and record the layers and parent tree.
3. Prove that a tree with n vertices has $n - 1$ edges.

4. Determine the chromatic number of an odd cycle and an even cycle.
5. Show that every planar simple graph with $n \geq 3$ vertices has at most $3n - 6$ edges.
6. In $G(n, p)$, compute the expected number of triangles.
7. For $G(n, c/n)$, use a branching-process heuristic to explain why $c = 1$ is a critical value.
8. Give one network question for which $G(n, p)$ is an inadequate model and identify the missing structural feature.

Selected hints and answers

1. Count incidences (v, e) in two ways.
2. Check that each non-source vertex receives a parent in the preceding layer.
3. Remove a leaf and use induction, or use connectedness plus acyclicity.
4. Odd cycles require three colors; even cycles are bipartite and require two.
5. Combine Euler's formula $n - m + f = 2$ with $3f \leq 2m$.
6. There are $\binom{n}{3}$ possible triangles, each present with probability p^3 .
7. The expected number of newly discovered neighbors is approximately c ; a branching process dies out for mean below 1 and may survive above 1.
8. Examples include social networks with heavy-tailed degrees, spatial networks with geometry, or networks with high clustering and community structure.

Chapter 9

Applied Mathematics: Probability Theory

Learning objectives

By the end of this chapter, you should be able to:

- construct probability spaces and compute conditional probabilities;
- work with discrete, continuous, and jointly distributed random variables;
- interpret expectation, conditional expectation, and independence structurally;
- compare modes of convergence and apply laws of large numbers and the central limit theorem;
- analyze introductory martingales, Markov chains, Poisson processes, and Brownian motion.

Probability theory is the mathematical language of uncertainty. It began as the study of games of chance, but its modern form is much broader: it describes random experiments, statistical inference, stochastic processes, information, finance, thermodynamics, and learning from data.

At the intuitive level, probability measures how likely an event is. At the rigorous level, probability is a special kind of measure defined on a sigma-algebra of events. This transition from counting favorable outcomes to measuring abstract event spaces is one of the major conceptual developments of modern mathematics.

The route of this chapter is:

classical probability $\rightarrow$ random variables $\rightarrow$ measure-theoretic probability
 $\rightarrow$ expectation $\rightarrow$ conditioning $\rightarrow$ limit theorems.

9.1 Classical Probability and Counting

Probability theory begins with uncertainty, but it becomes mathematics only after the uncertain outcomes are organized into a sample space. In finite classical problems, this organization is often hidden because the sample space is small and symmetric. We count favorable outcomes, count all equally likely outcomes, and take a ratio. This is the earliest and most concrete form of probability.

The limitation of the classical definition is equally important. It depends on a finite collection of equally likely outcomes. Many real probabilistic models do not have this form: a dart may land anywhere in a continuum, a lifetime may take any positive real value, and a stochastic process may produce an infinite path. The later measure-theoretic framework is designed to keep the successful parts of classical probability while removing this finite symmetry restriction.

9.1.1 Random Experiments and Sample Spaces

Definition 9.1.1 (Random Experiment). A **random experiment** is a procedure whose outcome is uncertain before it is performed, but whose possible outcomes can be described.

Definition 9.1.2 (Sample Space and Event). The **sample space**, denoted Ω , is the set of all possible outcomes of a random experiment. An **event** is a subset of Ω .

Example 9.1.1

If a fair coin is tossed once, then

$$\Omega = \{H, T\}.$$

The event that the coin lands heads is $\{H\}$. If two coins are tossed, then

$$\Omega = \{HH, HT, TH, TT\}.$$

The event that exactly one head appears is $\{HT, TH\}$.

9.1.2 Classical Definition of Probability

Definition 9.1.3 (Classical Probability). Suppose Ω is finite and all outcomes are equally likely. For an event $A \subseteq \Omega$, define

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

Example 9.1.2

When a fair six-sided die is rolled,

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

The probability of rolling an even number is

$$\mathbb{P}(\{2, 4, 6\}) = \frac{3}{6} = \frac{1}{2}.$$

Classical probability is simple and useful, but it depends on two strong assumptions: the sample space must be finite, and all outcomes must be equally likely. Many real problems violate at least one of these assumptions.

9.1.3 Basic Counting Principles

Counting is not a separate topic from probability; it is the computational engine behind finite probability models. The multiplication principle counts ordered choices made in stages. Permutations count arrangements where order matters. Combinations count selections where order is ignored. Many probability errors come from mixing these cases.

A useful habit is to describe explicitly what an outcome is before counting. Are we counting ordered tuples, unordered subsets, sequences with repetition, or sequences without repetition? Once the sample space is correctly specified, the probability calculation usually becomes straightforward.

Proposition 9.1.1: Addition Principle

If a task can be completed in either one of m ways or one of n ways, and these two collections of ways do not overlap, then the task can be completed in $m + n$ ways.

Proposition 9.1.2: Multiplication Principle

If a task consists of two stages, where the first stage can be completed in m ways and the second stage can be completed in n ways for each choice of the first, then the full task can be completed in mn ways.

Definition 9.1.4 (Permutation). The number of ways to arrange k objects chosen from n distinct objects, with order mattering and without replacement, is

$$P(n, k) = \frac{n!}{(n - k)!}.$$

Definition 9.1.5 (Combination). The number of ways to choose k objects from n distinct objects, with order not mattering, is

$$\binom{n}{k} = \frac{n!}{k!(n - k)!}.$$

Example 9.1.3

A five-card poker hand is a subset of size 5 from a deck of 52 cards. Since order does not matter, the number of possible hands is

$$\binom{52}{5}.$$

Theorem 9.1.1: Binomial Theorem

For $n \in \mathbb{N}$,

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Remark 9.1.1

The binomial coefficient $\binom{n}{k}$ appears because choosing which k factors contribute x uniquely determines a term $x^k y^{n-k}$ in the expansion.

9.2 Conditional Probability and Independence

Conditional probability changes the universe of discourse. When we condition on B , we restrict attention to the part of the sample space where B has occurred and then renormalize probabilities inside that smaller universe. The formula $P(A \mid B) = P(A \cap B)/P(B)$ is therefore not just an algebraic rule; it is a rule for updating the reference population.

Independence is often misunderstood as mutual exclusiveness. They are nearly opposite ideas. If two events are mutually exclusive and both have positive probability, the occurrence of one rules out the other, so they cannot be independent. Independence means that learning one event occurred does not change the probability of the other. In symbols, conditioning on B leaves the probability of A unchanged.

9.2.1 Conditional Probability

Often we want to update probability after receiving information. If event B has occurred, the relevant sample space is no longer Ω , but B .

Definition 9.2.1 (Conditional Probability). Let A and B be events with $\mathbb{P}(B) > 0$. The **conditional probability** of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Example 9.2.1

Roll a fair die. Let A be the event that the result is even, and let B be the event that the result is at least 4. Then

$$A = \{2, 4, 6\}, \quad B = \{4, 5, 6\}, \quad A \cap B = \{4, 6\}.$$

Therefore

$$\mathbb{P}(A \mid B) = \frac{2/6}{3/6} = \frac{2}{3}.$$

Proposition 9.2.1: Multiplication Rule

If $\mathbb{P}(B) > 0$, then

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B).$$

More generally,

$$\mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\mathbb{P}(A_2 \mid A_1) \cdots \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}),$$

provided all conditional probabilities are defined.

9.2.2 Independence

Definition 9.2.2 (Independence of Events). Two events A and B are **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

If $\mathbb{P}(B) > 0$, independence is equivalent to

$$\mathbb{P}(A \mid B) = \mathbb{P}(A).$$

That is, knowing B occurred does not change the probability of A .

Definition 9.2.3 (Mutual Independence). Events $A_1, \dots, A_n$ are **mutually independent** if for every non-empty subset $I \subseteq \{1, \dots, n\}$,

$$\mathbb{P}\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} \mathbb{P}(A_i).$$

Remark 9.2.1

Pairwise independence is not the same as mutual independence. It is possible for every pair of events to be independent while the full collection is not mutually independent.

9.2.3 Bayes' Theorem

Theorem 9.2.1: Law of Total Probability

Let $B_1, \dots, B_n$ be pairwise disjoint events whose union is Ω , and suppose $\mathbb{P}(B_i) > 0$. Then for any event A ,

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \mid B_i) \mathbb{P}(B_i).$$

Proof. Since Ω is the disjoint union of the B_i , the event A is the disjoint union of the events $A \cap B_i$. Hence

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \cap B_i) = \sum_{i=1}^n \mathbb{P}(A \mid B_i) \mathbb{P}(B_i).$$

□

Theorem 9.2.2: Bayes' Theorem

Under the assumptions of the law of total probability,

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j) \mathbb{P}(B_j)}{\sum_{i=1}^n \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)},$$

provided $\mathbb{P}(A) > 0$.

Proof. By the definition of conditional probability,

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \cap B_j)}{\mathbb{P}(A)}.$$

The numerator is $\mathbb{P}(A \mid B_j) \mathbb{P}(B_j)$, and the denominator is computed by the law of total probability. □

Remark 9.2.2

Bayes' theorem converts a forward probability $\mathbb{P}(A \mid B)$ into an inverse probability $\mathbb{P}(B \mid A)$. This is the mathematical foundation of Bayesian inference.

9.3 Discrete Random Variables

A random variable is not a variable that is random in isolation; it is a function that assigns a numerical value to each outcome. This functional viewpoint is important because it separates the underlying randomness from the numerical quantity we choose to observe. The same experiment may produce many random variables: from a dice roll we may record the face value, its parity, or whether it exceeds four.

Distributions allow us to forget irrelevant details of the sample space and focus on the probabilities of numerical values. Two different experiments can produce random variables with the same distribution. This is why probability theory often studies distributions directly: the distribution contains all information needed to compute probabilities and expectations involving that random variable alone.

9.3.1 Definition

Definition 9.3.1 (Discrete Random Variable). A **discrete random variable** is a function $X : \Omega \rightarrow \mathbb{R}$ whose range is finite or countable.

The event $\{X = x\}$ means

$$\{\omega \in \Omega : X(\omega) = x\}.$$

Definition 9.3.2 (Probability Mass Function). The **probability mass function** (PMF) of a discrete random variable X is

$$p_X(x) = \mathbb{P}(X = x).$$

It satisfies

$$p_X(x) \geq 0, \quad \sum_x p_X(x) = 1.$$

Definition 9.3.3 (Cumulative Distribution Function). The **cumulative distribution function** (CDF) of a random variable X is

$$F_X(x) = \mathbb{P}(X \leq x).$$

9.3.2 Expectation and Variance in the Discrete Case

Expectation is best understood as a weighted average, not as the value one necessarily expects to see in a single trial. A die roll has expectation 3.5, even though no face labeled 3.5 exists. The expectation describes long-run average behavior under repeated independent trials and gives a linear summary of the distribution.

Variance measures spread around the expectation. The use of squared deviation has two advantages: it prevents positive and negative deviations from canceling, and it interacts well with algebra. Standard deviation returns the scale to the original units, while variance is often easier to manipulate in proofs.

Definition 9.3.4 (Expectation). If X is a discrete random variable, its **expectation** is

$$\mathbb{E}[X] = \sum_x x \mathbb{P}(X = x),$$

provided the series converges absolutely.

Definition 9.3.5 (Variance). If $\mathbb{E}[X^2] < \infty$, the **variance** of X is

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Proposition 9.3.1: Computational Formula for Variance

If $\mathbb{E}[X^2] < \infty$, then

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Proof. Expand the square:

$$\mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu \mathbb{E}[X] + \mu^2,$$

where $\mu = \mathbb{E}[X]$. Thus the expression equals $\mathbb{E}[X^2] - \mu^2$. □

Proposition 9.3.2: Linearity of Expectation

For random variables X and Y with finite expectations, and constants $a, b \in \mathbb{R}$,

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

Remark 9.3.1

Linearity of expectation does not require independence. This is one of the most useful facts in elementary probability.

9.3.3 Common Discrete Distributions

The common discrete distributions are templates for recurring mechanisms. The Bernoulli distribution models one yes-or-no trial. The binomial distribution counts successes in a fixed number of independent Bernoulli trials. The geometric distribution records waiting time until the first success. The Poisson distribution often appears as an approximation for rare events occurring in a large number of opportunities.

Remembering the mechanism is more reliable than memorizing formulas alone. Once the story is clear, the probability mass function usually follows from independence, counting, or limiting approximation. This also helps decide which distribution is appropriate in an applied problem.

Definition 9.3.6 (Bernoulli Distribution). A random variable X has a **Bernoulli distribution** with parameter p , written $X \sim \text{Bernoulli}(p)$, if

$$\mathbb{P}(X = 1) = p, \quad \mathbb{P}(X = 0) = 1 - p.$$

Then

$$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p).$$

Definition 9.3.7 (Binomial Distribution). A random variable X has a **binomial distribution** with parameters n and p , written $X \sim \text{Bin}(n, p)$, if

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

Then

$$\mathbb{E}[X] = np, \quad \text{Var}(X) = np(1 - p).$$

Example 9.3.1

If a biased coin lands heads with probability p and is tossed n times independently, then the number of heads has distribution $\text{Bin}(n, p)$.

Definition 9.3.8 (Geometric Distribution). A random variable X has a **geometric distribution** with parameter p , written $X \sim \text{Geom}(p)$, if

$$\mathbb{P}(X = k) = (1 - p)^{k-1} p, \quad k = 1, 2, 3, \dots$$

It models the trial number of the first success in independent Bernoulli trials.

Definition 9.3.9 (Poisson Distribution). A random variable X has a **Poisson distribution** with parameter $\lambda > 0$, written $X \sim \text{Poisson}(\lambda)$, if

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Then

$$\mathbb{E}[X] = \lambda, \quad \text{Var}(X) = \lambda.$$

Remark 9.3.2

The Poisson distribution often appears as an approximation to rare-event counts. If n is large, p is small, and $np \approx \lambda$, then $\text{Bin}(n, p)$ is close to $\text{Poisson}(\lambda)$.

9.4 Continuous Random Variables

For continuous random variables, probabilities are assigned to intervals rather than individual points. A density function is not itself a probability; it is probability per unit length. The probability of an interval is obtained by integrating the density over that interval. This is why a density may exceed 1 at some points without contradiction, provided its total integral is 1.

The statement $P(X = a) = 0$ for a continuous random variable does not mean the value a is impossible in an informal physical sense. It means that single points carry no probability mass. Probability accumulates over sets with positive length, area, or more generally positive measure.

9.4.1 Density Functions

Classical finite probability cannot describe choosing a random point from an interval. If we choose uniformly from $[0, 1]$, the probability of any single point should be 0, yet the total probability of the interval should be 1. This requires a new language.

Definition 9.4.1 (Probability Density Function). A random variable X is said to have a **density** f_X if

$$\mathbb{P}(X \in A) = \int_A f_X(x) dx$$

for suitable subsets $A \subseteq \mathbb{R}$. The function f_X satisfies

$$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1.$$

Definition 9.4.2 (CDF for Continuous Variables). If X has density f_X , then

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt.$$

At points where F_X is differentiable,

$$F'_X(x) = f_X(x).$$

Definition 9.4.3 (Expectation). If X has density f_X , then

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx,$$

provided the integral of $|x|f_X(x)$ is finite.

9.4.2 Common Continuous Distributions

The standard continuous distributions also encode common mechanisms. The uniform distribution represents constant density over an interval. The exponential distribution models memoryless waiting time. The normal distribution arises as the universal limit shape for sums of many small independent effects. The gamma and beta distributions extend these ideas to waiting-time and proportion models.

In applications, choosing a distribution is a modeling decision. One asks which qualitative features are required: bounded support, symmetry, skewness, memorylessness, heavy tails, or additivity. The formula is then interpreted as a compact expression of those modeling assumptions.

Definition 9.4.4 (Uniform Distribution). A random variable X has a **uniform distribution** on $[a, b]$, written $X \sim U[a, b]$, if

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\mathbb{E}[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}.$$

Definition 9.4.5 (Exponential Distribution). A random variable X has an **exponential distribution** with rate $\lambda > 0$, written $X \sim \text{Exp}(\lambda)$, if

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Then

$$\mathbb{E}[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

Definition 9.4.6 (Normal Distribution). A random variable X has a **normal distribution** with mean μ and variance σ^2 , written $X \sim N(\mu, \sigma^2)$, if

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The special case $N(0, 1)$ is called the **standard normal distribution**.

Remark 9.4.1

The normal distribution is central because sums of many small independent random effects often become approximately normal. This phenomenon is made precise by the Central Limit Theorem.

9.5 The Measure-Theoretic Foundation

The measure-theoretic foundation is introduced because probability needs a precise language for events in infinite spaces. In a finite sample space, every subset can be assigned a probability. In uncountable spaces, assigning probabilities consistently to all subsets is impossible in general. Sigma-algebras specify which subsets are legitimate events.

A probability measure then assigns numbers to those events while respecting countable additivity. Countable additivity is the bridge between finite intuition and limiting processes. It allows probabilities of increasing unions, decreasing intersections, and infinite decompositions to be handled rigorously.

9.5.1 Why Classical Probability Is Not Enough

The classical definition works for finite equally likely sample spaces. But many important probability spaces are infinite:

1. choosing a random real number in $[0, 1]$;
2. observing the lifetime of a machine;
3. modeling the trajectory of a Brownian particle;
4. considering infinitely many coin tosses.

In such cases, not every subset of the sample space can be assigned a probability consistently. We therefore specify a collection of admissible events.

9.5.2 Sigma-Algebras

Definition 9.5.1 (Sigma-Algebra). Let Ω be a set. A collection $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ is a **sigma-algebra** if:

1. $\Omega \in \mathcal{F}$;
2. if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;
3. if $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

Elements of $\mathcal{F}$ are called **measurable sets** or **events**.

Proposition 9.5.1

If $\mathcal{F}$ is a sigma-algebra, then it is closed under countable intersections.

Proof. By De Morgan's law,

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c.$$

Since each A_n^c is in $\mathcal{F}$, the union is in $\mathcal{F}$, and its complement is also in $\mathcal{F}$. $\square$

Definition 9.5.2 (Borel Sigma-Algebra). The **Borel sigma-algebra** on $\mathbb{R}$, denoted $\mathcal{B}(\mathbb{R})$, is the smallest sigma-algebra containing all open subsets of $\mathbb{R}$.

9.5.3 Probability Spaces

Definition 9.5.3 (Probability Measure). Let $(\Omega, \mathcal{F})$ be a measurable space. A function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a **probability measure** if:

1. $\mathbb{P}(\Omega) = 1$;
2. if $A_1, A_2, \dots$ are pairwise disjoint events, then

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

Definition 9.5.4 (Probability Space). A **probability space** is a triple

$$(\Omega, \mathcal{F}, \mathbb{P}),$$

where Ω is a sample space, $\mathcal{F}$ is a sigma-algebra of events, and $\mathbb{P}$ is a probability measure.

Proposition 9.5.2: Basic Properties

For events $A, B \in \mathcal{F}$:

1. $\mathbb{P}(\emptyset) = 0$;
2. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;
3. if $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$;
4. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Proof. These follow from countable additivity. For example, $\Omega = A \cup A^c$ is a disjoint union, so $1 = \mathbb{P}(A) + \mathbb{P}(A^c)$. If $A \subseteq B$, then $B = A \cup (B \setminus A)$ disjointly, so $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$. $\square$

Proposition 9.5.3: Continuity of Probability

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

1. If $A_1 \subseteq A_2 \subseteq \dots$, then

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n).$$

2. If $A_1 \supseteq A_2 \supseteq \dots$, then

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n).$$

9.6 Random Variables as Measurable Functions

The definition of a random variable as a measurable function ensures that events of the form $\{X \in B\}$ are actually measurable events in the original probability space. Without this condition, the expression $P(X \in B)$ might not be well-defined. Thus measurability is the technical requirement that makes probability statements about X legitimate.

From this perspective, a distribution is the pushforward of the original probability measure through X . Instead of measuring subsets of the original outcome space directly, we measure their images on the real line by pulling them back through X . This is the conceptual link between abstract probability spaces and familiar probability distributions.

9.6.1 Measurability

In modern probability, a random variable is not fundamentally a variable. It is a measurable function from the sample space to a numerical space.

Definition 9.6.1 (Random Variable). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A **random variable** is a measurable function

$$X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R})),$$

meaning that for every Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}.$$

Remark 9.6.1

Measurability ensures that events such as $\{X \leq x\}$, $\{a < X < b\}$, and $\{X \in B\}$ have probabilities.

Definition 9.6.2 (Distribution). The **distribution** or **law** of a random variable X is the probability measure μ_X on $\mathbb{R}$ defined by

$$\mu_X(B) = \mathbb{P}(X \in B).$$

Definition 9.6.3 (Random Vector). A **random vector** is a measurable function

$$X : \Omega \rightarrow \mathbb{R}^n.$$

Equivalently, $X = (X_1, \dots, X_n)$ where each component X_i is a random variable.

9.6.2 Expectation as Integral

Definition 9.6.4 (Expectation). Let X be an integrable random variable. Its **expectation** is the Lebesgue integral

$$\mathbb{E}[X] = \int_{\Omega} X \, d\mathbb{P}.$$

The expectation exists as a finite number if

$$\mathbb{E}[|X|] < \infty.$$

Theorem 9.6.1: Law of the Unconscious Statistician

Let X be a random variable and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be measurable. If $g(X)$ is integrable, then

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) d\mu_X(x).$$

If X has density f_X , then

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

If X is discrete, then

$$\mathbb{E}[g(X)] = \sum_x g(x) \mathbb{P}(X = x).$$

9.7 Joint Distributions and Independence

Joint distributions describe several random variables at once. They contain not only the marginal behavior of each variable but also the dependence structure between them. Knowing the marginal distributions of X and Y separately is usually not enough to determine probabilities involving both variables, such as $P(X < Y)$ or $P(X + Y \leq 1)$.

Independence of random variables is stronger than pairwise intuition about individual events. It requires that events determined by one variable do not affect events determined by the other. In the discrete case this appears as factorization of joint probabilities; in the continuous case it appears as factorization of joint densities, when densities exist.

9.7.1 Joint Distributions

Definition 9.7.1 (Joint Distribution). For random variables X and Y , the **joint distribution** of (X, Y) is the probability measure on $\mathbb{R}^2$ defined by

$$\mu_{(X,Y)}(A) = \mathbb{P}((X, Y) \in A).$$

Definition 9.7.2 (Joint Density). If there exists a function $f_{X,Y}$ such that

$$\mathbb{P}((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy,$$

then $f_{X,Y}$ is called the **joint density** of X and Y .

Definition 9.7.3 (Marginal Density). If (X, Y) has joint density $f_{X,Y}$, then the marginal densities are

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx.$$

9.7.2 Independence of Random Variables

Definition 9.7.4 (Independence of Random Variables). Random variables X and Y are **independent** if for all Borel sets $A, B \subseteq \mathbb{R}$,

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A) \mathbb{P}(Y \in B).$$

Proposition 9.7.1: Factorization Criterion

If (X, Y) has joint density $f_{X,Y}$, then X and Y are independent if and only if

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

for almost every (x, y) .

Proposition 9.7.2

If X and Y are independent and integrable, then

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

Remark 9.7.1

The converse is false. The equality of expectations of products does not generally imply independence.

9.7.3 Covariance and Correlation

Definition 9.7.5 (Covariance). If X and Y have finite second moments, their **covariance** is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

Definition 9.7.6 (Correlation). If $\text{Var}(X) > 0$ and $\text{Var}(Y) > 0$, the **correlation coefficient** is

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$$

Proposition 9.7.3

$$|\rho_{X,Y}| \leq 1.$$

Proof. This follows from the Cauchy-Schwarz inequality applied to $X - \mathbb{E}[X]$ and $Y - \mathbb{E}[Y]$. $\square$

9.8 Conditional Expectation

Conditional expectation is the projection-like operation of probability theory. When conditioning on partial information, we replace a random variable by the best prediction that is measurable with respect to that information. The result is no longer necessarily a number; it may be another random variable depending on the information retained.

This viewpoint is essential in stochastic processes and martingales. As information grows over time, conditional expectations describe how predictions evolve. Many advanced probability theorems are built around the idea that fair games are processes whose future conditional expectation equals their present value.

9.8.1 Conditioning on Events

For an event B with $\mathbb{P}(B) > 0$, conditional expectation is defined by

$$\mathbb{E}[X \mid B] = \frac{1}{\mathbb{P}(B)} \int_B X \, d\mathbb{P}.$$

This is the average value of X restricted to the event B .

9.8.2 Conditioning on Random Variables

If X and Y have joint density, then a familiar formula is

$$f_{Y|X=x}(y) = \frac{f_{X,Y}(x, y)}{f_X(x)},$$

when $f_X(x) > 0$. Then

$$\mathbb{E}[Y | X = x] = \int_{-\infty}^{\infty} y f_{Y|X=x}(y) dy.$$

However, this formula is not the most general definition. It depends on densities and on conditioning on a point, which may have probability zero. Measure theory gives a more robust definition.

Definition 9.8.1 (Conditional Expectation Given a Sigma-Algebra). Let X be integrable and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-sigma-algebra. The **conditional expectation** of X given $\mathcal{G}$ is a random variable $\mathbb{E}[X | \mathcal{G}]$ such that:

1. $\mathbb{E}[X | \mathcal{G}]$ is $\mathcal{G}$ -measurable;
2. for every $G \in \mathcal{G}$,

$$\int_G \mathbb{E}[X | \mathcal{G}] d\mathbb{P} = \int_G X d\mathbb{P}.$$

It is unique up to equality almost surely.

Definition 9.8.2 (Conditional Expectation Given a Random Variable). We define

$$\mathbb{E}[Y | X] = \mathbb{E}[Y | \sigma(X)],$$

where $\sigma(X)$ is the sigma-algebra generated by X .

Theorem 9.8.1: Tower Property

If $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$ and X is integrable, then

$$\mathbb{E}[\mathbb{E}[X | \mathcal{G}] | \mathcal{H}] = \mathbb{E}[X | \mathcal{H}].$$

In particular,

$$\mathbb{E}[\mathbb{E}[X | \mathcal{G}]] = \mathbb{E}[X].$$

Remark 9.8.1

Conditional expectation is best understood as the best prediction of X using only the information contained in $\mathcal{G}$. The measurability condition says the prediction only uses that information; the integral condition says it is correct on average over every event visible to $\mathcal{G}$.

9.9 Inequalities in Probability

Probability inequalities are tools for converting moment information into tail bounds. Often we cannot compute an exact probability, but we can bound it using expectation, variance, or higher moments. Markov's inequality uses only non-negativity and expectation; Chebyshev's inequality uses variance; sharper inequalities use stronger structural assumptions.

These inequalities explain why moments matter. The expectation controls average size, while variance controls typical deviation from the mean. Even when a distribution is unknown, moment bounds can still guarantee that large deviations are rare in a quantitative sense.

Inequalities are essential because probability theory often studies estimates rather than exact values.

Theorem 9.9.1: Markov's Inequality

If $X \geq 0$ and $a > 0$, then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof. Since $X \geq a\mathbf{1}_{\{X \geq a\}}$,

$$\mathbb{E}[X] \geq a\mathbb{P}(X \geq a).$$

Dividing by a gives the result. $\square$

Theorem 9.9.2: Chebyshev's Inequality

If X has finite mean μ and variance σ^2 , then for every $\varepsilon > 0$,

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Theorem 9.9.3: Jensen's Inequality

If φ is convex and X is integrable, then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)],$$

provided the right-hand side exists.

Theorem 9.9.4: Cauchy-Schwarz Inequality

If $X, Y \in L^2$, then

$$|\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]}.$$

9.10 Generating Functions and Characteristic Functions

Generating functions convert probability questions into algebraic questions about functions. Instead of working directly with a sequence of probabilities or moments, we package that sequence into a single function. Operations on random variables then often become operations on generating functions.

Characteristic functions are especially important because they always exist and uniquely determine distributions. They turn sums of independent random variables into products, which is the same structural simplification provided by logarithms in multiplication problems. This is why characteristic functions are central in proofs of limit theorems.

9.10.1 Probability Generating Functions

Definition 9.10.1 (Probability Generating Function). If X is a non-negative integer-valued random variable, its **probability generating function** is

$$G_X(s) = \mathbb{E}[s^X] = \sum_{k=0}^{\infty} \mathbb{P}(X = k)s^k.$$

Proposition 9.10.1

If X and Y are independent non-negative integer-valued random variables, then

$$G_{X+Y}(s) = G_X(s)G_Y(s).$$

9.10.2 Moment Generating Functions

Definition 9.10.2 (Moment Generating Function). The **moment generating function** of X is

$$M_X(t) = \mathbb{E}[e^{tX}],$$

when this expectation is finite for t in a neighborhood of 0.

Remark 9.10.1

If M_X exists near 0, then differentiating gives moments:

$$M_X^{(n)}(0) = \mathbb{E}[X^n].$$

9.10.3 Characteristic Functions

Definition 9.10.3 (Characteristic Function). The **characteristic function** of a random variable X is

$$\varphi_X(t) = \mathbb{E}[e^{itX}], \quad t \in \mathbb{R}.$$

Unlike moment generating functions, characteristic functions always exist because $|e^{itX}| = 1$.

Proposition 9.10.2: Basic Properties

For every random variable X :

1. $\varphi_X(0) = 1$;
2. $|\varphi_X(t)| \leq 1$;
3. if X and Y are independent, then

$$\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t).$$

Theorem 9.10.1: Uniqueness Theorem

If $\varphi_X(t) = \varphi_Y(t)$ for every $t \in \mathbb{R}$, then X and Y have the same distribution.

9.11 Modes of Convergence

Different modes of convergence express different levels of probabilistic stability. Almost sure convergence follows individual outcomes except on a null set. Convergence in probability allows exceptional outcomes to occur but requires their probability to vanish. Convergence in distribution is weaker still: it only requires the distribution functions to converge at continuity points.

The hierarchy matters because limit theorems often naturally produce one mode rather than another. The law of large numbers is usually a statement about convergence in probability or almost surely. The central limit theorem is a statement about convergence in distribution after centering and scaling. Confusing these modes can lead to incorrect conclusions about sample paths or numerical approximations.

Probability theory has several notions of convergence. They are not interchangeable, because random variables carry both numerical and probabilistic structure.

Definition 9.11.1 (Almost Sure Convergence). A sequence X_n converges to X **almost surely**, written

$$X_n \xrightarrow{a.s.} X,$$

if

$$\mathbb{P}(\{\omega : X_n(\omega) \rightarrow X(\omega)\}) = 1.$$

Definition 9.11.2 (Convergence in Probability). A sequence X_n converges to X **in probability**, written

$$X_n \xrightarrow{\mathbb{P}} X,$$

if for every $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0.$$

Definition 9.11.3 (L^p Convergence). For $p \geq 1$, X_n converges to X in L^p , written

$$X_n \xrightarrow{L^p} X,$$

if

$$\mathbb{E}[|X_n - X|^p] \rightarrow 0.$$

Definition 9.11.4 (Convergence in Distribution). A sequence X_n converges to X **in distribution**, written

$$X_n \xrightarrow{d} X,$$

if

$$F_{X_n}(x) \rightarrow F_X(x)$$

for every continuity point x of F_X .

Theorem 9.11.1: Relations Between Convergence Modes

$$X_n \xrightarrow{a.s.} X \implies X_n \xrightarrow{\mathbb{P}} X \implies X_n \xrightarrow{d} X.$$

Also,

$$X_n \xrightarrow{L^p} X \implies X_n \xrightarrow{\mathbb{P}} X.$$

Proof of L^p Implies Probability. By Markov's inequality,

$$\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(|X_n - X|^p > \varepsilon^p) \leq \frac{\mathbb{E}[|X_n - X|^p]}{\varepsilon^p}.$$

If the numerator tends to 0, then the probability tends to 0. □

Remark 9.11.1

Convergence in distribution is the weakest among the common modes above. It describes convergence of laws, not necessarily convergence of random variables on the same sample space.

9.12 Limit Theorems

The law of large numbers formalizes the stabilization of averages. Under suitable assumptions, repeated independent observations average out randomness and converge toward the expected value. This theorem explains why empirical frequencies and sample means are meaningful estimators in statistics.

The central limit theorem describes the shape of fluctuations around that average. After subtracting the mean and scaling by the square root of the sample size, sums of many independent contributions often approach a normal distribution. The theorem is powerful because the limiting normal shape appears under broad conditions, even when the original variables are not normally distributed.

9.12.1 Law of Large Numbers

The Law of Large Numbers explains why averages stabilize. It is the mathematical justification for interpreting probability as long-run frequency.

Theorem 9.12.1: Weak Law of Large Numbers

Let $X_1, X_2, \dots$ be i.i.d. random variables with finite mean μ and finite variance σ^2 . Let

$$\bar{X}_n = \frac{1}{n} \sum_{k=1}^n X_k.$$

Then

$$\bar{X}_n \xrightarrow{\mathbb{P}} \mu.$$

Proof. By linearity of expectation,

$$\mathbb{E}[\bar{X}_n] = \mu.$$

By independence,

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{k=1}^n \text{Var}(X_k) = \frac{\sigma^2}{n}.$$

Chebyshev's inequality gives

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

□

Theorem 9.12.2: Kolmogorov Convergence Criterion

Let $Z_1, Z_2, \dots$ be independent random variables with $\mathbb{E}[Z_n] = 0$ and

$$\sum_{n=1}^{\infty} \text{Var}(Z_n) < \infty.$$

Then the series $\sum_{n=1}^{\infty} Z_n$ converges almost surely.

Proof. Set

$$r_m := \sum_{j=m+1}^{\infty} \text{Var}(Z_j),$$

so that $r_m \rightarrow 0$. Choose an increasing sequence of integers (n_k) such that $r_{n_k} < 2^{-3k}$. For every $N > n_k$, Kolmogorov's maximal inequality gives

$$\mathbb{P} \left(\max_{n_k < n \leq N} \left| \sum_{j=n_k+1}^n Z_j \right| > 2^{-k} \right) \leq 2^{2k} \sum_{j=n_k+1}^N \text{Var}(Z_j) \leq 2^{2k} r_{n_k} < 2^{-k}.$$

Letting $N \rightarrow \infty$ and using continuity of probability for increasing events, we obtain

$$\mathbb{P} \left(\sup_{n > n_k} \left| \sum_{j=n_k+1}^n Z_j \right| > 2^{-k} \right) < 2^{-k}.$$

These probabilities are summable. Hence the Borel–Cantelli lemma implies that, almost surely, for all sufficiently large k ,

$$\sup_{n > n_k} \left| \sum_{j=n_k+1}^n Z_j \right| \leq 2^{-k}.$$

Consequently, if $p, q > n_k$, then

$$\left| \sum_{j=1}^p Z_j - \sum_{j=1}^q Z_j \right| \leq 2^{1-k}.$$

Thus the sequence of partial sums is almost surely Cauchy and therefore converges almost surely. $\square$

Theorem 9.12.3: Kronecker's Lemma

If the numerical series

$$\sum_{n=1}^{\infty} \frac{a_n}{n}$$

converges, then

$$\frac{1}{n} \sum_{k=1}^n a_k \longrightarrow 0.$$

Proof. Let $s_n = \sum_{k=1}^n a_k/k$. Summation by parts gives

$$\frac{1}{n} \sum_{k=1}^n a_k = s_n - \frac{1}{n} \sum_{k=1}^{n-1} s_k.$$

If $s_n \rightarrow s$, then its Cesaro means also converge to s , so the difference tends to 0. $\square$

Theorem 9.12.4: Strong Law of Large Numbers

Let $X_1, X_2, \dots$ be i.i.d. random variables with $\mathbb{E}[|X_1|] < \infty$ and mean μ . Then

$$\bar{X}_n = \frac{1}{n} \sum_{k=1}^n X_k \xrightarrow{a.s.} \mu.$$

Proof. Define the truncated variables

$$Y_n = X_n \mathbf{1}_{\{|X_n| \leq n\}}.$$

The tail-sum formula implies

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n| > n) = \sum_{n=1}^{\infty} \mathbb{P}(|X_1| > n) \leq \mathbb{E}[|X_1|] < \infty.$$

By the first Borel–Cantelli lemma, $X_n = Y_n$ for all sufficiently large n , almost surely.

Next, the variables Y_n are independent and

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{\text{Var}(Y_n)}{n^2} &\leq \sum_{n=1}^{\infty} \frac{\mathbb{E}[X_1^2 \mathbf{1}_{\{|X_1| \leq n\}}]}{n^2} \\ &= \mathbb{E} \left[X_1^2 \sum_{n \geq |X_1|} \frac{1}{n^2} \right] \leq C \mathbb{E}[|X_1|] < \infty \end{aligned}$$

for an absolute constant C . Apply the Kolmogorov convergence criterion to

$$Z_n = \frac{Y_n - \mathbb{E}[Y_n]}{n}.$$

The series $\sum Z_n$ converges almost surely, so Kronecker's lemma yields

$$\frac{1}{n} \sum_{k=1}^n (Y_k - \mathbb{E}[Y_k]) \longrightarrow 0 \quad \text{almost surely.}$$

Because the X_n are identically distributed,

$$\mathbb{E}[Y_n] = \mathbb{E}[X_1 \mathbf{1}_{\{|X_1| \leq n\}}].$$

The integrand converges pointwise to X_1 and is dominated in absolute value by the integrable random variable $|X_1|$. Therefore dominated convergence gives

$$\mathbb{E}[Y_n] \rightarrow \mathbb{E}[X_1] = \mu.$$

Hence Cesaro averaging gives

$$\frac{1}{n} \sum_{k=1}^n \mathbb{E}[Y_k] \rightarrow \mu.$$

Finally, the averages of X_k and Y_k differ only in finitely many terms almost surely, so their difference tends to 0. Combining the three limits proves the theorem. $\square$

Remark 9.12.1

The weak law controls one probability for each n . The strong law controls an entire infinite sample path: outside one null set, the averages converge for all sufficiently large n . The truncation step is what allows the theorem to require only a finite first moment rather than a finite variance.

9.12.2 Central Limit Theorem

The Central Limit Theorem explains why the normal distribution appears so frequently.

Theorem 9.12.5: Levy's Continuity Theorem

If characteristic functions $\varphi_n(t)$ converge pointwise to a function $\varphi(t)$ that is continuous at 0 and is itself a characteristic function, then the corresponding distributions converge weakly to the distribution with characteristic function φ .

Theorem 9.12.6: Lindeberg–Levy Central Limit Theorem

Let $X_1, X_2, \dots$ be i.i.d. random variables with mean μ and variance $\sigma^2 > 0$. Let $S_n = X_1 + \dots + X_n$. Then

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0, 1).$$

Proof. Set

$$Y_k = \frac{X_k - \mu}{\sigma}.$$

Then $\mathbb{E}[Y_k] = 0$ and $\mathbb{E}[Y_k^2] = 1$. The elementary expansion

$$e^{iu} = 1 + iu - \frac{u^2}{2} + u^2\eta(u), \quad \eta(u) \rightarrow 0 \quad (u \rightarrow 0),$$

together with dominated convergence gives

$$\begin{aligned}\varphi_Y(t) &= \mathbb{E}[e^{itY}] \\ &= 1 + it\mathbb{E}[Y] - \frac{t^2}{2}\mathbb{E}[Y^2] + o(t^2) \\ &= 1 - \frac{t^2}{2} + o(t^2).\end{aligned}$$

Indeed, after dividing the remainder by t^2 , it is bounded by a constant multiple of Y^2 , which is integrable.

Now define

$$Z_n = \frac{1}{\sqrt{n}} \sum_{k=1}^n Y_k.$$

Independence gives

$$\varphi_{Z_n}(t) = \prod_{k=1}^n \varphi_Y\left(\frac{t}{\sqrt{n}}\right) = \left(\varphi_Y\left(\frac{t}{\sqrt{n}}\right)\right)^n.$$

Using the expansion at the scale $t/\sqrt{n}$,

$$\varphi_Y\left(\frac{t}{\sqrt{n}}\right) = 1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right).$$

Therefore

$$\log \varphi_{Z_n}(t) = n \log \left(1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right) \rightarrow -\frac{t^2}{2},$$

and hence

$$\varphi_{Z_n}(t) \rightarrow e^{-t^2/2}.$$

The limit is the characteristic function of $N(0, 1)$ and is continuous at 0. Levy's continuity theorem completes the proof. $\square$

Definition 9.12.1 (Lindeberg Condition). For each n , let $X_{n,1}, \dots, X_{n,k_n}$ be independent random variables with zero means. Put

$$s_n^2 = \sum_{k=1}^{k_n} \text{Var}(X_{n,k}).$$

The triangular array satisfies the **Lindeberg condition** if for every $\varepsilon > 0$,

$$\frac{1}{s_n^2} \sum_{k=1}^{k_n} \mathbb{E} \left[X_{n,k}^2 \mathbf{1}_{\{|X_{n,k}| > \varepsilon s_n\}} \right] \rightarrow 0.$$

Theorem 9.12.7: Lindeberg–Feller Central Limit Theorem

If the triangular array above satisfies the Lindeberg condition and $s_n^2 > 0$, then

$$\frac{1}{s_n} \sum_{k=1}^{k_n} X_{n,k} \xrightarrow{d} N(0, 1).$$

Remark 9.12.2

The Lindeberg condition says that no small collection of exceptionally large summands contributes a non-negligible fraction of the total variance. The i.i.d. central limit theorem is recovered by taking

$$X_{n,k} = \frac{X_k - \mu}{\sigma\sqrt{n}}, \quad 1 \leq k \leq n.$$

The Law of Large Numbers describes first-order behavior: averages approach the mean. The Central Limit Theorem describes second-order fluctuations: deviations of order $\sqrt{n}$ approach a universal Gaussian shape.

9.13 Borel-Cantelli Lemmas

The Borel-Cantelli lemmas answer a repeated-event question: if events $A_1, A_2, \dots$ occur over time, what can be said about the probability that infinitely many of them occur? The sum $\sum P(A_n)$ measures the total expected frequency of occurrence. If this sum is finite, infinitely many occurrences are impossible with probability one.

With independence, the converse direction becomes available: if the total probability mass diverges, then infinitely many of the independent events occur almost surely. These lemmas are powerful because they transform an infinite qualitative statement into a summability condition.

The Borel-Cantelli lemmas describe whether events happen infinitely often.

Definition 9.13.1 (Limit Superior of Events). For events $A_1, A_2, \dots$, define

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k.$$

This is the event that infinitely many of the A_n occur.

Theorem 9.13.1: First Borel-Cantelli Lemma

If

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty,$$

then

$$\mathbb{P}(\limsup A_n) = 0.$$

Proof. For every N ,

$$\limsup A_n \subseteq \bigcup_{n=N}^{\infty} A_n.$$

Hence, by countable subadditivity,

$$\mathbb{P}(\limsup A_n) \leq \sum_{n=N}^{\infty} \mathbb{P}(A_n).$$

Letting $N \rightarrow \infty$, the tail of the convergent series tends to 0. □

Theorem 9.13.2: Second Borel-Cantelli Lemma

If the events A_n are independent and

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty,$$

then

$$\mathbb{P}(\limsup A_n) = 1.$$

9.14 Introduction to Stochastic Processes

A stochastic process is a family of random variables indexed by time or another parameter. Instead of modeling one uncertain quantity, we model an evolving random system. Examples include random walks, stock prices, queues, branching populations, Brownian motion, and Markov chains.

The main new feature is dependence across time. One must describe not only the distribution at each time but also how values at different times are related. This is why filtrations, transition probabilities, covariance functions, and sample paths become central objects in the study of stochastic processes.

A stochastic process is a family of random variables indexed by time or space.

Definition 9.14.1 (Stochastic Process). A **stochastic process** is a collection

$$\{X_t : t \in T\}$$

of random variables defined on the same probability space. The index set T is often interpreted as time.

Example 9.14.1

If $T = \mathbb{N}$, the process is a discrete-time process. If $T = [0, \infty)$, it is a continuous-time process.

Definition 9.14.2 (Filtration). A **filtration** is an increasing family of sigma-algebras

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots.$$

The sigma-algebra $\mathcal{F}_n$ represents the information available up to time n .

Definition 9.14.3 (Martingale). A discrete-time stochastic process $\{X_n\}_{n \geq 0}$ is a **martingale** with respect to a filtration $\{\mathcal{F}_n\}$ if:

1. X_n is $\mathcal{F}_n$ -measurable;
2. $\mathbb{E}[|X_n|] < \infty$;
3. $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$.

Remark 9.14.1

A martingale formalizes the idea of a fair game: given all current information, the expected future value equals the present value.

Example 9.14.2: Simple Random Walk

Let $\xi_1, \xi_2, \dots$ be independent random variables with

$$\mathbb{P}(\xi_i = 1) = \mathbb{P}(\xi_i = -1) = \frac{1}{2}.$$

Define

$$S_n = \xi_1 + \dots + \xi_n.$$

Then $\{S_n\}$ is a martingale with respect to the natural filtration generated by $\xi_1, \dots, \xi_n$.

9.15 Distribution Functions and Transformations

A probability distribution can be described in several equivalent ways. Discrete laws are often specified by probability masses, continuous laws by densities, and general real-valued laws by cumulative distribution functions. The distribution function is the most universal of these descriptions because it exists even when a law is neither purely discrete nor purely continuous.

Definition 9.15.1 (Cumulative Distribution Function). For a real-valued random variable X , the **cumulative distribution function** is

$$F_X(x) = \mathbb{P}(X \leq x).$$

Theorem 9.15.1: Characterization of Distribution Functions

A function $F : \mathbb{R} \rightarrow [0, 1]$ is the distribution function of some real-valued random variable if and only if:

1. F is non-decreasing;
2. F is right-continuous;
3. $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.

Moreover,

$$\mathbb{P}(X = x) = F_X(x) - F_X(x^-),$$

so jumps of the distribution function are exactly point masses.

Example 9.15.1: A Mixed Distribution

Suppose $X = 0$ with probability $1/2$, while with probability $1/2$ it is uniformly distributed on $[0, 1]$. Then

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{2} + \frac{x}{2}, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

The jump at 0 records the atom, while the linear part records the continuous component. No single ordinary density describes the entire law.

9.15.1 Functions of One Random Variable

Theorem 9.15.2: Monotone Change of Variables

Let X have density f_X , and let $Y = g(X)$, where g is strictly monotone and continuously differentiable. Then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

on the range of g .

Proof. If g is increasing, then

$$F_Y(y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y)).$$

Differentiate and apply the chain rule. The decreasing case reverses the inequality and produces the same absolute-value formula. $\square$

Example 9.15.2: Squaring a Standard Normal Variable

Let $X \sim N(0, 1)$ and set $Y = X^2$. The map $x \mapsto x^2$ is not one-to-one, so both inverse branches contribute:

$$f_Y(y) = \frac{f_X(\sqrt{y})}{2\sqrt{y}} + \frac{f_X(-\sqrt{y})}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi y}} e^{-y/2}, \quad y > 0.$$

Thus X^2 has a chi-square distribution with one degree of freedom.

Theorem 9.15.3: Expectation by the Tail Formula

If $X \geq 0$, then

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > t) dt,$$

where both sides may be infinite.

Proof. Pointwise,

$$X(\omega) = \int_0^\infty \mathbf{1}_{\{X(\omega) > t\}} dt.$$

Tonelli's theorem allows the order of integration to be exchanged:

$$\mathbb{E}[X] = \int_0^\infty \mathbb{E}[\mathbf{1}_{\{X > t\}}] dt = \int_0^\infty \mathbb{P}(X > t) dt.$$

$\square$

9.16 Joint Densities, Convolution, and Conditioning

For several random variables, marginal distributions describe each coordinate separately, but they do not determine dependence. The joint distribution contains the complete information. Marginalization integrates out unwanted coordinates, conditioning renormalizes along a coordinate, and convolution describes sums of independent variables.

Definition 9.16.1 (Joint Density). A pair (X, Y) has joint density $f_{X,Y}$ if for every suitable Borel set $A \subseteq \mathbb{R}^2$,

$$\mathbb{P}((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy.$$

The marginal densities are

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx.$$

Theorem 9.16.1: Density Criterion for Independence

If (X, Y) has a joint density, then X and Y are independent if and only if

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

for almost every (x, y) .

Definition 9.16.2 (Conditional Density). If $f_Y(y) > 0$, define

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}.$$

For fixed y , this is a density in x and represents the distribution of X conditioned on $Y = y$.

Example 9.16.1: Uniform Distribution on a Triangle

Let (X, Y) be uniformly distributed on

$$T = \{(x, y) : 0 < y < x < 1\}.$$

Since the area of T is $1/2$, the joint density is 2 on T . For $0 < x < 1$,

$$f_X(x) = \int_0^x 2 dy = 2x.$$

For fixed x , the conditional density of Y given $X = x$ is

$$f_{Y|X}(y | x) = \frac{2}{2x} = \frac{1}{x}, \quad 0 < y < x.$$

Thus, conditioned on $X = x$, the variable Y is uniform on $(0, x)$.

Theorem 9.16.2: Convolution Formula

If X and Y are independent continuous random variables with densities f_X and f_Y , then $S = X + Y$ has density

$$f_S(s) = \int_{-\infty}^{\infty} f_X(x)f_Y(s-x) dx.$$

Proof. The change of variables $(x, y) \mapsto (x, s = x + y)$ has Jacobian determinant 1. Since independence gives joint density $f_X(x)f_Y(y)$, integrating out x yields the stated formula. $\square$

Example 9.16.2: Sum of Two Uniform Variables

Let X, Y be independent and uniform on $[0, 1]$. Then

$$f_{X+Y}(s) = \int_{\max(0, s-1)}^{\min(1, s)} 1 \, dx = \begin{cases} s, & 0 < s < 1, \\ 2 - s, & 1 \leq s < 2, \\ 0, & \text{otherwise.} \end{cases}$$

The triangular density reflects the increasing and then decreasing length of the line segment $x + y = s$ inside the unit square.

9.17 Convergence Theorems for Expectations

Limits of random variables are useful only if they can be passed through probabilities or expectations. Measure theory supplies precise hypotheses for doing so. Monotone convergence handles increasing non-negative approximations; dominated convergence handles signed variables controlled by one integrable bound.

Theorem 9.17.1: Monotone Convergence Theorem

If $0 \leq X_1 \leq X_2 \leq \dots$ and $X_n \rightarrow X$ almost surely, then

$$\mathbb{E}[X_n] \uparrow \mathbb{E}[X].$$

Theorem 9.17.2: Fatou's Lemma

If $X_n \geq 0$, then

$$\mathbb{E} \left[\liminf_{n \rightarrow \infty} X_n \right] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[X_n].$$

Theorem 9.17.3: Dominated Convergence Theorem

Suppose $X_n \rightarrow X$ almost surely and there exists an integrable random variable Y such that

$$|X_n| \leq Y$$

almost surely for every n . Then X is integrable and

$$\mathbb{E}[X_n] \rightarrow \mathbb{E}[X].$$

Example 9.17.1: Why Pointwise Convergence Is Not Enough

On $(0, 1)$ with Lebesgue probability measure, let

$$X_n = n \mathbf{1}_{(0, 1/n)}.$$

Then $X_n \rightarrow 0$ almost surely, but

$$\mathbb{E}[X_n] = 1$$

for every n . There is no single integrable random variable dominating all X_n . This example shows why an interchange of limit and expectation requires more than pointwise convergence.

Corollary 9.17.1: Continuity of Characteristic Functions

For every random variable X , its characteristic function

$$\varphi_X(t) = \mathbb{E}[e^{itX}]$$

is continuous in t .

Proof. If $t_n \rightarrow t$, then $e^{it_n X} \rightarrow e^{itX}$ almost surely, while

$$|e^{it_n X}| = 1.$$

Apply dominated convergence with the integrable bound 1. □

9.18 Conditional Expectation as Information

Conditioning on an event produces a number. Conditioning on a sigma-algebra produces a random variable, because the information available may lead to different predictions in different states. The conditional expectation $\mathbb{E}[X | \mathcal{G}]$ is the best $\mathcal{G}$ -measurable summary of X in an average sense.

Definition 9.18.1 (Conditional Expectation). Let X be integrable and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-sigma-algebra. A random variable Y is a version of $\mathbb{E}[X | \mathcal{G}]$ if:

1. Y is $\mathcal{G}$ -measurable;
2. Y is integrable;
3. for every $G \in \mathcal{G}$,

$$\int_G Y \, d\mathbb{P} = \int_G X \, d\mathbb{P}.$$

It is unique up to equality almost surely.

Theorem 9.18.1: Basic Properties

For integrable random variables X, Y and constants a, b :

1. **Linearity:**

$$\mathbb{E}[aX + bY | \mathcal{G}] = a\mathbb{E}[X | \mathcal{G}] + b\mathbb{E}[Y | \mathcal{G}].$$

2. **Taking out known factors:** if Z is bounded and $\mathcal{G}$ -measurable, then

$$\mathbb{E}[ZX | \mathcal{G}] = Z\mathbb{E}[X | \mathcal{G}].$$

3. **Tower property:** if $\mathcal{H} \subseteq \mathcal{G}$, then

$$\mathbb{E}[\mathbb{E}[X | \mathcal{G}] | \mathcal{H}] = \mathbb{E}[X | \mathcal{H}].$$

4. **Total expectation:**

$$\mathbb{E}[\mathbb{E}[X | \mathcal{G}]] = \mathbb{E}[X].$$

Example 9.18.1: Conditioning on a Finite Partition

Suppose $\mathcal{G}$ is generated by a partition $A_1, \dots, A_n$ with positive probabilities. Then

$$\mathbb{E}[X \mid \mathcal{G}] = \sum_{k=1}^n \frac{\mathbb{E}[X \mathbf{1}_{A_k}]}{\mathbb{P}(A_k)} \mathbf{1}_{A_k}.$$

The conditional expectation is constant on each information cell A_k , because $\mathcal{G}$ cannot distinguish outcomes within the same cell.

Theorem 9.18.2: Least-Squares Characterization

If $X \in L^2$ and $Y = \mathbb{E}[X \mid \mathcal{G}]$, then for every square-integrable $\mathcal{G}$ -measurable random variable Z ,

$$\mathbb{E}[(X - Y)^2] \leq \mathbb{E}[(X - Z)^2].$$

Proof. The defining property of conditional expectation implies

$$\mathbb{E}[(X - Y)(Y - Z)] = 0.$$

Therefore

$$\mathbb{E}[(X - Z)^2] = \mathbb{E}[(X - Y)^2] + \mathbb{E}[(Y - Z)^2],$$

which proves the claim. $\square$

Remark 9.18.1

This is the probabilistic analogue of orthogonal projection in an inner-product space. The sigma-algebra $\mathcal{G}$ determines which random variables are observable, and conditional expectation projects X onto that information.

9.19 Martingales and Stopping Times

A martingale models a process with zero conditional drift. The definition does not say that the process is constant or that its sample paths are stable. It says only that, given the present information, the expected next value equals the current value. This distinction is essential: a fair game may fluctuate dramatically even though no predictable profit is available.

Example 9.19.1: Centered Partial Sums

Let $X_1, X_2, \dots$ be independent integrable random variables with common mean μ , and define

$$M_n = \sum_{k=1}^n (X_k - \mu).$$

With the natural filtration $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$,

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n + \mathbb{E}[X_{n+1} - \mu] = M_n.$$

Hence (M_n) is a martingale.

Example 9.19.2: A Transformed Random Walk

For the simple symmetric random walk S_n , the process

$$M_n = S_n^2 - n$$

is a martingale. Indeed,

$$S_{n+1}^2 = (S_n + \xi_{n+1})^2 = S_n^2 + 2S_n\xi_{n+1} + 1,$$

and conditioning on the past gives

$$\mathbb{E}[S_{n+1}^2 - (n+1) \mid \mathcal{F}_n] = S_n^2 - n.$$

Definition 9.19.1 (Stopping Time). A random time $\tau : \Omega \rightarrow \{0, 1, 2, \dots, \infty\}$ is a **stopping time** with respect to $(\mathcal{F}_n)$ if

$$\{\tau \leq n\} \in \mathcal{F}_n$$

for every n . Thus, using information available at time n , one can determine whether stopping has already occurred.

Theorem 9.19.1: Optional Stopping for Bounded Times

Let (M_n) be a martingale and let τ be a stopping time bounded by a deterministic integer N . Then

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0].$$

Proof. Write

$$M_\tau = M_0 + \sum_{k=1}^N (M_k - M_{k-1}) \mathbf{1}_{\{\tau \geq k\}}.$$

The event $\{\tau \geq k\}$ belongs to $\mathcal{F}_{k-1}$, so

$$\mathbb{E}[(M_k - M_{k-1}) \mathbf{1}_{\{\tau \geq k\}}] = \mathbb{E}[\mathbf{1}_{\{\tau \geq k\}} \mathbb{E}[M_k - M_{k-1} \mid \mathcal{F}_{k-1}]] = 0.$$

Summing proves the theorem. $\square$

Note

Optional stopping is not valid for every unbounded stopping time. Additional hypotheses such as bounded increments, integrability, or uniform integrability are needed. The informal slogan “a fair game remains fair when one chooses when to stop” is correct only when the stopping rule satisfies appropriate conditions.

9.20 Markov Chains

A Markov chain is a stochastic process whose future depends on the past only through the present state. This is not the same as independence: successive states are usually dependent. The Markov property says that the current state contains all information from the past that is relevant for predicting the next state.

Definition 9.20.1 (Time-Homogeneous Markov Chain). A discrete-time process $(X_n)_{n \geq 0}$ on a countable state space S is a **time-homogeneous Markov chain** if

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = p_{ij}$$

whenever the conditional probability is defined. The matrix

$$P = (p_{ij})_{i,j \in S}$$

is the **transition matrix**.

Proposition 9.20.1: Chapman–Kolmogorov Equations

Let $p_{ij}^{(n)} = \mathbb{P}(X_n = j \mid X_0 = i)$. Then

$$p_{ij}^{(m+n)} = \sum_{k \in S} p_{ik}^{(m)} p_{kj}^{(n)}.$$

In matrix form,

$$P^{(m+n)} = P^{(m)} P^{(n)},$$

so for a time-homogeneous chain $P^{(n)} = P^n$.

Definition 9.20.2 (Stationary Distribution). A probability vector π on S is **stationary** if

$$\pi P = \pi.$$

If X_0 has distribution π , then every X_n has distribution π .

Example 9.20.1: Two-State Chain

Let

$$P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}, \quad a, b \in (0, 1).$$

Solving $\pi P = \pi$ with $\pi_0 + \pi_1 = 1$ gives

$$\pi = \left(\frac{b}{a+b}, \frac{a}{a+b} \right).$$

The larger the transition rate into a state relative to the rate out, the larger its long-run stationary weight.

Theorem 9.20.1: Finite Irreducible Aperiodic Chains

A finite Markov chain that is irreducible and aperiodic has a unique stationary distribution π , and for every initial state i ,

$$p_{ij}^{(n)} \rightarrow \pi_j$$

as $n \rightarrow \infty$.

Remark 9.20.1

Irreducibility means every state can eventually reach every other state. Aperiodicity excludes a forced cyclic rhythm. Together they ensure that the chain forgets its initial state and converges to equilibrium.

9.21 Poisson Processes and Brownian Motion

Two continuous-time processes appear throughout probability and applications. The Poisson process models randomly occurring events with a constant average rate. Brownian motion models continuous random fluctuation. Their sample paths look completely different: Poisson paths jump by integers, while Brownian paths are continuous but almost surely nowhere differentiable.

9.21.1 The Poisson Process

Definition 9.21.1 (Poisson Process). A process $(N_t)_{t \geq 0}$ is a **Poisson process with rate $\lambda > 0$** if:

1. $N_0 = 0$;
2. it has independent increments;
3. it has stationary increments;
4. for $s, t \geq 0$,

$$N_{t+s} - N_s \sim \text{Poisson}(\lambda t).$$

Proposition 9.21.1

For a Poisson process,

$$\mathbb{E}[N_t] = \lambda t, \quad \text{Var}(N_t) = \lambda t.$$

Moreover, the waiting time T_1 to the first event is exponential with parameter λ :

$$\mathbb{P}(T_1 > t) = \mathbb{P}(N_t = 0) = e^{-\lambda t}.$$

Example 9.21.1

If calls arrive according to a Poisson process at rate 3 per hour, then the probability of exactly five calls in two hours is

$$\mathbb{P}(N_2 = 5) = e^{-6} \frac{6^5}{5!}.$$

The probability of waiting more than twenty minutes for the next call is

$$e^{-3(1/3)} = e^{-1}.$$

9.21.2 Brownian Motion

Definition 9.21.2 (Standard Brownian Motion). A process $(B_t)_{t \geq 0}$ is a **standard Brownian motion** if:

1. $B_0 = 0$ almost surely;
2. it has independent increments;
3. $B_t - B_s \sim N(0, t - s)$ for $0 \leq s < t$;
4. its sample paths are continuous almost surely.

Proposition 9.21.2

For Brownian motion,

$$\mathbb{E}[B_t] = 0, \quad \text{Var}(B_t) = t, \quad \text{Cov}(B_s, B_t) = \min(s, t).$$

Proof. Assume $s \leq t$. Since $B_t = B_s + (B_t - B_s)$ and the increment is independent of B_s with mean zero,

$$\mathbb{E}[B_s B_t] = \mathbb{E}[B_s^2] + \mathbb{E}[B_s] \mathbb{E}[B_t - B_s] = s.$$

Thus the covariance is $s = \min(s, t)$. □

Example 9.21.2: A Brownian Martingale

The process (B_t) is a martingale with respect to its natural filtration. Another important martingale is

$$B_t^2 - t.$$

The subtraction of t compensates for the variance accumulated over time, just as $S_n^2 - n$ does for the simple random walk.

Remark 9.21.1

The scaling property

$$(B_{ct})_{t \geq 0} \stackrel{d}{=} (\sqrt{c}B_t)_{t \geq 0}$$

shows that time scales quadratically relative to space. This same scaling appears in the heat equation and explains the close relation between Brownian motion and diffusion.

9.22 Conclusion

Probability theory begins with simple counting, but its rigorous foundation is measure theory. Events are measurable sets, probabilities are measures, and random variables are measurable functions. This viewpoint unifies discrete and continuous probability and makes it possible to study infinite sequences, conditioning, convergence, and stochastic processes.

The major conceptual shift is from asking only “how many outcomes are favorable?” to asking “what measurable structure carries uncertainty, and how does probability behave on that structure?” Once this foundation is established, expectation becomes integration, independence becomes factorization of measures, conditioning becomes projection onto information, and the Law of Large Numbers and Central Limit Theorem explain why random systems exhibit stable macroscopic behavior.

9.23 Chapter Workshop

Worked example: Bayesian classification

A test has sensitivity 0.95 and specificity 0.90. Suppose the prevalence of the condition is 0.01. If D is the condition and $+$ is a positive result, then

$$\mathbb{P}(D \mid +) = \frac{\mathbb{P}(+ \mid D)\mathbb{P}(D)}{\mathbb{P}(+ \mid D)\mathbb{P}(D) + \mathbb{P}(+ \mid D^c)\mathbb{P}(D^c)}.$$

Substitution gives

$$\mathbb{P}(D \mid +) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.10 \cdot 0.99} \approx 0.0876.$$

A highly sensitive test can still have a modest positive predictive value when the base rate is small.

Application: Monte Carlo integration

If $U_1, \dots, U_n$ are independent uniform samples on $[0, 1]$, then

$$\hat{I}_n = \frac{1}{n} \sum_{k=1}^n f(U_k)$$

estimates $\int_0^1 f(x) dx$. The law of large numbers gives consistency, while the central limit theorem gives an approximate error scale

$$\frac{\sqrt{\text{Var}(f(U))}}{\sqrt{n}}.$$

The method extends to high-dimensional integrals where deterministic grids become prohibitively expensive.

Application: Markov models and ranking

A finite Markov chain evolves a probability vector by multiplication with a stochastic matrix. Stationary distributions describe long-run occupancy and underlie models of queues, population movement, and ranking on directed networks. Linear algebra determines invariant distributions; probability determines how sample paths approach them.

9.24 Exercises

- Two fair dice are rolled. Compute the conditional probability that the sum is 8 given that at least one die is 3.
- Let $X \sim \text{Binomial}(n, p)$. Derive $\mathbb{E}X$ and $\text{Var}(X)$ using indicator variables.
- If X has density $f(x) = 2x$ on $[0, 1]$, find its distribution function, mean, and variance.
- Give an example showing that pairwise independence need not imply mutual independence.
- Prove Markov's inequality and derive Chebyshev's inequality from it.
- Explain the difference between convergence in probability and almost-sure convergence.
- A Markov chain has transition matrix $P = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}$. Find its stationary distribution.
- For a Poisson process of rate λ , derive the distribution of the waiting time to the first arrival.

Selected hints and answers

- The conditioning event has 11 ordered outcomes; two of them, $(3, 5)$ and $(5, 3)$, have sum 8, so the probability is $2/11$.
- Write $X = I_1 + \dots + I_n$ with independent Bernoulli indicators.
- $F(x) = x^2$ on $[0, 1]$; $\mathbb{E}X = 2/3$ and $\text{Var}(X) = 1/18$.
- Let two fair bits A, B be independent and set $C = A \oplus B$.

5. Apply Markov to the nonnegative random variable $(X - \mu)^2$.
6. Almost-sure convergence controls entire sample paths outside one null set; convergence in probability controls the probability of a deviation separately for each n .
7. Solve $\pi P = \pi$ with $\pi_1 + \pi_2 = 1$; the answer is $(3/5, 2/5)$.

Afterword

A mathematical book usually ends at a point where mathematics itself does not.

The final theorem may have been proved, the last example may have been examined, and the notation may have reached a temporary resting place. Yet none of these provides a genuine conclusion. Every definition introduced in these pages admits further abstraction. Every theorem belongs to a larger theory. Every proof suggests questions about which assumptions are essential, which arguments can be generalized, and which structures remain undiscovered.

For this reason, the end of these notes should not be understood as the completion of a mathematical journey. It marks only the point at which one particular route has been recorded.

This manuscript began as an attempt to organize what I had learned. Like many students encountering university mathematics, I first met its subjects as separate courses. Analysis concerned limits and continuity. Linear algebra concerned vectors and matrices. Abstract algebra concerned groups, rings, and fields. Topology concerned open sets and continuous maps. Probability concerned random variables and distributions. Each subject appeared to possess its own terminology, techniques, and standards of argument.

Only gradually did their common structure become visible.

The same questions returned in different forms. What are the objects under consideration? Which relations between them matter? What transformations preserve the relevant structure? What assumptions are actually required for a conclusion? When can a local property be extended globally? When does an algebraic description correspond to a geometric one? When does convergence preserve the operations we wish to perform?

Once these questions are recognized, mathematics no longer appears as a collection of independent territories. It becomes a network of recurring ideas expressed through different languages.

Completeness appears first as a property of number systems and metric spaces, but later governs the existence of limits, fixed points, and solutions. Compactness begins as a topological condition, yet repeatedly converts local information into global control. Symmetry is encoded algebraically through groups, geometrically through transformations, and analytically through invariance. Linearity, initially encountered in systems of equations, becomes a principle for decomposing complicated objects into manageable components. Measure extends elementary notions of length and area, while probability gives those same structures a new interpretation.

To study mathematics is therefore not merely to collect results. It is to learn to recognize these patterns of thought as they reappear.

Writing these notes made that lesson more demanding. It is possible to believe that one understands a theorem while reading it. It is much harder to explain why its hypotheses are natural, reconstruct its proof without hidden steps, and place it within the surrounding theory. Exposition exposes the difference between familiarity and understanding. A sentence that seems harmless may conceal an undefined term. An “obvious” implication may require a lemma. A standard proof may depend on a convention established many pages earlier.

In this sense, writing became part of learning rather than a record produced after learning was complete.

The imperfections of this manuscript belong to the same process. Its subjects are unequal in scale, and no single volume can treat them with equal depth. Some chapters lead close to the frontier of a theory; others establish only the language required to begin. Certain arguments have been included because they illuminate a general method, while other important results have been omitted because their proper treatment would require a separate book. The selection reflects not a final map of mathematics, but the path by which one student attempted to see its landscape more clearly.

That path will inevitably change.

Definitions that once appeared artificial may later seem unavoidable. Proofs that once seemed difficult may become examples of a familiar technique. Conversely, statements that appear settled may reveal deeper questions when viewed from another subject. Mathematical maturity does not eliminate difficulty; it changes the level at which difficulty is encountered.

A beginning student asks how to prove a theorem. A more experienced student also asks why this theorem was formulated, how its assumptions were discovered, whether its conclusion is optimal, and what larger structure makes the proof possible. Progress is measured not only by the number of problems one can solve, but by the quality of the questions one learns to ask.

There is also a particular discipline in accepting incompleteness.

One may be tempted to postpone writing until every detail is mastered, or to delay sharing a set of notes until every chapter has reached the same degree of polish. But complete mastery is not a state that mathematics readily permits. Each answer enlarges the boundary of what remains unknown. Waiting for finality would mean never beginning.

The more honest goal is not perfection, but continued correction: to state assumptions more clearly, replace vague intuition with precise argument, discover counterexamples to careless claims, and revise explanations when a better structure becomes visible. Rigor is not achieved once and preserved automatically. It is a habit of returning to what has been written and asking whether it truly says what was intended.

The reader should approach these pages in the same spirit.

Do not treat the proofs as monuments that can only be observed from a distance. Take them apart. Reconstruct them. Search for shorter arguments and stronger conclusions. Ask what happens when an assumption is removed. Draw examples that fit a definition and counterexamples that nearly fit it. Compare different formulations of the same idea. A proof becomes part of one's mathematical understanding only after it can be recreated, adapted, and questioned.

Nor should these notes be read as an endpoint. Their natural continuation lies in the books, lectures, problems, and conversations from which mathematics is actually learned. Real analysis leads toward functional analysis, partial differential equations, and dynamical systems. Algebra leads toward representation theory, commutative algebra, and algebraic geometry. Topology leads toward manifolds, homotopy, and geometry. Measure and probability lead toward stochastic processes, statistics, and mathematical physics. The boundaries between these directions are themselves increasingly artificial.

The structure described in the preface as a mathematical edifice is therefore not a finished skyscraper. It is closer to a city under permanent construction. Some foundations are ancient; some towers are new. Different districts develop according to different needs, yet bridges continually appear between them. Old structures are strengthened, extended, or rebuilt when new perspectives demand it.

To enter mathematics is not simply to admire this city from above. It is to learn how to work within it.

One begins with small tasks: understanding a definition, checking an example, completing a proof. Over time, these acts develop into a way of thinking. Precision becomes less a formal obligation than a form of intellectual honesty. Abstraction becomes less an escape from concrete reality than a method for discovering what many concrete situations share. Proof becomes not merely a certificate of truth, but an explanation of why a statement could not have been otherwise under the given assumptions.

This is perhaps the deepest lesson I have taken from the process of writing these notes. Mathematics is not

defined solely by the certainty of its conclusions. It is defined by the care with which it moves from one idea to the next.

The manuscript ends here, but its purpose extends beyond its final page. It will have succeeded if it encourages the reader to continue independently: to question established arguments, to seek connections between distant subjects, and eventually to construct explanations that are clearer than those found here.

A preface looks forward to a path that has not yet been travelled. An afterword looks back and discovers that the path has changed the traveller.

What lies ahead is larger than what has been recorded.

Jinshuo Li
Shanghai Jiao Tong University